

**ABET
Self-Study Report**

For the

**Chemical Engineering
Program**

At

The University of Illinois at Chicago

Chicago, IL 60607



July 1, 2020

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**Program Self-Study Report
for
EAC of ABET
Accreditation or Reaccreditation**

BACKGROUND INFORMATION

A. Contact Information

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B. Program History

The College of Engineering was one of the original Colleges when University of Illinois at Chicago was founded in 1965 (originally named University of Illinois at Chicago Circle – renamed in 1982 after a merger with the University of Illinois at the Medical Center). Chemical Engineering started as a program within the Department of Energy Engineering. The other programs in the Department were Thermo-Mechanical Engineering and Fluids Engineering. In 1982 the College of Engineering was reorganized, and Chemical Engineering became one of its six constituent Departments. In 1986 the Department moved to 810 S Clinton Street (Chemical Engineering Building, CEB), with offices, laboratories and classrooms located conveniently in the same building. In 2001 the department began offering Chemical Engineering as a minor to students not enrolled in the Chemical Engineering undergraduate program. In 2003 the Department initiated an area of concentration in Biochemical Engineering. In 2019, the Department moved to the newly constructed Engineering Innovation Building, EIB at 945 W Taylor Street. The Chemical Engineering Department is the main tenant with all offices, laboratories, and classrooms located in EIB. The Civil and Materials Engineering and Mechanical and Industrial Engineering Departments also have some faculty offices and laboratories in the building.

The last general review took place October 5-7, 2014. Since then there has been some turnover in the faculty, but also an increase in the total number of faculty and affiliate faculty. Two senior faculty members (including the previous department head) retired and three faculty members moved to either another academic institution or to industry. Commensurately, and keeping in line

with increased student enrollment, 8 new faculty were hired: three associate professors, one who is now the head of the department (and professor), three assistant professors, two clinical associate professors, and one research assistant professor. Two adjunct professors teaching the senior design and laboratory courses were reappointed as clinical assistant professors. Another adjunct professor teaching two technical elective courses (Renewable Energy Technologies, Entrepreneurship in Chemical Engineering) was reappointed as a visiting research professor. The Department also has four Professor Emeriti, and 12 affiliate faculty from other departments in the University, including the UIC College of Medicine, and from industry. The UIC Chancellor, appointed in 2015, is also a senior chemical engineering professor in the Department.

Significant enhancements have been made to the curriculum. Three technical elective courses were added; one on the emerging role of nanostructures in drug development and delivery (CHE 425), the second on renewable energy technologies (CHE 451), and the third on the fundamentals of electrochemistry (CHE 453). A selective course on polymer science (CHE 330) has recently been added and will be taught for the first time in Fall 2020. Three more courses have recently been approved to the curriculum but have not yet been taught; a polymer science course (CHE 454 Molecular and Macromolecular Engineering), a nanotechnology course (CHE 455 Nanoscale Systems in Chemical Engineering), and an interfacial phenomena courses (CHE 457: Colloidal and Interfacial Phenomena). They will be part of a future improvement to the curriculum that is adding five new concentrations to the BS in Chemical Engineering degree; a concentration in Polymers and Molecular Engineering, Energy and Environment, Nanotechnology, Process Automation, and Entrepreneurship.

As part of the College of Engineering effort to improve the freshmen/transfer student Engineering Orientation course (ENGR100), the chemical engineering department added a four-week, major-specific, team-oriented, technical project to the curriculum. The technical project is also a requirement of the Guaranteed Paid Internship Program (GPIP) for freshmen students, a College initiative providing freshmen with an early research or industry internship experience. In addition, the senior-level professional development course (CHE 499) was improved with a seminar series of invited professional chemical engineers discussing careers in a wide variety of industries and non-traditional areas (patent law, for instance), as well as job searching tutorials from the Engineering Career Center. The Senior Design II (ChE 397) project experience is continually being improved by a team of industrial mentors who meet regularly with their student groups and suggest relevant and topical design projects each year. Ethics and safety learning has also been strengthened in the curriculum. In addition, the students have a more diverse set of research topics to choose from when taking the undergraduate research course (CHE 392). A CHE 392 course section has also been dedicated for the AIChE ChemE Car competition to allow students to obtain technical elective credit while performing research and developing the ChemE Car for the annual competition.

There has also been a focus on improving career development skills and learning activities. An annual, 1-day conference called “ChE Industry Day” was initiated with student input and provides round-table, informal discussions with practicing chemical engineers, an engineering expert panel discussion, and a networking event for the students to learn more about chemical engineering careers. In addition, the annual “ChemE Forum” has been added to the career development curriculum and consists of a one-day conference with tutorials on EIT/PE exam preparation, safety

tutorials from safety professionals, an alumni panel discussion, and a student/alumni networking event (alumni mixer).

C. Options

BS in Chemical Engineering — Biochemical Engineering Concentration

This concentration entails the completion of 8–10 semester hours distributed among CHE 422—Biochemical engineering (3 cr. hr.) as the technical elective and two elective courses in the non-major rubric category chosen from among BIOS 350—General Microbiology (3 cr. hr.), BIOS 351—Microbiology Laboratory (2 cr. hr.), CHEM 352—Introductory Biochemistry (3 cr. hr.) and CHEM 452—Biochemistry I (4 cr. hr.). (iii) Free elective (if needed) to make up 1 credit hour. Due to structure of the concentration and the prerequisites required for some of the courses, students in the concentration will be required to take a minimum of 130 semester hours for the degree.

Minor in Chemical Engineering (for non-ChE majors)

Students outside the Department of Chemical Engineering who wish to minor in Chemical Engineering must complete a specified sequence of classes totaling 16-18 semester hours (excluding prerequisite courses). This sequence includes three courses in Chemical Engineering:

- CHE 210—Material and Energy Balances (4 cr. hr.),
- CHE 301—Chemical Engineering Thermodynamics (3. cr. hr.),
- CHE 321—Chemical Reaction Engineering (3 cr. hr.),

One fluid mechanics course chosen from among

- CHE 311—Transport Phenomena I (3 cr. hr.)
- ME 211—Fluid Mechanics I (4 cr. hr.)

And one heat or mass transfer course chosen from among

- CHE 312—Transport Phenomena II (3 cr. ht.)
- ME 321—Heat Transfer (4 cr. hr.)
- CHE 313—Transport Phenomena III (3 cr. hr.)

Minors appear on a student's transcript, but are not listed on the UIC diploma. Among the small number of engineering minors, none have yet been minors in Chemical Engineering since 2004.

D. Program Delivery Modes

The Bachelor of Science in Chemical Engineering is offered to interested students in conventional day classes (lecture / laboratory / recitation). The program can be completed by full time students in four years by taking 15-18 credits per semester.

Program Locations

The UIC Chemical Engineering program is offered only on campus.

E. Public Disclosure

The Chemical Engineering Department's Program Educational Objectives (PEOs), Student Outcomes (SOs), annual student enrollment, and graduation data is available online to the public at the following URL.

<https://che.uic.edu/undergraduate-programs/che-major/>

F. Deficiencies, Weaknesses or Concerns from Previous Evaluation(s) and the Actions Taken to Address Them

For the 2014 accreditation cycle the ABET Final Statement (dated August 10, 2015) contained no Deficiencies, Weaknesses, or Concerns.

GENERAL CRITERIA

CRITERION 1. STUDENTS

The evaluation of Chemical Engineering students takes three forms:

1. Evaluation at the time of admission to the BSChE degree
2. Evaluation of progress towards the BSChE degree
3. Evaluation for completion of the BSChE degree

These evaluation processes are discussed in individual sections below for both incoming freshman and transfer students.

A. Student Admissions

Students are admitted directly into the Chemical Engineering Program in the College of Engineering as: a new freshman; an internal transfer student from another UIC college; or a transfer student from an institution other than UIC. The general evaluation procedures for new freshman and internal transfer students are described below. Evaluation of transfer students from outside UIC is discussed later in this section.

Freshman students

The two primary factors used to determine the admissibility of a new freshman applying to the University of Illinois at Chicago are the high school percentile rank (HSPR) and the American College Test (ACT) composite score. The University will also accept Scholarship Aptitude Test (SAT) scores. SAT scores are converted to an ACT composite equivalent. Should an applicant submit more than one set of entrance exam scores, then the Office of Admissions will use the highest score when processing an application.

Before making the admission decision, a student's high school course work is checked to meet the specified subject pattern. All applicants must have successfully completed the following high school courses in order to be eligible for admission to UIC.

- A. Four years of English
- B. Three years of Mathematics (must include algebra, geometry, advanced algebra/trigonometry)
- C. Three years of Science
- D. Three years of Social Science
- E. Two years of Language (other than English)
- F. One year of an elective.

The College of Engineering provides the Office of Admissions the guidelines to be used in admitting freshmen to the College. Freshman applications are evaluated by a College of Engineering Admission Committee consisting of the Director of Admission and Records, the Associate Dean for Undergraduate Affairs, and the Director of the Minority Engineering Recruiting and Retention Program. Admission is supported by high school grades, particularly in STEM courses, high composite ACT scores, and high scores on the mathematics subscale of the ACT. Table 1.1 shows the history of College of Engineering admissions standards for freshmen admissions for the past five years and indicates a fairly constant value of incoming student quality as measured by the composite ACT score.

All entering freshmen are admitted with the condition that they must take campus placement exams in math, chemistry and English composition. The results of these exams are critical in determining what courses students take during their first semester at UIC. If the results indicate the student needs a preparatory course in math, chemistry or English composition, then the student must register and pass the course.

Table 1.1 College of Engineering Admissions Standards for Freshmen Admissions

Academic Year	Composite ACT	Composite SAT	Percentile Rank in High School	Number of New Students Enrolled
	AVG.	AVG.	AVG.	
2014	27.35		84.14	340
2015	27.39		83.05	400
2016	27.41		86.49	392
2017	26.3		85.36	603
2018	27.21	1279	88.85	495
2019	26.96	1272	85.83	673

¹ Admission based on unweighted High School GPA and not Percentile Rank in High School

B. Evaluating Student Performance

Student progress is monitored both by the Office of the Associate Dean for Undergraduate Affairs, and by the Director of Undergraduate Studies (DUGS) of the Department of Chemical Engineering. The DUGS consults with the student's faculty advisor to discuss issues that may arise during such monitoring. During the ninth week of each semester each student must schedule an appointment with his/her adviser during the tenth week to plan an academic program for the following semester (see details on advising procedures in Section D), but especially to evaluate the student's progress during the past and current semester. During the advising sessions, students discuss with advisors their academic program and progress and may be advised to take remedial classes if needed, or rearrange their schedule to change their credit hour load to their advantage, or in extreme circumstances move to a program of study more suited to the student's academic strengths including changing their program of study. In the next two subsections, the processes for evaluating students' progress and probation procedures are described. These include departmental college and university policies.

Students are evaluated at the end of every semester during the College of Engineering Grade Review process. At that time, the Associate Dean for Undergraduate Administration (ADUA) reviews the file of each student who has either a term Grade Point Average (GPA) or cumulative UIC GPA below 2.00 (A=4.00).

In addition to monitoring the Grade Point Average of students, the College also uses the Deficit Point system to determine probation and drop decisions. The Deficit Point system has the advantage over the GPA system in that it clearly indicates the future academic performance that a student must achieve to return to clear status.

The following scale is on a per semester hour basis:

Letter grade	Grade points per credit hour	Deficit points
A	4	+2
B	3	+1
C	2 (2.0 minimum GPA at graduation)	0
D	1	-1
F	0	-2

If a student takes one 3-hour class and receives a grade of “D,” then this student would have a GPA of 1.00 and –3 deficit points. If another student takes four 3-hour classes and receives all D’s, then this student would also have a GPA of 1.00, but would have –12 deficit points. Although they both have the same GPA, the second student with –12 deficit points is having more serious academic difficulty. A student with negative deficit points must earn a positive number of deficit points in the future to bring the total back up to zero. Thus, a student with –12 deficit points must earn a combination of A’s and B’s for the number of hours necessary to achieve +12 deficit points to return to clear status.

Students may repeat a course to increase their knowledge of the subject matter. There are circumstances under which repeating a course is advisable and to a student’s advantage. There are also circumstances where repeating a course may disadvantage a student and narrow a student’s options. Some colleges require students to discuss any plan to repeat a course with their academic advisor before they register to repeat the course.

Courses with A or B grades may not be repeated. Normally, courses with a C grade may not be repeated. Courses with D or F grades may be repeated once without written permission. In all cases, the original grade for the course and the grade for each repeat will appear on the transcript. Only one registration for the course counts toward the total number of credits required for graduation. A course cannot be repeated after receiving credit in a course for which the repeat course is a prerequisite.

To repeat a course more than once requires written permission from the student’s college dean. Students who have been dismissed may not appeal on the grounds of intention to repeat courses. Certain courses may not be repeated; students should consult their college before repeating a course.

Undergraduate Grade Point Average Recalculation Policy

- Colleges may elect to implement this policy as early as the end of the junior year, or as late as the final semester during degree certification.
- The college will evaluate each student unable to graduate because of an academic grade point average deficit. The deficit may occur in the UIC GPA, the program GPA, or both.
- The academic record is examined so as to determine if courses, deemed appropriate by each college, with grades of F or grades of D, may be excluded from the student's grade point average.
- A total of four courses may be excluded, with one course excluded at a time until the necessary quality points are obtained to meet the UIC GPA requirement and/or degree program minimum GPA. The credit hours associated with excluded course grades will not count towards graduation requirements.
- Only the college can initiate course exclusions, and only in consultation with the evaluated student.
- Students may not request exclusions.
- Only courses taken at UIC may be converted as part of this policy.
- This policy only applies to undergraduate degree seeking students.
- All courses taken and all grades will remain on the official transcript.
- The grade exclusion policy does not apply to courses failed as the result of a student conduct hearing.
- In cases where these procedures would impact program accreditation, licensure or similar, colleges may use their discretion to allow the policy. Colleges, departments, and

Probation Rules

Rule 1. A less serious probation level is called Probation Level 1 or 2.00 Pro. This probation is for any student whose Term GPA is below 2.00, but whose Cumulative GPA is above 2.00. In the next semester, the student is expected to earn no grade less than C to continue.

Rule 2. Any student whose UIC Cumulative GPA falls below 2.00 is placed on Probation Level 2 or 2.25 Pro. In the next semester, the student is expected to earn no grade less than a C and at least one B in order to continue. A student is not required to return to clear status in one semester. For example, if a student finishes the first semester with –10 deficit points, then finishes the second semester with 3 hours of B and 3 hours of C, this student will have reduced the total deficit points to –7. Although the student is still on probation, the student satisfied the probation conditions and is allowed to continue. At this rate of +3 deficit points per term, it will

take the student 4 semesters to return to clear status, and the student is making progress towards that goal.

Rule 3. Graduation with a degree from the College of Engineering requires a minimum GPA of 2.00 in the major courses. Major courses are those required in the specific degree program as listed in the UIC Undergraduate Catalog. Rules 1 and 2 described above are also applied to the major courses.

Drop Rules

Rule 1. Any student who was on probation and did not satisfy the conditions of that probation may be dropped from the College of Engineering B.S. degree program.

Rule 2. A student who fails to make satisfactory progress toward a degree in the College of Engineering may be dropped. Examples of unsatisfactory progress are: -12 deficit points, excessive number of incomplete grades, failure to take courses required for the degree.

Rule 3. This rule applies to those students who had previously been dropped, were readmitted, and failed to meet the probation conditions necessary to continue in the College of Engineering. Students who have been dropped multiple times should pursue some other career goals. Only in rare cases will a student be readmitted after being dropped twice.

Prerequisite Monitoring and Enforcement

The importance and enforcement of prerequisite course requirements is advertised frequently during Advising Week and during registration periods. Early each semester, the Transfer Articulation and Degree Audit (TADA) office prepares a list of all students in the College that do not meet prerequisite course requirements for courses they are enrolled in. This list is sent to the Director of Engineering Admissions and Records, who then determines which students on the list have transfer credit that has not posted yet, or who have permission to take the course without the listed prerequisite courses (granted under rare circumstances via a General Petition form). The remaining students are notified via email (cc to DUGS, department head, course instructor, associate dean for undergraduate affairs) that they will be dropped from the course if they do not show proof of credit for the prerequisite course(s). The student generally has several days in which to respond to the email and/or make registration adjustments. In the third week, all remaining students are removed from the courses for which they are registered for improperly. Automatic, real-time prerequisite checking is implemented during registration for English, Math, Chemistry, and Physics.

C. Transfer Students and Transfer Courses

Transfer applicants are encouraged to complete 60 semester (90 quarter) hours of transferable work at the time of enrollment and are advised to complete English Composition I and II; Calculus I, II, III; Differential Equations; Physics I and II; Chemistry I. (Computer Science majors should have 12 semester hours (18 quarter hours) of lab science, which includes an 8 hour sequence). These are recommendations as opposed to rigid requirements, and a transfer

student can be admitted student without these courses if their GPA meets college standards. This eventuality would occur most commonly for second-degree students with non-technical backgrounds.

Students must have earned at least a cumulative GPA of 2.50 (A=4.00), including all UIC and all transfer work.

Transfer Credit Evaluation Criteria

The State of Illinois has formal articulation agreements between state community colleges and senior public institutions. Credit from a community college is limited in the sense that the last 60 hours of the degree must be completed at a senior institution. In addition, the University maintains a general provision that either the first 90 or the last 30 hours of the degree must be earned at UIC.

The College of Engineering maintains an extensive evaluation program for courses available at 32 Chicago area colleges that a majority of UIC transfer students come from. The articulation agreement transfer guide is available to prospective students on the COE website: <https://admissions.uic.edu/undergraduate/undergraduate-requirements-deadlines/transfer-requirements/transfer> . The transfer information is housed on a website (Transferology.com) which conveniently allows students to review course equivalency information between institutions participating in the Illinois Articulation Initiative. Transfer credit is determined based on course equivalency.

Work successfully completed in other fully accredited institutions (either those approved by one of the regional accrediting associations or those approved by one of the agencies recognized by the National Commission of Accrediting) is generally accepted by the University on an hour-for-hour basis, as shown on the official transcripts received from those institutions. For consistency, credit awarded on the quarter hour basis is converted to the semester hour system by multiplying the number of quarter hours by 2/3.

Credit from institutions with provisional accreditation is accepted on a deferred basis until it is validated by satisfactory completion of additional work taken in residence at the University or in another fully accredited institution. Credit from unaccredited institutions is not accepted. However, knowledge in courses taken at such institutions may be awarded credit for the equivalent UIC course by successfully passing a UIC proficiency exam. No transfer credit is awarded for any course work completed at an institution external to UIC with a grade of D.

Credit evaluation is done by the Office of the Associate Dean for Undergraduate Administration in the College of Engineering. In many instances, descriptions, syllabi, etc. are sent to a faculty member in the department offering similar courses for their recommendation as to the equivalency to a UIC course and acceptance for transfer credit. The general principle used in accepting work from another college or university is that of “equivalency.” Whether a course taken elsewhere is equivalent to a course at UIC is determined by the college office or in the appropriate department at UIC. Catalog descriptions that show course content and prerequisites for a course are the primary means for determining equivalency. If the catalog description is not

sufficient, syllabi, exams, computer programs, homework, textbooks, etc. may be requested from the student. Table 1.2 shows the Transfer students for the past five academic years. The data show a substantial increase in the number of transfer students entering the program during this period. This increase is driven largely by the increase in tuition costs, which encourages students to use lower cost community colleges to complete their pre-engineering course work.

Table 1.2 Transfer Students for Past Five Academic Years

Academic Year	Number of Transfer Students Enrolled
2014	336
2015	316
2016	359
2017	397
2018	384
2019	377

D. Advising and Career Guidance

The College of Engineering employs a mandatory advising system. Prior to the first semester of attendance, whether entering as freshmen or transferring from other schools, all students are advised by staff from the Office of the Dean for Undergraduate Administration. During their first semester, Chemical Engineering students are assigned to faculty advisors in the Department of Chemical Engineering, who will advise them every semester until graduation. No student is allowed to register without meeting with their advisor and receiving a signed registration form signed by their academic advisor. Registration holds are utilized to ensure that students meet with their advisors every semester prior to registering for the next term.

Faculty advisors help students with curriculum matters, selection of electives, career path input, and other curriculum and career advice. The CHE DUGS acts as an intermediary between faculty advisors and the Office of the Dean for Undergraduate Administration. Students first are advised to discuss curriculum issues with the DUGS for routine questions. The Office of the Dean for Undergraduate Administration maintains an academic counseling staff that helps students with questions related to degree requirements, transfer credits, course substitution, special programs and other related issues.

Based upon discussions at a “townhall” meeting between ECE students and the COE Dean, ECE department head and ECE DUGS, the COE implemented an advisor evaluation process starting in Fall 2006 to provide feedback to advisors on their performance in the advising process. Advisees rate their faculty advisors on a 5 point scale (1 lowest and 5 highest) as to their: 1) concern for the student, 2) knowledge of the curriculum, 3) availability during advising week and 4) encouragement of the student. Scores for all faculty members in each department are compiled by the COE and sent to all faculty within the department for feedback. The COE has also implemented an advisor award program for the top-rated advisor in each department as an incentive to improve advisor performance.

Figure 1.1 on the following page shows the form that is used for advisor evaluation. Table 1.3 shows average ratings for the 4 survey questions and an overall average for the CHE faculty

from Fall 2019. Table 1.4 shows the average ratings for all the faculty from 2015 to 2020. The ChE faculty ratings are high and we see from the advisor evaluation tables that the efforts of the CHE faculty are appreciated by our CHE students.

Table 1.3. Student evaluations of faculty advising, Fall 2019.

Professor	Concern	Knowledge	Availability	Encouragement	Average
Faculty 1	4.55	4.00	3.45	3.73	3.93
Faculty 2	4.74	4.74	4.37	4.21	4.51
Faculty 3	4.76	4.67	4.76	4.10	4.57
Faculty 5	4.33	4.17	4.72	4.11	4.33
Faculty 7	4.65	4.40	4.80	4.00	4.46
Faculty 8	4.25	4.17	4.58	4.00	4.25
Faculty 9	4.57	4.57	4.83	4.17	4.53
Faculty 12	3.90	3.95	4.19	3.62	3.92
Faculty 13	4.78	4.78	4.96	4.83	4.84
Faculty 14	4.92	4.88	4.83	4.71	4.83

Advising Evaluations Criteria: 5 = Strongly Agree, 4 = Agree, 3 = No Opinion, 2 = Disagree, 1 = Strongly Disagree

	F2015	Sp2016	F2016	Sp2017	F2017	Sp2018	F2018	Sp2019	F2019	Sp2020
Faculty 1	4.77	4.72	4.25	4.88	4.77	4.40	4.71	3.83	3.93	4.44
Faculty 2	4.67	4.73	4.43	4.50	4.01	4.07	4.53	4.59	4.51	4.68
Faculty 3	4.13	4.37	4.64	4.44	4.27	4.70	4.38		4.57	
Faculty 4	4.39			4.50	4.19		4.13	4.10		5.00
Faculty 5			4.43	4.67	4.39	4.31	4.58	4.31	4.33	4.15
Faculty 6	4.93	4.50	4.61	4.54	4.33	4.34				
Faculty 7	4.71	4.43	3.81	4.54	4.10	4.17	4.50	4.40	4.46	3.75
Faculty 8		4.51	4.45		4.38	4.42	4.22	4.70	4.25	4.58
Faculty 9		4.50	4.02	4.20	4.47	4.59	4.57	4.69	4.53	4.42
Faculty 10	4.72	4.59	4.67	3.00	3.00	4.33				
Faculty 11	4.56	4.50	4.54	5.00		4.47				4.75
Faculty 12			4.12	3.93	3.80	3.63	4.10	3.91	3.92	3.29
Faculty 13	4.83	4.82	4.73	4.82	4.85	4.60	4.59	4.95	4.84	4.71
Faculty 14	4.86	4.92	4.74	4.31	4.91	4.83	4.57	4.88	4.83	4.60
All Faculty	4.66	4.60	4.42	4.41	4.27	4.37	4.44	4.44	4.42	4.40

Advising Evaluations Criteria: 5 = Strongly Agree, 4 = Agree, 3 = No Opinion, 2 = Disagree, 1 = Strongly Disagree

Table 1.4. Averaged student advisor evaluations (2015-2020).

Figure 1.1. Advisor evaluation form and evaluation criteria.

Advisor Evaluation Form					
To the student: <i>In a continued effort by the college to recognize and reward faculty advising, we ask you that you fill out this form and drop-off the form in the ballot box in the student affairs office in your department (in order to maintain anonymity) along with your faculty advising form with your pre-registration information for the next semester.</i>					
Please select between 1 and 5 or each of the statements: 1 for strongly disagree and 5 for strongly agree.					
My Advisor is (Name_____)					
Signature of Advisor _____					
	Strongly Disagree	Disagree	No Opinion	Agree	Strongly Agree
1. Shows genuine concern for my academic and professional development	1	2	3	4	5
2. Has detailed knowledge of curriculum and provides good advice about courses	1	2	3	4	5
3. Was available at posted advising time during advising week	1	2	3	4	5
4. Encourages students to come at other times beyond the required meeting during advising week	1	2	3	4	5
Comments:_____ _____ _____ _____ _____ _____ _____					

Career Guidance

ABET Accreditation & UIC Engineering Career Center

Overview of services

- The Engineering Career Center (ECC) provides services and programs for both students enrolled in the College of Engineering and employers who recruit UIC engineering students, or hire engineers.
- The ECC is comprised of five full-time staff members and three part-time student assistants, for 5,000+ engineering students; meeting and working with students from freshmen – PhD candidates. The staff has grown from two full-time staff in 2014 to five staff in 2019.
- The ECC assists students with the internship and full-time search process, facilitating connections with employers nationally, regionally and locally.
- The University provides a free job platform for both students and employers. Employers and ECC staff post openings, students upload resumes and apply for positions.
- The ECC presents career outcomes information, staff updates and specific program updates to College of Engineering advisory board members biannually.

Services and programs for students

- Students have access to a well-developed website, with specific engineering career readiness information, as well as event listings.
- Students can schedule an appointment for an in-person one-on-one career advising meeting, they can also attend small group sessions for resume reviews, elevator pitch and interviewing skills.
- Each week the ECC delivers an electronic newsletter to all College of Engineering students, titled: The Spark. The newsletter contains an informative article along with featured employers, featured students and upcoming events.
- The ECC works closely with engineering student organizations presenting workshops to student members and arranging employer events, e.g. info sessions and tabling.
- Students attend two specific engineering and computer science career fairs per academic year put on by the ECC; students also receive marketing information on other career fairs conducted by campus-wide career offices.
 - New for 2018, the ECC conducted a two-day September career fair to accommodate the growing number of employers and students, this is now the standard format for the fall career fair.
 - New for 2019 the ECC implemented Career Fair Tours for students who were not yet ready to meet employers.
 - More than 1,600 students attended the Sept. 2019 career fair.
- ECC conducts presentation to classes including new student orientation course titled Engineering 100, Computer Science Ethics, Mechanical and Industrial Senior Seminar, Chemical Engineering Senior Seminar, and Professional Development for Bioengineers.
- The ECC is responsible for implementing the Guaranteed Paid Internship Program (GPIP) for new freshmen and new transfer students. This unique program offers an on-campus research internship to those students who meet program requirements. For freshmen the internship occurs after their freshmen year and for transfer students it occurs before their last year on campus.

Services and programs for employers

- Two career fairs per academic year; the ECC holds a two-day September career fair and a one-day February career fair; attendance by employers has grown over the past five years from 70 companies to more than 150.
- On-campus interview space immediately following each career fair and throughout the year is offered to employers with a cost to attend.
- The ECC initiated an Employer Roundtable event in 2018 to bring recruiters to campus who have supported UIC with hiring and others who have not yet recruited on campus. Employers were presented with background information on UIC, recruitment best practices from a company recruiter, a panel of company recruiters and a panel of students. The event was repeated in 2019 and is planned for 2020.
- Informational sessions and employer tabling – employers contact the ECC to come to campus to inform students on opportunities with their company, to communicate their company culture and collect resumes. These sessions have increased over the past five years with upwards of thirty sessions occurring during the academic year.

Employer feedback surveys

- Employers were surveyed after specific events such as the career fair and Employer Roundtable.
- Surveys are used to plan the upcoming year's events, to follow-up with specific employers, as well as to modify existing services in general.
- Ongoing feedback from employers is obtained through conversations at events, while interviewing students in the career center and through internship evaluations. This information is shared with staff on an ongoing basis.

Professional development for career center staff

- The ECC holds memberships in several professional organizations for conferences, workshops, meetings and resources for both personal development and to stay abreast of recruitment trends.
- Memberships in the following organizations currently exists among the staff:
 - Member of the National Association of Colleges and Employers
 - Career Leadership Collective
 - Cooperative Education & Internship Association
 - Chicago Career Professional Network
 - Chicagoland Chamber of Commerce
- In addition, two staff retreats are held each year for the purpose of planning for the upcoming year, and to discuss an educational topic.

First Destination Survey

- Graduating seniors with bachelor degrees are emailed a survey upon graduation to determine what their first destination is after their degree, e.g. industry or graduate school. The survey was created in collaboration with the UIC Office of Institutional Research and the campus career center.
- Follow-up surveying is implemented three months after graduation, and again six months after graduation with follow-up phone calls to those students who identify as “still seeking.”

- Results of the survey are provided in aggregate form on the ECC website and updated each year: <https://ecc.uic.edu/students/graduation-survey-results/>

Collaboration with other departments within the College of Engineering

- The ECC works directly with the Communications department on website graphics, website content, the electronic newsletter, and TV monitors in various engineering buildings.
- The Advancement and Development office cross-collaborate with the ECC on employers interested in hiring as well as giving.
- Academic advising office is often contacted for specific student issues/concerns.
- The ECC presents career services updates and First Destination survey results to the College of Engineering Advisory Board members twice per year.
- The Chemical Engineering and BioEngineering departments collaborate with the ECC to conduct an Industry Day each year; this event brings alumni back to campus to speak with undergraduate and graduate students on careers at their company; this event started in 2015 and still continues.

Collaboration with other departments on campus

- The ECC works closely with the central career services office, as well as the Business Career Center concerning the shared job platform, the hiring of staff, career fair volunteers, employer outreach, and general sharing of successful programs and services.
- The Director of the ECC and the Executive Director of the campus career center are co-chairs of a campus wide Career Council that meets twice per year. The purpose of the Council is to disseminate career trends concerning UIC students, as well as to learn from members of the Council.

E. Work in Lieu of Courses

The College of Engineering does not allow credit towards any degree based on work or life experience. However, students may establish credit toward an undergraduate degree through the following examinations, the most common being Advanced Placement:

- *ACT English/SAT Verbal*
- *Advanced Placement (AP)*
- *International Baccalaureate (IB)*
- *College-Level Examination Program (CLEP)*
- *State Seal of Biliteracy*
- *Proficiency Examinations for Enrolled Students*

Students must submit official grade reports/examination results to the Office of Admissions before credit can be awarded. UIC will not award transfer course credit based on another institution's evaluation of test results.

Advanced Placement (AP)

UIC will award course credit on the basis of scores earned on the Advanced Placement Examinations administered by the College Board. A list of applicable courses and corresponding

scores required for credit is provided in the Undergraduate Catalog:
<https://catalog.uic.edu/ucatalog/degree-programs/academic-standing/#AP>

Aside from liberal-arts courses satisfying General Education requirements, technical courses relevant to the Chemical Engineering major are as follows:

MATH 180
MATH 181
CHEM 112
CHEM 122/123
PHYS 141
PHYS 142

International Baccalaureate Exams (IB)

UIC will award credit on the basis of scores earned on the International Baccalaureate examinations. A list of applicable courses and corresponding scores required for credit is provided in the Undergraduate Catalog:

<https://catalog.uic.edu/ucatalog/degree-programs/academic-standing/#IB>

Aside from liberal-arts courses satisfying General education requirements, technical courses relevant to the Chemical Engineering major are as follows:

CHEM 122/123

ACT English/SAT Verbal

UIC will award three hours of passing credit for [ENGL 160](#) for an ACT English subscore of 27, an SAT Evidence-Based Reading and Writing score of 630 or more (students entering UIC Spring 2019 or after), 600 or more (students entering UIC 2016 to Fall 2018), or an SAT Critical Reading score of 610 or more (prior to 2016).

College-Level Examination Program (CLEP)

UIC awards credit based on scores earned on the College Level Examination Program (CLEP). When seeking credit for a CLEP exam taken at another institution, your official score will be evaluated on UIC's credit-granting policy.

CLEP credit will not count toward general education credit, university writing, quantitative reasoning, foreign language, residency, major, or minor requirements. A student cannot take a CLEP exam in a subject in which they have earned credit. Three elective semester hours will be awarded per exam as detailed below. A maximum of 30 semester hours is allowed if approved by the Engineering College and Chemical Engineering Department. The [full list of CLEP tests available](#) is available online. The Office of Testing Services administers CLEP tests on campus. More information is provided in the Undergraduate Catalog: <https://catalog.uic.edu/ucatalog/degree-programs/academic-standing/#CLEP>

State Seal of Biliteracy

UIC will award credit to students earning the State Seal of Biliteracy. Students who present appropriate documentation of the Seal to their advisor within three years of high school graduation will be awarded UIC proficiency credit as indicated below. More information is provided in the Undergraduate Catalog:

<https://catalog.uic.edu/ucat/degree-programs/academic-standing/#StateSealBiliteracy>

Proficiency Examinations for Enrolled Students

These are covered by general University rules, largely deferring to the department and college in question and delineated in detail in the Undergraduate Catalog:

<https://catalog.uic.edu/ucat/degree-programs/academic-standing/#ProficiencyExams>

Proficiency examinations may be offered in some subjects, such as English, chemistry and physics. Students should consult with the department to determine which examinations are offered. Before taking any proficiency examination, the student must obtain the consent of the college dean as well as the head or chairperson of the department concerned. The Chemical Engineering department does not offer proficiency exams for any of its courses.

F. Graduation Requirements

To earn a BS degree from the College of Engineering at UIC, students need to complete University, college, and department degree requirements. University and college degree requirements for all College of Engineering students are outlined below.

1.F.1. Semester Hour Requirement

Degree Program/ Department	Degree Conferred	Total Hours
Chemical Engineering	BS in Chemical Engineering	128

1.F.2. Course Requirements

General Education Core.

General Education at UIC is designed to serve as a foundation for lifelong learning. Students are required to complete a minimum of 24 semester hours in the General Education Core with at least one course from each of the following categories:

- I. Analyzing the Natural World
- II. Understanding the Individual and Society
- III. Understanding the Past
- IV. Understanding the Creative Arts
- V. Exploring World Cultures
- VI. Understanding U.S. Society

A description and list of courses for each General Education Core category is provided in the *General Education* section of the Undergraduate Catalog.

<https://catalog.uic.edu/ucat/degree-programs/general-education/#c>

Information on meeting the General Education requirements for each degree program is provided in the College of Engineering department sections. Chemical Engineering majors satisfy the General Education Core category I through other requirements of the major.

General Education Proficiencies—University Writing Requirement.

College of Engineering students meet the requirement by achieving a passing grade in English 160 and 161 (6 hours total) or by achieving proficiency through the AP English Language Exam, ACT English subscore, or SAT verbal score. Credit for English 160 may be earned on the basis of a score of 3–5 on the AP English Language and Composition exam, an ACT English subscore of 27 or more, or an SAT Verbal score of 630 or more. The Department of English reserves the right to require a student to take a preparatory course as a prerequisite for [ENGL 160](#) if the student's score on the Writing Placement Test reveals the need for such a course. Whenever questions arise with regard to the fulfillment of the University Writing requirement through transfer courses, a writing portfolio, or standard examinations, the Department of English will determine whether to grant the student an exemption from the requirement. More information is provided in the Undergraduate Catalog link: <https://engl.uic.edu/programs/first-year-writing-program/policies/>

Orientation Course Requirement.

All incoming freshmen and transfer students must take an engineering orientation course ENGR 100, as appropriate, during their first or second term at UIC. Satisfactory completion of the engineering orientation course is a graduation requirement.

Student Evaluation

All CHE students are continuously evaluated throughout their tenure in the CHE Department. This effort is aided by the Degree Audit Reporting System (DARS). This appraises students of what courses they have taken and which courses remain to be taken for the degree. DARS communicates with the Course Registration System (CRS) software, which checks that course pre-requisites have been taken by the registering student. If the pre-requisite has not been taken, the student is not allowed to register for that class.

During the student's senior year, the student is required to fill in a request to graduate checklist with the COE. The checklist includes the formal course requirements, as well as several additional requirements, including participation in Engineering EXPO in conjunction with CHE 396 (Senior Design I) or CHE 397 (Senior Design II). In addition to specific course requirements which are detailed in Section 5, the BS in Chemical Engineering also entails two other activities.

1. Completion of the Senior Exit Survey and Senior Written Exam
2. Completing CHE 499 (Professional Development Seminar) during the Fall semester of their senior year.

Residency Requirements

In addition, a minimum residency requirement stipulating the number of consecutive credits that must be taken at UIC is in place and must be satisfied for graduation. The policy (which was most recently revised in Fall 2007) states:

“Either the first 90 semester hours or the last 30 semester hours of University work must be taken at UIC. In addition, at least one-half of the semester hours required in the student's major area of study must be completed at UIC. Concurrent attendance at the University of

Illinois at Chicago and another collegiate institution or enrollment during the summer at another institution when approved by the student’s college, does not interrupt the UIC enrollment residence requirement for graduation. Under exceptional circumstances, the enrollment residence requirement may be waived by the dean of the student’s college upon petition of the student.”

The following table gives a summary of COE enrollments over the past five years. The table shows that the enrollment decline of a few years ago has reversed. In part, this is due to extensive recruiting efforts that all departments in the college have initiated over the past three years, while ACT requirements were not lowered as seen in Table 1.4.

Table 1.4 Enrollment Trends for Past Five Academic Years

	Year (2014)	Year (2015)	Year (2016)	Year (2017)	Year (2018)	Year (2019)
Full-time Students	2465	2671	2798	3120	3382	3716
Part-time Students	259	284	296	314	332	287
Student FTE ¹	3182	3464	3626	4078	4480	4850
Graduates	484	511	601	629	696	724

¹ FTE = Full-Time Equivalent

G. Transcripts of Recent Graduates

Student transcripts will be provided as requested by the EAC Team Chair.

CRITERION 2. PROGRAM EDUCATIONAL OBJECTIVES

A. Mission Statement

2.A.1. Scope and Mission Statement of the University of Illinois at Chicago (UIC)

The largest university in one of the world's most vibrant cities:

The University of Illinois at Chicago is an acclaimed research institution with 16 colleges dedicated to the discovery and distribution of knowledge.

“UIC's mission is:

To create knowledge that transforms our views of the world and, through sharing and application, transforms the world.

To provide a wide range of students with the educational opportunity only a leading research university can offer.

To address the challenges and opportunities facing not only Chicago but all Great Cities of the 21st century, as expressed by our Great Cities Commitment.

To foster scholarship and practices that reflect and respond to the increasing diversity of the U.S. in a rapidly globalizing world.

To train professionals in a wide range of public service disciplines, serving Illinois as the principal educator of health science professionals and as a major healthcare provider to underserved communities.”

The UIC mission statement was most recently ratified by the faculty senate of the University of Illinois at Chicago on April 27, 2006. This statement is prominently displayed to all the UIC's constituencies at the following URL.

<http://www.uic.edu/uic/about/scope.shtml>

2.A.2. Mission Statement of the College of Engineering (COE)

The UIC College of Engineering brings together two properties that are often disconnected in higher education: excellence and accessibility. We seek to provide world-class preparation to students of all backgrounds in an urban setting, creating new opportunities for young engineers and contributing to diversity in the field.

Mission of the College

UIC Engineering's mission is **access to excellence and success**. We educate students to innovate and to lead. We expand global knowledge of the engineering field through excellence in original research. In addition, as a public institution we prioritize our relationship with the city of Chicago

and the state of Illinois. We foster strong local partnerships, provide crucial engineering expertise and facilities, and serve as a major economic driver for the region.

Our current priorities include:

- Recruiting bright students with unique stories and talents
- Attracting and retaining world-class teachers and researchers
- Building top-tier facilities
- Creating a strong alumni community
- Establishing strong partnerships with government, industry, and academia

Undergraduate Study in Engineering

A primary goal of the UIC College of Engineering is to ensure that its students are well prepared for:

Practice in the engineering profession;

Continued formal education at the graduate level; and

Continued education to adapt to evolving technologies and changing markets.

College faculty and administration are continually reevaluating and revising curricula so that engineering and computer science degree programs consistently incorporate the changes that are occurring in technology and society.

The college attracts students and faculty who represent a broad spectrum of nationalities, cultures, races, ages, and genders. Diversity is also reflected in the number and types of employment opportunities available to students. Area corporate partnerships support co-op and internship experiences that are tailored to individual student needs.

The College of Engineering mission statement is available online to students and all other College constituencies at the following URL:

<https://engineering.uic.edu/about/>

It also appears in the undergraduate catalog at the following URL:

<https://catalog.uic.edu/ucatalog/colleges-depts/engineering/#a1>

Consistent with this mission statement, the College of Engineering publishes the following Educational Objectives for students in the Undergraduate catalog at the following URL.

<https://catalog.uic.edu/ucatalog/colleges-depts/engineering/#a3>

Educational Objectives

The UIC College of Engineering offers undergraduate and graduate students opportunities to join faculty in cutting-edge research. In the classroom, students become familiar with the fundamental mathematical and scientific principles that are common to engineering and computer science disciplines, and they learn to apply these principles to current engineering and computer science problems of analysis, design, and experimentation. Through individual and group projects, students make use of current techniques, instruments, equipment, and computers, and gain proficiency in communicating the results of their work. Study in other disciplines provides students with an understanding of the professional ethical responsibilities of practicing engineers. Students also have the opportunity to participate in a number of the many on-campus student chapters of national engineering professional organizations as a way to supplement their classroom experiences.

In the first two years, each student will be required to complete courses in mathematics, chemistry and physics (or other science requirements, for computer science majors), and University Writing. Beginning in the second year, the student will start course work in a particular major that represents the technical phase of the student's academic career and constitutes a cohesive program of advanced work in a chosen field. Although the course work in the major becomes progressively specialized in the junior and senior years, each student is also required to take engineering or computer science courses outside of his or her chosen field.

A student must also complete course work in the general fields of humanities and social sciences. Because engineers and computer scientists are no longer narrow specialists, they must recognize the effects of their work on the general welfare of society. The humanities/social sciences phase of their education helps them to become serious contributors to the quality of life. Requirements for the degrees often include free electives that introduce flexibility into the curricula.

2.A.3. Mission statement of the Department Chemical Engineering

“The mission of the Department of Chemical Engineering is to serve society through excellence in education, research, and public service. We provide for our students an education in chemical engineering, and we aspire to instill in them the attitudes, values, and vision that will prepare them for lifetimes of continued learning and leadership in their chosen careers. Through scholarship, the Department strives to generate new knowledge and technology for the benefit of the State of Illinois, the nation, and beyond.”

The department mission statement is available online to the students and all other College constituencies at the following URL:

<https://che.uic.edu/undergraduate-programs/abet-accreditation/>

B. Program Educational Objectives

PEO	Category	Description
1	Depth	In engineering practice, advanced study or original research, UIC Chemical Engineering bachelor's degree graduates will be effective in applying fundamental principles, scientific knowledge, rigorous analysis, creative design, and conceptual innovation.
2	Breadth	Based upon mastery of engineering in a broad, societal context, UIC Chemical Engineering graduates will have successful careers in the public or private sectors, or in the pursuit of graduate education.
3	Professionalism and Service	In the service of industry, government, the engineering profession and society at large, graduates will function effectively in the complex modern work environment with clear communication, responsible teamwork, and high standards of ethics, professionalism, safety and protection of the environment.
4	Long Learning and Leadership	Based upon a rigorous undergraduate program that is innovative, challenging, open and supportive, graduates will enhance their skills and knowledge through life-long learning and demonstrate professional leadership.

The Program Educational Objectives are prominently listed in the Undergraduate-ChemE Major link on the home page of the departmental website: www.che.uic.edu, or <https://che.uic.edu/undergraduate-programs/che-major/>. Additionally, the PEOs are included in all undergraduate course syllabi, and class time is devoted to explaining how the knowledge, skills and principles developed by the course in question contribute to the professional capacities represented by the PEOs.

C. Consistency of the Program Educational Objectives with the Mission of the Institution

Component of UIC mission	Contributing PEO
To create knowledge that transforms our views of the world and, through sharing and application, transforms the world.	1
To provide a wide range of students with the educational opportunity only a leading research university can offer.	1, 4
To address the challenges and opportunities facing not only Chicago but all Great Cities of the 21st century, as expressed by our Great Cities Commitment.	2, 3
To foster scholarship and practices that reflect and respond to the increasing diversity of the U.S. in a rapidly globalizing world.	3
To train professionals in a wide range of public service disciplines, serving Illinois as the principal educator of health science professionals and as a major healthcare provider to underserved communities.	2

The healthcare component of UIC's mission (arising from the specific history of UIC and its medical school within the Chicago Metropolitan Area) is not directly served by the Chemical Engineering program, and would not be expected among the objectives of typical chemical engineering programs. Recently however, there has been a strong collaboration between the Chemical Engineering Department and the College of Medicine on research studies such as nanomaterials, drug delivery, biocompatible materials, cancer detection instrumentation and pharmaceuticals. Students have contributed directly by performing undergraduate research in these areas. Also, graduates could be employed in related sectors, and hence the Chemical Engineering bachelor's degree program could contribute indirectly to medical technology and healthcare.

D. Program Constituencies

1.D.1. Program Constituencies.

Program Constituency	Vehicles of Consultation
Students	Course Update Surveys (each semester) Academic advising (each semester) Senior Exit Survey (semester of graduation)
Faculty	Faculty meetings
Alumni	Alumni feedback surveys (ongoing) Alumni at large (department sponsored events, guest lectures, LinkedIn UIC Chemical Engineering group, email contact, telephone contact) Engineering Expo judging (each spring semester)
Industrial and Professional	External Advisory Board meeting AIChE Chicago Section (ongoing) Industrial mentors in Senior Design II (CHE 397) class Engineering Expo judging (each spring semester)
University Administration	Educational Policy Committee meetings (monthly)

Our students are critically important stakeholders in the curriculum and policies of the Chemical Engineering department, and we consult them regularly as summarized above. However, students are not in as strong a position as other constituents to determine the appropriate aspects and levels of professional accomplishment several years after graduation. Surely their feedback forms part of the conceptual milieu in which the department formulates policy. But we do not regard students formally as “constituents” in the review and updating of PEOs.

1.D.2. Meeting the Needs of Program Constituents and Other Stakeholders

Students. The Chemical Engineering program exists to educate and promote young engineers and launch them on successful careers in industry, post-graduate education and government service. In order that students understand the broader professional context toward which they are being educated, the PEOs are listed on the syllabus of each undergraduate course and class discussion is devoted to putting the performance indicators of the course in the professional and career context of PEOs. With this motivation, PEOs guide the professional development of our students and set targets for their knowledge, skills and principles.

Faculty. PEOs represent the ultimate goals of undergraduate teaching and learning, and thereby serve to organize our efforts in design, development, implementation and improvement of the curriculum toward professional success and effectiveness of our graduates.

Alumni. PEOs represent the cornerstone for professional success, effectiveness and leadership of our alumni.

Industry. For local and National industry that seeks to hire, develop and promote professional engineers, PEOs represent the spectrum of professional abilities and capacities of UIC Chemical Engineering graduates that make them desirable and enable them to contribute and lead. The industrial constituency is represented by the External Advisory Board (EAB) and also members of the AIChE Chicago Section.

The EAB has been assembled from among prominent industrial and corporate engineers from various sectors and localities, who complement each other's technical expertise.

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Anastasia Vaia

Chief Of Staff to the Petrochemicals COO
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United Kingdom

The EAB is helping to steer the department's direction, structure and mission.

A number of practicing chemical engineers, members of the AIChE Chicago Section, partner with the department on (i) improving pedagogy (industrial mentors and guest lecturers in the Senior Design sequence, CHE 396, 397 – Senior Design), (ii) outreach activities (high schools, middle schools), and (iii) interactions with the AIChE Student Chapter. These industrial constituents include:

Bipin Vora: Retired from UOP, UOP Fellow and AIChE fellow

Adam Brostow: Honeywell UOP

Dennis O'Brien: Jacob's Consultancy

Jon Prudhom: Honeywell UOP

Jerome Palmer: Retired from Ambitech

Patrick Shannon: Retired from Middough

Belma Demirel: BP

Lance Baird: Retired from UOP

Gavin Palmer: Brindley Engineering

David Bahr: Carbon Clean Solutions

Mike Armstrong: Jacob's Consultancy

Annette Johnson: Abbvie

E. Process for Review of the Program Educational Objectives

Periodic review and updating of the PEOs involves all constituencies of the department. The PEOs were discussed by faculty and our industrial constituents at the meetings of the EAB on September 30, 2016, July 22, 2019 and at the most recent EAB meeting to be held in July 2020. The PEOs were considered acceptable and no further changes were deemed necessary.

Alumni feedback is obtained where possible in person at return visits, such as at our annual Spring ChemE Forum and alumni mixer event, judging at Engineering Expo, invited class lectures and student seminars, and other points of contact with the faculty.

Finally, input from University administration is obtained from the Office of the Associate Dean for Undergraduate Affairs at regular meetings of the COE Educational Policy Committee (EPC). This last avenue is an avenue for maintaining consistency between departmental PEOs and the UIC and COE mission statements.

For the ABET site visit we will present the following documentation on PEOs:

- Relevant agendas, notes and/or minutes from the EAB meetings.
- Correspondence, notes and/or minutes on faculty and EPC discussion of PEOs.
- Correspondence, surveys from the alumni.
- Industry Day Survey of professional engineer panel discussion speakers, October 2019.

CRITERION 3. STUDENT OUTCOMES

A. Student Outcomes

Student Outcomes are prominently listed in the “ChemE Major” link on the home page of the departmental website: www.che.uic.edu and can also be accessed via the direct link; <https://che.uic.edu/undergraduate-programs/che-major/>. Additionally, relevant Student Outcomes are included in all undergraduate course syllabi, and class time is devoted to explaining how the knowledge, skills and principles developed by the course in question contribute to the understanding and abilities represented by the Student Outcomes. The processes (Section 2.E.) for revisiting and updating PEOs also afford opportunities to reconsider Student Outcomes and amend them as necessary. The Student Outcomes from 2014 were revised and coincide directly with the ABET Student Outcomes 1-7, as shown in Table 3-1. As a reference, Table 3-2 shows the mapping of the previous 2014 Student Outcomes (1-13) to the current Student Outcomes.

Table 3-1. Student Outcome relationships to ABET Student Outcomes and the Program Educational Objectives.

SO	ABET	Description	PEO
1	1	An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.	1
2	2	An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors	1, 2,3
3	3	An ability to communicate effectively with a range of audiences.	3
4	4	An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	2, 3
5	5	An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	3
6	6	An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	1
7	7	An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.	4

Table 3-2 Mapping of Student Learning Outcomes: 2014 to 2020.

Current Student Outcomes	Student Outcomes (ABET 2014)
(1) An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.	(1) Ability to apply knowledge of mathematics, science and engineering (2) Ability to identify, formulate and solve engineering problems
(2) An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	(3) Ability to use techniques, skills, and modern chemical engineering tools necessary for engineering practice (7) Ability to design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability
(3) An ability to communicate effectively with a range of audiences.	(12) Ability to communicate effectively
(4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	(8) Broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context (9) Knowledge of contemporary issues (13) Understanding of professional and ethical responsibilities
(5) An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	(11) Ability to function on multidisciplinary teams
(6) An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	(4) Ability to design experiments (5) Ability to conduct experiments (6) Ability to analyze and interpret data
(7) An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.	(10) Recognition of the need for and ability to engage in life-long learning

B. Relationship of Student Outcomes to Program Educational Objectives

Table 3-1 also shows how the Student Learning Outcomes relate to the Program Education Objectives. Student Outcomes 1, 2 and 6 represent the core of fundamental engineering knowledge, skills and tools that prepare graduates to enact PEO 1 (depth) in their subsequent careers. Student Outcome 4 embodies the social, economic and political context in which graduates advance their careers and impact society (PEO 2). Student Outcomes 3, 4, and 5 establish the abilities of graduates to attain high professional and ethical standards in their careers and advance public safety and environmental protection, with clear communication and teamwork (PEO 3). Student Outcome 7 reflects the capacity of graduates to renew their knowledge and skills, enhance their innovation and lead (PEO 4).

C. Coverage of Student Outcomes in Courses

Coverage of Student Outcomes in any given course is classified using the following four-point scale:

- 4 = topic is main focus of course, grade largely dependent on demonstrated knowledge.
- 3 = course grade partially dependent on demonstrated knowledge on this topic.
- 2 = topic introduced in course, course grade not dependent on demonstrated knowledge.
- 1 = topic not covered specifically in course.

Tables 3-3 and 3-4 indicate the coverage of Student Outcomes in each of the Chemical Engineering courses. Overall coverage of SOs by ChE courses is summarized in Table 3-5: all SOs are covered at levels 3 or 4 in at least two ChE courses. SOs #1, #2 and #3 receive this level of emphasis in eighteen, nine, and eleven courses, respectively. Table 3-6 indicates the coverage of SOs in each of the technical courses outside the Chemical Engineering department.

Table 3-3. Coverage of Student Outcomes 1-7 by Chemical Engineering Courses.

Course	(1) An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.	(2) An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	(3) An ability to communicate effectively with a range of audiences.	(4) An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	(5) An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	(6) An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	(7) An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.
CHE 201	3	2	2	2	2	1	2
CHE 205	4	2	1	1	1	2	1
CHE 210	4	2	1	1	1	1	1
CHE 301	3	2	3	2	1	1	2
CHE 311	4	4	1	1	1	1	1
CHE 312	4	3	1	2	1	1	3
CHE 313	4	4	3	2	3	3	4
CHE 321	4	4	2	3	3	2	3
CHE 341	4	4	3	1	1	1	1
CHE 381	4	1	4	3	4	4	3
CHE 382	4	1	4	3	4	4	3
CHE 396	4	4	3	4	4	3	3
CHE 397	4	4	4	4	4	3	3
CHE 422	2	3	3	1	1	1	2
CHE 425	4	4	3	3	3	2	4
CHE 433	4	1	1	1	1	1	1
CHE 451	3	4	4	4	4	1	4
CHE 453	4	4	2	1	1	2	1
CHE 499	1	1	3	3	1	1	3

4 = topic is main focus of course, grade largely dependent on demonstrated knowledge;

3 = course grade partially dependent on demonstrated knowledge on this topic;

2 = topic introduced in course, course grade not dependent on demonstrated knowledge;

1 = topic not covered specifically in course.

Abbreviated, descriptive course titles

CHE 201 Intro Eng. Thermo

CHE 301 Chem Eng Thermo

CHE 313 Transport Phenom III

CHE 381 Unit Ops Lab I

CHE 397 Senior Design II

CHE 433 Process Simulation: Aspen

CHE 499 Professional Development Seminar

CHE 205 Comp. Meth. In ChE

CHE 311 Transport Phenom I

CHE 321 Reaction Engineering

CHE 382 Unit Ops Lab II

CHE 422 Biochem Engineering

CHE 451 Renewable Energy

CHE 210 Mat. Energy Balance

CHE 312 Transport Phenom II

CHE 341 Process Control

ChE 396 Senior Design I

CHE 425 Nano Pharm App

CHE 453 Fund Electrochem

Table 3-4. Coverage of Student Outcomes 1-7 by CHE 494 Selected Topic Courses.

Course	(1) An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.	(2) An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	(3) An ability to communicate effectively with a range of audiences.	(4) An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	(5) An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	(6) An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	(7) An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.
CHE 494-A	1	3	4	4	4	1	4
CHE 494-B	2	1	3	4	4	4	4
CHE 494-C	2	1	3	4	4	4	4
CHE 494-D	4	3	4	1	1	3	4
CHE 494-E	4	3	2	3	2	4	3
CHE 494-F	4	4	4	1	3	3	4
CHE 494-G	3		3				2

4 = topic is main focus of course, grade largely dependent on demonstrated knowledge;

3 = course grade partially dependent on demonstrated knowledge on this topic;

2 = topic introduced in course, course grade not dependent on demonstrated knowledge;

1 = topic not covered specifically in course.

CHE 494 A Entrepreneurship in Chemical Engineering

CHE 494 B Graphene and 2D Materials

CHE 494 C Materials for Energy

CHE 494 D Molecular and Macromolecular Engineering

CHE 494 E Nano and Micro Engineered Membranes

CHE 494 F Nanotechnology and Bio-Nanotechnology

CHE 494 G Polymer Processing and Technology

Table 3-5. Summary of the Coverage of Student Outcomes in Chemical Engineering Courses.

Student Outcomes		Number of courses at the indicated level of coverage			
#	Description	4 Topic is main focus of course, grade largely dependent on demonstrated knowledge;	3 Course grade partially dependent on demonstrated knowledge on this topic;	2 Topic introduced in course, course grade not dependent on demonstrated knowledge;	1 Topic not covered specifically in course.
1	An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.	18	4	3	1
2	An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	9	6	4	6
3	An ability to communicate effectively with a range of audiences.	7	11	4	4
4	An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	6	7	4	7
5	An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	8	5	2	9
6	An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	5	5	4	10
7	An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.	8	8	4	6

Table 3-6. Coverage of Student Outcomes 1-7 by non-ChE technical courses.

Course	(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.	(2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	(3) (an ability to communicate effectively with a range of audiences.	(4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	(5) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	(6) an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	(7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.
MATH 180	4	2	2	1	1	1	1
MATH 181	4	2	2	1	1	1	1
MATH 210	4	2	2	1	1	1	1
MATH 220	4	2	2	1	1	1	1
PHYS 141	4	2	2	3	3	3	1
PHYS 142	4	2	2	3	3	3	1
CHEM 122/123	4	2	1	3	3	4	1
CHEM 124/125	4	2	1	3	3	4	1
CHEM 222	4	2	1	3	3	4	1
CHEM 232	4	2	1	1	1	1	1
CHEM 233	4	2	1	4	4	4	1
CHEM 234	4	2	1	1	1	1	1
CHEM 342	4	2	1	1	1	1	1
CHEM 346	4	2	1	1	1	1	1
CS 109	2	2	4	1	1	1	1
CME 260	4	4	3	3	3	3	1
ECE 210	3	3	3	1	1	4	1

4 = topic is main focus of course, grade largely dependent on demonstrated knowledge;

3 = course grade partially dependent on demonstrated knowledge on this topic;

2 = topic introduced in course, course grade not dependent on demonstrated knowledge;

1 = topic not covered specifically in course.

Abbreviated, descriptive course titles:

MATH 180 Calculus I

MATH 220 Intro. Differential equations

CHEM 122/124 Gen Chem I

CHEM 232 Organic Chemistry I

CHEM 342 Physical Chemistry I

CME 260 = Properties of Materials ECE 210 = Elect Circuit Analysis

MATH 181 Calculus II

PHYS 141 General Physics I

CHEM 124/125 Gen Chem II

CHEM 233 Organic Chem. Lab I

CHEM 346 Physical Chemistry II

MATH 210 Calculus III

PHYS 142 General Physics II

CHEM 222 Analytical Chemistry

CHEM 234 Organic Chemistry II

CS 109 Prog with MatLab

ENGR 100 = Engineering Orientation

CRITERION 4. CONTINUOUS IMPROVEMENT

A. Student Outcomes

Oversight and conduction of assessment is ultimately the responsibility of the Undergraduate Committee (UGC) and its chair, the Director of Undergraduate Studies (DUGS). However, the overall assessment methodology has been developed in such a way that all faculty and some industrial representatives are involved.

4.A.1. Assessment Methods and Frequency of Administration.

The overall plan and schedule of Student Outcome assessment is summarized in Table 4-1. Direct assessment rests on a triad of methods, each of which probes the outcomes in different ways, each semester or annually, by different evaluators: Method #1- Grading rubrics in CHE courses, Method #2- Senior Written Examination, and Method #3- Engineering Expo judging rubrics. Table 4-2 shows how these three assessment methods cover the various Student Outcomes (SOs) and also map to the Program Educational Objectives (PEO). In summary, SOs 1-5 and 7 are covered by two or three direct assessment methods and SO 6 is covered by one assessment method (Method #1). Indirect assessment of the SOs is achieved via an online Exit Survey that is filled out by graduating seniors toward the end of the spring semester (Method #4). Finally, in each course the regular student evaluations of teaching (administered centrally by the Office of the Vice Provost for Faculty Affairs) are augmented by a much more detailed and penetrating online feedback survey that was developed in-house to facilitate continuous improvement of courses by individual faculty (Method #5). These assessment and feedback methods and results are now described in detail.

Table 4-1. Student outcome assessment methods and frequency of administration.

Method	Description (Abbreviation)	Category	Schedule	Responsibility
1	Grading rubrics in CHE courses (Course Rubrics)	Direct / Formative	Selected courses each year	All faculty
2	Senior Written Examination	Direct / Summative	Fall semester before graduation	DUGS/Undergraduate Committee
3	Engineering Expo judging rubrics (Expo Rubrics)	Direct / Summative	Spring semester	Technical Judges
4	Senior Exit Survey	Indirect / Summative	Spring semester	DUGS/Undergraduate Committee
5	Course Update Surveys (Course Survey)	Feedback	Each semester	All faculty

Table 4-2. Coverage of Student Outcomes by Direct Assessment Methods.

ABET SO	Description	PEO	<u>Method #1</u> Course Rubrics	<u>Method #2</u> Senior Written Exam	<u>Method #3</u> Expo Rubrics
1	An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.	1	X	X	
2	An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors	1	X	X	X
3	An ability to communicate effectively with a range of audiences	3	X		X
4	An ability to recognize ethical And professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts	2,3	X	X	
5	An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives	3	X	X	
6	An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions	1	X		
7	An ability to acquire and apply new knowledge as needed, using appropriate learning strategies	4	X	X	

Assessment Method #1 (Direct / Formative): Rubric-based assessment of a major assignment in required courses. In the courses selected to be assessed that semester, the professor formulates a detailed rubric to assess specific performance indicators (PIs) among the students on a major assessment assignment (e.g., exam or portion of exam, term paper, lab report, milestone presentation). These PIs are mapped to some or all of the relevant SOs for that course as specified in the syllabus (see Appendix A). Table 4-3 indicates the coverage and schedule of SO assessment by this mechanism. We cannot claim to cover every relevant SO for every course every semester, because the coverage in any given course hinges on the major assessment assignment being formulated by the professor. But the partially overlapping “collage” of SO “footprints” in all the courses does ultimately provide good coverage of the SOs overall. Furthermore, this approach regularly engages the entire faculty in the process of assessment, and keeps assessment uppermost in our minds. The SO assessment rubrics certainly do require more time and effort from professors than does the typical academic grading process for the assignments involved, but at a manageable level that encourages buy-in among the faculty. Significantly, we thereby also avoid the burden of additional assessment activities for students. The assessment information compiled by the professor may include (1) major assessment assignment, (2) assessment rubric, (3) calculations (e.g., Excel spreadsheet, when appropriate), and (4) summary of results. Exhibit 4-1 (Section 4.C) shows these documents for CHE 201 (Introduction to Engineering Thermodynamics) in Fall 2018.

Table 4-3. Schedule of direct assessment of Student Outcomes by Method #1 (Course Rubrics).

SO	2020	2019	2018	2017	2016	2015
1	301 312 341	205 210 313 321 381 422	201 205 210 381	201 382	201 205 382	201 205 381
2		396 397 433	396	396	205	205
3 (oral)		381 397	381 396 425			381
3 (written)		381 396 397	381 396	382 396	382	381
4 (safety& environmental)	382 201	381 396 201	381 396	382 396	382	381
4 (Global, economic, societal, ethics)	499	396 499	397	396		
5		397	396	382 397	382	381
6		381	381	382	382	381
7		381	381	382	382	381

Assessment Method #2 (Direct / Summative): Senior Written Exam. During the last ABET accreditation cycle (2008-2014), Assessment Method #2 consisted of a compulsory oral exit interview and oral exam administered to all graduating seniors in the spring semester of their graduation year. However, student enrollment has increased substantially to the point that in 2015, the senior graduating class became too large to efficiently administer the oral assessments in a timely fashion (each interview takes 40-60 minutes). The Undergraduate Committee decided to move back to a Senior Written Examination for Assessment Method #2. It is part of the CHE 499 (Professional Development Seminar) course which was moved to the Fall semester of the academic year that the students are graduating. The Senior Written Examination questions are taken directly from the previous oral exit interview and oral exam: (1) numerical methods and computational/software proficiency, (2) engineering estimates and problem solving, (3) career planning and potential for life-long learning, (4) professional ethics and broader impacts of technology, (5) process and laboratory safety, (6) interdisciplinary teamwork. Scoring is done with a detailed rubric that maps to the SOs and achievement of the PEOs. A sample question sheet and the associated blank rubrics are included as Exhibit 4-2 (Section 4.C). It should be emphasized that the senior written exam portion constitutes a rigorous **direct** assessment of student knowledge, abilities and proficiencies, and is not based on self-assessment by students (i.e., indirect assessment). The schedule of direct assessment by Method #2 is shown in Table 4-4.

Assessment Method #3 (Direct / Summative): Judging rubrics from the annual Engineering Expo. Participation in the annual Engineering Expo has been a formal course requirement in CHE 397 (Senior Design II) since before the ABET accreditation cycle in 2008. At this event the design groups present posters on their projects to the UIC community, and their work and communication skills are evaluated by a panel of judges drawn from industry (including alumni) and faculty. To augment the College-wide, standardized judging sheets (used to award prizes), the Chemical Engineering department developed a rubric with rating categories that map to our specific Student Outcomes. There is also a general comments section so that free-format feedback can be cycled back to the students. Sample rubrics forms appear in Exhibit 4-3 (Section 4.C). The schedule of direct assessment by Method #3 is shown in Table 4-4.

Assessment Method #4 (Indirect / Summative): Senior Exit Survey. A survey was developed by the Undergraduate Committee in consultation with the department faculty to obtain self-assessment of graduating seniors on all the SOs as well as feedback for improvement of the curriculum. This survey methodology progressed from a paper form to an online system programmed in-house in PHP code in 2008 and was also used during this accreditation cycle (2015-2020). Upon submission of the filled-in survey, each student receives an automatically generated, anonymous code number that he/she is required to the CHE 499 instructor as proof of completion. The system automatically compiles a summary report that (i) calculates averages and (Satisfactory, Developing, Unsatisfactory) vectors for the Student Outcomes, and (ii) aggregates student responses for each of the feedback questions. A sample printout of the survey is provided in Exhibit 4-4 (Section 4.C). A schedule of the assessment by Method #4 is shown in Table 4-4.

Assessment Method #5 (Feedback): Course Update Survey. A survey was developed by the Undergraduate Committee to enable student feedback toward professors improving courses. After earlier paper versions, this survey was implemented in online form (programmed in-house in PHP) on the departmental assessment website. The system automatically generates summary reports that aggregate student responses to each question. The program emails each professor the report(s) for his/her course(s). An interesting and useful feature is the labeling of each response with the number of hours per week that the student spent on the course (ascertained in a preliminary multiple-choice question). This puts the feedback in the context of student effort. A sample printout of the survey is provided in Exhibit 4-5 (Section 4.C). The schedule of assessment by this method is shown in Table 4-4.

Table 4-4. Schedule of assessment by Methods #2, #3, #4, and #5.

Method	2020	2019	2018	2017	2016	2015
#2 (direct) Senior Written Exam		Fall	Fall	Fall	Spring Fall	Spring
#3 (direct) Eng Expo Judging Rubrics		Spring	Spring			Spring
#4 (indirect) Senior Exit Survey	Spring				Fall	Spring
#5 (feedback) Course Update Survey	Spring Fall	Spring Fall	Spring Fall	Spring Fall	Spring Fall	Spring Fall

4.A.2. Expected levels of Attainment for the Student Outcomes.

Based upon our formulation of the course rubrics (Method #1), the Senior Written Examination (Assessment Method #2), and the Engineering Expo judging rubrics (Method #3), we align the rubric assessment levels of S (Satisfactory), D (Developing) and U (Unsatisfactory) roughly with letter grades A/B, C and D, respectively. The rubric assessment levels are given numerical values of: Satisfactory = 3, Developing = 2, and Unsatisfactory = 1. We are looking to achieve a rating score above about 2.30, 65% or better attainment of Satisfactory, and 85% or better attainment for Satisfactory or Developing for each Student Outcome among our students.

4.A.3. Evaluation of the Attainment of Student Outcomes.

Reduction of assessment data and their synthesis into higher-level information on attainment of SOs proceeds via four levels of analysis. The calculations are carried out by the faculty member(s) conducting the assessment and/or the Undergraduate Committee (assisted by an assessment TA).

Level 1 analysis:

Reduction of data from assessment Rubrics.

For each of the three direct assessment methods (Table 4-1), individual students are assessed using Performance Indicators on rubric item such as exams questions, lab reports, oral presentations, individual and team projects (including computer coding, process simulation and process design), technical papers, safety quizzes, safety tutorials, and ethics assignments. Performance Indicators are given a “3” for Satisfactory, “2” for Developing, and “1” for Unsatisfactory. Each PI rubric score is then averaged over all the students, and a total PI average for the rubric item is also calculated. The results for individual PI rubric items are then mapped onto the applicable SOs. One rubric item may (but need not) relate to more than one SO, and each SO may have contributions from several rubric items. This procedure produces direct assessment results on attainment of SOs for; (i) students in a particular course in a given semester (Assessment Method #1 – Course Rubrics), and (ii) a graduating senior class (Assessment Method #2 – Senior Written Examination; and Assessment Method #3 – Engineering Expo Judging rubrics).

Level 2 analysis:

Average and (S, D, U) attainment assessed with 4 methods by Student Outcome.

For each SO we combine the (S, D, U) results of all Level 1 analysis into a one-page summary report. This allows tracking of trends for each assessment method over time and also exposes similarities and differences between the methods. The summary reports for all SOs appear in Exhibit 4-6 (Section 4.C).

Level 3 analysis:

Average and satisfactory attainment for all Student Outcomes by method.

For each method we combine the average attainment rating and the percentage of students attaining between “Satisfactory” and “Developing” for all SOs year by year into a one-page summary report. This facilitates tracking SOs over time and also comparisons between SOs to determine which aspects of the curriculum are most in need of improvement. The summary reports for all methods appear in Exhibit 4-7 (Section 4.C.). The results of indirect assessment are shown in Table 4-8.

Level 4 analysis:

Satisfactory attainment for all Student Outcomes.

For each SO we aggregate the Level 3 results from all three direct assessment methods into an overall percentage of students attaining; 1) Developing or higher, 2) Satisfactory. We also aggregate from all three direct assessment methods the rating scores for each SO. The Level 2 results from indirect assessment (Method #4) are carried over from the Level 3 analysis. The three summaries are shown in Tables 4-5, 4-6 and 4-7, and give us an overall, “bird’s eye” view of SO attainment and which informs which outcomes need improvement.

The direct assessment method summary shown in Tables 4-5, 4-6, and 4-7 indicate that, by and large, the UIC Chemical Engineering undergraduates are sufficiently attaining the departmental Student Outcome criteria of 1) reaching a 65% or better attainment of a Satisfactory score, 2) achieving a 85% or better attainment of a Satisfactory or Developing score, and realizing a rating score above 2.30/3.00. The results from SO #1, SO #2, SO #3, SO #5, and SO #6 are steady and

indicate that our students are sufficiently learning the core engineering, mathematics, science, experimentation, teamwork, communication and design skills.

Safety/environmental (S &E) and global, economic, societal and ethics (GESE), as covered by SO #4, have shown increasing values from 2015 to 2020 and reflects our efforts to bolster these two student learning outcomes. Student outcome #2 which encompasses the application of engineering design also has high values and reflects the effort to by our industrial mentors to instruct the students in chemical engineering industry design practice with an emphasis on following industry standards in their senior design projects.

Table 4-8 presents the average scores from the Senior Exit Survey, which is a self-evaluation by the students to rate how they feel they have achieved learning the skills as given in the 7 student outcomes. All the ratings are between 1 = Excellent and 2 = Strong and correlate well with the three direct assessment method evaluations summarized in Tables 4-5, 4-6 and 4-7.

Table 4-5. Percentage of students attaining a rating of Satisfactory for all the SO's as determined by direct assessment (aggregate of Methods #1, #2, #3).

SO	2020	2019	2018	2017	2016	2015
1	67.2	64.8	69.0	66.8	59.6	64.9
2	64.9	81.2	74.8	74.3	56.0	67.9
3	76.2	82.5	80.8	76.6	70.0	81.3
4	95.2	78.7	90.3	77.4	84.6	78.6
5		91.9	73.8	86.2	88.9	73.4
6		73.9	79.3	66.9	66.7	69.6
7		77.1	72.6	69.6	78.3	87.2

Table 4-6. Percentage of students attaining a rating of Satisfactory or Developing for all the SO's as determined by direct assessment (aggregate of Methods #1, #2, and #3).

SO	2020	2019	2018	2017	2016	2015
1	86.6	87.2	91.9	91.2	90.7	91.8
2	99.3	97.9	97.1	97.1	84.8	94.0
3	98.9	98.4	97.8	98.8	99.3	99.8
4	98.4	98.5	96.7	97.6	96.4	92.1
5		99.5	96.8	99.1	98.3	97.7
6		99.1	98.3	100.0	98.6	98.7
7		99.2	95.9	99.3	95.5	100.0

Table 4-7. Average rating score for all the SO's as determined by direct assessment (aggregate of Methods #1, #2, and #3).

SO	2020	2019	2018	2017	2016	2015
1	2.51	2.44	2.56	2.52	2.33	2.54
2		2.76	2.72	2.72	2.13	2.52
3	2.86	2.76	2.77	2.69	2.72	2.80
4	2.94	2.74	2.84	2.72	2.76	2.71
5		2.79	2.73	2.76	2.73	2.79
6		2.76	2.72	2.69	2.68	2.72
7		2.56	2.55	2.76	2.70	2.80

Table 4-8. Average rating score for all the SO's as determined by indirect assessment Method #4 (Senior Exit Survey)

Note: a different rating system is used than Methods #1- #3:

(1 = Excellent, 2 = Strong, 3 = Moderate, 4 = Weak, 5 = Poor)

SO	2020	2016	2015
1	1.47	1.50	1.44
2	1.61	1.62	1.60
3	1.40	1.45	1.46
4	1.30	1.26	1.50
5	1.45	1.39	1.60
6	1.54	1.63	1.54
7	1.51	1.57	1.75

B. Continuous Improvement

The loop of assessment, evaluation and feedback from constituencies has been closed in various ways to improve the curriculum and student learning experience. These cases are now described in detail.

4.B.1. Ongoing Feedback-Based Improvement of Courses.

The Course Update Surveys (Assessment Method #5, see Exhibit 4-5 for example) are detailed and give useful information to individual professors for improving their courses. This survey procedure was initiated in 2008 as an automated online system programmed in-house with PHP and has been used in the current 2014-2020 cycle. The university also conducts a standard student evaluation of teaching for each course at the university level. The average course evaluations from these student evaluations for each CHE faculty for the last five years are shown in Figure 4.1. All chemical engineering faculty have ratings higher than 3.25 which indicates greater than satisfactory to excellent ratings

We maintain an ongoing feedback loop for continuous improvement of existing courses by the following procedure. At the end of every semester, each professor administers the Course Update Survey in his applicable courses toward the end of the semester. (This is in addition to the standard student evaluations of teaching conducted centrally at the university level.) Within two weeks or so the professor receives by email the compiled survey results. The professor reflects on this student feedback and the feedback from the university student teaching evaluations and formulates self-study statements on how he/she intends to improve that course for the next year (or semester). The Course Update Surveys and university teaching evaluations represent a regular and extensive consultation with our student constituency with tangibly positive effects on their courses and learning experiences. In addition, faculty respond to their course evaluations with recommendations and actions to improve their courses which are captured through the Faculty Performance Profiles done every July.

A sample self-study statement for CHE 301 (Chemical Engineering Thermodynamics), CHE 341 (Chemical Process Control) and CHE 433 (Process Simulation Using Aspen Plus) are shown in Exhibit 4-8 (section 4.C). Self-study statements written by faculty members and FPP course

recommendations and actions will be available for inspection during the upcoming ABET site visit.

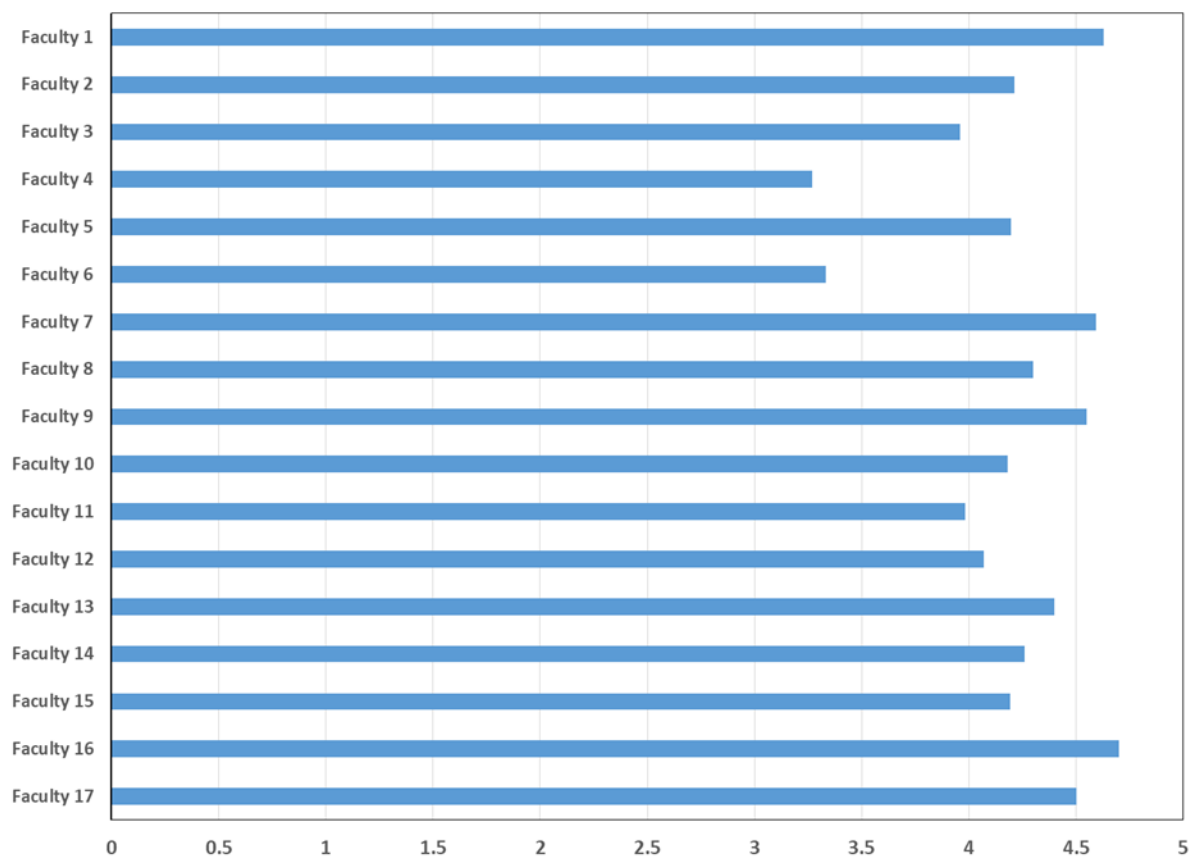


Figure 4.1. Average course evaluations for CHE faculty for the last five years (2015-2020).

4.B.2. Ongoing Feedback-Based Improvement of CHE 381/382 Chemical Engineering Laboratory I and II.

Based on student feedback from the Course Evaluation Surveys, improvements to the Chemical Engineering Laboratory courses (CHE 381 and 382) has focused on three areas since 2015; new laboratory experiments and repair of existing lab equipment, revisions of lab manuals and procedures, and increasing the number of lab sections to allow for smaller lab teams. Three new lab experiments were introduced into the lab curriculum; Gas Absorption, Distillation and Process Control. A new distillation apparatus was purchased using a teaching-equipment procurement budget from the Dean's Office in the College of Engineering in 2015. The new distillation column was a considerable upgrade from the previous distillation column, allowing the students to conduct more precise and meaningful experiments. A process control system (level control) was also purchased using the same teaching-equipment procurement budget to provide hands-on experience with process control principles and as a supplement to the Process Control course taught senior year. It now gives our students a unique advantage as many chemical engineering programs do not have process control experimentation in their curriculum. In addition, an existing but dormant gas absorption experiment was made operational to allow two unit operation principles to be taught in the same experiment; packed bed and gas

absorption. Secondly, the lab manuals for all 12 experiments were re-written and modernized to improve the lab procedures, and improve lab safety and sustainability. For example, a lab procedure was revised to reduce mineral oil usage in the Liquid Extraction Experiment. Due to increasing enrollment, the one-section lab course was first split into two sections and then into four sections to allow for smaller lab teams of 4 members each. This has allowed for more personalized instruction and a better student learning experience.

4.B.3. Ongoing Feedback-Based Improvement of CHE 499- Professional Development Seminar course.

Based on student feedback from the Course Evaluation Surveys, the CHE 499 Professional Development Seminar curriculum was revised to create a seminar series of 8-10 speakers throughout the semester. Invited speakers give talks on topics related to chemical engineering practice, professional development and job searching. They come from a variety of industries such as oil/gas, pharmaceutical, fine chemicals, food, patent law, alternative energy and semiconductors. The UIC Engineering Career Center also gives two seminars on job-related topics. Due to increasing enrollment, the oral exam and interview were changed to a written examination that is given to the students as a take-home assignment. In addition, an ethics quiz was added to the course based on discussions and feedback from the College Education Policy Committee to enhance student learning of ethics in the engineering curriculum. This will be discussed in more detail in the following section 4.B.5.

4.B.4. Ongoing Improvements to Extracurricular Student Learning Experiences: Career Development, Industry/Student Networking, Alumni/Student Connections

In 2015, two undergraduate students approached the Engineering Career Center to ask how the College might improve the amount of companies attending the Engineering Career Fair hiring chemical engineers. This initial discussion led to the formation of an ad-hoc committee to brainstorm ideas on improving industry/student networking for undergraduate chemical engineering students. The ad-hoc committee consisted of the two undergraduate students, the director of the Engineering Career Center, and the Director of Undergraduate Studies in Chemical Engineering. After many discussions over a six-month period, the committee recommended expanding our industrial liaisons by creating an annual “Chemical Engineering Industry Day” event. The faculty voted to approve this event and it debuted in Fall 2015 as round-robin table discussions between the students and 18-20 professional chemical engineers working in a variety of industries. Industry Day has been a popular event and has continued each year in the Fall semester. After receiving feedback from the industry attendees and students in 2018, the event was improved to include a 6-speaker panel discussion and a high-top table networking hour at the end of the afternoon to promote further mingling and discussions between the students and the professional engineers. This activity fits into the department policy of providing the students with a learning experience to obtain the necessary skills and opportunities to become successful in their careers.

Another initiative that was spearheaded by the student chapter of AIChE in Spring 2016 is the ChemE Forum. The ChemE Forum consists of an afternoon of seminars on a broad area of topics that are pertinent to chemical engineering and professional development. Seminar topics have

included a safety tutorial by a safety consultant firm, a seminar on FE test preparation, and short presentations by the faculty on their research areas. The Forum is open to all undergraduates and we also obtained faculty buy-in so that classes on that day are canceled as well as any quizzes or exams. Another unique aspect of the ChemE Forum is that an alumni mixer event occurs at the end where current students are encouraged to stay and talk to the UIC chemical engineer alumni. This event has also been very popular, and has continued each Spring semester since 2016. After student and alumni feedback, we added an alumni panel discussion before the alumni mixer to discuss a myriad of pertinent topics, with questions from the students. During the last ChemE Forum in 2019, we added a Jeopardy game with chemical engineering and UIC related questions after the alumni panel discussion. The game has been a huge hit with the alumni and students. This activity fits into the department goal of increasing our sense of community and student identity as chemical engineers and to provide a more enriching educational experience to our students.

Based on our collaborations with the Engineering Career Center, we have also enhanced industry/student and alumni/student interactions by initiating a LinkedIn UIC Chemical Engineering group, and becoming more involved in industry activities such as setting up a UIC Chemical Engineering booth at the ProFood Expo, and working closely with the Engineering Career Center to boost companies hiring ChE's at the UIC Career Fair.

4.B.5. Improvement in the Ethics Content of the Curriculum

Based on feedback from the industrial mentors in the senior design course, an assignment on ethics was added to the topics in CHE 396 and CHE 397 in 2017. Ethics is also covered in several other courses (CHE 201, CHE 341, CHE 433), but on a more informal level, where ethical situations related to engineering practice are given and discussion in class. An ethics quiz was also added to the CHE 499 course in 2019 based on discussions and feedback from the College Education Policy Committee to enhance student learning of ethics in the engineering curriculum. Ethics as a rubric item is assessed in SO #4 and has shown a steady improvement of scores both in the course assessment and senior written examination assessment. In addition, the indirect assessment from the Senior Exit Survey of SO #4 indicates that the student feel very strongly that they have achieved the skills for this rubric item.

4.B.6. Ongoing Improvements to the Safety Content of the Curriculum.

The incorporation of safety-related material into the Chemical Engineering curriculum was initiated in the last ABET cycle (2008-2014) to emphasize the importance of safety training for students and is now widely incorporated into many courses. The safety rubric is specifically tracked and assessed using SO #4-Safety (see Exhibit 4-6: Level 2 Analysis of Assessment Data). Specific SaChE tutorials from the AIChE website are now used regularly in several courses (CHE 201, 381, 382, 396 and 397) and include the following titles; "Introduction to Process Safety", "Hazard Recognition", "Dust Explosions", and "Chemical Reactivity Hazards". After suggestions from the senior design industry mentors, we also incorporated additional safety assignments into the CHE 396 and CHE 397 Senior Design I and II courses. These safety-related assignments discuss the hazards associated with the specific processes that the student

groups are designing and include a HAZOP analysis in the form of a report and presentation. In addition, self-study reflection from the CHE 381 and 382 Course Evaluation Surveys led to the addition of two safety assignments in 2015; an industrial R&D Chemical Hygiene Plan, and a Hazardous Chemical Safety Course reading assignment with a quiz to grade comprehension. Finally, many of our undergraduates perform research with the faculty in their research laboratories through the CHE 392 Undergraduate Research course. These students are now required to complete a UIC Environmental, Health and Safety Office (EHSO) Laboratory Training seminar in order to work in the labs.

4.B.7. Improvement of the ENGR100- Freshmen Engineering Success Program (FESP) Chemical Engineering Breakout Session

In 2014, the College of Engineering initiated a program in the freshmen engineering orientation course (ENGR 100) called the ‘Freshmen Engineering Success Program’ (FESP). The program is designed to increase retention of the freshmen engineering students, help them start identifying as engineers, and provide opportunities to meet faculty and upperclassmen in their majors. In addition, it is a requirement to be considered for the Guaranteed Paid Internship Program (GPIP) that is funded by the College. One part of FESP are 5-week breakout sessions within each specific major that include a team-based technical project. The FESP committee, consisting of faculty from each engineering major, periodically meet to discuss how to improve the breakout sessions and other aspects of FESP. Based on this feedback and discussions with the chemical engineering faculty, we have improved the technical project by making it a hands-on activity rather than a research term paper project. Currently the students build a solar cell, perform experiments, analyze the data, and write a team report. We are currently working to make the further improve the technical project so that it stays as a more personalized experience for the freshmen in their departments, but is sustainable as the engineering student enrollment increases to record levels.

4.B.8. Future Improvements to the Curriculum: New Concentrations to the BS major and New Courses Aligned with the Concentrations.

Based on discussions and feedback from our key constituencies and our faculty, a proposal has recently been approved by the chemical engineering faculty and the College, and is now being considered for approval by the university to modify our undergraduate program by adding five optional concentrations to the BS degree:

1. Concentration in Polymers and Molecular Engineering
2. Concentration in Energy and Environment
3. Concentration in Nanotechnology
4. Concentration in Process Automation
5. Concentration in Entrepreneurship

The BS in chemical engineering major currently has one optional concentration, “Biochemical Engineering”. The addition of the five concentrations will strengthen the curriculum by providing more in-depth studies in key areas of chemical engineering; polymers, energy, environment and process automation. The department will also be more competitive with other

chemical engineering programs by providing additional topic choices in emerging and multi-disciplinary fields of study; molecular engineering, nanotechnology and entrepreneurship.

In addition to the proposed concentrations, several new courses have been proposed to provide the curriculum for the 5 new concentrations. The proposed courses, CHE 230 Molecular Systems in Chemical Engineering, and CHE 330 Polymer Science, will introduce several topics that are important to traditional and emerging fields in chemical engineering; polymers, molecular engineering, macromolecular systems, and nanomaterials. The proposed courses; CHE 454 Molecular and Macromolecular Engineering, CHE 455 Nanoscale Systems in Chemical Engineering, and CHE 457: Colloidal and Interfacial Phenomena will expand the list of technical electives in chemical engineering and will provide more comprehensive knowledge of the above topics; polymers, molecular and macromolecular systems, and nanomaterials. A technical elective in colloids and interfacial phenomena will also be added to provide comprehensive knowledge in this important field of study. The addition of these 5 courses will strengthen our core curriculum in key areas to modern chemical engineering practice where they play a central role in biological, chemical/industrial, environmental and pharmaceutical processes and technologies. Ultimately, they will provide the curriculum for the 5 new chemical engineering concentrations that will be proposed to our core program; Polymer and Molecular Engineering, Nanotechnology, Energy and Environment, Process Automation, and Intellectual Property

Continuous Improvement Documentation:

In addition to the self-study statements written by faculty members, for the ABET site visit we will also present the following documentation to support the continuous improvement cycles described above (4.B.1 to 4.B.8):

- Minutes of departmental faculty meetings.
- Minutes of undergraduate committee meetings.
- Minutes of College EPC meetings.
- Agenda/minutes from the Chemical Engineering Curriculum Day in 2018.
- Notes/minutes from the advisory board meetings.
- Results from the advisory board survey in 2019.
- Notes/minutes from the Industry Day adhoc committee, 2015-2019.
- Results from the Industry Day panel speaker survey in 2019.
- Results from the alumni survey in 2019.
- Notes/minutes from the FESP committee meetings.
- Notes/minutes from the chemical engineering education committee.

C. Additional Information

The following documents provide samples of the assessment, evaluation and continuous improvement procedures described in Sections 4.A and 4.B. A complete set of documents will be provided at the ABET site visit.

Exhibit 4-1.

Assessment Method #1 (Direct / Formative):

Rubric-based assessment of a major assignment in required courses.

Sample Description: Spring 2020 Assessment assignment, rubric and results for CHE 301.

ChE 301 – Chemical Engineering Thermodynamics –Spring 2020

Student Learning Outcome 1

Rubric for assessment problem on final exam (Problems #1, 2, 4, 5)

- 1. Tank Pressurization Calculations with Redlich-Kwong Equation of State.**
- 2. Air Dew Point Calculations**
- VLE Data Reduction with the Peng-Robinson Equation of State
- 4. Excess Enthalpy Models for Isopropanol-Water Mixtures**
- 5. VLE Data Reduction with the NRTL Model**

SLO 1 Performance Indicator (PI)	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
1. Apply mathematical and scientific principles to formulate models and systems relevant to the subject Problem 1A,1B 100 pts	Does not understand the connection between mathematical models and the system or process to be analyzed or designed (<50 pts)	Chooses a mathematical model or scientific principle that applies to an engineering problem, but has trouble in model development (50-70 pts)	Able to successfully combine mathematical and/or scientific principles to formulate models and systems relevant to chemical engineering (>70 pts)
2. Solve engineering problems by using the concepts of algebra, calculus, differential equations and/or linear algebra Problem 4A,4B,4C 100 pts	Does not understand the application of algebra, calculus, differential equations and/or linear algebra in solving chemical engineering problems (<50 pts)	Shows nearly complete understanding of applications of calculus and/or linear algebra in problem-solving (50-70 pts)	Applies concepts of algebra, calculus, differential equations and/or linear algebra to solve chemical engineering problems (>70 pts)
4. Translates academic theory into engineering applications Problem 2,5A,5B,5C 100 pts	Does not appear to grasp the connection between theory and the problem (<50 pts)	Some gaps in understanding the application of theory to the problem and expects theory to predict reality (50-70 pts)	Translates academic theory into engineering applications and accepts limitations of mathematical models of physical reality (>70 pts)

ChE 301 – Chemical Engineering Thermodynamics –Spring 2020

Student Learning Outcome 1

RESULTS BY PERFORMANCE INDICATOR			
Rubric #	1	2	4
Mapped SO	1	1	1
Average	2.56	2.78	2.44
Std Dev	0.78	0.65	0.86

Average scores were fairly uniform although there was some variation among students. More students were strong in solving engineering problems using theoretical thermodynamic concepts, and concepts of algebra, calculus, differential equations and/or linear algebra. Overall performance was satisfactory for all performance indicators.

Problem 1: Tank Pressurization Calculations with the Redlich-Kwong EOS

A cylindrical tank with volume $\{V = 25\text{ m}^3\}$ is originally filled with nitrogen at conditions of $\{T_o = 25^\circ\text{C}$ and $P_o = 1\text{ bar}\}$. The tank must be pressurized to a pressure of $\{P = 25\text{ bar}\}$ using a high pressure line which supplies nitrogen at a constant temperature and pressure equal to $\{T = 25^\circ\text{C}$ and $P = 50\text{ bar}\}$.

- A. Prove that by combining the dynamic energy and mass balances we obtain the equation:

$d(MU) = H dM$ where M is the mass(in moles) of nitrogen inside the tank at any time, U is the molar internal energy of nitrogen inside the tank and H is the molar enthalpy of the nitrogen in the high pressure line feeding the tank

- B. Integrate the equation obtained above and calculate the amount of nitrogen added to the tank during the pressurization process using (a) the ideal gas law and (b) the Redlich-Kwong equation of state. What is the error in assuming that the gas is ideal?

Exhibit 4-2.

Assessment Method #2 (Direct / Summative):

Senior Written Exam

Sample Description: Spring 2015 Senior Written Examination and rubrics.

(See next 4 pages)

BSCHE Senior Written Examination- Spring 2015

Scoring Rubrics Mapped to Student Learning Outcomes and Program Educational Objectives

Professionalism and Broader Knowledge

Question #	Performance Indicator (PI)	SLO	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
1	1 Background research on career options and plans	7	No significant research. Unfamiliar with application deadlines, interview procedures and employer policies.	Has checked some information on specific universities or companies on the internet. Actively seeking job interviews or knows relevant deadlines for graduate schools.	Can quote employer policies (e.g., career tracks, promotions, sponsorship of advanced degrees). Has submitted applications, has job interviews scheduled, or already accepted for job or graduate school.
2a	2. Ethical and societal ramifications of technology	4	Unable to think of an example or cannot identify ethical ramifications of supplied technological topic.	Can independently come up with an example or identify an ethical ramification of supplied example.	Clearly explains ethical impact or social issues related to technological topic.
2c	3. Approach to ethical dilemmas	4	Unable to recognize ethical dilemma in hypothetical scenario. Purely ad hoc or “gut feel” approach toward ethical decisions. Does not know where or to whom to turn for guidance. Confounds fact-finding with ethical consideration.	Recognizes ethical dilemma in hypothetical scenario (or own experience). Logical considerations toward resolution. Can formulate steps to seek guidance.	Articulates resolution of dilemma as outcome of systematic thought process or ethical framework. Familiar with channels to air ethical concerns and/or cognizant officers.
2d	4. AIChE Code of Ethics	4	Unfamiliar with existence or content of code.	Knows where to find code and is familiar with some provisions.	Has read code more than once and can quote main provisions.
3	5. Multidisciplinary teamwork	5	Cannot name any collaborating engineering fields or explain context of collaboration.	Names one other type of engineer and articulates some aspects of collaboration.	Names two other types of engineer and clearly articulates example or specific contexts in which his/her work could intersect.

Technical Assessment

Question #	Performance Indicator (PI)	SLO	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
4	6. Description of numerical method and formulation of algorithm	1,2	Does not know how to proceed and/or unable to get started.	Describes most of numerical method but shows some gaps in understanding or is cannot formulate algorithm.	Describes numerical method correctly and successfully formulates algorithm.
4	7.Proficiency with Excel or MATLAB	1	Unfamiliar with Excel, MATLAB (and any other math package). Unable to get started or needs extensive consultation of on-line manual to get started. Incomplete or incorrect solution.	Able to do significant portion or most of problem with some quick consultation of on-line help.	Fluent in finishing problem completely with little or no consultation of online help.
5	8.Identification of system, surroundings and applicable flow streams	1,6	Does not mention system or surroundings or falters seriously in implementation.	Provides many details of system, surroundings and flow streams correctly	Complete and correct formulation of engineering system.
5	9.Effective problem solving skills	1	Unable to translate physical ideas into mathematical form. Ineffective solution.	Formulates mathematical problem substantially and makes substantial progress toward correct solution	Correct mathematical formulation and completion of correct solution, including units

RESULTS BY PERFORMANCE INDICATOR:

Rubric item (PI)	1	2	3	4	5	6	7	8	9
Mapped SLO	7	4	4	4	5	1,2	1	1,6	1
Average	2.76	2.54	2.57	2.13	2.63	2.57	2.59	2.37	2.41
Std Dev	0.42	0.72	0.54	0.88	0.53	0.62	0.58	0.68	0.69

RESULTS BY STUDENT OUTCOME ALL ASSESSMENTS

	1	2	4	5	6	7
Average Score	2.48	2.57	2.41	2.63	2.37	2.76

BSCHE Graduate Written Examination – Spring 2015

Written examination conducted with graduating seniors in Chemical Engineering

Name:

Question 1

Career plans. (Scoring rubric was designed to assess potential for life-long learning.)

Where do you see yourself in 5-10 years, and what steps will you take to get there?

How have you researched your career plans and options (companies, graduate schools, PE certification, etc.)?

Does your prospective employer sponsor master's degree or MBA? Under which terms?

Question 2

Professional ethics and ethical ramifications of technology.

Can you name a recent news item or significant piece of legislation that impacts on technology or engineering, and comment on its ethical ramifications?

Have you experienced an ethical dilemma in your educational or work (e.g., co-op) experience?

How would you approach an ethical dilemma? Where or to whom would you turn for guidance? Can ethical problems be approached in terms of a systematic methodology, or are you guided more by gut reaction, personal experience, upbringing, etc.?

Are you familiar with the AIChE Code of Ethics? Can you name specific provisions in it?

Which courses at UIC have contributed most to your views on professional ethics? How?

Question 3

Working in multidisciplinary teams

Can you name two other types of engineers that you might find yourself working with?

In what context or application would you be collaborating with them?

Question 4

Numerical methods and computational proficiency (Excel, MATLAB or Maple): choose one problem.

(a) Generate a graph of the solution of the following ODE initial value problem.

$$\frac{dx}{dt} = \frac{1}{\tau} [x_{\text{in}}(t) - x(t)] - kx(t)^2, \quad x(0) = 2$$
$$x_{\text{in}}(t) = 1 + \varepsilon \sin(\omega t)$$

with the constants $\tau = 3.4$, $k = 1.2$, $\varepsilon = 0.1$ and $\omega = 5$. Can you think of a chemical engineering problem (or, at least, a course) in which such an ODE would arise?

(b) Find a numerical value for the following definite integral.

$$\int_1^7 \frac{dx}{1 + \cos^3 x + \ln x}$$

Question 5

Engineering estimate.

The waterwheel shown in the following three pictures was observed to make one complete revolution every 15 seconds. Estimate the power output (in units of horsepower)?

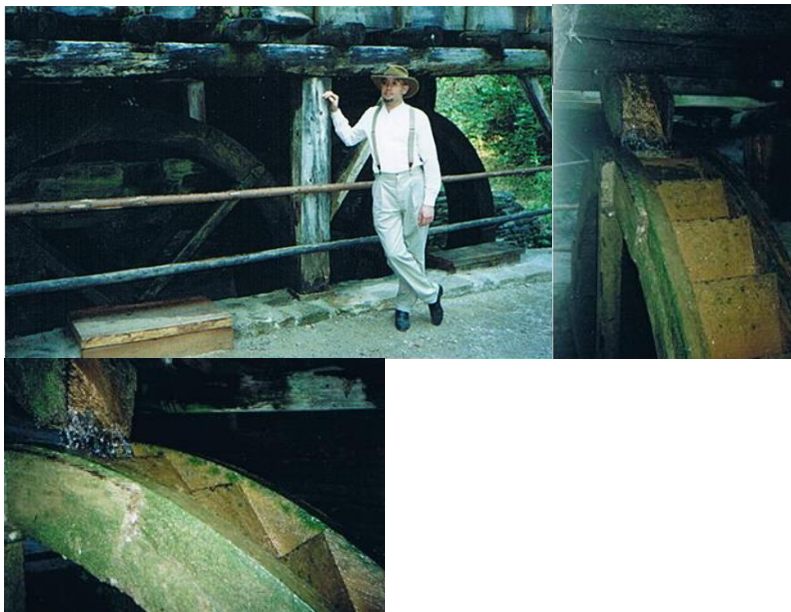


Exhibit 4-3.*Assessment Method #3 (Direct / Summative):**Judging rubrics from the annual Engineering Expo.***Sample Description: BSCHE Project Team: Judge Score Assessment- Spring 2019**

Scoring Rubrics Mapped to Student Learning Outcomes and Program Educational Objectives

Technical Quality

Performance Indicator (PI)	SLO	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
1. Identification of project objective(s)	2	No design strategy; haphazard approach	Uses a design strategy with guidance	Develops a design strategy, decomposition of work into subtasks
2. Evaluation of design alternatives with realistic constraints (economic, environmental, social, ethical)	2	No consideration of economics, safety, and environment	Can use prior knowledge to design individual pieces of equipment competently when guided to do so	Understands how areas interrelate and demonstrates ability to integrate prior knowledge into a new problem
3. Consideration of factors influencing the design (cost, manufacturability, sustainability, maintainability)	2	Cannot design processes or individual pieces of equipment without significant amounts of help	Can follow a previous example competently	Suggests new approaches and improves on what has been done before
4. Overall fulfillment of project objectives	2	No consideration of economics, safety, and environment	Design is done, but procedures and equations are not documented or referenced	Develops a solution that includes economic, safety, environmental and other realistic constraints
5. Project innovation	2	Unable to relate prior knowledge to the design problem	Can use prior knowledge to design individual pieces of equipment competently when guided to do so	Understands how areas interrelate and demonstrates ability to integrate prior knowledge into a new problem
6. Project practicality in view of contemporary engineering standards/practice	2	No application of engineering and/or scientific principles	Includes only minor or cursory consideration of economic, safety, and environmental constraints	Applies engineering and/or scientific principles correctly to design practical processes

Presentation Quality

Performance Indicator (PI)	SLO	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
7. General professionalism and oral communication	3	Talk is poorly organized, e.g. no clear introduction or summary of talk is presented	Presents key elements of an oral presentation adequately, but "tell them" not clearly applied	Plans and delivers an oral presentation effectively; applies the principle of "(tell them)" -- well organized
8. Knowledge of the background material	3	Unable to relate prior knowledge to the design problem	Can use prior knowledge to design individual pieces of equipment competently when guided to do so	Understands how areas interrelate and demonstrates ability to integrate prior knowledge into a new problem
9. Organization & quality of presentation material	3	Organizes materials in a logical sequence to enhance the reader's comprehension	Material organized well, but some sections and sub-sections are not identified clearly	Graphs, tables or diagrams are not easy to understand
10. Project description (see Program Book)	3	Text rambles, points made are only understood with repeated reading, and key points are not organized	Articulates ideas, but writing is somewhat disjointed, superfluous or difficult to follow	Articulates ideas clearly and concisely
11. Group dynamics/member participation	3	No interaction between team members. One member only talks	Interaction between team members, but one person leads talk	Discussion by all team members; all understand project
12. Quality, effectiveness, & use of visual aids	3	Multiple slides are unclear or incomprehensible. Presents well mechanically: · Makes eye contact · Can be easily heard · Speaks comfortably with minimal prompts (notecards) · No distracting nervous habits	Visual aides have minor errors or are not always clearly visible. Has some minor difficulties with the mechanical aspects of the presentation · Eye contact is sporadic · Occasionally difficult to hear or understand speaking	Uses visual aids effectively. Presents well mechanically: · Makes eye contact · Can be easily heard · Speaks comfortably with minimal prompts (notecards) · No distracting nervous habits

Results

RESULTS BY PERFORMANCE INDICATOR: Prelab, Final Lab Reports												
Rubric Item (PI)	1	2	3	4	5	6	7	8	9	10	11	12
Mapped SLO	2	2	2	2	2	2	3	3	3	3	3	3
Average	2.96	2.68	2.88	2.92	2.64	2.88	2.88	2.88	2.84	2.88	2.80	2.76
Std Dev	0.20	0.48	0.33	0.28	0.49	0.33	0.33	0.33	0.37	0.33	0.41	0.44

RESULTS BY STUDENT OUTCOME ALL ASSESSMENTS	2	3
Average Score	2.83	2.84
Standard dev.	0.13	0.05

Exhibit 4-4.

*Assessment Method #4 (Indirect / Summative):
Senior Exit Survey.*

Sample Description: Fall 2016, PDF printout of online survey (PHP/CSS file).

(1.) Based upon your experiences as a Chemical Engineering student at UIC, please rate your skills in the following categories, using the following criteria:

(1 = Excellent, 2 = Strong, 3 = Moderate, 4 = Weak, 5 = Poor)

- 1.50 Ability to apply knowledge of mathematics, science and engineering.
- 1.63 Ability to design and conduct experiments, as well as to analyze and interpret data.
- 1.67 Ability to design a system, component, or process to meet desired needs.
- 1.39 Ability to function on multi-disciplinary teams.
- 1.50 Ability to identify, formulate, and solve engineering problems.
- 1.26 Understanding of professional and ethical responsibility.
- 1.45 Ability to communicate effectively.
- 1.46 Broad education necessary to understand the impact of engineering solutions in a global and societal context.
- 1.40 Recognition of the need for, and an ability to engage in life-long learning.
- 1.85 Knowledge of contemporary issues.
- 1.57 Ability to use the techniques, skills, and modern engineering tools necessary for engineering practice.
- 1.73 Overall, do you feel that your undergraduate degree will prepare you for your career?

(2.) Which courses or topics were the most interesting or useful?

(3.) Which courses or topics were the least interesting or useful?.

(4.) What courses/topics should we consider deleting or adding to the curriculum?

(5.) Do you have any additional comments or suggestions?.

(6.) How many years/months did it take for you to complete your B.S. degree in chemical engineering?

(7.) Did you work part-time while you were a student? If yes, then approximately how many hours per week?

(8.) On average, how much many hours did you spend on each ChE course outside of class?

(9.) Did you participate in any summer research/internship program at UIC or at other schools? Did you complete an internship at a company? What company? Was this a useful experience? What specifically did you learn that was most useful?

(10.) Have you found a job? If so, in which company and industry? Please indicate your job title or specialization.

(11.) Are you planning to attend a graduate or a professional school? If so please list the name of the school and the specific program (M.S./Ph.D/M.D.) that you will enroll in, or those programs that you have looked into.

(12.) Do you have a strategy in mind for continued professional development and/or continue learning? For example, do you intend to take the FE/PE exam at some point in your career? If so, when?

Exhibit 4-5.**Assessment Method #5 (Feedback):****Course Update Survey.****Sample Description:** Spring 2020, PDF printout of online survey (PHP/CSS file).

6/11/2020

Course Feedback Survey

**UIC Department of Chemical Engineering
Anonymous Student Feedback Survey for Updating and Improving Courses
Spring 2020**

ChE Course

201	205	210	301	311	312	313	321	341	382	397	422	425
☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

(1.) How many courses did you take this term?

1	2	3	4	5	6
☐	☐	☐	☐	☐	☐

(2.) Approximately how many hours per week did you devote to **all of your courses** outside of lectures?

1-5	6-10	11-15	16-20	31-30	31-40
☐	☐	☐	☐	☐	☐

(3.) Approximately how many hours per week did you devote to **this course** outside of lectures?

1-3	4-6	7-9	10-12	13-15	15-20
☐	☐	☐	☐	☐	☐

(4.) Please rate usage, relevance and/or applicability of the following subjects to this course.

	None	Minor	Significant	Extensive
Calculus and vector calculus	☐	☐	☐	☐
Differential equations	☐	☐	☐	☐
Chemistry	☐	☐	☐	☐
Excel spreadsheets	☐	☐	☐	☐
Math packages (Maple / Matlab / Mathcad)	☐	☐	☐	☐
Computer programming (Fortran / C++)	☐	☐	☐	☐
Aspen Plus	☐	☐	☐	☐
Social/societal and geopolitical issues	☐	☐	☐	☐
Environmental, regulatory, health and safety issues	☐	☐	☐	☐
Energy and natural resource issues	☐	☐	☐	☐
Professional ethics and responsibilities	☐	☐	☐	☐
Analysis and interpretation of data	☐	☐	☐	☐
Design of system, component or process to meet desired needs	☐	☐	☐	☐
Application of engineering principles to solve engineering problems	☐	☐	☐	☐
Formulation of engineering problems	☐	☐	☐	☐

(5.) Please comment on the balance of coverage between theoretical derivations versus practical problems and examples.

(6.) Were there any problematic or unfamiliar mathematical concepts utilized in this course that would have benefited from preparation or exposure in previous classes?

(7.) Were there any computational methods utilized in this course that would have benefited from preparation or exposure in previous classes?

(8.) How useful was the class text? Would you prefer an alternative textbook (suggestions and comments)?

www.uic-che.org/assessment/students/courses_2020_spring/survey_course.php

1/2

6/11/2020

Course Feedback Survey

(9.) How well did the class follow the text? Can you make any suggestions in this regard?

(10.) Was the pace of lectures appropriate? Too fast? Too slow?

(11.) Was the difficulty and quantity of homework appropriate to the material?

(12.) Comment on the difficulty and applicability of exams to the course material.

(13.) Please list three most beneficial aspects of this course?

(14.) Please comment on REMOTE LEARNING; quality of the online platform (Zoom), clarity and accessibility of instructor and lectures, use of other features (breakout rooms, etc).

(15.) Please comment on how well the online assignments were administered; exams, quizzes, homework, other assignments

(16.) How can we improve REMOTE LEARNING for this course? Additional comments.

Did this course have a teaching assistant(s)?

Yes No
☐ ☐

If yes, please give his/her names.

If yes, rate the teaching assistant with regard to the following aspects.

	Excellent	Very Good	Good	Fair	Poor
Approachability	☐	☐	☐	☐	☐
Helpfulness in office hours and/or with questions in general	☐	☐	☐	☐	☐
Knowledge of and background in the course material	☐	☐	☐	☐	☐
Grading of homeworks (helpful comments on papers, fairness) - if applicable	☐	☐	☐	☐	☐
Grading of exams and/or quizzes (helpful comments on papers, fairness) - if applicable	☐	☐	☐	☐	☐
Overall effectiveness	☐	☐	☐	☐	☐

Exhibit 4-6.*Level 2 analysis of assessment data.***Attainment of Student Outcome #1****An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.**

Assessment Method #1 (Direct / Formative): Course Rubrics

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)	67.2	67.8	66.9	70.9	69.6	73.2
% (2 = Developing)	19.3	17.8	23.2	18.8	20.9	18.6
% (1 = Unsatisfactory)	13.4	14.4	9.9	10.3	9.5	8.2
% > Developing	86.6	85.6	90.1	89.7	90.5	91.8
Average Rating	2.51	2.48	2.52	2.57	2.37	2.59

Assessment Method #2 (Direct / Summative): Senior Written Exam

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)		61.7	71.0	62.7	49.5	56.5
% (2 = Developing)		27.1	22.6	30.0	41.4	35.3
% (1 = Unsatisfactory)		11.3	6.3	7.3	9.1	8.2
% > Developing		88.8	93.7	92.7	90.9	91.8
Average Rating		2.40	2.59	2.46	2.29	2.48

Assessment Method #3 (Direct / Summative): Engineering Expo Rubrics

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)	N/A					
% (2 = Developing)						
% (1 = Unsatisfactory)						
% > Developing						
Average Rating						

Assessment Method #4 (Indirect / Summative): Senior Exit Survey

*Note different rating system than above Methods #1- #3:**(1 = Excellent, 2 = Strong, 3 = Moderate, 4 = Weak, 5 = Poor)*

Year	2020	2019	2018	2017	2016	2015
Average Rating	1.47				1.50	1.44

Attainment of Student Outcome #2

An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.

Assessment Method #1 (Direct / Formative): Course Rubrics

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)		82.5	80.1	84.0	55.6	64.4
% (2 = Developing)		14.5	12.9	13.1	19.4	24.0
% (1 = Unsatisfactory)		2.9	7.0	2.8	25.0	11.5
% > Developing		97.1	93.0	97.2	75.0	88.5
Average Rating		2.75	2.73	2.81	1.75	2.25

Assessment Method #2 (Direct / Summative): Senior Written Exam

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)		78.3	71.4	64.6	56.4	63.0
% (2 = Developing)		18.3	27.0	32.3	38.2	30.4
% (1 = Unsatisfactory)		3.3	1.6	3.1	5.5	6.5
% > Developing		96.7	98.4	96.9	94.5	93.5
Average Rating		2.70	2.70	2.62	2.51	2.57

Assessment Method #3 (Direct / Summative): Engineering Expo Rubrics

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)	64.9	82.7	72.8			76.2
% (2 = Developing)	34.4	17.3	27.2			23.8
% (1 = Unsatisfactory)	0.7	0.0	0.0			0.0
% > Developing	99.3	100.0	100.0			100.0
Average Rating	2.64	2.83	2.73			2.74

Assessment Method #4 (Indirect / Summative): Senior Exit Survey

Note different rating system than above Methods #1- #3:

(1 = Excellent, 2 = Strong, 3 = Moderate, 4 = Weak, 5 = Poor)

Year	2020	2019	2018	2017	2016	2015
Average Rating	1.61				1.62	1.60

Attainment of Student Outcome #3**An ability to communicate effectively with a range of audiences.**

Assessment Method #1 (Direct / Formative): Course Rubrics

Oral

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)		75.0	82.6			77.4
% (2 = Developing)		23.4	15.7			22.6
% (1 = Unsatisfactory)		1.6	1.7			0.0
% > Developing		98.4	98.5			100.0
Average Rating		2.72	2.75			2.63

Written

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)		88.6	76.9	76.6	70.0	73.5
% (2 = Developing)		8.4	18.8	22.2	29.3	25.8
% (1 = Unsatisfactory)		3.1	4.3	1.2	0.8	0.7
% > Developing		96.9	95.7	98.8	99.3	99.3
Average Rating		2.86	2.73	2.75	2.69	2.72

Assessment Method #2 (Direct / Summative): Senior Written Exam

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)	N/A					
% (2 = Developing)						
% (1 = Unsatisfactory)						
% > Developing						
Average Rating						

Assessment Method #3 (Direct / Summative): Engineering Expo Rubrics

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)	76.2	84.0	82.8			92.9
% (2 = Developing)	22.7	16.0	16.4			7.1
% (1 = Unsatisfactory)	1.1	0.0	0.8			0.0
% > Developing	98.9	100.0	99.2			100.0
Average Rating	2.75	2.84	2.82			2.90

Assessment Method #4 (Indirect / Summative): Senior Exit Survey

*Note different rating system than above Methods #1- #3:**(1 = Excellent, 2 = Strong, 3 = Moderate, 4 = Weak, 5 = Poor)*

Year	2020	2019	2018	2017	2016	2015
Average Rating	1.40				1.45	1.46

Attainment of Student Outcome #4

An ability to recognize ethical, safety, and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.

Assessment Method #1 (Direct / Formative): Course Rubrics

Safety and Environment (S&E)

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)	100.0	90.1	99.2	89.7	100.0	100.0
% (2 = Developing)	0.0	8.89	0.8	9.5	0.0	0.0
% (1 = Unsatisfactory)	0.0	1.1	0.0	0.8	0.0	0.0
% > Developing	100.0	98.9	100.0	99.2	100.0	100.0
Average Rating	3.00	2.90	2.99	2.89	3.00	3.00

Global, economic, societal, ethics (GESE)

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)	90.3	63.1	88.5	65.6		
% (2 = Developing)	6.5	36.9	1.8	29.5		
% (1 = Unsatisfactory)	3.2	0.0	9.8	4.9		
% > Developing	96.8	100.0	90.2	95.1		
Average Rating	2.87	2.62	2.79	2.61		

Assessment Method #2 (Direct / Summative): Senior Written Exam

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)		82.8	83.1	76.9	69.1	57.2
% (2 = Developing)		13.9	16.9	22.1	23.6	26.8
% (1 = Unsatisfactory)		3.3	0.0	1.0	7.3	15.9
% > Developing		96.7	100.0	98.5	92.7	84.1
Average Rating		2.71	2.74	2.65	2.52	2.41

Assessment Method #3 (Direct / Summative): Engineering Expo Rubrics

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)	N/A					
% (2 = Developing)						
% (1 = Unsatisfactory)						
% > Developing						
Average Rating						

Assessment Method #4 (Indirect / Summative): Senior Exit Survey

Note different rating system than above Methods #1- #3:

(1 = Excellent, 2 = Strong, 3 = Moderate, 4 = Weak, 5 = Poor)

Year	2020	2019	2018	2017	2016	2015
Average Rating	1.30				1.26	1.50

Attainment of Student Outcome #5

An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.

Assessment Method #1 (Direct / Formative): Course Rubrics

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)		90.5	66.6	83.2	88.7	81.5
% (2 = Developing)		8.5	31.8	16.3	7.9	16.1
% (1 = Unsatisfactory)		1.0	1.6	0.4	3.4	2.4
97.6% > Developing		99.0	98.4	99.6	96.6	97.6
Average Rating		2.94	2.87	2.90	2.85	2.75

Assessment Method #2 (Direct / Summative): Senior Written Exam

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)		93.3	81.0	89.2	89.1	65.2
% (2 = Developing)		6.7	14.3	9.2	10.9	32.6
% (1 = Unsatisfactory)		0.0	4.8	1.5	0.0	2.2
% > Developing		100.0	95.2	98.5	100.0	97.82
Average Rating		2.70	2.56	2.67	2.71	2.63

Assessment Method #3 (Direct / Summative): Engineering Expo Rubrics

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)	N/A					
% (2 = Developing)						
% (1 = Unsatisfactory)						
% > Developing						
Average Rating						

Assessment Method #4 (Indirect / Summative): Senior Exit Survey

Note different rating system than above Methods #1- #3:

(1 = Excellent, 2 = Strong, 3 = Moderate, 4 = Weak, 5 = Poor)

Year	2020	2019	2018	2017	2016	2015
Average Rating	1.45				1.39	1.60

Attainment of Student Outcome #6

An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.

Assessment Method #1 (Direct / Formative): Course Rubrics

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)		73.9	79.3	66.9	66.7	69.6
% (2 = Developing)		25.2	19.0	33.1	31.9	29.0
% (1 = Unsatisfactory)		0.9	1.7	0.0	1.4	1.3
% > Developing		99.1	98.3	100.0	98.6	98.7
Average Rating		2.72	2.76	2.72	2.69	2.68

Assessment Method #2 (Direct / Summative): Senior Written Exam

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)	N/A					
% (2 = Developing)						
% (1 = Unsatisfactory)						
% > Developing						
Average Rating						

Assessment Method #3 (Direct / Summative): Engineering Expo Rubrics

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)	N/A					
% (2 = Developing)						
% (1 = Unsatisfactory)						
% > Developing						
Average Rating						

Assessment Method #4 (Indirect / Summative): Senior Exit Survey

Note different rating system than above Methods #1- #3:

(1 = Excellent, 2 = Strong, 3 = Moderate, 4 = Weak, 5 = Poor)

Year	2020	2019	2018	2017	2016	2015
Average Rating	1.54				1.63	1.54

Attainment of Student Outcome #7

An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

Assessment Method #1 (Direct / Formative): Course Rubrics

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)		84.1	70.6	53.0	89.3	96.0
% (2 = Developing)		14.3	21.1	47.0	11.0	4.0
% (1 = Unsatisfactory)		1.6	8.3	0.0	0.0	0.0
% > Developing		98.4	91.7	100.0	100.0	100.0
Average Rating		2.83	2.62	2.53	2.89	2.96

Assessment Method #2 (Direct / Summative): Senior Written Exam

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)		70.0	74.6	86.2	67.3	78.3
% (2 = Developing)		30.0	25.4	12.3	23.6	21.7
% (1 = Unsatisfactory)		0.0	0.0	1.5	9.1	0.0
% > Developing		100.0	100.0	98.5	90.9	100.0
Average Rating		2.50	2.56	2.63	2.44	2.76

Assessment Method #3 (Direct / Summative): Engineering Expo Rubrics

Year	2020	2019	2018	2017	2016	2015
% (3 = Satisfactory)						
% (2 = Developing)						
% (1 = Unsatisfactory)						
% > Developing						
Average Rating						

Assessment Method #4 (Indirect / Summative): Senior Exit Survey

Note different rating system than above Methods #1- #3:

(1 = Excellent, 2 = Strong, 3 = Moderate, 4 = Weak, 5 = Poor)

Year	2020	2019	2018	2017	2016	2015
Average Rating	1.51				1.57	1.75

Exhibit 4-7.*Level 3 analysis of assessment data.***Direct Assessment Method #1: Direct / Formative
Course Rubrics****3 = S = Satisfactory****2 = D = Developing****1 = U = Unsatisfactory**

Percentage of students attaining S

SO	2020	2019	2018	2017	2016	2015
1	67.2	67.8	66.9	70.9	69.6	73.2
2	82.5	80.1	84.0	55.6	64.4	82.5
3 oral		75.0	82.6			77.4
3 written	88.6	76.9	76.6	70.0	73.5	88.6
4 (S&E)	100.0	90.1	99.2	89.7	100.0	100.0
4 (GESE)	90.3	63.1	88.5	65.6		
5		90.5	66.6	83.2	88.7	81.5
6		73.9	79.3	66.9	66.7	69.6
7		84.1	70.6	53.0	89.3	96.0

Average rating among students

SO	2020	2019	2018	2017	2016	2015
1	2.51	2.48	2.52	2.57	2.37	2.59
2		2.75	2.73	2.81	1.75	2.25
3 oral		2.72	2.75			2.63
3 written	2.86	2.73	2.75	2.69	2.72	2.86
4 (S&E)	3.00	2.90	2.99	2.89	3.00	3.00
4 (GESE)	2.87	2.62	2.79	2.61		
5	2.94	2.87	2.90	2.85	2.75	2.94
6	2.72	2.76	2.72	2.69	2.68	2.72
7	2.83	2.62	2.53	2.89	2.96	2.83

(S&E): Safety and environmental**(GESE): Global, economic, societal, ethics**

Direct Assessment Method #2: Direct / Summative
Senior Written Examination

3 = S = Satisfactory

2 = D = Developing

1 = U = Unsatisfactory

Percentage of students attaining S

SO	2020	2019	2018	2017	2016	2015
1		61.7	71.0	62.7	49.5	56.5
2		78.3	71.4	64.6	56.4	63.0
3						
4		82.8	83.1	76.9	69.1	57.2
5		93.3	81.0	89.2	89.1	65.2
6						
7		70.0	74.6	86.2	67.3	78.3

Average rating among students

SO	2020	2019	2018	2017	2016	2015
1		2.40	2.59	2.46	2.29	2.48
2		2.70	2.70	2.62	2.51	2.57
3						
4		2.71	2.74	2.65	2.52	2.41
5		2.70	2.56	2.67	2.71	2.63
6						
7		2.50	2.56	2.63	2.44	2.76

**Assessment Method #3 – Direct / Summative
Engineering Expo Rubrics**

3 = S = Satisfactory

2 = D = Developing

1 = U = Unsatisfactory

Percentage of students attaining S

SO	2020	2019	2018	2017	2016	2015
1						
2	64.9*	82.7	72.8			76.2
3	76.2*	84.0	82.8			92.9
4						
5						
6						
7						

* Expo judging performed remotely due to pandemic

Average rating among students

SO	2020	2019	2018	2017	2016	2015
1						
2		2.83	2.73			2.74
3		2.84	2.82			2.90
4						
5						
6						
7						

Assessment Method #4 – Indirect / Summative
Senior Exit Survey

Average rating among students

Note: Different rating system than above Methods #1- #3:

(1 = Excellent, 2 = Strong, 3 = Moderate, 4 = Weak, 5 = Poor)

SO	2020	2019	2018	2017	2016	2015
1	1.47				1.50	1.44
2	1.61				1.62	1.60
3	1.40				1.45	1.46
4	1.30				1.26	1.50
5	1.45				1.39	1.60
6	1.54				1.63	1.54
7	1.51				1.57	1.75

Exhibit 4-8.

Continuous improvement – closing the feedback loop on assessment:

Sample Dates: Fall 2019, Fall 2019, Spring 2020.

Sample Description: Faculty self-studies for CHE 301, 341 and 433.

Self-Study Report for CHE301, CHE433 and CHE341

For the past five years that I have been teaching at the Chemical Engineering Department at UIC, my evaluations have been fairly consistent mostly in the range 4-5 with an overall score of Excellent.

The principal object or goal of my teaching method is to find a point of view from which the subject appears in its greatest simplicity. I strongly believe that the ultimate goal of teaching is to enhance learning without focusing on the testing aspect which tends to cause a lot of anxiety and uneasiness in students. As such, without sacrificing the fundamental aspects of my courses (Thermodynamics, Process Control and Simulation) I design problems not from available textbooks but from my own personal experience of over 30 years in the chemical, petrochemical and refining industries.

Although these problems are extremely challenging and sometimes with no obvious solutions, the students tend to enjoy them enormously and they always give positive feedback in their evaluations. (although they do complain about the overwhelming load)

One aspect of my teaching that seems to upset some (but not all) students is that I do not follow a traditional textbook. I recommend a number of textbooks in the beginning of each semester but I do not endorse or adopt any of them. It find it incredulous that I would have to exclusively follow someone else's idea on a particular engineering subject and especially Thermodynamics and Process Control. So I write my own notes from a variety of sources and my personal experience in industry and distribute them to the students for free. Most students enjoy the notes and give positive feedback but there is always a small number that prefers the traditional way of teaching. Furthermore I find it unconscious that the textbooks are so expensive and I don't expect my students to have the means to pay for these textbooks. As a result I will continue on my method of writing and improving and distributing my own notes. Hopefully in a short time I will be able to combine all my notes in a manuscript that I can distribute for free.

Another aspect of my teaching that seems to annoy some students is that I do not post solutions. I explain to them that the reason for that is, if I post the solutions most of the students copy them or take pictures with their smart devices and end up learning absolutely nothing. Instead I encourage the students, if they don't understand parts of the assignment to come to me (my door is always open) and we can have a conversation in order to enhance their learning.

Finally another major aspect from the evaluations appears to be my stringent requirements for attendance, tardiness and work ethics. Simply I cannot tolerate laziness and lack of respect from those who walk in late interrupting my lectures as they navigate through occupied seats to find an empty one. Attendance is mandatory in my lectures, late students are not allowed in the classroom and outstanding work ethic is rewarded handsomely in terms of improving the grades when I finally assign them. Some students seem to have a problem with that, but overall most of them support my methods.

To close up my self-study report I would like to stress again that my methods of teaching are designed to enhance learning without placing an unfair burden on the students through testing.

CRITERION 5. CURRICULUM

A. Program Curriculum

5.A.1. Study Plan.

Details of our Bachelors of Science in Chemical Engineering curriculum can be found in Table 5-1. The program is a semester based program that requires 8 semesters (4 years) to complete. A typical degree plan is shown in Figure 5-1.

5.A.2. Alignment of the Program Curriculum with Program Educational Objectives.

The Chemical Engineering PEOs (Section 2.B) describe career success of our graduates based upon depth, breadth, professionalism and skills in life-long learning — acquired in a rigorous and innovative learning environment.

Depth. Table 5-1 demonstrates that students are required to take 43 credit hours in advanced chemical engineering core courses plus 9 credit hours of supporting engineering courses. These core courses cover a wide range of topics beginning with an Introduction to Thermodynamics (CHE 201, 3 hours) typically taken by the fall of the sophomore year. This is followed by Computational Methods in Chemical Engineering (CHE 205, 3 hours) and the Material and Energy Balances course (CHE 210, 4 hours). CHE 205 is a distinctly chemical engineering mathematics and computation course, carefully formulated to preview key concepts from the core courses and develop the relevant problem-solving skills. CHE 210 introduces the critical concepts of control-volume balances as well as basic problems-solving skills for problems that may contain a large number of variables and unknowns. The introduction of these tools in the sophomore year allows the students to build a base of engineering design and computational expertise that they will further develop as they advance through the junior level courses and ultimately apply during their senior year in the Unit Operations Laboratory (CHE 381, 382) and Senior Design sequence (CHE 396, 397). In the junior year, students take the first two courses in Transport Phenomena (CHE 311 and 312, 6 hours), which cover fluid mechanics (momentum transfer), heat transfer, and mass transfer. In addition, students take an advanced course in Chemical Engineering Thermodynamics (CHE 301, 3 hours) in which students learn to apply equilibrium criteria to multiphase systems. The course in Chemical Reaction Engineering (CHE 321, 3 hours) which deals with the design of reactors and a third course in Transport phenomena (CHE 313, 3 hours) that covers separation processes complete the junior-level core in Chemical Engineering.

Senior-level courses complete the major in chemical engineering subject areas and complement the engineering science courses which dominate the junior year. These courses are: a two course sequence of Unit Operations Laboratory (CHE 381 and 382, 4 hours) in which transport, separation, and process control experiments are performed, data are analyzed and laboratory reports are written. The chemical process control course (CHE 341, 3 hours) introduces control theory and uses examples from all of the previous courses. Finally, the capstone two course Senior Design sequence (CHE 396 and 397, 8 hours) focuses on design of chemical processes and plants. In these courses, students demonstrate their ability to independently apply chemical engineering

principles, cost estimation, and economic evaluation methodologies, flow sheet synthesis and process simulation, optimization techniques, and report writing and oral presentation skill in order to address contemporary chemical engineering design problems. The Senior Design II course (CHE 397, 4 hours) involves a team oriented design project which is supervised by an industrial mentor. Senior Design and Unit Operations Laboratory require our students to demonstrate and synthesize a depth of knowledge in many important chemical engineering areas, including troubleshooting equipment, designing experiments, plant design, economics, process safety, and ethics, as well as building teamwork, communication and presentation skills. In addition, students are required to take a technical elective course (3 hours) which may be any 400-level chemical engineering graduate course or an independent research course (CHE 392, 3 hours) in which the student can pursue in more depth a particular topic of personal interest under the direct supervision of a departmental faculty member. These topics range from Process Simulation with Aspen Plus, a key skill used in Senior Design II, Renewable Energy Technologies, Nanotechnology for Pharmaceutical Applications, and a host of other topics. Students may also pursue a concentration in biochemical engineering which requires CHE 422 (3 hours) and supporting courses from outside of Chemical Engineering. Furthermore, it is worth mentioning that the department is in the process of adding five additional concentrations to the major; Polymers and Molecular Engineering, Energy and Environment, Nanotechnology, Process Automation, and Entrepreneurship. It is anticipated that the concentration will be approved in 2021. New courses will be developed and offered as requirements for the concentrations; Molecular Systems in Chemical Engineering, Polymer Science, Fundamentals of Electrochemistry, Molecular and Macromolecular Engineering, Nanoscale Systems in Chemical Engineering, Colloidal and Interfacial Phenomena.

The rigorous sequence of courses described above attests to the depth of knowledge with which Chemical Engineering emerge upon graduation and upon which they build their professional careers.

Breadth. The curriculum also requires students to demonstrate a breadth of knowledge. This includes a thorough grounding in chemistry and a working knowledge of advanced topics in chemistry such as organic, physical, and analytical chemistry. The courses that address the Program Educational Objective of breadth begin in the freshman year with an introduction to General Chemistry (CHEM 122/123 and 124/125, 10 hours). In the sophomore year students take a sequence of three courses in organic chemistry (CHEM 232, 233, and 234, 10 hours) which consist of two lecture courses and one laboratory course. In the junior year the chemistry courses are completed with a combination lecture/laboratory course in Analytical Chemistry (CHEM 222, 4 hours) and a two-course sequence in physical chemistry (CHEM 342 and 346, 6 hours). Students must also demonstrate a high level of mathematical skills in a four course sequence (MATH 180, 181, 210, and 220: 14 hours) taken during the freshman and sophomore years that involves differential and integral calculus, vector and multivariable calculus, differential equations and other advanced topics in mathematics. These mathematics skills are used throughout the curriculum. Also during the freshman and sophomore years a two-course sequence in physics (PHYS 141 and 142: 8 hours with a lab element) requires our students to demonstrate a basic grounding in mechanics and electromagnetism.

In addition to the basic science courses described above, our students must also demonstrate their breadth of technical knowledge by taking engineering courses outside of Chemical Engineering.

These are a course in Properties of Materials (CME 260: 3 hours), a course in Computer Programming (CS 109: 3 hours with a lab element) and a course in Electrical Circuit Analysis (ECE 210: 3 hours with a lab element). These courses add breadth by giving our students insight into other related engineering disciplines and prepares them for the multidisciplinary career environment in which they will work.

Professionalism. In the freshman year our students take a general Engineering Orientation course (ENGR 100: 0 hours) course that gives an introduction to the development of a career in engineering. The students learn about the engineering curriculum at UIC, including the various student organizations available to them. They also are introduced to the Engineering Career Center and learn about engineering careers of all varieties, and good professional skills and habits. Another important part of ENGR 100 are the major-specific breakout sessions where the students meet faculty and upperclassmen in their major and also participate in a technical project related to their discipline. This activity was added to bring them into the engineering community during their first semester, increase the retention of the students and also to help them begin to identify as engineers. The design course sequence described above (CHE 396 and 397: 7 hours) is also integral to our students' development of professionalism. The design courses bring in speakers from industry who discuss current issues and describe the requirements of a professional career in chemical engineering. Topics concerning safety, professional ethics, design strategies, business strategies, are discussed. In addition, the students are required to take a professional development seminar (CHE 499, 0 hours) which includes presentations by practicing chemical engineers and career development professionals. The students are exposed to the diverse career paths and industries where chemical engineers are employed, such as food processing, pharmaceuticals, oil & gas, specialty chemicals, and business/law. In addition, a variety of issues such as engineering ethics are covered and culminates in a written senior examination. Two other activities have been added that promote professionalism; ChE Industry Day and the ChemE Forum. Students interact with professional engineers in a panel and round-table discussion as well as a networking event. The ChemE Forum is an afternoon of engineering-related seminars that culminates in an alumni mixer that the students are invited so that they can interact with chemical engineering alumni on an informal level. Finally, all of our students are required to participate in the College of Engineering's Engineering Expo where the final design projects are presented and judged by engineers from industry. This requires our students to connect their design projects and the basic engineering science and design principles applied to the projects to the complex modern world and the many issues that must be addressed in any real world application of engineering.

Life-Long Learning. We address this Program Educational Objective with a curriculum that forces students to develop the knowledge, skills and mental orientation that will equip them to be life-long learners. For example, in the Senior Design sequence, students learn to write a literature review, which requires that they begin to explore the vast research literature available. They must synthesize knowledge from the literature and apply it to their specific design project. During the course of the semester, the students are asked to give presentations detailing the progress on their designs and thereby develop their oral communication skills. Finally they receive feedback not only from the course instructor and industrial mentor, but also from professional engineers who participate as judges at the Engineering Expo. Also, students discuss many current issues in chemical engineering including development of new energy sources and products. These topics often become the major theme for many of the design projects. Finally, our students are required

to participate in extensive one-on-one advising with a faculty member every semester. During the advising sessions, students are asked to plan their career or academic objectives and to consider how to best prepare to achieve the goals. This one-on-one mentoring is a valuable part of providing a learning environment that is suited to the individual student and to promote the skills and knowledge needed to pursue life-long learning. As mentioned above, the Professional Development Seminar (CHE 499) brings in industrial and professional speakers to open up our students' vistas on unexpected career paths, real-life experience in subfields of chemical engineering, and interaction with other types of engineers. Also, the ChemE Forum introduces the students to topics such as process safety, the FE/PE license, six-sigma certification and professional time management.

Finally, career options and advanced degrees relevant to pharmaceutical and medically relevant fields is enhanced by the Biochemical Engineering Concentration. To fulfill this option (which appears on the transcript), students are required to complete 9–10 semester hours in elective courses by taking the following courses: (i) CHE 422—Biochemical Engineering, (ii) two electives in non-major rubric category (i.e., outside Chemical Engineering) from among BIOS 350—General Microbiology (3 cr. hr.), BIOS 351—Microbiology Laboratory (2 cr. hr.), CHEM 352—Introductory Biochemistry (3 cr. hr.) or CHEM 452—Biochemistry I (4 cr. hr.), (iii) Free elective (if needed) to make up 1 credit hour. As mentioned above, the department is also in the process of adding five additional concentrations to the major; Polymers and Molecular Engineering, Energy and Environment, Nanotechnology, Process Automation, and Entrepreneurship.

5.A.3. Support of the Attainment of Student Outcomes by the Curriculum and Prerequisites.

Section 3.C has detailed how attainment of Student Outcomes is facilitated by both Chemical Engineering courses (Tables 3-3, 3-4, and 3-5) and technical courses outside our department (Tables 3-6). Within the ChE program each Student Outcome receives significant coverage by multiple courses, and this is bolstered by further treatment in the non-ChE portion of the curriculum.

5.A.4. Prerequisite Structure of Required Courses.

Figure 5-2 shows the prerequisite structure of required courses within the ChE curriculum.

5.A.5. Meeting the Requirements of Depth of Study for General and Program Criteria.

It is shown in Table 5.1 that our programs course requirements meet or exceed the minimum credit hours and distribution. How the Chemical Engineering program stands in relation to ABET requirements is detailed as follows.

- (a) **ABET Requirement.** *“A minimum of 30 semester credit hours (or equivalent) of a combination of college-level mathematics and basic sciences with experimental experience appropriate to the program.*

Our program features 52 semester hours of college level mathematics and basic sciences. This is distributed among mathematics (4 courses, 14 hours), physics (2 courses, 8 hours),

chemistry (8 courses, 30 hours). Laboratory experience is integrated into the General Chemistry (CHEM 122/123 and 124/125, 10 hours) and General Physics (PHYS 141, 142, 8 hours) and accounts for half of Analytical Chemistry (CHEM 222, 4 hours). In addition, CHEM 233 (2 hours) is a dedicated Organic Chemistry Lab course.

- (b) ABET Requirement.** *“A minimum of 45 semester credit hours (or equivalent) of engineering topics appropriate to the program, consisting of engineering and computer sciences and engineering design, and utilizing modern engineering tools.*

Our program has 52 hours of engineering topics, distributed as follows.

Sophomore engineering courses outside major (6 hours): ECE 210, CME 260, CS 109.

Sophomore Chemical Engineering courses (10 hours): CHE 201, 205, 210.

Junior Chemical Engineering Courses (15 hours): CHE 301, 311, 312, 313, 321.

Senior Chemical Engineering Courses (18 hours): CHE 341, 381, 382, 396, 397, 4XX.

- (c) ABET Requirement.** *“A broad education component that complements the technical content of the curriculum and is consistent with the program educational objectives.*

Our program features 6 hours of English composition, 15 hours of General Education courses divided amongst five broad categories, and 3 additional hours of elective credit that can also be used for general education. Many students use the latter courses to study a foreign language or take business courses. We also have many students who opt for a minor in Mathematics since only two additional courses (one in the form of the free elective) are necessary to complete that requirement.

- (d) ABET Requirement:** *“A culminating major engineering design experience that 1) incorporates appropriate engineering standards and multiple constraints, and 2) is based on the knowledge and skills acquired in earlier course work.*

5.A.6. Major Design Experience.

The major design experience in our program is composed of CHE 396 and 397, which are the two capstone design courses. The first course (CHE 396) in the sequence focuses on the design and economic analysis of chemical processes. This includes cost estimation and economic evaluation methodologies and well as design of individual processing units. Students become more familiar with process flow simulation, although they have already had experience with the Aspen Plus in several previous courses (at a minimum in CHE 205, CHE 301 and CHE 313), and also in the technical elective, CHE 433 Process Simulation with Aspen Plus. The design of unit operations relies on the Chemical Engineering core and basic engineering science courses from the junior year. These include chemical engineering thermodynamics where the students are taught the basics of equation of state models and activity coefficient models upon which to base their design simulations. In addition student utilize the transport phenomena sequence (CHE 311, 312, and 313) which supports the design of heat exchangers, piping systems, and separation processes by supplying the estimates of the transport properties and the insight provide by the engineering analysis of these unit operations. The reactor design course (CHE 321) furnishes an understanding of basic reactor modeling which is relevant for many process units. CHE 396 begins the conceptual

transition from theory to practice, from “ideal” to “real”. Operational consequences of connecting process units together are explored, as are pumping and piping, fouling, instrumentation, controls, process safety, occupational health, sustainability, optimization with regard to energy and raw materials, and engineering ethics. Since 2008, we have moved to define the spring semester design projects late in CHE 396, in order to enable better time flow in CHE 397.

In the second design course (CHE 397) students are assigned to work on a major design project in small teams of three or four students each. Each team is assigned an industrial mentor, a practicing design engineer who provides feedback and guidance on engineering practice and industry standards in mandatory weekly meetings carried out on-site. Within a well-structured framework of report milestones, the teams give multiple presentations describing their coalescing designs throughout the semester in front of their classmates and the team mentors to trace their progress and build detail into the final design. The overall design of a process flow sheet and the control and optimization of the process is supported by the previous design course (CHE 396), the mass and energy balance course (CHE 210) and the process control course (CHE 341). The milestones culminate in the final written and oral reports (co-judged by the industrial mentors) as well as presentation of the completed project at the annual Engineering Expo, where it is judged by practicing industrial engineers. Projects are reviewed by judges and the instructor to be such that the designs meet modern and realistic design constraints and standards. It is essential that projects are reviewed by practicing engineers. Throughout the semester, guest speakers are invited to give presentations on several aspects of the chemical industry and career development. Students also have an opportunity to meet informally with visitors to discuss their own interests. One advantage of extensive industrial contact and observation of individual students is the opportunity for networking. Senior Design often leads to job offers for our graduating seniors.

With 52 hours of basic mathematics and science, 52 hours of engineering topics and 21 hours of general education with an additional 3 hours of free electives, UIC’s Chemical Engineering program satisfies ABET requirements.

5.A.7. Cooperative Education

Although it is not required, many of our students do participate in co-op and internship programs. Our location in a major metropolitan center with extensive chemical, materials, energy, food, healthcare, consulting, electronics industries, to name a few, offers our students many opportunities for a meaningful industrial experience during their undergraduate education. However, course credit is not offered for these activities.

5.A.8. Material to be Available for Review at the Site Visit.

The Program Evaluator will have access to an extensive sampling of course materials (syllabi, handouts, copies of textbooks, homework assignments, quizzes, and exams) and corresponding student work. We will also provide extensive documentation of assessment methods/data, evaluation procedures, feedback and continuous improvement, along with detail information on how course content supports attainment of Student Outcomes. This material will be organized in a web format that facilitates efficient examination and cross-checking of the material.

B. Course Syllabi

Appendix A provides the syllabi of all courses used to satisfy the mathematics, science, and chemical engineering discipline-specific requirements required by Criterion 5.

Table 5-1 Curriculum

Chemical Engineering

Course (Department, Number, Title) List all courses in the program by term starting with the first term of the first year and ending with the last term of the final year.	Indicate whether course is Required, Elective or a Selected Elective by an R, an E or an SE. ¹	Subject Area (Credit Hours)			Last Two Terms the Course was Offered: Year and, Semester, or Quarter	Maximum Section Enrollment for the Last Two Terms the Course was Offered ²
		Math & Basic Sciences	Engineering Topics; Check if Contains Significant Design (✓)	Other		
Freshman Year, First Semester						
ENGL 160 English Composition I	R			3	Fall 2019 Spring 2020	24 24
MATH 180 Calculus I	R	4			Fall 2019 Spring 2020	120:25 120:25
CHEM 122/123 General Chemistry Lecture and Lab I	R	5			Fall 2019 Spring 2020	252:21:24 252:21:24
ENGR 100 Engineering Orientation	R		0		Fall 2018 Fall 2019	240:256 185:350
General Education Core II: Understanding Individual & Society	SE			3	Fall 2019 Spring 2020	
Freshman Year, Second Semester						
CS 109 Programming for Engineers with MatLab	R		3		Fall 2019 Spring 2020	240:22 240:22
MATH 181 Calculus II	R	4			Fall 2019 Spring 2020	126:23 126:23
CHEM 124/125 General Chemistry Lecture and Lab II	R	5			Fall 2019 Spring 2020	210:24 210:24

PHYS 141 General Physics I	R	4			Fall 2019 Spring 2020	120:24 120:24
Sophomore Year, Third Semester						
MATH 210 Calculus III	R	3			Fall 2019 Spring 2020	120:20 120:20
CHEM 222 Analytical Chemistry	R	4			Fall 2019 Spring 2020	96:24 96:24
PHYS 142 General Physics II	R	4			Fall 2019 Spring 2020	60:24 60:24
ENGL 161 English Composition II	R			3	Fall 2019 Spring 2020	24 24
CHE 201 Intro to Engineering Thermodynamics	R		3		Fall 2019 Spring 2020	53 22
Sophomore Year, Fourth Semester						
MATH 220 Intro to Differential Equations	R	3			Fall 2019 Spring 2020	120:20 120:20
CHE 210 Material & Energy Balances	R		4√		Fall 2019 Spring 2020	22 45
CHE 205 Computational Methods in Chemical Engineering	R		3		Fall 2019 Spring 2020	47 38
CHEM 232 Organic Chemistry I	R	4			Fall 2019 Spring 2020	264:22 264:22
CHEM 233 Organic Chemistry Laboratory I	R	2			Fall 2019 Spring 2020	22 22
Junior Year, Fifth Semester						
CHE 311 Transport Phenomena I	R		3√		Fall 2019 Spring 2020	51 19
CHE 301 Chemical Engineering Thermodynamics	R		3		Fall 2019 Spring 2020	46 21

CHEM 346 Physical Chemistry I	R	3			Spring 2019 Fall 2019	78 78
ECE 210 Electrical Circuit Analysis	R		3		Fall 2019 Spring 2020	100:36 100:36
CHEM 234 Organic Chemistry II	R	4			Fall 2019 Spring 2020	189:21 189:21
Junior Year, Sixth Semester						
CHE 312 Transport Phenomena II	R		3√		Fall 2019 Spring 2020	19 49
CHE 313 Transport Phenomena III (Separation Processes)	R		3√		Fall 2019 Spring 2020	24 48
CHE 321 Chemical Reaction Engineering	R		3√		Fall 2019 Spring 2020	19 50
CHEM 346 Physical Chemistry II	R	3			Fall 2019 Spring 2020	34 34
CME 260 Property of Materials	R		3		Fall 2019 Spring 2020	103 116
General Education Core III: Understanding the Past	SE			3	Fall 2019 Spring 2020	
Senior Year, Seventh Semester						
CHE 381 Chemical Engineering Laboratory I	R		2		Fall 2018 Fall 2019	14:15:16:16 17:16:18:15
CHE 396 Senior Design I	R		4√		Fall 2018 Fall 2019	29:30 31:36
CHEM 499 Professional Development Seminar	R		0		Fall 2018 Fall 2019	60 65
CHE Technical Elective, e.g. CHE 422 Biochemical Engineering, CHE 433 Process Simulation with Aspen Plus, CHE 494-Selected Topics in Chemical Engineering	R		3√		Fall 2019 Spring 2020	38 10

General Education Core IV: Understanding the Creative Arts		SE			3	Fall 2019 Spring 2020	
General Education Core V: Exploring World Cultures		SE			3	Fall 2019 Spring 2020	
Senior Year, Eighth Semester							
CHE 382 Chemical Engineering Laboratory II		R		2		Spring 2019 Spring 2020	12:14:16:18 11:13:10:20
CHE 397 Senior Design II		R		4√		Spring 2019 Spring 2020	25:34 25:38
CHE 341 Chemical Process Control		R		3√		Spring 2019 Spring 2020	59 63
Elective Outside the Major Rubric		SE			3	Fall 2019 Spring 2020	
General Education Core VI: Understanding U.S. Society		SE			3	Fall 2019 Spring 2020	
Add rows as needed to show all courses in the curriculum.							
TOTALS (in terms of semester credit hours)			52 Hours	52 Hours	24 Hours		
OVERALL TOTAL CREDIT HOURS FOR COMPLETION OF PROGRAM		128					
Total must satisfy minimum credit hours	Minimum Semester Credit Hours		30 Hours	45 Hours			

1. **Required** courses are required of all students in the program, **elective** courses (often referred to as open or free electives) are optional for students, and **selected elective** courses are those for which students must take one or more courses from a specified group.
2. For courses that include multiple elements (lecture, laboratory, recitation, etc.), indicate the maximum enrollment in each element. For selected elective courses, indicate the maximum enrollment for each option.

Instructional materials and student work verifying compliance with ABET criteria for the categories indicated above will be required during the campus visit.

FRESHMAN		Total Hours = 15			Total Hours = 16
	ENGL 160	3 English Composition I		CS 109 (= MATH 180)	3 Programming for Eng w/ MatLab
	MATH 180	4 Calculus I		MATH 181 (MATH 180)	4 Calculus II
	CHEM 122	4 General Chemistry I Lecture		CHEM 124 (CHEM 122)	4 Gen. Chemistry II Lecture
	CHEM 123	1 General Chemistry I Lab		CHEM 125 (CHEM 123)	1 Gen. Chemistry II Lab
	ENGR 100	0 Engineering Orientation		PHYS 141 (MATH 180)	4 Gen. Physics - Mech.
	HUM/SOC	3 General Education Course			
SOPHOMORE		Total Hours = 17			Total Hours = 16
	MATH 210 (MATH 181)	3 Calculus III		MATH 220 (MATH 210)	3 Differential Equation I
	CHEM 222 (CHEM 124/125)	4 Analytical Chemistry		CHE 205 (=CHE 210 & MATH 210)	3 Comput. Methods in ChE
	PHYS 142 (MATH 181, PHYS 141)	4 Gen. Physics II-Elec. & Mag		CHE 210 (= CHE 205)	4 Material & Energy Balances
	CHE 201 (MATH 181, PHYS 141)	3 Intro to Thermodynamics		CHEM 232 (CHEM 124/125)	4 Organic Chemistry I
	ENGL 161	3 English Composition II		CHEM 233 (= CHEM 232)	2 Organic Chemistry Lab I
JUNIOR		Total Hours = 16			Total Hours = 18
	CHE 311 (CHE 201, 205, CHE 210)**	3 Transport Phenomena		CHE 312 (CHE 311)	3 Transport Phenomena II
	CHE 301 (CHE 201 & 205)	3 ChE Thermodynamics		CHE 313 (CHE 301)	3 Transport Phenomena III
	CHEM 342	3 Physical Chemistry I		CHE 321 (CHE 301, 210, MATH 220)	3 Chem. Reaction Eng.
	(CHEM124/125, PHYS142, =MATH 210)			CHEM 346 (CHEM 342)	3 Physical Chemistry II
	CHEM 234 (CHEM 232)	4 Organic Chemistry II		CME 260	3 Property of Materials
	ECE 210 (PHYS 142, =MATH 220)	3 Elec. Circuit Analysis		(CHEM 122/123, MATH 181, PHYS 141)	
				HUM/SOC	3 General Education Course
SENIOR		Total Hours = 15			Total Hours = 15
	CHE 396 (CHE 312, 313, 321)	4 Senior Design I		CHE 397 (CHE 396)*	4 Senior Design II
	CHE 381 (CHE 312)	2 Chemical Eng. Lab I		CHE 341 (CHE 312, 313, 321 & MATH 220)	3 Chemical Process Control
	HUM/SOC	3 General Education Course		CHE 382 (CHE 381, 313)	2 Chemical Eng. Lab II
	HUM/SOC	3 General Education Course		ELECTIVE	3 Outside Major
	CHE ELECTIVE	3 400 Level ChE course or Equiv.		HUM/SOC	3 General Education Course
	CHE 499	0 Prof. Development Seminar			
Total Hours in Major = 128					
Legend:					
 Milestone Courses: Must be taken in the specified semester to be able to graduate in 4 yrs.					
() Prerequisites		(=) Concurrent registration	(**) Credit or concurrent registration with MATH 220		

Figure 5-1. Typical degree plan for the Chemical Engineering Program.

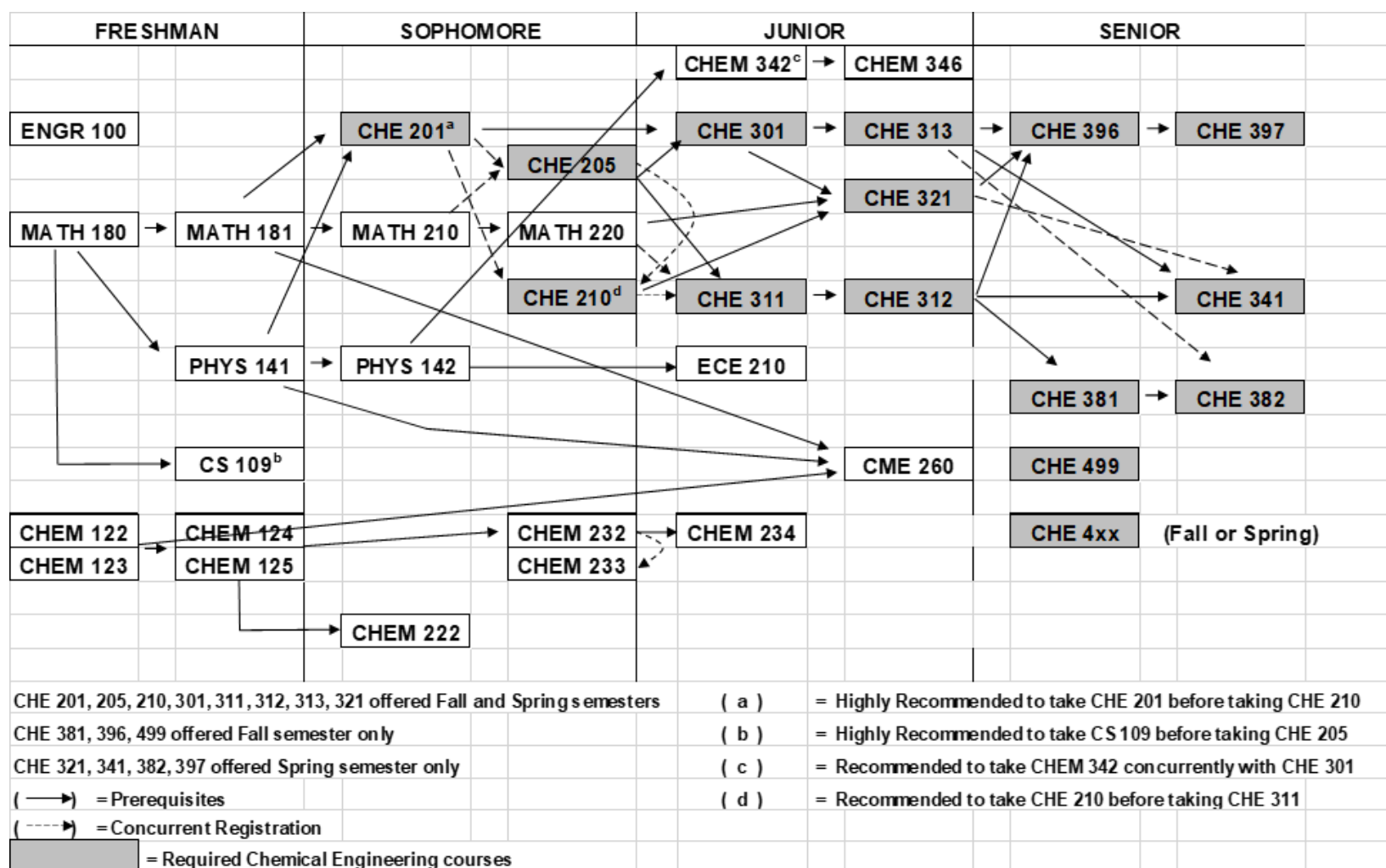


Figure 5-2. Pre-requisite structure of required courses within the ChE curriculum.

CRITERION 6. FACULTY

A. Faculty Qualifications

The majority of professors in the department have PhD in Chemical Engineering with the rest having PhD's in closely aligned disciplines such as polymer engineering and environmental engineering. Five professors have industrial or national lab experience and three professors have established their own companies. Several have had post-doctoral experience outside the US, and are also very familiar with the educational system in Europe and Asia. In addition, during the past five years the Department has appointed three clinical professors with teaching as their main responsibility (from 50% to 100% each semester – 1 or two classes), and two research professors. Three of these professors have spent most of their careers in Industry.

The normal course load of a research active faculty member is four classes per year. However, under College rules faculty who actively participate in the training of PhD students get a reduced course load of three classes and in some cases for faculty members with heavy research activities, the load is two classes. Overall, we are moving towards a 2.5 courses per year for all research active faculty (2 classes one year, and 3 classes the next). Faculty are allowed to “buy out” a class, but in the past five years it has not happened in the Department.

B. Faculty Workload

Expertise among the faculty is sufficiently diverse that any course in the undergraduate curriculum could be taught by at least two of the current faculty members. This teaching versatility has been fostered by the diverse educational backgrounds of faculty members both in traditional chemical engineering and related fields, in addition to both long and short teaching experiences. Research in support of teaching is also encouraged and pursued by most of the faculty through the undergraduate research course (CHE 392) and through the University Honors College. A summary of the distribution of faculty workload summary and Activity Analysis are given in Tables 6.1 and 6.2. We would like to point out that while Table 6.2 shows the actual time being spent by the faculty in different activities as per university policies, Table 6.1 also takes into consideration the actual impact of the activity in judging its level, so there may not necessarily be a direct correspondence between the two Tables.

C. Faculty Size

The Department currently has 265 undergraduate students. In academic year 2019-2020 there were eleven, tenured/tenure-track (TT) faculty (one is Chancellor of UIC and one is 25% in the Chemical Engineering department, with the Bioengineering department being the main appointment). There are also three, non-tenure track (NTT) clinical professor faculty, and two research professors. Tables 6.1 lists these faculty members and their teaching and other responsibilities.

The current faculty workload as described earlier in Section 6.B permits more than adequate advising and educational interactions with students. It also provides enough depth of expertise to permit faculty sabbatical leaves for professional development without adverse effects on teaching. For example, one faculty member went on sabbatical in Fall 2019. Given below is a short summary of the faculty involvement with students, university service, professional development, and interactions with industry and professional practitioners.

6.C.1. Assistant Professors.

Dr. Shafigh Mehraeen Dr. Shafigh Mehraeen is an active member of the ACS (American Chemical Society), AIChE (American Institute for Chemical Engineers), and ASME (American Society of Mechanical Engineers). He has presented in AIChE, Midwest AIChE, MRS (Materials Research Institute), and ASME. At UIC, he is currently a member of the graduate admission committee of Chemical Engineering, which reviews applications for M.S. and Ph.D. programs. He is a co-PI on an NIH grant, focusing on cell-free drug delivery to heal damaged retina. He has been a panelist for the National Science Foundation in CBET (Chemical, Bioengineering, Environmental, and Transport systems) for review of proposals. He is also an active reviewer for several journals including Physical Review Letters, Physical Chemistry Letters, and Langmuir.

Dr. Meenesh Singh is the director of the Materials and Systems Engineering Lab (MaSEL) at UIC. Prior to this appointment at UIC, he was a Postdoctoral Fellow at the Joint Center for Artificial Photosynthesis with a joint affiliation to UC Berkeley and Lawrence Berkeley National Lab. Dr. Singh obtained his B.E degree in chemical engineering from Sardar Patel University in 2005, M.Tech degree in chemical engineering from Indian Institute of Technology Bombay in 2008 and Ph.D in chemical engineering from Purdue University in 2013. He has designed various artificial photosynthetic systems for water-splitting and CO₂ reduction to make renewable fuels. His research resulted in ~30 publications in high impact journals including PNAS and Energy & Environmental Science, ~70 presentations in international conferences, and ~10 invited talks. He currently serves on the editorial board of Challenges, Sustainability, and Current Trends in Chemical Engineering and Process Technology. He has chaired and organized several sessions in the separation division and chemical reaction engineering division of AIChE. He is a reviewer for >15 leading journals. His research has received several awards, including AIChE George Klinzing Best Ph.D. Award, AIChE Process Development Division best student paper award, AIChE Separation Division graduate student research award, and the Outstanding Research Award from Purdue and McDonnell Douglas fellowship. His current research has been covered by leading news agencies such as CNN, IEEE Spectrum, Chemistry World, Digital Trends, Futurism, Quanta Magazine, Cosmos Magazine, TechXplore, SciTechDaily, AzoNano, and he has also appeared on The Dana Barrett Show and Pulse 95 Radio. Dr. Singh has also been active as the faculty advisor for the UIC AIChE ChemE car team, which placed nationally in the November 2019 competition.

Dr. Sangil Kim is an active member of the Electrochemical Society, the American Institute of Chemical Engineers (AIChE), and the North American Membrane Society. He has been a panelist for the National Science Foundation (NSF) and Department of Energy (DOE) for review of proposals and is an active reviewer for several journals such as the Journal of Membrane Science, ACS Energy Letters, and Industrial & Engineering Chemistry Research. Dr. Kim's research has focused on developing new membrane materials, understanding functioning mechanisms of the

nanostructured device, and applying for biomedical engineering, energy-efficient water purification and gas separation, and energy storage. He has led as a PI or co-PI multidisciplinary research projects sponsored by DTRA, DOE, ARPA-E, and NSF. He currently serves as a member of the CHE Graduate Committee and the CHE Advisory Committee.

6.C.2. Associate Professors.

Dr. Brian Chaplin is an active member of the American Chemical Society, and organizes research symposia annually. He frequently attends the North American Membrane Society (NAMS) conference. He has also developed several teaching modules that were targeted at increasing scientific literacy of middle school and elementary school students. Dr. Chaplin has served on proposal review panels for the National Science Foundation and Environmental Protection Agency, and is a frequent reviewer of several professional journals. Dr. Chaplin has several externally funding research grants focused on water treatment and is recognized as a leader in the field of electrochemical water treatment. He has utilized this expertise to form meaningful collaborations with other academic researchers, consultants, municipalities, and industrial endusers. For example, he currently serves on the science leadership council for Current, a Chicago-based non-profit organization focused on water innovation.

Dr. Gang Cheng is an active member of the American Institute of Chemical Engineers (AIChE) and the American Chemical Society (ACS). He is currently a member of the university senate and the director of graduate studies at the Department of Chemical Engineering. He is an awardee of the NSF Faculty Early Career Development award. He is currently advising one Ph.D. student, one MS student, and one post-doc. He has been a panelist for the National Science Foundation for review of proposals and is an active reviewer for journals such as Science Advances, Nature Communications, and Advanced Materials.

Dr. Ying Liu is an executive member of the Sino-American Pharmaceutical Professionals Association (SAPA) Midwest. She regularly attends the national meetings for the American Institute of Chemical Engineering, the American Chemical Society, and the American Physical Society. She has close collaborations with researchers in the medical school at UIC as well as researchers in the nation and out of the country, and Argonne National Laboratory. Dr. Liu's research is currently funded by National Science Foundation (NSF)-CAREER award and Department of Defense (DoD)-Army grant. Dr. Liu has been an active reviewer for more than 20 journals (including Nature Physics, JACS and ACS Nano) and panel reviewer for NSF proposals.

Dr. Vivek Sharma is an active member of Society of Rheology (SOR), American Institute of Chemical Engineers (AIChE), American Physical Society (APS) and American Chemical Society (ACS). He organizes and chairs sessions at several national meetings: regularly at SOR Meeting, ACS Colloids, AIChE (2013-2020), and APS (2014-2020), and also at ACS, Adhesion Society, and U. S. National Congress on Theoretical and Applied Mechanics. Dr. Vivek Sharma regularly attends the meetings for these professional societies, and presents papers and posters at these conferences. In addition, Dr. Sharma has participated in Gordon Research Conferences on Protein Unfolding, Polymer Physics, Adhesion, and on Colloid, Macromolecular and Polyelectrolyte solutions. Dr. Sharma has served on several panels for the National Science

Foundation (NSF), and actively reviews papers for over a dozen scientific journals. Dr. Sharma has given over sixty invited talks at conferences, in industry and in universities, won the 3M Non-tenured Faculty Award (2019), TA Distinguished Young Rheologist Award (2017) and UIC's 2017 COE Teaching Award. Dr. Sharma's students, have won eight prizes and recognitions at GRC meetings, APS March Meeting (including Padden Award), AIChE and ACS Colloids meetings.

Dr. Lewis E. Wedgewood is an active member of the American Institute of Chemical Engineers (AIChE) and the Society of Rheology (SOR) and has participated in the Midwest AIChE conferences. He is currently a member of the department's Advisory Committee which reviews department academic policy and financial expenditures. He is also on the department Undergraduate Committee and the college Awards Committee which awards student scholarships and faculty teaching awards. He has been a panelist for the National Science Foundation for review of proposals and is an active reviewer for journals such as the *Journal of Fluid Mechanics* and *Applied Physics Letters*.

6.C.3. Full Professors.

Dr. Michael Amiridis is Professor and Chancellor of the University of Illinois at Chicago, where he joined the university in 2015. Prior to his becoming UIC chancellor, Amiridis spent over two decades as a professor, researcher and senior administrator at the University of South Carolina. He received his PhD in Chemical Engineering at the University of Wisconsin, Madison, in 1991. As UIC's chancellor, Amiridis heads Chicago's largest university campus, with more than 33,300 students. The university, which holds the distinguished R1 research status in the Carnegie Classification of Institutions of Higher Education, ranks among the top 65 out of more than 650 national universities in federal research funding. He has earned many honors, including the NSF-CAREER Award in 1996, Top Cited Article in *Catalyst Today* in 2004, and a Research Achievement Award from the University of South Carolina in 2005. His research interests include heterogeneous catalysis and chemical kinetics, synthetic strategies, catalytic destruction of chlorinated aromatic compounds over supported transition metal oxides, and heterogeneous synthesis of flavanones.

Dr. Vikas Berry is Professor and Head of UIC's Department of Chemical Engineering, where he joined the faculty in 2015. He received his PhD in chemical engineering from Virginia Tech in 2006. He obtained his master's degree from the University of Kansas in 2003 and his bachelor's degree from the Indian Institute of Technology-Delhi in 1999. Berry has made pioneering contributions in the fields of graphene quantum materials and bio/nano technologies. He has earned many honors, including the NSF-CAREER Award in 2011, the Sigma Xi Outstanding Junior Scientist Award in 2010, and the Big 12 Fellowship in 2009, and a chaired-professorship appointment (the William H. Honstead Professorship at Kansas State University) from 2012 to 2014. His work on graphene biointerfaces is considered a nodal point in the evolution of graphene's biological applications. Berry is a member of the editorial board of *Scientific Reports* (Nature Publishing Group) and the *Journal of Nanoscience Letters*. Research results from Berry's group have been published in several high-impact journals, including *Nature Photonics*, *Nature Communications*, *Nano Letters*, *Advanced Materials*, and *Small*, and have been featured in *Nature*, *Science News*, *Washington Post*, *The Economist*, the *Wall Street Journal*, *Discover*,

Chemical Engineering Progress, and *Physics World*, among other news outlets. Berry's research is funded by the NSF, DoD, and industry.

Dr. Christos Takoudis (25%, Home Department: Bioengineering) is an active participant in meetings of the American Institute of Chemical Engineers, American Chemical Society, and the Materials Research Society. Has active research projects with several local industrial organizations (Motorola, Air-Liquide, etc), in addition to several federal agencies. He is active in consulting activities, and as a panelist for funding agencies and reviewer for professional journals.

6.C.4. Clinical and Research Professors.

Dr. Said Al-Hallaj is active in corporate research on alternative energy and battery technology. He serves as chairman and CEO of AllCell Technologies, LLC and is an experienced entrepreneur with a demonstrated history of working in the energy storage R&D and industry. He has developed two technical elective courses, CHE 451 Renewable Energy Technologies, and "Entrepreneurship for Chemical Engineers". He is also active in the UIC Electrochemistry Collaborative.

Dr. Sanjay K. Behura is a Research Assistant Professor and Graduate Faculty at University of Illinois at Chicago. His inter-disciplinary research program is focused on '2D Quantum Materials for Next Generation Solar Cells and Energy Harvesting'. Dr. Behura has received \$450K research grants as PI or Co-PI from federal and industrial sources and was awarded with DOE user-research-program at Argonne. He has published 35 peer-reviewed journal articles including *Nature Photonics*, *ACS Nano*, *JACS*, *Nano Letters*, *Renewable & Sustainable Energy Reviews* and he is a co-inventor of 3 worldwide patent applications. Dr. Behura has received several awards including DST Young Scientist Travel Award, Canada Commonwealth Graduate Scholarship, and Best Science Graduate Award. Dr. Behura serves as an Editorial Board Member of *Scientific Reports* and chairs/co-chairs sessions on Graphene and 2D Materials at the MRS and AIChE Meetings. His research work has been featured in *Nature India*, *NDTV*, and *Times of India*.

Dr. Betul Bilgin is Clinical Assistant Professor and has been teaching the *Senior Design I and II* courses for 6 years and *Introduction to Thermodynamics* for two years. Since her appointment in 2014 she has been exploring active learning, peer instruction, team-based, hands-on, application-based techniques in her classes to fully engage her students. Dr. Bilgin has improved the *Senior Design* course by integrating ethical, economical, safety, environmental and societal aspects of chemical engineering into her course and introducing structured requirements. As a result of her efforts, she was selected as a **UIC Teaching Scholar** for Spring 2017, named as an American Institute of Chemical Engineers (AIChE) "**35 under 35**" winner in the education category for 2017 and named as American Society for Engineering Education (ASEE) "**20 under 40**" awardee for 2018. Since her arrival in 2014, Dr. Bilgin has helped to create a community within the department to bridge the gap between the industry and academia. Hence, she has been involved in both the Student Chapter, the Chicago Local sections of AIChE, AIChE Education Division and ASEE. She was able to increase the number of industrial mentors from six to fifteen since her arrival. This led to a very effective communication between the industry and department, which soon led to other collaborations. She attended the *ASEE Chemical Engineering Summer School* in July 2017 and the *National Effective Teaching Institute (NETI) Workshop* in May 2017.

Dr. Michael Caracotsios is currently a Clinical Associate Professor in the Department of Chemical Engineering after retiring from UOP/Honeywell in 2015. He teaches and develops academic material and computational tools for courses in Thermodynamics (CHE 301), Process Simulation and Design (CHE 433) and Process Control (CHE 341). He is also the President and Founder of AthenaVisual, Inc., a company that promotes the use of Athena Visual Studio, a state-of-the-art engineering software for mathematical modeling and data analysis. Michael has supervised a number of undergraduate research programs that have resulted in state-of-the-art computational software for Thermodynamics, Process Control and Heat Exchanger design. Most recently he supervised an undergraduate research project that resulted in the development of an Excel based tool used for cost estimation of chemical plants as well as financial calculations. All computational tools are seamlessly integrated into the lectures that allow the students to solve complex real-life program that Michael brings into the classroom from his 30 years of industrial experience. Dr. Caracotsios graduated from the University of Wisconsin-Madison with a Ph.D. in Chemical Engineering. Michael is the author of two patents, twelve publications, and numerous presentations at National and International conferences. He is also co-author of the book “Computer-Aided Design of Chemically Reactive Systems” published by John Wiley and Sons, Inc. in 2006. Michael is a strong supporter of the idea to give students free textbooks. Towards this goal he has been writing and distributing for free lecture notes on the subjects of Thermodynamics, Simulation and Process Control. Michael frequently invites experts from the industry to give seminars on a variety of subjects ranging from process design, and control to safety and leadership themes. He also visits local high schools and promotes Chemical Engineering to young students stressing the importance of the profession to the welfare of the society. He participates in the Freshman Engineering Success Program (FESP) as well as all undergraduate student activities in the department. Michael is an active member of the American Institute of Chemical Engineers.

Dr. Alan D. Zdunek is a Clinical Associate Professor and the Director of Undergraduate Studies in the Chemical Engineering department. He serves as Director of the Freshmen Engineering Success Program (FESP) in the College of Engineering, and is active in several university committees; the COE Education Policy Committee, the UIC Career Council, and the UIC Electrochemistry Collaborative. He is also the faculty advisor for the UIC AIChE student chapter and is involved in all the undergraduate student activities in the department. He was a co-PI on two research projects; an NSF grant developing advanced materials for solid-oxide fuel cells, and a UIC Proof-of-Concept grant studying the feasibility of a water desalination technology. He has also supervised several undergraduate research projects. Community service activities include giving “What is Chemical Engineering” presentations at local high schools and community colleges, and volunteering for the North Lawndale College Prep High School Senior Mentor program. Utilizing his 17 years of industrial experience, Dr. Zdunek consults in the areas of corrosion engineering and electrochemical technology. He is an active member of the American Institute of Chemical Engineers (AIChE), American Chemical Society (ACS) and The Electrochemical Society (ECS), and has served as the Chairman of the ECS Chicago section.

6.C.5. Current/Past National Science Foundation CAREER/NYI Awardees Among the Faculty.

Dr. Brian Chaplin 2015
Dr. Gang Cheng. 2015
Dr. Ying Liu, 2014
Dr. Vikas Berry, 2011
Dr. Michael Amiridis, 1996

6.C.6. Faculty Teaching, Student Advising, and Other Notable Awards.

- Dr. Vivek Sharma
 - COE Faculty Teaching Award (2017)
- Dr. Betul Bilgin
 - Teaching Scholar, Selected for UIC Teaching Scholar Program (Spring 2017)
 - AIChE “35 Under 35” awardee (August 2017)
 - ASEE “20 Under 40” awardee, (June 2018)
- Dr. Alan Zdunek
 - COE Faculty Teaching Award (2016, 2017, 2018)
- Dr. Lewis Wedgewood
 - UIC Award for Excellence in Teaching (2013-2014)
 - UIC Teaching Recognition Program Award (2000-2001)
 - COE Harold A. Simon Award for Excellence in Teaching (2003-2004)
 - COE Faculty Teaching Award (2009, 2007)
 - COE Faculty Advising Award (2007, 2008)
- Michael Amiridis
 - AAAS Fellow

Other awards given to the faculty are listed in their Faculty Vitae in Appendix B.

6.C.7. Changes in Faculty.

Dr. Vikas Berry was hired in Fall 2014, becoming Interim Department Head in Fall 2015 and Department Head in Fall 2016. Dr. Michael Amiridis was appointed as Chancellor of the University in 2015, with Chemical Engineering as his home department. Dr. Sangil Kim was hired in Fall 2015 as an Assistant Professor. Three professors were hired in Fall 2016; Dr. Gang Cheng as an Associate Professor, and Drs. Meenesh Singh and Shafigh Mehraeen as Assistant Professors. The College of Engineering established a Clinical Professor (non-tenure teaching) track in 2014.

Since that time, two professors were promoted to Clinical Assistant Professor in 2014; Dr. Betul Bilgin and Dr. Alan Zdunek, and one professor was hired as a Clinical Associate Professor, Dr. Michael Caracotsios. In addition, Dr. Sanjay Behura as hired as a Research Assistant Professor in Fall 2016. In Fall 2020, Dr. Anh Ngo will start as an Associate Professor. Further, the department is working on recruiting a professor (Endowed Professorship: Satish Saxena Professor). Effective the end June 2019, Dr. Ludwig Nitsche also retired from UIC.

D. Professional Development.

The professional development activities of each faculty in the Department are provided in the short bio summaries given in sections 6.C.1-6.C.4 above. The department places much emphasis on the development of all faculty members, especially junior and mid-career faculty, as well as the clinical (non-tenure teaching track) faculty. A formal mentor is appointed for each Assistant Professor appointed in the Department. The mentor works closely with the faculty member in developing better and more effective teaching strategies, as well as developing exciting new research ideas and provides continuous feedback on the faculty member's progress towards tenure. A special program has been setup in the department to support CAREER proposals from assistant professors (biweekly meetings with mentors, mock panel review). For mid-career faculty, although a formal mentor is not appointed, the Department Head holds regular meetings with them to enhance both their teaching and research effectiveness. This includes arranging meetings with representatives of both industrial and governmental funding agencies, and alerting them when funding is available for these activities through university wide-programs as well as from external agencies. The department also provides seed funding to both early career and mid-career faculty to initiate research projects. In addition, the department provides funding to faculty to visit funding agencies and present talks at professional meeting. As part of its contribution to Faculty Development, the University has provided a new personal computer to all faculty on a 3/4 year cycle as well as to new faculty members when they first join. The University additionally shares the cost of startup costs of new faculty with the College and the Department through various programs. The department encourages faculty members to be active in professional societies, and supplements the cost of attending their meetings when they do not have travel funds available in grants. In addition the department provides funds to faculty members to visit granting agencies, and covers other related costs of grant proposal development. For faculty members with 25% appointments in the department, the primary responsibility of faculty development is that of their home departments. Finally, to ensure that all faculty have adequate computing resources to ensure their professional development, the Department provides software needed to fulfill their teaching and professional responsibilities.

E. Authority and Responsibility of Faculty

Every new course developed (or modified) is initiated by an individual faculty member in the Department. The faculty member in consultation with other faculty with similar expertise generally develops a new outline, which after discussion is approved by the Undergraduate Committee and presented to the entire faculty for approval. After such approval, the faculty member/s prepares a detailed outline of the course with hours to be spent on each subject/topic, suggests an appropriate text book, and submits such a detailed proposal to the Undergraduate Committee. Once approved

by the Undergraduate Committee, it is discussed by the entire faculty during a regularly scheduled or special faculty meeting. After approval by the Department, it is sent to the College of Engineering, where it is considered by the College Education Policy Committee (which has one member from each department and is chaired by an elected faculty member). The Associate Dean of Undergraduate Affairs is an ex-officio member of this Committee. After approval by the College Educational Policy Committee it is considered and approved by the entire faculty of the College of Engineering. After such College approval it is forwarded to the Office of the Provost and Vice-Chancellor for Academic Affairs. That office seeks advice of the Senate Educational Policy Committee. After such approval it is sent to the University Senate for final approval for the following academic year.

Table 6-1. Faculty Qualifications

Chemical Engineering

Faculty Name	Highest Degree Earned- Field and Year	Rank ¹	Type of Academic Appointment ² T, TT, NTT	FT or PT ³	Years of Experience			Professional Registration/ Certification	Level of Activity ⁴ H, M, or L		
					Govt./Ind. Practice	Teaching	This Institution		Professional Organizations	Professional Development	Consulting/summer work in industry
V. Berry	PhD (ChE) 2006	P	T	FT	1	13	6		M	M	M
B. Bilgin	PhD (ChE) 2014	ASC	NTT	FT	0	6	6		M	M	L
S. Behura	PhD(Solar Energy) 2014	ASC	NTT	FT	1	4	4		M	M	L
M. Caracotsios	PhD(ChE) 1986	AST	NTT	FT	30	35	5		M	M	M
G. Cheng	PhD(ChE), 2009	AST	T	FT	3	11	4		M	M	M
S. Kim		ASC	TT	FT							
S. Mehraeen	PhD (MAE), 2011	ASC	TT	FT	2	8	4		M	H	M
M.R. Singh	PhD (ChE), 2013	ASC	TT	FT	3	4	4		M	M	L
S. Al-Hallaj	PhD (ChE), 1999	A	NTT	PT	15	3	3		M	M	H
B. Chaplin	PhD (ChE), 2007	AST	T	FT	4	10	7		M	H	M
Y. Liu	PhD (MAE), 2007	AST	T	FT	0	12	12		M	H	M
V. Sharma	PhD (Polym. Sci. & Eng.), 2008	AST	T	FT	0.5	7	7		H	H	H
C. Takoudis	PhD (ChE), 1982	P	T	FT	1	39	24		M	H	H
L. E. Wedgewood	PhD (ChE), 1988	ASC	T	FT	0	30	22		M	H	L
A. Zdunek	PhD (ChE), 1990	A	NTT	FT	17	12	10		M	M	M

1. Code: P = Professor ASC = Associate Professor AST = Assistant Professor I = Instructor A = Adjunct O = Other

2. Code: TT = Tenure Track T = Tenured NTT = Non Tenure Track

3. At the institution

4. The level of activity, high, medium or low, should reflect an average over the year prior to the visit plus the two previous years.

Table 6-2. Faculty Workload Summary**Chemical Engineering**

Faculty Member (name)	PT or FT ¹	Classes Taught (Course No./Credit Hrs.) Term and Year ²	Program Activity Distribution ³			% of Time Devoted to the Program ⁵
			Teaching	Research or Scholarship	Other ⁴	
V. Berry	FT	CHE 321 (3) Spring-2020	30%	40%	30% (Head)	100
B. Bilgin	FT	CHE 396 (4) Fall 2014, 2015, 2016, 2017, 2018, 2019 CHE 397 (4) Spring 2015, 2016, 2017, 2018, 2019, 2020 CHE 201 (3) Fall 2018, 2019, Spring 2019, 2020 , CHE 205 (3) Spring 2020	70%	10%	20%	100
S. Behura	FT	ChE 494 2D Materials (4) Spring 2017, Fall 2017, Spring 2018, Fall 2018, Fall 2019 and ChE 494 Materials for Energy (4) Spring 2019, Spring 2020	40%	60%		100
M. Caracotsios	FT	CHE 301/3hrs, CHE 433/3hrs Fall 2019 CHE 301/3hrs, CHE 341/3hrs Spring 2020	70%	10%	20%	80
G. Cheng	FT	CHE 494(3) (F-17), CHE 494(3) (S-18), CHE 311(3) (F-18), CHE 422(3) (S-19, S-20)	35%	35%	30% (DGS)	100
S. Kim	FT		40%	40%	20%	100
S. Mehraeen	FT	CHE 205 (4) Fall 2019 CHE 445 (4) Fall 2019 CHE 431 (4) Spring 2020	40%	40%	20%	100
M.R. Singh	FT	CHE 321 (4) Fall 2019 CHE 527 (4) Spring 2020 CHE 392 (3) Fall 2019, Spring 2020	40%	40%	20%	100
S. Al-Hallaj	PT					10

B. Chaplin	FT	CHE494 Fundamentals of Electrochemistry (3) Spring-2020	40%	40%	20%	100
Y. Liu	FT	CHE311 (3) (S18, F19), CHE341 (3) (S15), CHE527 (4) (S15, S16, S17, S18), CHE425 (3/4) (F17, F18, F19, S20), CHE494 (3) (F16)	40%	40%	20%	100
V. Sharma	FT	CHE 312 (F2019, S2013-2019) CHE 311 (S2020), CHE 512 (S2020)< CHE494 S2019	40%	40%	20%	100
C. Takoudis	FT	CHE 312 (3) Spring- 2020	40%	40%	20%	100
L. E. Wedgewood	FT	CHE 210 (F2019) CHE 410 (F2019) CHE 210 (S2020) CHE 440 (S2020) CHE 392 (S2020)	40%	40%	20%	100
A. Zdunek	FT	CHE 381/2hrs, CHE 382/2hrs, CHE 499/0hrs	60%	10%	30% (DUGS)	100

1. FT = Full Time Faculty or PT = Part Time Faculty, at the institution
2. For the academic year for which the Self-Study Report is being prepared.
3. Program activity distribution should be in percent of effort in the program and should total 100%.
4. Indicate sabbatical leave, etc., under "Other."
5. Out of the total time employed at the institution.

CRITERION 7. FACILITIES

A. Offices, Classrooms and Laboratories

The chemical engineering department moved into its new home in the Engineering Innovation Building (EIB) in summer 2019. The 57,500-square-foot-building is at the corner of Morgan and Taylor Streets on UIC's east campus and is shared by the Chemical Engineering, Civil and Materials, and Mechanical Engineering Departments; the Chemical Engineering department is the largest tenant. The EIB is a state-of-the-art facility designed by Dewberry architects and built by Berglund Construction and is in the process of being LEED Gold certified. The new building features high-tech classrooms and laboratories, including vibration-dampened nanotechnology labs, biomedical research rooms, molecular engineering research labs, and a highly attuned darkroom for complex fluid imaging. Following is a summary of each of our program's facilities in terms of their ability to support the attainment of the student outcomes and to provide an atmosphere conducive to learning.

7.A.1. Offices.

The Engineering Innovation Building contains 16 offices for chemical engineering faculty located on the first and second floors, two offices for teaching assistants (EIB 264 and 276), and one central office for the administrative staff (EIB 226). Faculty offices have a 43" LED 4KTV monitor mounted on the wall and one whiteboard for student discussions. There is secure WIFI connectivity throughout the building.

7.A.2. Classrooms.

There are two classrooms in EIB, Rooms 124 and 271. Room 124 has a state-of-the-art, lecture board containing a MultiTouch Video Wall System of 8 X 55" LCD video wall displays. Room 271 is a state-of-the-art Computational Teaching Lab equipped with 21 Versa table Revolution Flip Top Computer Consoles for enhanced instruction and problem-based learning in numerical and computer courses. There is also a 75" LED 4KTV monitor for lectures as well as a whiteboard. Both rooms are equipped with moveable tablet armchairs, an Integrated Multimedia lectern with an installed computer, laptop connectivity, a document camera, a Blu-Ray player and internet connectivity. Most chemical engineering courses are taught in these two classrooms, but other university classrooms with similar features are also utilized in order to provide the best class schedule for the students.

7.A.3. Laboratory Facilities.

Unit Operations Laboratory. The Unit Operations Laboratory for CHE 381 and 382 is located on the first floor of EIB in Room 128. A unique feature is that can be viewed from classroom EIB 124 by a large viewing window. The unit operations laboratory contains 12 functional chemical engineering process demonstrators supporting hands-on instruction of chemical engineering principles such as heat, mass and momentum transfer, fluid properties, and separations processes such as distillation, evaporation, absorption, filtration and extraction. The laboratory space is specifically designed as an instructional chemical engineering laboratory with dedicated workspaces for the process equipment and easy access to utilities such as 120V and 220V electricity, industrial hot and chilled water, purified water,

compressed air, and vacuum lines. Eight moveable lab stations and two permanent lab bench areas are located throughout the lab and for the analytical equipment used in the process experiments. A new 48kW electric steam generator was installed in December 2019 and is plumbed directly to the appropriate process equipment. There are two fume hoods and four dropdown snorkel lines for chemical vapor control. The laboratory is furnished with two (2) computers connected to the distillation column and process control demonstrators, respectively. High-speed WiFi and ethernet connections to the UIC network are both available in the lab. Most students use personal laptop computers, tablets, or other devices. One section of the lab contains conference tables and chairs for the lab teams to work on their calculations and lab reports.

Engineering Makerspace. The College of Engineering Makerspace is a workspace dedicated to support engineering undergraduate students in prototyping and manufacturing for class work and student organizations. Makerspace hosts a range of facilities and tools essential for building physical prototypes such as 3D printers, laser cutters, CNC mills, an electronics bench, textile bench, and printing, vinyl cutting and scanning equipment. Other equipment includes a band saw, drill press, grinder, belt sander, lathe and miter saw. Software used include AutoCAD, Solidworks, Autodesk Inventor, CoreIDRAW, EagleCAD and Onshape. Makerspace offers three courses; ENGR 111 Additive Manufacturing, ENGR 112 Subtractive Machining and ENGR 214 Advanced Manufacturing Methods. Makerspace is in the Roosevelt Building.

7.A.4. Additional Facilities.

A Student Lounge (Room 232) is located on of the 2nd floor in the northeast corner of the building. The lounge has floor-to-ceiling windows on two sides with excellent ambient lighting and features a kitchenette with sink, microwave and refrigerator, several work/lunch tables, a sofa, a 75” LED 4KTV, and an electric piano. A library/reading room (Room 276) with a 43” LED 4KTV monitor and moveable whiteboard, a theory lab (Room 230), and a meeting/teleconferencing room (Room 224) with a 75” LED Multi-touch Display are also used by our students. There is an all-gender bathroom on the second floor (Room 238), and a lactation room located on the second floor (Room 262) that also serves as a meditation/prayer room. A large foyer in the front of the building has five, 75” LED 4KTV monitors and is used periodically for student and alumni events. An outside area behind the building features a grass lawn and three picnic tables with chairs. Computers and laboratories are serviced by a small machine and electronics shop on site (Room 170). The College of Engineering machine shop and a glassblower are in the Science and Engineering Laboratory building (SEL) located across the street from EIB.

B. Computing Resources

The Chemical Engineering department has 18 computers running Windows 7 Professional for student/teaching use and one laser printer (Room 271) for student use. Additionally, our Faculty Research Laboratories include 39 computer workstations and PC clusters. These machines range from desktop PCs to Unix Workstations to a pair of 8 node LINUX clusters. These computers are utilized for both data analysis as well as simulations. All student/teaching computers have installed Aspen, MATLAB, Microsoft Office 365 as well as other software specific to classes. The department of Chemical Engineering funds and maintains

an exclusive 150 user license for the worldwide standard Aspen Chemical Simulation Software. Students can download the software onto their laptops as well as use the software in the computer lab (EIB 271) on the 18 Versa table Revolution Flip Top Computer Console workstations. In addition to the above, the students have access to an Epson Brightlink Pro interactive whiteboard in EIB 271.

In the Unit Operations Laboratory, there are 2 computers that are dedicated to running the distillation column and process control demonstrators. High-speed WIFI and Ethernet connections to the UIC network are both available in the lab. Most students use personal laptop computers, tablets, or other devices.

Aside from computers dedicated to student use in the Chemical Engineering Building (CEB), computer labs are distributed throughout the campus – although the vast majority are concentrated on the east campus within easy walking distance of EIB. General information and a schedule for specific computer labs can be found in the following website:

<http://accc.uic.edu/service/computer-labs>

All these computers are installed with Excel and MATLAB and Microsoft Office 365, which are the main software packages that Chemical Engineering students would need.

For advanced computing needs and high-performance computing (HPC) on campus, the students have access to two clusters, Extreme and SABER. Extreme is a powerful machine with 3,500 cores, 24.5 Terabytes of memory and 1.25 Petabytes of local raw storage, of which 275 Terabytes is raw fast scratch storage. The cluster is composed of traditional, high memory and GPU computing nodes: 200 HPC worker nodes, 160 x Generation I: Intel Xeon E5-2670 @ 2.60 GHz, cache size of 20MB with 16 cores in each node, 128GB RAM and 1TB local storage; 40 x Generation II: Intel Xeon E5-2670 @ 2.50 GHz, cache size of 20MB with 20 cores per node, 128 GB RAM and 1TB local storage; 3 x Generation I High Memory Worker Nodes: Intel Xeon E5-4650L @ 2.60GHz, cache size of 20MB with 32 cores per node with 1TB RAM; 50 x Generation II GPU Worker Nodes; 3 x Login Nodes; and 2 x Admin Nodes. The traditional HPC and high memory worker nodes currently run CentOS 6.9. The GPU worker nodes currently run CentOS 7.5.

The Shared Analytics & Big-data Enterprise Resource (SABER) was introduced into the Advanced Cyberinfrastructure for Education and Research (ACER) HPC resource offerings in May 2017. In addition to traditional HPC workloads, this NSF-funded cluster provides access to various Big-Data/Analytics software packages. The SABER cluster consists of traditional high-performance computing nodes; 76 x HPC Nodes with Intel Xeon E5-2650 @ 2.20GHz; 24 x Cores/Node with 30 MB cache, 128GB RAM, 1TB Local Storage. Software on each node is CentOS 7.3.

C. Guidance

Undergraduates performing research in faculty laboratories are required to attend a UIC Environmental Health and Safety Office (EHSO) laboratory safety training course before working in the laboratory. Information on the safety courses available can be found at: <https://ehso.uic.edu/training/introduction-to-laboratory-safety-training/>

Undergraduates enrolled in the unit operations laboratory (CHE 381 and 382) are provided training and guidance by the instructor and teaching assistants on proper usage and safety precautions before they begin their experimental work on the chemical process demonstrators. This is initially done in the first lab period of the semester, and thereafter during ‘pre-lab’ laboratory periods. Additional guidance and oversight is provided while the lab teams are performing the experiments and afterwards to ensure proper equipment shut-off procedures.

Makerspace staff provide guidance to students before they can use the machining tools and equipment. Three courses for more detailed instruction are also offered; ENGR 111 Additive Manufacturing, ENGR 112 Subtractive Machining and ENGR 214 Advanced Manufacturing Methods.

7.C.1. Support Policies.

All computing support policies for computer account usage, network access, are other support policies can be found at: <https://accc.uic.edu/about/policies/>

All ChE students, faculty and staff should read and agree with all these policies before using computing resources.

7.D.2. Accessibility.

ACCC also work closely with the Disability Resource Center to ensure accessibility resources are available to the UIC community. More information can be found following this link:

<http://accc.uic.edu/service/learning-environments/accessibility>

D. Maintenance and Upgrading of Facilities

7.D.1. Maintenance.

Routine maintenance and an ongoing improvement program for the Chemical Engineering Unit Operations Laboratories (ChE 381 and 382) are conducted on a continuous basis throughout the year. Suggestions for improvement are taken from student assessment (e.g., Senior Exit Interview and CHE 381/382 course evaluations), and faculty input. Maintenance and improvements are funded by the student laboratory fee, and if needed, by department or college funds. It has included the following:

7.D.2. Selected Ongoing Improvements.

Year	Experiment	Status
2015 Armfield	Process Control	Purchased process control demonstrators: PCT 50, 53, 54, 55

2015	Distillation	Purchased a continuous distillation column (Armfield UOP 3CC) to replace existing column
2015	Fluidized Beds	introduced a new particle medium (ion-resin) to replace sand particles
2015-19 (annual)	Fluid flow in pipes	Replaced leaking valves and pressure ports
2016	Distillation	Developed a new distillation experiment using new distillation column
2016 bed	Gas Absorption	Created a new experiment combining packed with gas absorption experiments using the Armfield UOP7 apparatus
2016	Fluid flow in pipes	Replaced pressure tubing fittings for better measurements
2017	Liquid extraction	Established oil reuse procedure to reduce waste
2017	Process Control	Created a process control experiment (Level Control) to lab schedule
2017	Liquid extraction	Replaced faulty centrifugal oil feed pump
2018 units	Membrane Separation	Replaced non-functional O ₂ sensors with new
2018	Liquid Extraction	Purchased new titration equipment (burettes, Burette holders)
2018	Distillation	Purchased digital refractometer, supervised undergraduate research project to create new methodology
a	Humidification	Purchased a new digital hygrometer with better accuracy for measuring humidity in
flowing air		
2019	Facilities	Acquired a new digital balance (3kg max)

Improvements performed after moving to new building, August 2019-present:

The move to the Engineering Innovation Building in August 2019 provided a unique opportunity to assess the condition of the chemical process equipment and to make improvements and upgrades to improve equipment operation and enhance the student learning experience. One significant upgrade to the lab was the purchase and installation of a new electric steam generator, including steam lines installed to the equipment requiring steam for heat transfer. In addition, as the chemical process demonstrators were re-connected to the utilities, all components; valves, flowmeters, vessels, piping, mechanical components, and electric wiring were examined and replaced if needed. For instance, new piping was added, or existing piping was replumbed to the humidification, evaporation, heat transfer, liquid extraction, fluidized bed, ultrafiltration and gas absorption experiments. New electric wiring was added to the humidification, liquid extraction, and ultrafiltration units. The water manifold to the heat transfer experiment was redesigned and a new flowmeter added to the external heat exchanger for more accurate measurements. Additional safety items were also added such as safety hazards signage, and flooring to reduce slippage around the process equipment.

7.D.3. Routine and Periodic Maintenance.

Weekly: Students rinse, clean and dry all labware (beakers, flasks, graduated cylinders, buckets, pipettes, test tubes, racks, trays). Students clean up common lab bench areas, especially around scales/balances, and put away any equipment used during lab period. Students shutdown their experiments, clean up the apparatus and bench area, and leave the experiment ready for the next lab team. Water, air, steam, pumps and individual electric power boxes are turned off. Instructor/TA performs an inspection at the end of each lab period to make sure all experiments are shutdown correctly; chemicals are properly stored and equipment is put away.

Semester: Faculty/staff perform a run-through of the lab experiments at the beginning of the semester before the first lab period to make sure the experiments are working properly. An inventory is performed at the end of the semester to determine if lab supplies/chemicals/gases need to be ordered for the next semester. Equipment is periodically cleaned, recalibrated, refilled or replaced by faculty/staff during the semester. For example, capillary tubes in the viscometer experiment are cleaned out with a solvent to remove Carbopol polymer residue left inside. Manometers are checked to make sure they are filled correctly. The boiler is turned on in the morning before the lab commences and is checked periodically to make sure it is working correctly. Safety equipment is maintained in good working order; safety spill kits and chemical spill pads are inspected and replaced when needed, the laboratory safety manual is updated annually, MSDS binder is updated when needed, first-aid kits and eye showers are inspected weekly, an ample supply of nitrile lab gloves, heat-resistant gloves, chemical-resistant gloves and sharps gloves are maintained in the lab, and safety glasses are checked for breakage and new glasses ordered if necessary.

The departmental Safety Committee which includes the professor teaching the Unit Operations Laboratory, and the staff responsible for lab and computer maintenance, regularly assess laboratories and equipment, keep the faculty and departmental administration apprised, and make updates as necessary.

7.D.4. Upgrading Facilities.

The Chemical Engineering department move into the new Engineering Innovation Building has been a major upgrade to our classroom, instructional laboratory, research laboratory, office, and conference facilities. Here is a brief summary of some of the significant upgrades related to student instruction.

Classrooms: The two classrooms in the Engineering Innovation Building are equipped with Integrated Multimedia lecterns with an installed computer, laptop connectivity, a document camera, a Blu-Ray player and internet connectivity to facilitate the use of software demonstrations and computer visuals during instruction. Room 124 has a state-of-the-art, lecture board containing a MultiTouch Video Wall System of eight, 55" LCD video wall displays. This new technology allows instructors to illustrate complex and abstract concepts much easier than can be accomplished using the white board. We anticipate that instructors will be able to achieve the desired learning outcomes on a wider group of students, as some students learn most effectively through visualization.

Offices: Faculty offices include a 43-inch 4KTV LED displays mounted to the wall that can be connected to the faculty's laptop or desktop computer. This allows better communication

with students during in-office meetings when discussing coursework, undergraduate research projects and advising.

Laboratory: Instructional laboratory upgrades have been described in the previous sections.

7.D.5. Update on Laboratory Development Plan.

A laboratory development plan was initiated during the previous ABET review cycle. The plan was discussed with the Engineering Advisory Board (EAB) and the Senior Design Industrial Mentors in order to solicit input on ways we can improve the laboratory experiments that we offer to our students. Based on the feedback from these groups and from student suggestions, we purchased a *Process Control* demonstrator and a new continuous distillation column for the Unit Operations Laboratory course. Experimental procedures were developed and incorporated into the lab schedule in academic year 2015-2016 (Distillation) and academic year 2016-2017 (Process Control). The process control experiment has been a great success since the students are taking CHE 341 Process Control concurrently with CHE 382. The process control laboratory provides a hands-on experience and reinforces the concepts that are being taught in this CHE 341. The new distillation column has also provided a more enhanced experience as well. The unit is considerably easier to operate in the timeframe of the lab period and allows better operational control; the distillation column can be run in various modes such as a batch distillation, a continuous feed with varying feed positions, and reflux ratios that can also be varied. More accurate measurements are now made using the computer-controlled data acquisition system and the digital refractometer.

Further improvements are also planned. Feedback to further improve the unit operations laboratory was obtained from a department faculty curriculum meeting. One suggestion was to add a chemical reaction engineering experiment. The department has obtained a

E. Library Services

A public research facility, the University Library at the University of Illinois at Chicago serves an urban campus of more than 33,000 students and 11,000 faculty and staff members, as well as the Chicago metropolitan area and the State of Illinois.

The Richard J. Daley Library is located at 801 South Morgan on UIC's East Campus and is the main university library. The Richard J. Daley Library serves nine colleges, including Engineering. Patrons have access to more than 2.2 million volumes, 634 databases, 504,341 e-books, and over 60,000 current journal titles. 381 journals are included in the category of Chemical Engineering, such as AIChE journal, Chemical Engineering Science, Colloid and Polymer Science, Colloids and Surfaces, Combustion and Flame, Journal of Catalysis.

The Library of the Health Sciences (LHS) located at 1750 West Polk is one of the largest health sciences libraries in the nation and continues to support education, research, and clinical practice. With more than 725,000 volumes and 5,200 journals.

The availability of the books and journals can be checked on the website of the library. The available books can be borrowed from the library and the journal articles can be either copied in the library or downloaded if the e-copies available. If an item is not available from UIC library, it may be accessed by requesting interlibrary loan. The items can be renewed in the library or online after signing up an account.

The faculty members can submit course reserves and recommend a purchase of books or subscription of journals by contacting the reserves staff or filling up the online form.

Besides UIC libraries, the Chemical Engineering has a small library located in our building. The reference books can be used in the library. The library can also be reserved as a study room.

Overall Comment on Facilities

Aside from the above specifics on facilities dedicated to the Chemical engineering program, it is worth mentioning that since transitioning to our new building, we have been able to keep the strong community feeling between the faculty, students and staff that was present in our previous location. Faculty are never more than a few feet away from classrooms, the labs and the student lounge, and our student community benefits from a significant esprit de corps that translates into a more complete educational experience for our students.

CRITERION 8. INSTITUTIONAL SUPPORT

A. Leadership

The Undergraduate Program in Chemical Engineering is administered by a three-member Undergraduate Committee chaired by a faculty member designated as the Director of Undergraduate Studies (DUGS). The DUGS is responsible for the monitoring of the effectiveness of the curriculum to meet the declared objectives and outcomes of each course, as well as the overall curriculum. This responsibility also entails oversight of all assessment procedures and coordination. The department has implemented a four-year cycle to update every course taught in our undergraduate program. Each year a subset of the courses taught is chosen by this committee for evaluation. The committee then conducts student surveys as well as obtains input from other stake holders such as alumni, recruiters, external reviewers (members of the Industrial Advisory Board, Judges of our annual Engineering Expo) and suggests changes in the contents of a class chosen for such evaluation. These changes are then discussed during a Departmental Faculty Meeting, and after discussion and sometimes modifications to the recommendation, changes are approved and become effective from the following academic year. Similar process is also employed for improvements in the overall program. The committee also considers the approval for requests for modifications in the curriculum of individual students under special circumstances. In addition, the committee also approves new courses developed by individual faculty members, before they are submitted to the entire faculty for their approval. We have weekly curriculum meetings between the department head, director of undergraduate studies and the director of graduate studies. Any ideas developed in these meetings are vetted by the Alumni via a survey, discussed in the undergraduate or graduate committees before going to the faculty for inputs.

The entire curriculum of the Department is also discussed as a whole during the annual meetings of the Industrial Advisory Board, and regular meetings with Alumni Groups, as well as representatives of organizations that hire significant number of our graduates. If issues concerning either courses or the overall curriculum are raised in these meetings, these issues are handled by the Undergraduate Committee outside the two-year cycle mentioned above. Such Assessment meetings have helped develop several improvements to our curriculum.

One such improvement that is being planned for 2021 is the addition of 5 new concentrations that will be added soon to our curriculum. These include (a) Energy and Environment, (b) Polymer and Molecular Engineering, (c) Nanotechnology, (d) Process Automation, and (e) Entrepreneurship. Four new courses have also been added (a) Molecular Systems in Chemical Engineering, (b) Polymer Science, (c) Nanoscale Systems in Chemical Engineering, and (d) Molecular and Macromolecular Engineering.

B. Program Budget and Financial Support

The institutional budget for the Department of Chemical Engineering consists of two parts, personnel and the operating costs. The personnel costs are those for salaries of faculty, union staff members, academic professionals and teaching assistants. The budgeted amount for these salaries increases each year by an average percentage mandated by the University. These increases have so far been to retain qualified personnel in the department. The budget of the College is currently in part tied to the enrolment in Engineering and the tuition revenue generated by the enrolled students. While the Department budget is not tied directly to enrolments, it is heavily influenced by it. The department has had stable student enrolments in the last few years. A few years ago, we increased the number of TAs in the department from 9 to 16 to reflect an increase in our enrollments. The budget for the operating costs has not increased for the last few years and due to poor statewide economic conditions and has in fact

decreased slightly in the last fiscal year. The department has three accounts where it generates additional revenue: (a) The Indirect Cost account (the fraction of the overhead funds available from research grants that is returned to the department), (b) Gift account (gift funds given by benefactors of the department) and (c) Educational Program account (tuition revenue from other educational programs, such as China 3+2, India-MS programs). These additional funds have therefore also been used to ensure that the program objectives are met.

Some examples of how these funds have been used in meeting program objectives are given below:

- (i) Updating Departmental Computing Services: The funds have been used to purchase site licenses for software needed by our students and faculty for their classroom activities (ASPEN, CHEMCAD. Etc.).
- (ii) New Unit Operations Equipment: Distillation column, process control equipment, digital balance, digital hygrometer, lab computers.
- (iii) Professional Development Supplement: The faculty receive a professional development funds from the university (~\$900 for non-tenure-track and \$1200 for tenure-track). The above mentioned funds have been used to supplement the funds from university for teaching workshops, and conference visits.
- (iv) Startup Packages: The department hired three assistant professors and one associate professor in 2016. The departmental share of the startup funds is covered by these funds.

Financial Framework

The current sources of financial support for the Department are the State of Illinois with additional revenue mentioned above. The state funding covers faculty and staff salary, as well as some additional non-permanent positions such as Teaching Assistants and part-time student workers.

This portion of the budget is fixed each year by the University -- changes made are generally consistent with the support provided to the University by the State of Illinois. In addition the Department receives a portion of the Indirect Cost Recovery (ICR) from research grants housed in the department, or those in which the department faculty participate. Currently the University returns 47.5% of the overhead cost to the College of Engineering, and this is then divided equally between the College and the Department. The Department actively works with the Development Office of the College and University to raise gifts from alumni and local and national businesses with an interest in our Department and University. These gifts when they are discretionary are also used to fund special projects in the department including sometimes emergency repairs to equipment in the UG Laboratories that has not been budgeted. Our development activities have resulted in one major gift and several smaller gifts. The major gift is being used for an endowed professorship position in the department.

While there has been no increase in state budget over a period of time, the alternate revenue streams have seen an increase driven by a steady increase in our student enrollments and our international programs. This together with careful budgetary management in general by the Dean's office has allowed us to providing a high quality education for the undergraduate students.

Further, in 2018, we started the process on constructing a \$45M building (Engineering Innovation Building) to house the entire department. The building has a state-of-the-art classroom, unlike any other with a large touch-screen. Further, the room is annexed to our unit-operations laboratory via a large window through which students in the lecture room can see the various equipment; a model that was inspired by the medical observatory classrooms. The building also has student learning space spread throughout the building, including computational room, library, lounge, and atrium seating space. College alumni donor funding for the scholarships and computer software is another example of support for our program. Strong support enables our effectiveness in teaching and learning. The College conducts surveys of teaching and advising every semester, and the Chemical Engineering department usually ranks the highest in these surveys. Another measure of the quality of instruction we have provided to our students is the Annual Engineering Expo. In the past years, our students have won a disproportionately higher share of awards. This year, due to COVID-19 stay-at-home guidelines, we had a virtual Expo, and although no awards were given, the chemical engineering teams received high judge scores for their projects.

While there has been no increase in state budget over a period of time, alternate revenue streams have seen an increase driven by a steady increase in our international programs. This has allowed us to persist in providing a quality education for the undergraduate students.

C. Staffing

The support staff in the department consists of a Business/Office Manager, an Undergraduate and Graduate Services Coordinator, an Assistant to the Head, Lab/Computer Manager and Communication Officer. The Chemical Engineering organizational chart is shown in Figure 8.1. In addition, the department generally hires a graduate student for operational support. By way of professional development, the Business/Office Manager attends approximately 6-8 university workshops per year and this number is roughly 3-4 for the Undergraduate/Graduate Coordinator. Our staffers have won a number of service and recognition awards.

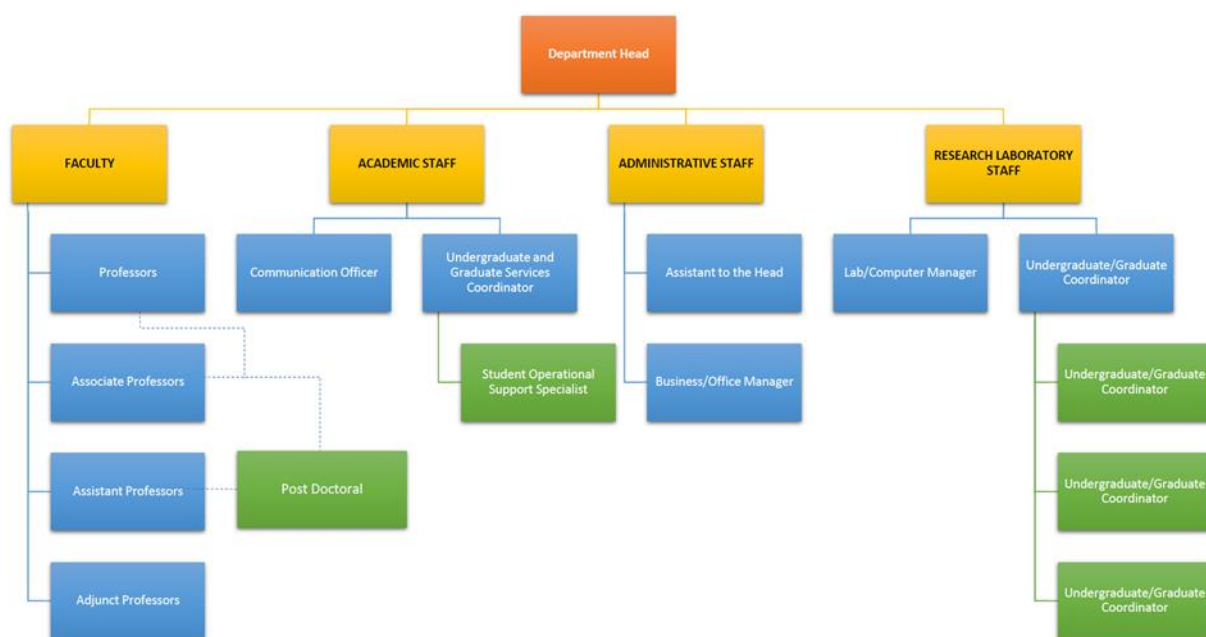


Figure 8.1. Chemical Engineering Faculty and Staff Organizational Chart.

D. Faculty Hiring and Retention

The Department has been successful with support from the College and the University to recruit several excellent additions to our Faculty. The following table shows the faculty hiring conducted over the last six years in our department:

Year	Name	Position	More Information
2014	Vikas Berry	Associate Professor	CAREER Awardee
2014	Betul Bilgin	Clinical Assistant Professor	AIChE 35 under 35
2014	Michael Caracotsios	Clinical Associate Professor	
2016	Sangil Kim	Assistant Professor	
2016	Meenesh Singh	Assistant Professor	
2016	Shafigh Mehraeen	Assistant Professor	
2016	Gang Cheng	Associate Professor	CAREER Awardee
2020	Anh Ngo	Associate Professor	Starting Date: Aug 2020
2020		Satish Saxena Professor	Starting Date: Aug 2020

These new faculty members have broadened the teaching and research programs of the department already. Since 2014, we have had one retirement of Prof. Ludwig Nitsche and one move of Prof. Belinda Akpa. We currently have 5 NSF CAREER awardees and 1 AAAS fellow on our faculty.

E. Support of Faculty Professional Development

8.E.1. Promotion and Tenure Seminars.

The university conducts several professional development seminars, hosted by the Office of Faculty Affairs, and coordinated by the vice-provost for faculty affairs, Nancy Freitag. Some examples of such seminars over the last year include:

1. New Faculty Orientation
2. Expanding our Global Connections: International Research, Collaboration, Study Abroad and Campus Experiences
3. Innovation at UIC
4. Navigating the Research Landscape
5. Mentoring in Research Education and Career Development
6. Faculty Search Committee Training
7. Diversity Education
8. Mid-Career Writing
9. Research Training

8.E.2. Mentoring Program.

There are university-wide and department-wide mentoring programs. Mentoring is important to retain faculty, improve morale and enhance performance generally. The university wide mentoring program supplements the efforts of individual departments to provide assistance and guidance to their probationary faculty members. The Office of Faculty Affairs coordinates the process by which a junior faculty may select a mentor from a list of senior faculty who have volunteered their services. In general junior faculty can ask for and receive guidance on how to get things done within the university and how to chart a productive research path.

The Chemical Engineering department appoints a faculty mentor for all new Assistant Professors as part of the support program. The mentor/mentee are expected to meet at least once in two weeks in a formal manner; which most of the mentoring occurs in an informal manner; the mentor does not have any extraordinary influence on personnel matters concerning the faculty member. The mentor is expected to help the faculty member in writing proposals, papers and any other help he may wish to seek. In addition, the department has a CAREER committee that thoroughly mentors assistant professors submitting NSF-CAREER proposals from May to July. For mid-career faculty a mentor is generally not appointed, but senior faculty members have open discussions with faculty to suggest research topics and upcoming research funding opportunities, to enhance their professional development.

8.E.3. Workshops on increasing teaching effectiveness.

Several workshops are offered each year for professional development in teaching through the Center for Teaching Excellence. The Center for Teaching Excellence (CTE), composed of faculty members who have been recognized for their demonstrated teaching excellence, provides a mechanism by which faculty, staff, and administrators can work collaboratively toward the improvement of instruction and the advancement of learning in UIC's culturally pluralistic community. Through the development of programs, policies, resources and rewards, CTE strives to encourage every member of the academic community to develop interests, abilities, and achievements as both teachers and learners. CTE offers various programs and resources to support excellent teaching and learning throughout the UIC community including:

- Teaching Recognition Program (TRP) and Award for Excellence in Teaching (AET) are programs that recognize and reward documented excellence of UIC faculty in instruction.
- Departmental Teaching Excellence Award (DTEA) recognizes the commitment to teaching and documented instructional excellence of departments and other academic units.
- CTE provides publications, reports, links, and other instructional resources for faculty, staff, and administrators. Center seminars are coordinated by the Vice-Provost for Faculty Affairs. **Registration is free for these seminars and they are widely publicized. In most cases faculty members decide on their own which seminars would enhance their teaching abilities, while often a colleague will suggest one of these to a junior colleague.**

8.E.4. Sabbatical leaves for professional development.

Faculty may apply and expect to receive sabbatical leaves for the purpose of study or research, as recommended by the head of the department and with the concurrence of the Dean of the College of Engineering.

PROGRAM CRITERIA

The Chemical Engineering Program Criteria require that the curriculum must:

- a.) Provide a thorough grounding in the basic sciences including chemistry, physics, and/or biology, with some content at an advanced level, appropriate to the objectives of the program.
- b.) Include the engineering application of these basic sciences to the design, analysis, and control of chemical, physical, and/or biological processes, including economic, environmental, global, and societal issues as well as the hazards associated with chemical processes.

Coverage of these areas in the Chemical engineering curriculum was discussed above under Criterion 5. The course work that addresses the program criteria are briefly recapped in this section. Examples of student work will be available during the on-site visit that will demonstrate a working knowledge of these topics. Chemical engineering students begin their chemistry courses during their freshman year with general chemistry (two lecture/laboratory courses: Chem 122/123 and 124/125: 10 credit hours). Organic chemistry (two lecture courses: Chem. 232 and 234: 8 credit hours, and one laboratory course: Chem 233: 2 credit hours) is typically taken in the sophomore year, while physical chemistry (two lecture courses: Chem 342 and 346(selective): 6 credit hours) and analytical chemistry (one lecture course: Chem 222: 4 credit hours) fall in the junior year.

Chemical engineering courses begin with an introduction to thermodynamics (ChE 201: 3 credit hours) in the fall semester of the sophomore year. This is followed by a computational methods course (ChE 205: 3 credit hours) and the material and energy balance course (ChE 210: 4 credit hours). The former develops mathematical background specifically addressing chemical engineering applications and general problem-solving skills using Excel, VBA, MATLAB and Aspen Plus process simulation. The latter introduces the critical concept of balances as well as basic problem-solving skills to be applied and further developed throughout the curriculum. In the junior year, students take chemical engineering thermodynamics (ChE 301: 3 credit hours) and three transport phenomena courses (ChE 311, 312 and 313: 9 credit hours) which cover fluid mechanics, heat and mass transfer and separation processes and. In the spring semester of junior year, the students take chemical reaction engineering (ChE 321: 3 credit hours) which includes design work. Senior level courses complete the major chemical engineering subject areas and complement the design experience. These courses are: the two chemical engineering laboratory courses (ChE 381: 2 credit hours and ChE 382: 2 credit hours); chemical process control (ChE 341: 3 credit hours), and the two senior design courses. Preparation for engineering practice culminates in the design course sequence (ChE 396 and 397: 8 credit hours) taken in the senior year. In these courses the students demonstrate their ability to independently apply chemical engineering principles, cost estimation and economic evaluation methodologies, flow sheet synthesis and process simulation, optimization techniques, and report writing and oral presentation skills, to contemporary chemical engineering design problems. Industrial mentors provide guidance to the design teams and instruct the students in engineering practice and industrial standards. Content in the above courses is further augmented by a host of technical electives such as Process Simulation with Aspen Plus, Biochemical Engineering, Renewable Energy Technologies, and Nanotechnology for Pharmaceutical Applications, and Electrochemistry. Syllabi for all required and elective courses are given in Appendix A.

Coverage of ethics was discussed in Section 4.B.5. Assignments on ethics are covered in several courses in the curriculum; CHE 396, 397 as well as CHE 201, 341 and 433. Safety and hazards associated with chemical processes was discussed in Section 4.B.6. Mandatory student

completion of SaChE and relevant class discussion are integrated into the Chemical Engineering curriculum in CHE 381, 382, 396 and 397. Student knowledge and proficiency in laboratory and chemical process safety are assessed directly the curriculum in CHE 381, 382, 396 and 397.

APPENDICES

Appendix A – Course Syllabi

A.1. Chemical Engineering Courses.

CHE 201

1. Course number and name

CHE 201. Introduction to Thermodynamics

2. Credits, contact hours and categorization

Credit hours: 3; Contact hours: 3 lecture hours per week; Engineering

3. Instructor's or course coordinator's name

Betul Bilgin, Alan Zdunek

4. Text book, title, author, publisher, year

Fundamentals of Engineering Thermodynamics, Moran/Shapiro/Boettner/Bailey, 9th Edition (2018)

5. Specific course information

a. Brief description of the content of the course (catalog description)

Work and energy; conversion of energy; theory of gases and other states of matter; applications to energy conversion devices. Second law of thermodynamics, entropy, and equilibrium, with applications.

b. Prerequisites or co-requisites

Math 181, Calculus II and Phys 141, General Physics I (Mechanics).

c. Indicate whether a required, elective, or selected elective.

Required.

6. Specific goals for the course

a. Specific outcomes of instruction.

Students will learn to:

- Apply energy balances for closed systems and control volumes.
- Evaluate thermodynamic efficiencies of compressors, turbines, power plants, refrigeration and heat pump cycles.
- Identify phases at a given state and find thermodynamic properties such as internal energy, specific volumes from steam tables.
- Draw and interpret pressure-enthalpy (P-H), pressure-specific volume, temperature specific volume, temperature-entropy charts.
- Relate thermodynamic properties via partial derivate, Maxwell's relations.
- Interpret phase diagrams of binary systems.
- Apply the entropy balance and identify if the process is reversible or irreversible.

Project

One, semester long project related to thermodynamics concept. Students will be working with groups and are expected to design a calorimeter. Project includes research phase, modeling phase, prototype phase and testing phase. Design competition will be held among all the groups. All the groups are required to create Adobe Spark webpage for their design.

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome 1 Not Covered, 2 Introduced, 3 Reinforced, 4 Emphasized	1	2	3	4
(8) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.			X	
(9) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.		X		
(10) an ability to communicate effectively with a range of audiences.		X		
(11) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.		X		
(12) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.		X		
(7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.		X		

7. Brief list of topics to be covered

- I. Introduction
- II. Fundamentals (Chapter 1)
- III. The First Law (Chapter 2)
- IV. Properties of Pure Fluids (Chapter 3)
- V. Control Volume Analysis (Chapter 4)
- VI. The Second Law of Thermodynamics (Chapter 5)
- VII. Entropy (Chapter 6)
- VIII. Thermodynamic Relations (Chapter 11 Section 1 through 5)
- IX. Power Cycles (Chapter 9; select sections)

SaChE safety modules:

Students are expected to complete SaChE safety modules online and submit their certificates of completion to the instructor. There are only two relevant SaChE modules for CHE 201:

- The importance of Process Safety
- Hazard Recognition

CHE 205

1. Course number and name

CHE 205. Computational Methods in Chemical Engineering

2. Credits, contact hours and categorization

Credit hours: 3; Contact hours: 3 lecture hours per week; Engineering

3. Instructor's or course coordinator's name

Shafiq Mehraeen, Betul Bilgin, Brian Chaplin

4. Textbook, title, author, publisher, year

Numerical Methods for Chemical Engineers Using Excel, VBA, and MATLAB, by V.J. Law.

ISBN: 978-1-4665-7534-9, CRC Press, 2013.

a. Other supplemental materials

VBA programming; tutorials on the internet.

Consult with the following websites:

http://excelvbatutor.com/vba_tutorial.html

http://www.homeandlearn.org/basic_mathematical_operators.html

<http://www.automateexcel.com/vba-code-examples>.

5. Specific course information

a. Brief description of the content of the course (catalog description)

Computational methods and software relevant to unit operations. Excel spreadsheets (curve fitting, heat conduction), Matlab, Aspen Plus (process simulation), algorithms and object-oriented concepts in chemical engineering.

b. Prerequisites or co-requisites

Credit or concurrent registration in CHE 201 and MATH 210.

c. Indicate whether a required, elective, or selected elective.

Required.

6. Specific goals for the course

a. Specific outcomes of instruction.

- Formulate a chemical engineering problem as a mathematical model and select appropriate solution method.
- Write and modify VBA scripts for specific computational purposes and run them.
- Use Excel functionality to tabulate thermophysical property correlations and equations of state.
- Analyze 1d (linear) and 2d (bilinear) interpolation from data tables with VBA macros.
- Calculate vector coordinates for changes of basis vectors with Excel.
- Solve ordinary differential equations (ODE) with Excel.
- Solve optimization design problems with Excel Goal Seek and Solver functionalities and VBA macros.
- Solve partial differential equations (PDE).
- Carry out error analysis of data and regression analysis.

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome 1 Not Covered, 2 Introduced, 3 Reinforced, 4 Emphasized	1	2	3	4
(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.		X		
(6) an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.		X		

7. **Brief list of topics to be covered**

- Ch. 1: Roots of a single nonlinear equation
- Ch. 2: Visual Basic for applications programming
- Ch. 3: Linear algebra and systems of linear equations
- Ch. 4: Numerical differentiation and integration
- Ch. 5: Ordinary differential equations (initial value problems)
- Ch. 6: Ordinary differential equations (boundary value problems)
- Ch. 8: Partial differential equations
- Ch. 7: Regression analysis and parameter estimation
- Ch. 9: Linear programming, nonlinear programming, nonlinear equations, and nonlinear regression using Solver

CHE 210

1. Course number and name

CHE 210. Material and Energy Balances

2. Credits and contact hours

Credit hours: 4; Contact hours: 150 min/wk. lecture, 50 min/wk. discussion; Engineering

3. Instructor's or course coordinator's name

Lewis Wedgewood

4. Textbook, title, author, publisher, year

Introduction to Chemical Processes: Principles, Analysis, Synthesis by Regina M Murphy, McGraw-Hill, 1st edition, New York, 2007

5. Specific course information

a. Brief description of the content of the course (catalog description)

Material and energy balances applied to chemical systems. Introduction to chemical and physical properties. Introduction to the use of computers for chemical process calculations.

b. Prerequisites or co-requisites

Credit or concurrent registration in CHE 201 and CHE 205.

c. Indicate whether a required, elective, or selected elective.

Required.

6. Specific goals for the course

a. Specific outcomes of instruction.

- Synthesize a block flow diagram from a process description
- Identify and modify reaction pathways to achieve specific production while minimizing waste and occurrence of hazardous compounds
- Perform a degree of freedom analysis to determine degree of process specification
- Perform component, element, and total mass (or molar) material balances
- Quantify and make appropriate use of system performance specifications for reactors, splitters, separators
- Correctly identify modes of operation for process units (steady-state, transient, batch, continuous) and simplify the general material balance equation appropriately.
- Perform unit conversions as necessary.
- Calculate reaction equilibrium constants and determine fractional conversion and selectivity in equilibrium reactions.
- Use model equations and graphical data to determine composition in vapor-liquid phase equilibria
- Formulate process analysis problems as a system of linear equations
- Use tabulated data or model equations to evaluate changes in state properties
- Perform energy balances to determine basic heating/cooling requirements of a reactive process

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome 1 Not Covered, 2 Introduced, 3 Reinforced, 4 Emphasized		1	2	3	4
(1)	an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(2)	an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.		X	X	
(3)	an ability to communicate effectively with a range of audiences.	X			
(4)	an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	X			
(5)	an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	X			
(6)	an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	X			
(7)	an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.	X			

7. Brief list of topics to be covered

- Raw materials, reaction path synthesis
- Generation-consumption analysis
- Material balance equations
- Process flow sheeting
- Degree of freedom analysis
- Reactor materials balance
- Stream composition and system performance (conversion, selectivity, yield)
- Recycle, purge
- Chemical reaction equilibrium and chemical kinetics
- Separation technologies
- Phase equilibria
- Energy balances, Energy flows
- Energy data and model equations
- Energy balances on reactive processes

CHE 301

1. Course number and name

CHE 301. Chemical Engineering Thermodynamics

2. Credits, contact hours and categorization

Credit hours: 3; Contact hours: 3 lecture hours per week; Engineering

3. Instructor's or course coordinator's name

Michael Caracotsios

4. Textbook, title, author, publisher, year

Fundamentals of Chemical Engineering Thermodynamics, Themis Matsoukas, Prentice Hall, 2013.

a. other supplemental materials

- Chemical Engineering Thermodynamics Lecture Notes, will be distributed to all students free of charge via the UIC Blackboard, Michael Caracotsios, 2020
- J. M. Smith, H. C. Van Ness and M. M. Abbott, Introduction to Chemical Engineering Thermodynamics, 7th Edition, McGraw-Hill, 2005.
- S. I. Sandler, Chemical and Engineering Thermodynamics, 3rd Edition, John Wiley & Sons, Inc., 1999.

5. Specific course information

a. Brief description of the content of the course (catalog description)

Review of first and second laws of thermodynamics; Equations of state; Multi-component systems and phase equilibria; Chemical reaction thermodynamics and reaction equilibria.

b. Prerequisites or co-requisites

CHE 201 and CHE 205.

c. Indicate whether a required, elective, or selected elective.

Required.

6. Specific goals for the course

a. Specific outcomes of instruction.

Understand the basic concepts of Internal Energy, Gibbs Energy, Entropy and Enthalpy.

Apply control volume analysis to a variety of situations related the phase and chemical equilibrium and solve problems utilizing computer software.

Understand the theory of thermodynamics of ideal and non-ideal solutions

Understand the applicability and use of various equations of state in describing the thermodynamic state of pure fluids and mixtures thereof.

Determine equilibrium conditions between phases using equations of state and liquid activity coefficients.

Determine equilibrium conditions between reactants and products using equilibrium constants and Gibbs energy minimization concepts.

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome 1 Not Covered, 2 Introduced, 3 Reinforced, 4 Emphasized	1	2	3	4
(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(3) an ability to communicate effectively with a range of audiences.		X		

7. **Brief list of topics to be covered**

REVIEW AND BACKGROUND: Introduction to Thermodynamics in Chemical Engineering,
Elements of Differential and Integral Calculus, Numerical Methods for Algebraic and Differential Equations, Mass and Energy Balances in the Context of Thermodynamics, Tutorials on Programming with VBA and Athena Visual Studio

THERMODYNAMIC PROPERTIES OF SINGLE-COMPONENT FLUIDS: PVT Behavior of Single-component Fluids, Gibbs Phase Rule for Single-component Fluids, The Ideal Gas Behavior, Cubic and Virial Equations of State, Fugacity and Fugacity Coefficients from Equations of State, Thermodynamic Properties of Single-Phase and Two-Phase Systems

THERMODYNAMIC PROPERTIES OF MULTI-COMPONENT MIXTURES: Partial Molar Properties Excess and Residual Thermal Properties, the Gibbs Phase Rule for Multi-component Mixtures, PVT Behavior of Multi-component Mixtures, Cubic Equations of State and Mixing Rules, Fugacity and Fugacity Coefficients from Equations of State, Liquid Activity Coefficients for Non-ideal Solutions, Thermodynamic Properties of Single-Phase and Two-Phase Systems

MULTICOMPONENT PHASE EQUILIBRIUM: Criteria for Phase Equilibrium, Vapor-Liquid Equilibrium: The Fundamental Problem, Flash Calculations: PT, PH, PV, TH, TV and HV Calculations, VLE Data Analysis: Activity Coefficients from Experimental Data, Liquid-Liquid Equilibrium: The Fundamental Problem

CHEMICAL REACTION EQUILIBRIUM: Criteria for Equilibrium in Chemical Reactions, Gibbs Free Energy and Equilibrium Constants, Evaluation of Equilibrium Constants from Thermodynamic Data, Calculation of Equilibrium Conversion for Single and Multi-reaction Systems

CHE 311

1. Course number and name

CHE 311. Transport Phenomena I

2. Credits, contact hours and categorization

Credit hours: 3; Contact Hours: 150 min/week lecture; Engineering

3. Instructor's or course coordinator's name

Ying Liu, Vivek Sharma, Gang Cheng

4. Text book, title, author, publisher, year

R.B.Bird, W.E. Stewart & E. N. Lightfoot, Transport Phenomena, 2nd Ed., Wiley (2006)

5. Specific course information

- a. Brief description of the content of the course (catalog description)
Momentum transport phenomena in chemical engineering. Fluid statics. Fluid mechanics; laminar and turbulent flow; boundary layers; flow over immersed bodies.
- b. Prerequisites or co-requisites
Credit or concurrent registration in CHE 205, CHE 210, and MATH 220.
- c. Indicate whether a required, elective, or selected elective.
Required.

6. Specific goals for the course

- a. Performance indicators of student learning.
 - Calculate scalar, vector, and tensor operations.
 - Understand the essential physical principles of momentum transport and definition of viscosity.
 - Set up differential momentum balance and analyze boundary conditions.
 - Solve constant-coefficient ODEs and apply boundary conditions to determine steady state one-dimensional flow velocity and stress profiles.
 - Solve PDEs by using the methods of separation of variables and combination of variables to result the velocity profiles of the flow field which has two independent variables.
 - Formulate and solve the mathematical models to show how viscosity of a Newtonian fluid can be measured by various types of viscometers. Analyze the advantage and disadvantage of each viscometer.
 - Apply time-smoothed equations of change for turbulent flow.
 - Analyze the time-smoothed velocity profile near a wall.
 - Understand the physical meaning and mathematical expression of the empirical expressions for the turbulent momentum flux.
 - Understand the definition of friction factor. Use Moody diagram to solve various pipe-design problems.

- Set up macroscopic mass, momentum, and mechanical energy balances for isothermal flow systems.

a. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome 1 Not Covered, 2 Introduced, 3 Reinforced, 4 Emphasized	1	2	3	4
(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.			X	
(2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.		X		
(3) an ability to communicate effectively with a range of audiences.			X	
(4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.		X		
(7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.		X	X	

7. **Brief list of topics to be covered**

Week #1: Introduction and mathematical preliminaries

Scalars, vectors, and tensors—operations involving them

Week #2: Transport coefficients (Chap. 1, 9, 17)

Fourier's law of heat conduction

Fick's law of binary diffusivity

Newton's law of viscosity

Week #3: Shell Momentum balances and velocity distribution in laminar flow

Week #4: Apply shell momentum balances on more representative laminar flows

Week #5: Conservation equations

Differential equations for mass and linear momentum

Week #6: Apply differential mass and linear momentum balance on representative flow problems

Week #7: Conservation equations for angular momentum* and energy balance

Week #8: Macroscopic balances for mass and momentum

Week #9: Macroscopic balances for mechanical energy

Week #10: Velocity distributions with more than one independent variable

Time dependence

Stream function

Week #11: Similarity analysis

Week #12: Introduction of turbulent flow

Week #13: Time-smoothed velocity profile for turbulent flow near a wall

Week #14: Friction factors for flow in a circular tube

Week #15: Friction factors for flow around a sphere

CHE 312

1. Course number and name

CHE 312. Transport phenomena II

2. Credits, contact hours and categorization

Credit hours: 3; Contact hours: 3 lecture hours per week; Engineering

3. Instructor's or course coordinator's name

Vivek Sharma, Christos Takoudis

4. Textbook, title, author, publisher, year

Required –Transport Phenomena, Revised 2nd Edition by R. B. Bird, W. E. Stewart and E. N. Lightfoot. John Wiley & Sons, Inc (2006).

5. Specific course information

a. Brief description of the content of the course (catalog description)

Heat and mass transport phenomena. Heat conduction, convection and radiation. Heat exchanger design. Diffusion. Mass transfer coefficients.

b. Prerequisites or co-requisites

CHE 311

c. Indicate whether a required, elective, or selected elective.

Required.

6. Specific goals for the course

a. Specific outcomes of instruction.

The broad objectives of this course are:

(a) To develop clear understanding of the basic mechanisms of heat and mass transfer in physical processes encountered in daily living experiences and in chemical engineering practice; and

(b) To develop the skills to describe heat and mass processes in terms of mathematical models, make appropriate assumptions to simplify the models, and perform calculations to obtain qualitative and/or quantitative information which can lead to better design or better understanding of the processes.

a. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome	1	2	3	4
(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.			X	
(3) an ability to communicate effectively with a range of audiences.	X			
(4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.		X		
(5) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	X			
(6) an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	X			
(7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.			X	

7. Brief list of topics to be covered

- Introduction to Conduction
- One-Dimensional Steady-State Conduction
- Two-Dimensional Steady-State Conduction
- Transient Conduction
- Introduction to Convection
- Heat Transfer in External and Internal Flows
- Free Convection
- Heat Exchangers
- Radiation Heat Transfer, Radiation Exchange Between Surfaces
- Diffusion Mass Transfer
- Diffusion – Homogeneous Chemical Reaction Systems

- Diffusion – Heterogeneous Chemical Reaction Systems
- Transient Diffusion and Applications
- Diffusion and Chemical Reaction Inside a Porous Catalyst
- Mass Transfer Design Calculations
- Multimode Heat and Mass Transfer

CHE 313

1. Course number and name

CHE 313. Transport phenomena III

2. Credits, contact hours and categorization

Credit hours: 3; Contact hours: 3 lecture hours per week; Engineering

3. Instructor's or course coordinator's name

Sangil Kim, Christos Takoudis, Gang Cheng

4. Textbook, title, author, publisher, year

Wankat, P. C., Separation Processes Engineering: Includes Mass Transfer Analysis, 4th edition, Prentice Hall, 2017. -OR-

Christie John Geankoplis, Allen Hersel, Daniel H. Lepek, Transport processes and Separation Process Principles, 5th edition, Prentice Hall, 2018

a. Other supplemental materials

Seader, J.D et al., Separation Processes Principles, 3rd edition, John Wiley & Sons, New Jersey, 2011.

5. Specific course information

a. Brief description of the content of the course (catalog description)

Mass transfer and phase equilibria. Multistage separations; applications in distillation; extraction; absorption and drying.

b. Prerequisites or co-requisites

CHE 301

c. Indicate whether a required, elective, or selected elective.

Required.

6. Specific goals for the course

a. Specific outcomes of instruction.

- Be familiar with separation principles, design, separation mechanisms, processes, and applications
- Understand the underlying thermodynamic and mass transfer principles
- Understand the operation of simple and multistage batch distillation systems, and McCabe-Thiele method
- Discuss the differences between batch and continuous operation

- Explore use of various types of separation techniques
- Explore use of alternative separation techniques to the existing ones

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome	1	2	3	4
(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.				X
(3) an ability to communicate effectively with a range of audiences.			X	
(4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.		X		
(5) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.			X	
(6) an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.			X	
(7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.				X

7. Brief list of topics to be covered

Introduction to separation processes
 Flash Distillation
 Binary distillation
 Multicomponent distillation
 Batch distillation
 Stage and packed column design
 Absorption and stripping
 Liquid-liquid extraction
 Membrane separations

CHE 321

1. Course number and name

CHE 321. Chemical Reaction Engineering

2. Credits, contact hours and categorization

Credit hours: 3; Contact hours: 3 lecture hours per week; Engineering

3. Instructor's or course coordinator's name

Vikas Berry, Meenesh Singh

4. Textbook, title, author, publisher, year

"Elements of Chemical Reaction Engineering",
Fifth, Fourth or Third Edition, by Fogler, H. Scott

5. Specific course information

a. Brief description of the content of the course (catalog description)

Kinetics of homogeneous single reactions. Ideal reactors: batch, stirred tank and plug flow systems. Conversion and yield in multiple reactions. Design and optimization of reactors. Non-isothermal reactors.

b. Prerequisites or co-requisites

CHE 210, CHE 301, MATH 220.

c. Indicate whether a required, elective, or selected elective.

Required.

6. Specific goals for the course

a. Specific outcomes of instruction.

1. Compare batch reactors, continuous stirred tank reactors (CSTR), and plug flow reactors (PFR).
2. Derive, solve the mass balance for isothermal reaction in a CSTR, PFR, or batch reactor.
3. Calculate the equilibrium conversion in reactor type for isothermal or adiabatic conditions.
4. Solve the governing mass and energy balances for a CSTR, PFR, packed bed, or batch reactor.
5. Estimate reaction orders, reaction rate constants, and Arrhenius parameters from experimental kinetic data.
6. Solve the governing mass balances for multiple reactions in a CSTR, PFR, or batch reactor
7. Suggest which type of continuous reactor (CSTR or PFR) is advantageous for maximizing product selectivity for series or parallel reactions
8. Derive a Langmuir-Hinshelwood kinetic rate law
9. Incorporate a mass transfer coefficient into a reaction rate expression to account for external mass transfer limitations
10. Calculate the effectiveness factor for a catalyst particle given physical properties of the particle, the reaction rate constant, and the diffusion coefficient for the reactant
11. Calculate the residence time distribution from a tracer study.
12. Draw similarity between electron transport through nanoparticles and chemical reactions.

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome	1	2	3	4
(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.				X
(3) an ability to communicate effectively with a range of audiences.		X		
(4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.			X	
(5) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.			X	
(6) an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.		X		
(7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.			X	

7. **Brief list of topics to be covered**

Reaction fundamentals

Reactor Types, General Mole Balances, Conversion, Reactor Sizing, Stoichiometric Tables

Isothermal Reactor Design

SOFTWARE - Polymath and MaTLAB for Reactor Design

Pressure Drop in Reactors

Unsteady State Operation, Rate Analysis

Multiple Reactions

General Energy Balance & CSTR, PFR Energy Balance, Batch Energy Balance

Equilibrium Conversion

Non-Isothermal Reactor Example

Multiple Steady States in a CSTR/Multiple Steady States/Multiple Reactions

Summary of Non-Isothermal Reactors

Catalytic Reactions, Langmuir-Hinshelwood Rate Laws

Langmuir-Hinshelwood Rate Laws, Heterogeneous Reactors, Mass Transfer

External and Internal Mass Transfer Limitations

Effectiveness Factor and Falsified Kinetics

Residence Time Distribution

Zero-Parameter Flow Models, Non-Ideal Flow and combination of ideal reactors

CHE 341

1. Course number and name

CHE 341. Chemical Process Control

2. Credits, contact hours and categorization

Credit hours: 3; Contact hours: 3 lecture hours per week; Engineering

3. Instructor's or course coordinator's name

Michael Caracotsios

4. Textbook, title, author, publisher, year

Process Dynamics and Control, D.E. Seborg, T.F. Edgar, D.A. Mellichamp, and F.J. Doyle, Wiley, 2011.

a. other supplemental materials

- Chemical Process Control Lecture Notes, will be distributed to all students free of charge via the UIC Blackboard, Michael Caracotsios, 2020
- T.E. Marlin, Process Control: Designing Processes and Control Systems for Dynamic Performance, McGraw-Hill, 2000; B.A. Ogunnaike and W.H. Ray, Process Dynamics, Modeling and Control, Oxford, 1994; J.B. Riggs and M. N. Karim, Chemical and Bio-Process Control, Ferret Publishing, Austin, Texas, 2016; G. Stephanopoulos, Chemical Process Control: An Introduction to Theory and Practice, Prentice Hall, 1984.
- SOFTWARE: Athena Visual Studio, EXCEL based Thermodynamic Calculator CHE301 Calculator.xlsm; EXCEL based Process Control tool CHE341 Loop Simulator.xlsm

5. Specific course information

a. Brief description of the content of the course (catalog description)

Analysis and design of chemical processes and control systems. Feedback and feedforward controllers. Stability, tuning, and simulation of P-I-D controllers. Introduction to the control of chemical plants and digital process control.

b. Prerequisites or co-requisites

MATH 220 and CHE 312 and CHE 313 and CHE 321.

c. Indicate whether a required, elective, or selected elective.

Required.

6. Specific goals for the course

a. Specific outcomes of instruction. Students will:

- Understand the basic concepts of process dynamic modeling for a variety of chemical engineering systems.
- Know how to develop empirical models from process data using the step response methodology
- Understand the theory of feedback and feed forward control and determine optimal controller settings for stable and robust performance.
- Know when to select and how to implement cascade and feed forward controllers
- Apply available computer software (Matlab, Excel or Athena Visual Studio), or use Laplace transformation theory to solve process dynamic

models and investigate their open and closed loop response to various external disturbances.

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome		1	2	3	4
(1)	an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(2)	an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.				X
(3)	an ability to communicate effectively with a range of audiences.			X	

7. Brief list of topics to be covered

INTRODUCTION TO PROCESS CONTROL IN CHEMICAL ENGINEERING: Elements of Differential and Integral Calculus, Numerical Methods for Algebraic and Differential Equations, Mass and Energy Balances in the Context of Process Control, Tutorials on Programming with VBA and Athena Visual Studio

CONTROL OF INDIVIDUAL UNITS: Reactors, Heat Exchangers, Distillation Columns, Compressors, Other Unit Operations

CONTROL CONCEPTS AND DYNAMIC MODELING: Development of Dynamic Process Models, Solution of Dynamic Models and Open Loop Response, Laplace Transforms and Transfer Functions, Linearization of Process Dynamic Models, Process Identification from Plant Data

FEEDBACK CONTROL FUNDAMENTALS: P, PI and PID Controllers, Block Diagrams and Routh-Hurwitz Stability Criterion, Instrumentation Hardware and Representation

FEEDBACK CONTROLLER DESIGN AND TUNING: Criteria for Controller Tuning, PID Controller Tuning Methods, Guidelines and Troubleshooting, Common Control Loops

ADVANCED CONTROL SYSTEMS AND MIMO SYSTEMS: Feedforward and Cascade Control, Interaction and Loop Pairing and Decoupling, Relative Gain Array Control of Common Industrial Process Units, Model Predictive Control of Linear Processes, Advanced Process Control and Real-Time Optimization

CHE 381

1. Course number and name

CHE 381: Chemical Engineering Laboratory I

2. Credits, contact hours and categorization

Credit hours: 2; Contact hours: 4 lab hours per week; Engineering

3. Instructor's or course coordinator's name

Alan D. Zdunek

4. Textbook, title, author, publisher, year

Unit Operations of Chemical Engineering, W.L. McCabe, J.C. Smith, P. Harriott, 7th Edition, McGraw-Hill, 2005.

Other supplemental materials: Provided online in the UIC Blackboard course folder to help aid in successfully preparing and carrying out the experiments. Material includes; Lab report templates, lab presentation templates, laboratory safety information, technical writing guidelines, error analysis methodology, specific articles and files related to the various experiments including vendor information on the experimental apparatus and components, theory and empirical correlations, engineering charts, and material and physical properties.

5. Specific course information

- a. Brief description of the content of the course (catalog description) Heat and momentum transfer operations associated with chemical processes. These include heat exchangers, fluid properties and fluid flow. Technical report writing, computer calculations.
- b. Prerequisites or co-requisites CHE 312.
- c. Indicate whether a required, elective, or selected elective.
Required.

6. Specific goals for the course

Provide students with a hands-on opportunity to work with physical equipment in a laboratory environment.

- a. Specific outcomes of instruction.
 - 1 Solve complex, open-ended engineering problems creatively, in a team setting.
 - 2 Integrate knowledge and skills from prior chemical engineering courses.
 - 3 Design and conduct realistic experiments, analyze data, draw concrete conclusions and make relevant recommendations.
 - 4 Communicate effort and findings effectively through the written and oral word.
 - 5 Master and apply safety analysis and safety practices in a laboratory environment.

- b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome		1	2	3	4
(1)	an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(2)	an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	X			
(3)	an ability to communicate effectively with a range of audiences.				X
(4)	an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.			X	
(5)	an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.				X
(6)	an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.				X
(7)	an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.			X	

7. **Brief list of topics to be covered**

Specific lab experiments performed are:

- 1) Viscosity: Non-Newtonian fluids
- 2) Distillation
- 3) Fluid Flow in Pipe Systems
- 4) Fluidized Beds
- 5) Process Control
- 6) Membrane Separation

List of Topics:

Experimental verification of theoretical principles, familiarization with engineering equipment and exposure to an engineering sense, teamwork skills; participation, leadership and team member assessment, time value; experimental planning, implementation, and completion, experimental skills; design, process safety analysis, data collection, error analysis, written and oral technical communication, integrated problem-solving.

Other topics:

Week #1: General lab safety information, technical writing and technical presentation tips and guidelines.

Week # 2: Industrial Chemical Hygiene Plan review

Week #12-14: SaChE Module: Chemical Reactivity Hazards

CHE 382

1. Course number and name

CHE 382: Chemical Engineering Laboratory II

2. Credits, contact hours and categorization

Credit hours: 2; Contact hours: 4 lab hours per week; Engineering

3. Instructor's or course coordinator's name

Alan D. Zdunek

4. Textbook, title, author, publisher, year

Unit Operations of Chemical Engineering, W.L. McCabe, J.C. Smith, P. Harriott, 7th Edition, McGraw-Hill, 2005.

Other supplemental materials: Provided online in the UIC Blackboard course folder to help aid in successfully preparing and carrying out the experiments. Material includes; Lab report templates, lab presentation templates, laboratory safety information, technical writing guidelines, error analysis methodology, specific articles and files related to the various experiments including vendor information on the experimental apparatus and components, theory and empirical correlations, engineering charts, and material and physical properties.

5. Specific course information

- a. Brief description of the content of the course (catalog description) Heat, momentum and mass transfer operations associated with chemical processes; these include distillation columns, reactors, humidifiers, and evaporators. Technical report writing, computer calculations.
- b. Prerequisites or co-requisites CHE 381 and credit or concurrent registration in CHE 313.
- c. Indicate whether a required, elective, or selected elective.
Required.

6. Specific goals for the course

Provide students with a hands-on opportunity to work with physical equipment in a laboratory environment.

- a. Specific outcomes of instruction.
 - 1 Solve complex, open-ended engineering problems creatively, in a team setting.
 - 2 Integrate knowledge and skills from prior chemical engineering courses.
 - 3 Design and conduct realistic experiments, analyze data, draw concrete conclusions and make relevant recommendations.
 - 4 Communicate effort and findings effectively through the written and oral word.
 - 5 Master and apply safety analysis and safety practices in a laboratory environment.

- b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome		1	2	3	4
(1)	an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(2)	an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	X			
(3)	an ability to communicate effectively with a range of audiences.				X
(4)	an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.			X	
(5)	an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.				X
(6)	an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.				X
(7)	an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.			X	

7. **Brief list of topics to be covered**

Specific lab experiments performed are:

- 1) Gas Absorption
- 2) Continuous Distillation or Ultrafiltration
- 3) Heat Transfer
- 4) Evaporation
- 5) Humidification
- 6) Liquid-Liquid Extraction

List of Topics:

Experimental verification of theoretical principles, familiarization with engineering equipment and exposure to an engineering sense, teamwork skills; participation, leadership and team member assessment, time value; experimental planning, implementation, and completion, experimental skills; design, process safety analysis, data collection, error analysis, written and oral technical communication, integrated problem-solving.

Other topics:

Week #1: General lab safety information, technical writing and technical presentation tips and guidelines.

Week # 2: JT Baker Hazardous Chemical Safety Course; Toxic Chemicals, Flammables, Corrosive Chemicals, Reactive Chemicals, Insidious Hazards, Compressed Gases

Week #9: SaChE Module: Dust Explosion Control

CHE 396

1. Course number and name

CHE 396: Chemical Engineering Design I

2. Credits and contact hours

Credit hours: 4. Contact hours: Two, 70 min lecture periods; Engineering.

3. Instructor's or course coordinator's name

Betul Bilgin

4. Textbook, title, author, publisher, year

Towler, G., Sinnott, R., Chemical Engineering Design: Principles, Practice and Economics of Plant and Process Design, Second Edition, Elsevier, 2012.

a. Other supplemental materials

Seider, Seader, Lewin, Widagdo, Product and Process Design Principles, 3rd Edition, Wiley, 2009

Writing Style and Standards in Undergraduate Reports, 3e, by Jeffrey Donnell, Sheldon Jeter, Colin MacDougall, and Jacqueline Snedeker

5. Specific course information

a. Brief description of the content of the course (catalog description)

Introduction to modern, process design and development, engineering economics, and report writing. Design and cost of equipment relating to materials handling to heat transfer, mass transfer, and reactors.

b. Prerequisites or co-requisites

CHE 312, CHE 313, CHE 321

c. Indicate whether a required, elective, or selected elective.

Required

6. Specific goals for the course

a. Specific outcomes of instruction.

Provide a real-world perspective on process design

Introduce design and selection of equipment

Give insight into practical constraints that govern design decisions

Help you translate knowledge into practice & start solving real problem

Improve presentation, analytical thinking and team working skills

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome		1	2	3	4
(1)	an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(2)	an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.				X
(3)	an ability to communicate effectively with a range of audiences.			X	
(4)	an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.				X
(5)	an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.				X
(6)	an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.			X	
(7)	an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.			X	

7. Brief list of topics to be covered

1. Chemical Design Introduction
2. Flowsheet development
3. Utilities and Energy Recovery
4. Process simulation
5. Process control
6. Materials of construction
7. Capital cost
8. Operating cost
9. Economic Analysis
10. Process Hazard, Professional ethics
11. Professional Ethics Presentations
12. General Site Consideration

CHE 397

1. Course number and name

CHE 397: Chemical Engineering Design II

2. Credits and contact hours

Credit hours: 4. Contact hours: Two, 70 min lecture periods; Engineering.

3. Instructor's or course coordinator's name

Betul Bilgin

4. Textbook, title, author, publisher, year

Towler, G., Sinnott, R., Chemical Engineering Design: Principles, Practice and Economics of Plant and Process Design, Second Edition, Elsevier, 2012.

a. Other supplemental materials

Seider, Seader, Lewin, Widagdo, Product and Process Design Principles, 3rd Edition, Wiley, 2009

Writing Style and Standards in Undergraduate Reports, 3e, by Jeffrey Donnell, Sheldon Jeter, Colin MacDougall, and Jacqueline Snedeker

5. Specific course information

a. Brief description of the content of the course (catalog description)

Application of principles and design methodology of chemical engineering to the design of large-scale chemical processes and plants. A major design project is assigned for solution and presentation at the College Engineering Expo by students working in small groups with industry mentors. Extensive computer use required.

b. Prerequisites or co-requisites

CHE 396

c. Indicate whether a required, elective, or selected elective.

Required

6. Specific goals for the course

a. Specific outcomes of instruction.

This course involves the application of all prior CHE concepts to the chemical process design. Design needs to be analyzed from safety, environmental impact, feasibility, risk and economics perspectives. Course aims to equip students with the skill to evaluate open-ended design problems conceptually, analytically and practically.

Students will be able to:

- Develop design plan and project charter for each design phase
- Work effectively with team members
- Present effectively technical material based on audience
- Identify key components in market analysis and construct design basis
- Analyze competing processes
- Write technical content in a way that is appropriate to the audience
- Apply economic analysis
- Identify potential safety concerns in the process and formulate strategies to mitigate those
- Simulate full process with Aspen Plus or SuperPro
- Size process equipment

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome		1	2	3	4
(1)	an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(2)	an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.				X
(3)	an ability to communicate effectively with a range of audiences.				X
(4)	an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.				X
(5)	an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.				X
(6)	an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.			X	
(7)	an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.			X	

7. Brief list of topics to be covered

- 13. Process description, Design basis, BFD, PFD Flowsheet development
 - 14. Equipment list and sizing
 - 15. Utilities demand summary for entire plant
 - 16. Process simulation
 - 17. Process control strategy, process parameters
 - 18. Materials of construction
 - 19. Plant Cost Analysis (NPV, sensitivity analysis, payback period, capital cost, Aspen economic analysis
- Environmental and safety concerns, process hazards, mitigation.

CHE 422

1. Course number and name

CHE 422. Biochemical Engineering

2. Credits, contact hours and categorization

Credit hours: 3. Contact hours: 3 lecture hours per week.

3. Instructor's or course coordinator's name

Gang Cheng

4. Text book, title, author, publisher, year

Bioprocess Engineering: Basic Concepts by Michael L Shuler, Fikret Kargi, Matthew DeLisa, Prentice Hall, 3rd Edition, 2017.

a. Other supplemental materials

Molecular Biology of the Cell, 6th Edition, Bruce Alberts, Alexander D. Johnson, Julian Lewis, David Morgan, Martin Raff, Keith Roberts, Peter Walter, Garland Science, 2014.

5. Specific course information

a. Brief description of the content of the course (catalog description)

Enzyme-catalyzed and microbially-mediated processes. Free and immobilized enzymes. Batch and continuous cell cultures. Transport phenomena in microbial systems and fermentation processes. Design of biological reactors.

b. Prerequisites or co-requisites

Consent of the instructor.

c. Indicate whether a required, elective, or selected elective.

Elective.

6. Specific goals for the course

a. Specific outcomes of instruction.

To be able to describe and understand recombinant DNA, fermentation and separation methods such as PCR, filtration and centrifugation, cell rupture, and purification techniques such as adsorption, solvent extraction, membrane filtration, chromatography (affinity, ion, hydrophobic, reverse phase), precipitation, and crystallization. To apply fundamental principles for the separation of biopharmaceuticals via chemical engineering unit operations.

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcomes		1	2	3	4
(1)	an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.		x		
(2)	an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.			x	
(3)	an ability to communicate effectively with a range of audiences.			x	
(4)	an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	x			
(5)	an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	x			
(6)	an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	x			
(7)	an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.		x		

7. **Brief list of topics to be covered**

The course is aimed at senior undergraduate and graduate students and is designed to move engineers and other non-biologists past the initial language and knowledge barriers of the Biochemical Engineering field.

- Fundamental recombinant DNA techniques
- Genetically engineered biomacromolecules,
- General molecular biology techniques; Polymerase Chain Reaction (PCR) basics, site-directed and random mutagenesis, display techniques, DNA sequencing, DNA and protein gel electrophoresis, cutting and splicing DNA, protein expression, DNA and RNA hybridization techniques, and Western blots.
- Protein engineering and protein design approaches; knowledge-based design, computational protein design, and directed evolution.
- Examples of engineered proteins with novel structural and functional properties
- Fermentation (basic principles of fermentation processes, scaling-up),
- Product recovery and separation (cell disruption, sedimentation, filtration/centrifugation, solvent extraction, adsorption, precipitation), product purification (adsorption, ultrafiltration, crystallization) and
- Final separation and advances in bioseparation strategies.

Process design; creating engineered strains and cell culture/fermentation. Purification of the end product for energy, biotechnology or biomedical applications.

CHE 425

1. Course number and name

CHE 425 Nanotechnology for Pharmaceutical Applications

2. Credits, contact hours and categorization

Credit hours: 3. Contact hours: 3 lecture hours per week.

3. Instructor's or course coordinator's name

Ying Liu

4. Text book, title, author, publisher, year

Drug Delivery: Engineering Principles for Drug Therapy Topics in Chemical Engineering, W. Mark Saltzman, Oxford, 2001.

a. Other supplemental materials

Selected reading materials from high-impact journals and a few textbooks, including but not limited to, Science, Nature, JACS, PRL, Biomacromolecules, ACS Nano, Nano Letters, Journal of Colloids and Interfacial Science.

5. Specific course information

a. Brief description of the content of the course (catalog description)

Emerging role of nanostructures in drug development and delivery. Principles of nanostructure formation, characterization, surface functionalization, in vitro and in vivo transport, and visualization.

b. Prerequisites or co-requisites

CHE 301 and CHE 311 and CHE 312; or consent of the instructor.

c. Indicate whether a required, elective, or selected elective.

Elective.

6. Specific goals for the course

a. Specific outcomes of instruction. Students will be able to:

- Understand the fundamental and essential concepts used in nanotechnology, synthesis, fabrication and characterization.
- Analyze thermodynamic and kinetic control of the nanoparticle formation and stabilization
- Analyze drug loading, encapsulation efficiency, and transport
- Discuss nanomaterial selection, design and fabrication.
- Discuss nanomaterial impact on the environment
- Apply written and oral communication skills to a final term paper and presentation

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcomes		1	2	3	4
(1)	an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(2)	an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.				X
(3)	an ability to communicate effectively with a range of audiences.			X	
(4)	an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.			X	
(5)	an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.			X	
(6)	an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	X			
(7)	an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.				X

7. Brief list of topics to be covered

- Factors that impact the development ability of drug candidates.
- Evolution of controlled drug delivery systems.
- Advanced Nanotechnologies and Micro technologies for Drug Delivery
- Self-assembled delivery vehicles: basic theoretical considerations and modeling concepts.
- Nanoparticle characterization
- Liposomes as intravenous solubilizers.
- Polymeric nanoparticle drug delivery systems.
- Nano suspensions with enhanced drug dissolution rates of poorly water-soluble drugs.
- Pharmaceutical Fundamentals
- Physiological, Biochemical, and Chemical barriers to Oral Drug Delivery.
- Cell culture models for drug transport studies.

CHE 433

1. Course number and name

CHE 433. Process Simulation with Aspen Plus

2. Credits, contact hours and categorization

Credit hours: 3; Contact hours: 3 lecture hours per week; Engineering

3. Instructor's or course coordinator's name

Michael Caracotsios

4. Textbook, title, author, publisher, year

Process Simulation with Aspen Plus Lecture Notes, will be distributed to all students free of charge via the UIC Blackboard, Michael Caracotsios, 2020.

a. other supplemental materials

- Ralph Schefflan, Teach yourself the Basics of Aspen Plus, 2nd Edition, Wiley. (2016).
- 2. Gavin Towler and Ray Sinnott, Chemical Engineering Design, 2nd Edition, Elsevier Ltd. (2013).
- Lorenz T. Biegler, Ignacio E. Grossmann and Arthur W. Westerberg, Systematic Methods of Chemical Process Design, Prentice Hall International Series, December (1999).
- T. F. Edgar and D. M. Himmelblau, Optimization of Chemical Processes, McGraw-Hill, Inc.(1988).
- A. W. Westerberg, H. P. Hutchison, R. L. Motard and P. Winter, Process Flowsheeting, 1st Edition, Cambridge University Press (1979).
- SOFTWARE: Athena Visual Studio from www.AthenaVisual.com; EXCEL, ASPEN Plus and optionally MATLAB and ASPEN Hysys.

5. Specific course information

a. Brief description of the content of the course (catalog description)

Main principles of Process Simulation (Material and Energy Balances) relevant to chemical and biological processes. Process Simulation or Flowsheeting is defined as the use of commercially available products (like ASPEN Plus, ASPEN Hysys, UniSim, VMG, ChemCad, gProms, Pro II) to perform steady-state Heat and Weight Balances, Sizing and Cost calculations for chemical, biological, petrochemical and refining processes.

b. Prerequisites or co-requisites

CHE 312 and CHE 313 and CHE 321; or consent of the instructor.

c. Indicate whether a required, elective, or selected elective.

Elective.

6. Specific goals for the course

a. Specific outcomes of instruction. Students will:

- Select and justify choices for physical property models required for accurate heat and weight balance calculations.
- Design and model unit operations such as chemical reactors, heat exchangers, heaters and coolers, pumps, compressors and turbines, mixers, distillation and packed columns and user specific operations.
- Combine units operations in a cohesive Flowsheet that describes existing or conceptual processes.

- Familiarize themselves with computational aspects such as design specs, calculator blocks, sensitivity analyses, flow sheet convergence
 - Perform optimization studies.
- b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome	1	2	3	4
(1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.			X	
(3) an ability to communicate effectively with a range of audiences.			X	
(4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.			X	
(5) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.			X	

7. **Brief list of topics to be covered**

INTRODUCTION TO PROCESS FLOWSHEETING IN CHEMICAL ENGINEERING: Elements of Differential and Integral Calculus, Numerical Methods for Algebraic and Differential Equations, Mass and Energy Balances in the Context of Process Simulation, Tutorial on Programming with ASPEN Plus and Athena Visual Studio

THERMODYNAMICS AND PROCESS UNIT OPERATIONS: Vapor-Liquid Equilibrium and Property Methods, VLE Data Regression, Mixers and Splitters, Two and Three Phase Flash Blocks, Heaters, Coolers and Condensers, Distillation, Pumps, Compressors and Turbines, Example Flowsheets and Design Specifications

PROCESS FLOWSHEETING FUNDAMENTALS: Recycle Stream Selection and Tear Equations, Convergence Methods for Tear Equations, Calculator and Transfer Function Blocks, Simultaneous Recycle Convergence and Design Specifications, Sensitivity Analysis and Case Studies, Sequential Modular vs. Equation Based Solution Methods

CHEMICAL KINETICS AND REACTOR UNIT OPERATIONS CSTR: Plug Flow and Fixed Bed Reactors, Conversion and Equilibrium Reactors, Example Flowsheets with Design Specifications, Excel Interface for Custom Reactor and Physical Models

COMPLEX UNIT OPERATIONS AND ADVANCED SIMULATION TOPICS: Packed columns for Absorption and Stripping Operations, Rate based vs. Equilibrium Models, Homogeneous and Heterogeneous Azeotropic Distillation, Process Optimization and Pinch Analysis for Energy Optimization, Heat Exchangers and Heat Exchanger Network Optimization, Solids Handling and Electrolyte Systems

CHE 451

1. Course number and name

CHE 451 Renewable Energy Technologies

2. Credits, contact hours and categorization

Credit hours: 3; Contact hours: One, 165 min/week lecture; Engineering

3. Instructor's or course coordinator's name

Said Al-Hallaj

4. Text book, title, author, publisher, year

"Hybrid Hydrogen Systems for Stationary and Transportation Applications", Said Al-Hallaj and Kristofer Kiszynski, 1st Edition, 2011, Springer Publishing, ISBN 978-1-84628-466-3.

5. Specific course information

a. Brief description of the content of the course (catalog description)

The course will cover fundamentals of renewable energy technologies such as solar, wind and biomass. In addition, course will provide an introduction to various energy storage technologies such as batteries and fuel cells. Finally, the hydrogen economy promises and challenges such as hydrogen storage and transportation will be discussed and analyzed.

b. Prerequisites or co-requisites

Consent of the instructor.

c. Indicate whether a required, elective, or selected elective.

Elective.

6. Specific goals for the course

a. Specific outcomes of instruction. Students will be able to:

- Understand the fundamentals and essential concepts used in renewable energy technologies such as solar, wind, biomass.
- Understand and analyze energy storage technologies such as batteries, fuel cells and supercapacitors.
- Address key ideas, as well as technical and economic issues related to renewable energy in a final project term paper.

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcomes		1	2	3	4
(1)	an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.			X	
(2)	an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.				X
(3)	an ability to communicate effectively with a range of audiences.				X
(4)	an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.				X
(5)	an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.				X
(6)	an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	X			
(7)	an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.				X

7. Brief list of topics to be covered

- 1 Introduction (Renewable Energy and Sustainability)
- 2 Solar Energy Calculations and Solar Photovoltaics (PV)
- 3 Building Integrated Photovoltaics
- 4 Wind Energy Basics and Applications
- 5 Bioenergy
- 6 Hydrogen Economy and Hydrogen Storage
- 7 Fuel Cells
- 8 Energy Storage for Smart Grid Applications
- 9 Geothermal Energy
- 10 Hydrogen Economy: Production and Infrastructure
- 11 Optimization of Renewable Hydrogen Energy Systems
- 12 Control of Hybrid Fuel Cell/Battery Systems
- 13 Solar Desalination and Hybrid Fuel Cell/Desalination

CHE 494

Selected Topics in Chemical Engineering

1. Course number and name

CHE 494 Two-Dimensional (2D) Nanomaterials: Science, Processing, and Applications

2. Credits, contact hours and categorization

Credit hours: 3; Contact hours: Two, 75 min/wk. lectures; Engineering.

3. Instructor's or course coordinator's name

Sanjay Behura

4. Text book, title, author, publisher, year

2D Materials: Properties and Devices, P. Avouris, T. F. Heinz, and T. Low, Cambridge University Press, 2017.

a. Other supplemental materials

Raman Spectroscopy in Graphene Related Systems, A. Jorio, M. S. Dresselhaus, R. Saito, G., Dresselhaus; Selected Journal Articles from Nature, Science, Nano Letters, ACS Nano, JACS, PRL, PRB, and More.

5. Specific course information

a. Brief description of the content of the course (catalog description)

The goal of this course is to prepare students with foundational understanding of advanced two-dimensional (2D) nanomaterials including graphene, and the tools required to characterize such atomically-thin materials, and their potential applications.

b. Prerequisites or co-requisites

Consent of the instructor.

c. Indicate whether a required, elective, or selected elective.

Elective.

6. Specific goals for the course

a. Specific outcomes of instruction. Students will:

- Understand what a 2D material is and how are they different from their bulk 3D counterparts.
- Learn how atomically-thin 2D materials and their quantum heterostructures are produced.
- Learn what are the unique and superlative properties of 2D materials and how to characterize them
- Understand the range of potential applications of the 2D materials
- Produce 2D materials in the lab

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcomes		1	2	3	4
(1)	an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.		X		
(2)	an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	X			
(3)	an ability to communicate effectively with a range of audiences.			X	
(4)	an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.				X
(5)	an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.				X
(6)	an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.				X
(7)	an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.				X

7. Brief list of topics to be covered

Science of 2D Materials: Low-Dimensional Materials; Layered Materials; 2D Materials; 3D vs. 2D Materials; Modifications of Materials Properties: Doping in Bulk and 2D Materials; Fundamentals of Graphene, Hexagonal Boron Nitride, and Transition Metal Dichalcogenides; and Science of van der Waals 2D Heterostructures. (This part will focus on physics and chemistry of 2D materials, their unique structure and intriguing properties).

Processing of 2D Materials: Synthesis Techniques for Advanced 2D Materials (Graphene, Hexagonal Boron Nitride, and Transition Metal Dichalcogenides) and their van der Waals bound 2D Heterostructures. (This part will focus on both top-down and bottom-up strategies for the synthesis of 2D materials and understanding the ensuing growth mechanisms).

Characterization and Application of 2D Materials: Analysis of 2D Materials by Raman Spectroscopy, X-Ray Photoelectron Spectroscopy, Scanning Electron Microscopy, Transmission Electron Microscopy, and Scanning Probe Microscopy. Important Applications of 2D Materials including Nanoelectronics (Field Effect Transistors), Energy Conversion (Photovoltaic Solar Cells), and Energy Storage (Batteries). (This part will focus on basic working principles of advanced characterization tools, detailed analysis of 2D materials, and their potential applications).

2D Lab Work: Synthesis of Graphene on Metal Films via Chemical Vapor Deposition (CVD), Its Chemical Transfer, and Raman Spectroscopic and AFM Characterization.

CHE 494

Selected Topics in Chemical Engineering

1. Course number and name

CHE 494 Entrepreneurship in Chemical Engineering

2. Credits, contact hours and categorization

Credit hours: 3; Contact hours: One, 165 min/wk. lecture; Engineering.

3. Instructor's or course coordinator's name

Said Al-Hallaj

4. Text book, title, author, publisher, year

Selected publications, reports and handouts.

5. Specific course information

a. Brief description of the content of the course (catalog description)

The course provides an introductory view into the various aspects involved in technology commercialization (i.e. taking an idea from the lab to the marketplace).

b. Prerequisites or co-requisites

Consent of the instructor.

c. Indicate whether a required, elective, or selected elective.

Elective.

6. Specific goals for the course

a. Specific outcomes of instruction. Students will:

- Understand fundamentals and basic concepts of intellectual property, confidentiality of Non-Disclosure Agreements (NDA), patent applications, copyrights, Know-how
- Learn about UIC resources for entrepreneurship; Office of Technology Management, Invention Center, Business Plan Competition, Illinois Venture
- Learn how to develop a business development strategy, business plan, marketing strategies, finance and accounting
- Understand the fundamentals and basic concepts of starting a business, legal structure (LLC, Inc., operating agreements, governance
- Learn concepts and practical strategies for fundraising, grants, seed/angel investment, venture capital, strategic partnerships
- Present a Case Study with final project term paper and presentation

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcomes		1	2	3	4
(1)	an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.	X			
(2)	an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.			X	
(3)	an ability to communicate effectively with a range of audiences.				X
(4)	an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.				X
(5)	an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.				X
(6)	an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	X			
(7)	an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.				X

7. Brief list of topics to be covered

- Anderson Entrepreneurship Quiz
- Protecting your intellectual property (IP) (Guest Lecturer: Mark Krivchenia, UIC-OTM)
- UIC Resources Overview (Guest: Mark Krivchenia, UIC-OTM)
- Business Development Strategy (Mike Pintar, COO NETenergy)
- Why Marketing Study is Important
- Financials Basics Steve Wolff, CFO AllCell Technologies, LLC
- Starting Your Own Business (is it a good idea?)
- How to build an effective team
- Corporate Entrepreneurship
- Pitching your idea

Fundraising Strategy (are you ready to pitch your idea?)

CHE 494
Selected Topics in Chemical Engineering

1. Course number and name

CHE 494 Materials for Energy

2. Credits, contact hours and categorization

Credit hours: 3; Contact hours: Two, 75 min/wk. lectures; Engineering.

3. Instructor's or course coordinator's name

Sanjay Behura

4. Text book, title, author, publisher, year

Introduction to Solid State Physics, 8th Edition, Charles Kittel, Wiley, 2004.

a. Other supplemental materials

2D Materials: Properties and Devices, P. Avouris, T. F. Heinz, and T. Low;
Physics of Solar Cells, Peter Würfel;
Semiconductor Material and Device Characterization, Dieter K. Schroder, 3rd
Edition, Wiley.

5. Specific course information

a. Brief description of the content of the course (catalog description)

The goal of this course is to prepare students with foundational understanding of the advanced materials including low-dimensional nanostructures, tools required to synthesize and characterize such materials, and their potential applications in photovoltaic energy conversion.

b. Prerequisites or co-requisites

Consent of the instructor.

c. Indicate whether a required, elective, or selected elective.

Elective.

6. Specific goals for the course

a. Specific outcomes of instruction. Students will:

- Understand the science of advanced materials
- Learn methods and techniques for advanced materials processing
- Learn 2D material characterization methods
- Understand principles of solar energy harvesting
- Synthesize and characterize advanced materials in the lab
- Write final project term paper and give a presentation

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcomes		1	2	3	4
(1)	an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.		X		
(2)	an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	X			
(3)	an ability to communicate effectively with a range of audiences.			X	
(4)	an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.				X
(5)	an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.				X
(6)	an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.				X
(7)	an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.				X

7. Brief list of topics to be covered

Science of Advanced Materials: Quantum Theory of Solids; Semiconductor Physics; Low-Dimensional Materials Science; Modifications of Materials Properties. (This part will focus on physics and chemistry of materials, their unique structure and intriguing properties).

Advanced Materials Processing: Synthesis Techniques for Advanced Material Thin Films: Physical Vapor Deposition and Chemical Vapor Deposition. (This part will focus on various strategies for the nucleation of materials and understanding the ensuing growth mechanisms).

Material Characterization: Analysis of Materials by Raman Spectroscopy, X-Ray Photoelectron Spectroscopy, Scanning Electron Microscopy, Transmission Electron Microscopy, Scanning Probe Microscopy, and Electrical Characterization. (This part will focus on basic working principles of advanced characterization tools).

Principles of Solar Energy Harvesting: Properties of Sunlight; Solar Cell Operation; Design and Development of Advanced Solar Cells: Silicon Homojunction and Graphene-Based Schottky-Junction Solar Cells, Organic and Dye-Sensitized Molecular Solar Cells, and Hybrid Organic-Inorganic Perovskite Photovoltaic Cells.

Energy Lab Work: (i) Synthesis and Characterization of Low-Dimensional Materials (2D-Graphene and 1D-ZnO) and (ii) Design and Development of Dye-Sensitized Molecular Cells.

CHE 499

1. Course number and name

CHE 499. Professional Development Seminar

2. Credits, contact hours and categorization

Credit hours: 0; Contact hours: 10-12 seminars/lectures per semester; Engineering

3. Instructor's or course coordinator's name

Alan Zdunek

4. Text book, title, author, publisher, year

None

5. Specific course information

a. Brief description of the content of the course (catalog description)

Seminar-based course. Invited speakers provide information on industries hiring chemical engineers, job-related experience, traditional and non-traditional chemical engineering career paths, and career advice. Career development information from Engineering Career Center. Student evaluation of their educational experience in the Chemical Engineering Department including senior exit survey and ethics exam.

b. Prerequisites or co-requisites

Open only to chemical engineering seniors, Must be taken in the last (Fall) semester of study.

c. Indicate whether a required, elective, or selected elective.

Required.

6. Specific goals for the course

a. Specific outcomes of instruction. Students will:

- Interact with practicing chemical engineers
- Learn about traditional and non-tradition chemical engineering career paths, both in academia and industry
- Learn job searching skills; networking, resume writing, interviewing skills, salary negotiation
- Learn what industries hire chemical engineers
- Learn about their role as UIC Chemical Engineering alumni
- Relate their UIC educational experience in an exit survey

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome	1	2	3	4
(3) an ability to communicate effectively with a range of audiences.			X	
(4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.			X	
(7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.			X	

7. Brief list of topics to be covered

- Where chemical engineers work: Chemical engineering-related industries
- Traditional and non-traditional chemical engineering career paths
- Chemical engineering job-related experience
- Career development information: job searching, resume writing, interviewing skills, salary negotiation, networking, alumni contacts.
- Ethics in the workplace

Selected Seminar Speakers from past years:

Job-Searching Skills: Jonna McHugh, UIC Engineering Career Center

Consulting: Kevin Meaux, Harbour Contractors

Safety/Environment: Lissa Aquino, Stepan Chemicals

Research & Development: Dr. George Foutou, Cabot Microelectronics

Business Management: Ryan Neris, ErgoSeal

Process Engineering-Food: Rick Burns, Tetrapak, Food process engineering

Entrepreneurship: Dr. Said Al-Hallaj, AllCell Technology, Entrepreneurship

Chemical Engineering Role in Sustainability: Dr. Gavin Towler, UOP

Oil/Gas Process Engineering: Elizabeth Carter, UOP

Food Industry: Brian Guthrie, Cargill

Food Industry: Lynette Hernandez, Nestle USA

Petrochemicals: Anastasia Vaia, BP Aromatics Technologies

Process Control-Food: Thomas Scheller, ProLeiT

Patent Agent: Daniel Hernandez

Automotive: Arwa Hasan, Navistar

Sales: Pedro Lopez, Spraying Systems, Inc.

A.2. Other Engineering Courses.

CS 109

1. Course number and name

C/C++ Programming for Engineers with Matlab

2. Credits, contact hours, and categorization.

3 credit hours, 100 min/week lecture, 110 min/week laboratory, Engineering

3. Instructor's or course coordinator's name

Shannon Reckinger

4. Text book, title, author, and year

We are using an online textbook called a Zybook. Follow these instructions to get access:

Log on to Blackboard. On the left panel, click on the "Zybooks (HW)". Click on the first homework assignment that will take you to the Zybooks website; Subscribe

a. other supplemental materials

MATLAB is the commercial software used in this course and is available on all campus computers, including the lab computers.

5. Specific course information

a. Brief description of the content of the course (catalog description)

Program design. Data types and operators, control structures, functions, file I/O, arrays and structures. Engineering applications: Matrices, equation solution, MatLab. Programming assignments.

b. Prerequisites or co-requisites

Co-requisite: MATH 180

c. Indicate whether a required, elective, or selected elective

Required

6. Specific goals for the course

a. Specific outcomes of instruction

This class is specifically designed as an introduction to computer science for engineering students. It will teach you to *think* like a computer and *act* like a computer scientist. It will change how you approach problems and provide you with powerful tools that you *can* use to change the world. You *will* learn how to program in this course.

b. Explicitly indicate which of the student outcomes are addressed by the course.

SLO # 1: An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

7. Brief list of topics to be covered

MATLAB Basics

Arrays

Plotting and Strings

Branches

Loops

Lab Proficiencies

Functions

2D Arrays

Image Processing

Data Analysis

Animations

Calculus

Sound

C++

CME 260

1. Course number and name

Properties of Materials

2. Credits, contact hours, and categorization.

3 credit hours, 3 contact hours per week, Engineering

3. Instructor's or course coordinator's name

Michael McNallan, Matthew Daly

4. Text book, title, author, and year

"Introduction to Materials Science for Engineers", 8th Edition, 2015 James F. Shackelford, Pearson Higher Education, Upper Saddle River, NJ

a. other supplemental materials

None

5. Specific course information

a. Brief description of the content of the course (catalog description)

Introduction to the relationships between composition and microstructure; correlation with physical and mechanical behavior of metals, ceramics, and polymers. Manufacturing methods. Service performance. Materials selection.

b. Prerequisites or co-requisites

CHEM 112 and MATH 181 and PHYS 141

c. Indicate whether a required, elective, or selected elective

Required

6. Specific goals for the course

a. Specific outcomes of instruction

This is an introductory level course in materials science and engineering. It is intended to acquaint the student with the structure and properties of materials used in engineering systems.

b. Explicitly indicate which of the student outcomes are addressed by the course.

SLO # 1: An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

7. Brief list of topics to be covered

1. Introduction, Atomic Bonding
2. Crystallography
3. Crystal Defects and Diffusion
4. Mechanical Properties of Metals
5. Phase Diagrams
6. Phase transformations
7. Processing of Metals
8. Ceramics
9. Polymers and Composites
10. Electrical and Magnetic Materials

ECE 210

1. Course number and name

ECE 210, Electrical Circuit Analysis

2. Credits and contact hours

3 credit hours, 100 min/week lecture, 170 min/week laboratory, Engineering

3. Instructor's or course coordinator's name

Kimberly Fitzgerald

4. Text book, title, author, and year

Fundamentals of Electric Circuits, 5th Edition, by Charles K. Alexander and Matthew N. O. Sadiku, McGraw-Hill Companies, Inc., 2013.

4a. Other supplemental materials

None.

5. Specific course information

5a. Brief description of the content of the course (catalog description)

Linear circuit analysis: networks, network theorems, dependent sources, operational amplifiers, energy storage elements, transient analysis, sinusoidal analysis, frequency response, filters.

5b. Prerequisites or co-requisites

PHYS 142, General Physics II, Electricity and Magnetism; credit or concurrent registration in MATH 220, Introduction to Differential Equations.

5c. Indicate whether a required, elective, or selected elective

Required.

6. Specific goals for the course

6a. Specific outcomes of instruction

This course has two main objectives: to introduce fundamental circuit theorems and to introduce circuit analysis methods and strategies in the time and frequency domains. Laboratory experiments, in addition to introducing electronic measurement techniques, have been designed to verify theorems and circuit analysis methods.

6b. Indicate which of the student outcomes are addressed by the course

SLO #1: An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

SLO #3: An ability to communicate effectively with a range of audiences.

SLO #6: An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.

7. Brief list of topics to be covered

1. Basic electric circuit variables and elements, Ohm's Law, current vs. voltage characteristics.
2. DC power and power balance circuit solution strategy.
3. Kirchhoff's Laws and basic analysis strategy.
4. Transformations to equivalent circuits (series, parallel resistors and sources).
5. Voltage and current division rules, double dividers.
6. Circuits with dependent sources.
7. Superposition principle.
8. Thevenin's and Norton's Theorems, source transformations, maximum power transfer.
9. Formal Methods (mesh current and node voltage).
10. Capacitor and Inductor time domain characteristics, RC and LR transient analysis.
11. Review of signals (sinusoidal, exponential, etc.).
12. Review of complex numbers and complex arithmetic.
13. AC steady state, phasors, impedances, frequency characteristics of circuits

ENGR 100

1. Course number and name

Engineering Orientation- ENGR 100

2. Credits and contact hours

0 credit hours, 75 min/week contact hours, Engineering

3. Instructor's or course coordinator's name

Simona Narubin (Full session instructor); Department Break-out Sessions: Miiri Kotche (BioE), Alan Zdunek (ChE), Hossein Ataei (CME), Patrick Troy (CS), Vahe Caliskan (ECE), Saeed Manafzadeh (MIE)

4. Text book, title, author, and year

Rosie Revere, Engineer, Andrea Beaty, 2013.

4a. Other supplemental materials

Sample resumes, handouts and presentation materials available on Blackboard

5. Specific course information

5a. Brief description of the content of the course (catalog description)

General Orientation course on skills pertinent to students for successful completion of their undergraduate degree. Required of all engineering students. Course Information: Satisfactory/Unsatisfactory grading only. No graduation credit. Should be taken in the first semester after acceptance into the College of Engineering.

5b. Prerequisites or co-requisites

Admission to the College of Engineering.

5c. Indicate whether a required, elective, or selected elective

Required.

6. Specific goals for the course

6a. Specific outcomes of instruction

1) Establish a critical and ongoing connection with the College of Engineering and departments; 2) Guide incoming freshmen and transfer students through their first semester in the College of Engineering; 3) Provide enhanced communication utilizing social media and networking opportunities via guest speakers.

Educational outcome: Students become familiar with Engineering education, the engineering profession, the various fields of engineering covered by the COE, engineering professional societies, interviews and resumes, ethics, the role of alumni, and academic advising. Students participate in a department-specific technical project. Students meeting certain criteria are eligible for the Guaranteed Paid Internship Program (GPIP).

6b. Indicate which of the student outcomes are addressed by the course

SLO #3: An ability to communicate effectively with a range of audiences.

SLO #5: An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.

SLO #6: An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.

SLO #7: An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

7. Brief list of topics to be covered

1. Student organization presentations
2. Attending student organization meetings
3. Counseling Center
4. Women's Leadership
5. Gender and Sexuality
6. Consulting (Branden Irvin, Jensen & Hughes)
7. Engineering Career Center presentations
8. Academic advising tutorial
9. Lynda Skill tutorial/playlist
10. Individual Course planning and scheduling
11. "Get to Know Your Department", Upper-class student interviews
12. Hands-on Technical Project; Department-specific

A.3. Mathematics and Basic Science Courses.

CHEM 116

8. Course number and name

Honors and Majors General and Analytical Chemistry I

9. Credits, contact hours and categorization

5 credit hours, 250 min/week, lecture/discussion, 180 min/week lab period: Basic science

10. Instructor's or course coordinator's name

Scott Shippy

11. Text book, title, author, and year

ONE REQUIRED From this list:

A) Zumdahl, CHEMICAL PRINCIPLES (8th edition) ISBN: 9781305581982

B) Zumdahl, CHEMICAL PRINCIPLES (7th edition) ISBN: 9781111580650

ONE REQUIRED From this list:

A) Harris, QUANTITATIVE CHEM.ANALYSIS (9th edition) ISBN:

9781464135385

B) Harris, QUANTITATIVE CHEM.ANALYSIS (8th edition) ISBN:

9781429218153

a. other supplemental materials

On-line homework: The on-line homework for this class will be through the OWLv2 system with Cengage.com

CHEM 123 Laboratory Manual is needed for the laboratory work.

12. Specific course information

a. Brief description of the content of the course (catalog description)

Chemical Engineering Honors Students: First semester of general and analytical chemistry with laboratory. Coverage of the fundamentals of chemistry including stoichiometry and equilibrium. Coverage of the principles of analytical chemistry, including the use of instrumentation.

b. Prerequisites or co-requisites

Superior performance on the UIC chemistry placement examination

c. Indicate whether a required, elective, or selected elective

Required for Honors students

13. Specific goals for the course

a. Specific outcomes of instruction

Students will learn stoichiometry, periodicity, reaction types, the gaseous state, solution, chemical equilibria, acid-base equilibria, dissolution-precipitation equilibria. Students will learn principles and practices of analytical chemistry. The laboratory portion of the course is balanced by an in-depth review of general chemistry principles (in particular, stoichiometry, equilibrium, and titrimetry) as applied to the problem of separating and quantifying substances from mixtures.

- b. Explicitly indicate which of the student outcomes are addressed by the course.

SLO # 1: An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

SLO # 6: An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.

Brief list of topics to be covered

1. Introduction, Atoms, Molecules, and Ions, Periodicity, Nomenclature
2. Lab 1: Calibration of Volumetric Glassware (BB) and Data Analysis Lab (LM3)
3. Stoichiometry
4. Lab 2: Glucose in "Blood Serum"(BB)
5. Lab 3: pH Dependence of the Course of a Redox Reaction (LM8)
6. Chemical Reactions
7. Lab 4: Calibration of a pH meter and K_a of an indicator (LM1 & LMX)
8. Lab 5: Gravimetric Determination of Ca as Calcium oxalate monohydrate (BB)
9. Lewis Structures, VSEPR and Gases
10. Lab 6: Gravimetric Determination of Fe as rust (BB)
11. Liquids
12. Lab 7: Determination of NaHCO_3 /Molar Mass of CO_2 /Value of R (LM7)
13. Solutions
14. Lab 8: Statistical Evaluation of Acid/Base Indicators (BB)
15. Lab 9: Preparing Standard Acid & Base (BB)
16. Chemical Equilibrium , Acids and Bases
17. Lab 10: Using a pH electrode for an Acid/Base Titration(BB)
18. Lab 11: Analysis of a Mixture of Carbonate and Bicarbonate (BB)
19. Lab 12: Stabilization of pH with Buffers (LM12)

CHEM 118

14. Course number and name

Honors and Majors General and Analytical Chemistry II

15. Credits, contact hours and categorization

5 credit hours, 250 min/week, lecture/discussion, 180 min/week lab period: Basic science

16. Instructor's or course coordinator's name

Scott Shippy

17. Text book, title, author, and year

ONE REQUIRED From this list:

A) Zumdahl, CHEMICAL PRINCIPLES (8th edition) ISBN: 9781305581982

B) Zumdahl, CHEMICAL PRINCIPLES (7th edition) ISBN: 9781111580650

ONE REQUIRED From this list:

A) Harris, QUANTITATIVE CHEM.ANALYSIS (9th edition) ISBN:

9781464135385

B) Harris, QUANTITATIVE CHEM.ANALYSIS (8th edition) ISBN:

9781429218153

a. other supplemental materials

On-line homework: The on-line homework for this class will be through the OWLv2 system with Cengage.com

CHEM 123 Laboratory Manual is needed for the laboratory work.

18. Specific course information

a. Brief description of the content of the course (catalog description)

Chemical Engineering Honors students: Second semester of general and analytical chemistry with laboratory. Coverage of the fundamentals of chemistry including atomic and molecular structure, thermodynamics, and kinetics. Coverage of principles of analytical chemistry, including the use of instrumentation.

b. Prerequisites or co-requisites

Grade of C or better in CHEM 116

c. Indicate whether a required, elective, or selected elective

Required for Honors students

19. Specific goals for the course

a. Specific outcomes of instruction

Students will learn phase transitions, thermochemistry, spontaneity/equilibrium, electrochemistry, kinetics, bonding, and order/symmetry in condensed phases, coordination compounds, descriptive chemistry the problem of separating and quantifying substances from mixtures. The laboratory portion of the course is balanced by an in-depth review of general chemistry principles as applied to the problem of separating and quantifying substances from mixtures.

- b. Explicitly indicate which of the student outcomes are addressed by the course.

SLO # 1: An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

SLO # 6: An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.

Brief list of topics to be covered

1. Energy , Enthalpy and Thermochemistry
2. Lab 1: Calorimetry and Heat of Vaporization of Liquid Nitrogen, Regression Analysis
3. Energy , Enthalpy and Thermochemistry
4. Lab 2: Calorimetry and Hess' Law (LM)
5. Lab 3: Enthalpy and Entropy Borax dissolution (LM)
6. Spontaneity, Entropy and Free Energy
7. Lab 4: Statistical Evaluation of Acid/Base Indicators (BB)
8. Lab 5: Mn in Steel (BB)
9. Electrochemistry and Potentiometry
10. Lab 6: Iodide with ISE (BB)
11. Lab 7: Ascorbic Acid iodometry (BB)
12. Chemical Kinetics
13. Lab 8: Iodide-catalyzed decomposition of hydrogen peroxide (LM)
14. Lab 9: Concentration and Temperature effects on Reaction rates (LM)
15. Quantum Mechanics and Atomic Theory
16. Lab 10: Kinetics of the Reaction of Coumarin-102 with Base using Fluorescence (LM)
17. Lab 11: Stoichiometry of a metal-ligand complex (LM)
18. Covalent Bonding and Introduction to Spectroscopy
19. Lab 12: Ion Exchange Separation (BB)
20. Lab 13: EDTA Titration of Ca and Mg (BB)
21. Coordination Compounds and Structures of Solids
22. Lab 14: DNA Electrophoresis (BB)

CHEM 122

20. Course number and name

General College Chemistry I

21. Credits, contact hours and categorization

4 credit hours, 150 min/week, lecture: Basic science

22. Instructor's or course coordinator's name

Department of Chemistry staff

23. Text book, title, author, and year

Chemistry: The Molecular Nature of Matter and Change with advanced topics, 8th edition by Silberberg and Amateis, 2017

a. other supplemental materials

McGraw-Hill Connect on-line homework purchased through Blackboard

24. Specific course information

a. Brief description of the content of the course (catalog description)

This course is the first-semester of a two-semester comprehensive general chemistry sequence that explores the behavior of matter quantitatively by linking basic chemical concepts with problem solving. Students will demonstrate a good understanding of the basic structure of atoms, periodic properties of the elements, bonding concepts of atoms forming compounds and their resulting shapes, naming of compounds and formulas, stoichiometry of chemical reactions, different types of chemical reactions, properties of solution and gaseous phase matter, and role of intermolecular forces on physical properties. Furthermore, students will learn to apply chemical equilibrium principles to both gas-phase reactions and aqueous solution chemistry with emphasis on acids and bases, dissolution-precipitation, and precipitation reactions.

b. Prerequisites or co-requisites

Grade of C or better in CHEM 101 or adequate performance on the UIC chemistry placement examination, and concurrent registration or Grade of C or better in CHEM 123

c. Indicate whether a required, elective, or selected elective

Required

25. Specific goals for the course

a. Specific outcomes of instruction

A first-semester course in general chemistry, including stoichiometry, periodicity, reaction types, the gaseous state, solution, chemical equilibria, acid-base equilibria, dissolution-precipitation equilibria.

b. Explicitly indicate which of the student outcomes are addressed by the course.

SLO # 1: An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

Brief list of topics to be covered

20. Definitions, Units, Problem Solving, Precision and Accuracy, Components Of Matter, Atoms, Elements and the Periodic Table
21. Introduction to Valence Electrons, Periodic Table Trends for Electronegativity and Atomic Size, Bonding, Ionic and Covalent Compounds, Formulas, Masses and Naming, Mixtures
22. Stoichiometry: Moles, Mass Percent, Chemical Formulas, Combustion Writing and Balancing Chemical Equations Calculating Quantities of Reactant and Product
23. Introduction to Solution Terms, Molarity and other Concentration Terms, Writing Equations for Aqueous Ionic Reactions
24. Precipitation Reaction and Solubility Rules, Introduction to Acids and Bases, pH concept, Arrhenius definition, Bronsted-Lowry concept, Acid-Base Reactions
25. Oxidation-Reduction Reactions and Related Reaction Types, The Reversibility of Reactions, Gases: Overview, Pressure and Measurement, Laws, Ideal Gas Law
26. Finish Ideal Gas Law, Rearrangements of The Ideal Gas Law Kinetic Molecular Theory, Real Gases
27. Chemical Equilibrium: Equilibrium Constant, Reaction Quotient and Pressure Terms
28. Chemical Equilibrium: Determining Direction, Solving Equilibrium Problems, Le Chatelier's Principle, Acid-Base Equilibria: Introduction to K_a
29. Revisiting Bronsted-Lowry, Solving Weak-Acid Equilibria Problems, pH and Auto ionization of Water, Acid Strength
30. Acid-Base Equilibria: Weak Bases and Their Relation to Weak Acids, Acid-Base Properties of Salt Solutions
31. Generalizing Bronsted-Lowry: The Leveling Effect, Lewis Acids and Bases Aqueous Ionic Equilibrium: Acid-Base Buffers
32. Acid-Base Titration Curves and Indicators Equilibria of Slightly Soluble Ionic Compounds, Predicting Precipitation
33. Equilibria of Slightly Soluble Ionic Compounds, Predicting Precipitation, Henry's Law, Raoult's Law, Introduction to colligative properties and review of relevant concentration terms
34. Review Colligative properties, calculations with osmotic pressure, colloids

CHEM 123

1. **Course number and name**

General College Chemistry Laboratory I

2. **Credits, contact hours and categorization**

1 credit hour, 2 hr. 50 min lab session per week, basic science

3. **Instructor's or course coordinator's name**

Department of Chemistry staff

4. **Text book, title, author, and year**

Lab Manual/notebook: The lab manual used for this course is a combined lab manual with lab notebook pages included. It is available from the UIC bookstore.

a. other supplemental materials

Labflow Software

5. **Specific course information**

a. **Brief description of the content of the course (catalog description)**

This is a first-semester laboratory course in general chemistry which involves experiments in stoichiometry, periodicity, reaction types, the gaseous state, solution stoichiometry, chemical equilibria, acid-base equilibria and dissolution-precipitation equilibria.

b. **Prerequisites or co-requisites**

Grade of C or better in CHEM 101 and concurrent registration in CHEM 122 or Grade of C or better in CHEM 122

c. **Indicate whether a required, elective, or selected elective**

Required

6. **Specific goals for the course**

a. **Specific outcomes of instruction:**

- Increased awareness of safe handling and disposal of chemicals in the laboratory,
- Learn proper use of bench-top instrumentation and lab techniques for chemical analysis,
- Document and analyze scientific data from chemical systems to draw conclusions.

The course is designed to enrich knowledge of general chemistry topics through experimentation.

b. **Explicitly indicate which of the student outcomes are addressed by the course.**

SLO # 1: An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

SLO # 6: An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.

7. Brief list of topics to be covered:

1. Safety orientation, Lab expectations, Review
2. Intro to Labflow
3. Calibration of pH meter
4. Lab 1: Use of Volumetric Glassware: The Floating Egg Problem
5. Lab 2: A Cycle of Chemical Reactions of Copper
6. Beer's Law and Solution Concentration with Dilution
7. Lab 3: Determination of Serum Albumin
8. Lab 4: Determination of NaHCO_3 , Molar Mass of CO_2 , and the Value of R
9. Lab 5: Tree Leaves and the Global Carbon Cycle, Part II
10. Lab 6: Acid-Base Titration
11. Lab 7: Analysis of Glass Cleaners, Part I
12. Lab 7: Analysis of Glass Cleaners, Part II (Practical)
13. Lab 8: Stabilization of pH with Buffers
14. Lab 9: Salt Solubility Lab

CHEM 124

1. **Course number and name**

General College Chemistry II

2. **Credits, contact hours and categorization**

4 credit hours, 150 min/week, lecture; Basic science

3. **Instructor's or course coordinator's name**

Department of Chemistry staff

4. **Text book, title, author, and year**

Chemistry: The Molecular Nature of Matter and change with advanced topics, 8th edition by Silberberg and Amateis, 2017

a. other supplemental materials

McGraw-Hill Connect on-line homework purchased through Blackboard

5. **Specific course information**

a. **Brief description of the content of the course (catalog description)**

A second-semester course in general chemistry including phase transitions, thermochemistry, spontaneity/equilibrium, electrochemistry, kinetics, bonding, and order/symmetry in condensed phases, coordination compounds, descriptive chemistry.

b. **Prerequisites or co-requisites**

Grade of C or better in CHEM 112; or Grade of C or better in CHEM 122 and Grade of C or better in CHEM 123; or Grade of C or better in CHEM 116; and concurrent registration or Grade of C or better in CHEM 125.

c. **Indicate whether a required, elective, or selected elective**

Required

6. **Specific goals for the course**

a. **Specific outcomes of instruction**

The emphasis of the course is in thermochemistry, spontaneity/equilibrium, electrochemistry, and kinetics, an introduction to quantum concepts, bonding, coordination compounds, and nuclear chemistry.

b. **Explicitly indicate which of the student outcomes are addressed by the course.**

Student Learning Outcome # 1: An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

7. Brief list of topics to be covered

1. Thermochemistry I (Energy)
2. Thermochemistry II (Energy)
3. Thermodynamics I (entropy/free energy)
4. Thermodynamics II (entropy/free energy)
5. Redox Reactions, Electrochemistry I
6. Electrochemistry II
7. Chemical Kinetics I
8. Chemical Kinetics II
9. Quantum Mechanics, Hydrogen Atom, Many-Electron Atoms
10. Bonding Theories I: Valence Bond
11. Bonding Theories II: Molecular Orbital
12. Coordination Chemistry I
13. Coordination Chemistry II
14. Nuclear Chemistry

CHEM 125

1. Course number and name

General College Chemistry Laboratory II

2. Credits, contact hours and categorization

1 credit hour, 2 hr. 50 min lab session per week, basic science

3. Instructor's or course coordinator's name

Department of Chemistry staff

4. Text book, title, author, and year

Lab Manual/notebook: The lab manual used for this course is a combined lab manual with lab notebook pages included. It is available from the UIC bookstore.

- a. other supplemental materials

Labflow Software

5. Specific course information

- a. Brief description of the content of the course (catalog description)

This is a second-semester laboratory course in general chemistry which involves experiments in phase transitions, thermochemistry, spontaneity/equilibrium, electrochemistry, kinetics, bonding, order/symmetry in condensed phases, coordination compounds, and descriptive chemistry.

- b. Prerequisites or co-requisites

Grade of C or better in CHEM 112; or Grade of C or better in CHEM 122 and Grade of C or better in CHEM 123; or Grade of C or better in CHEM 116 and concurrent registration or Grade of C or better in CHEM 124.

- c. Indicate whether a required, elective, or selected elective

Required

6. Specific goals for the course

- a. Specific outcomes of instruction:

- Increased awareness of safe handling and disposal of chemicals in the laboratory,
- Learn proper use of bench-top instrumentation and lab techniques for chemical analysis,
- Record and analyze scientific data from chemical systems to draw conclusions.

The course is designed to enrich knowledge of general chemistry topics listed above through experimentation.

- b. Explicitly indicate which of the student outcomes are addressed by the course.

SLO # 1: An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

SLO # 6: An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.

7. Brief list of topics to be covered:

1. Safety orientation, Lab Review
2. Significant Figures, Solution Concentrations, and Dilutions
3. Regression Analysis
4. Lab 1: Calorimetry and Hess's Law
5. Lab 2: Enthalpy and Entropy Changes for Borax Dissolution
6. Lab 3: pH Dependence of a Redox Reaction
7. Lab 4: Electrochemical Measurement and Batteries
8. Chemical Kinetics
9. Lab 5: $I^-(aq)$ Catalyzed Decomposition of H_2O_2 : An Initial Rate Kinetics Approach
10. Lab 6: Concentration and Temperature Effects on Reaction Rates
11. Lab 7: The Emission Spectra of Some Elements
12. Lab 8: A Kinetics Experiment of the Reaction of Coumarin-102 with Base Using Fluorescence
13. Lab 9: Fluorescence, Quenching, and DNA
14. Lab 10: Metal-Ligand Stoichiometry

CHEM 222

1. Course number and name

Analytical Chemistry

2. Credits, contact hours and categorization

4 credit hours, Lecture 150 min/week; two, 3 hr. lab sessions per week; Basic science

3. Instructor's or course coordinator's name

Department of Chemistry staff

4. Text book, title, author, and year

Daniel C. Harris, *Quantitative Chemical Analysis*, 9th Ed (W. H. Freeman and Company, 2016).

- a. other supplemental materials

Solutions Manual (optional); on-line homework, www.saplinglearning.com
Laboratory: UNUSED bound lab notebook with carbon (carbonless) copies

5. Specific course information

- a. Brief description of the content of the course (catalog description)

Theory and application of chemical equilibria and instrumentation in quantitative analysis. Includes two weekly three-hour laboratories.

- b. Prerequisites or co-requisites

Course Information: Prerequisite(s): Grade of C or better in CHEM 114; or Grade of C or better in CHEM 124 and Grade of C or better in CHEM 125; or Grade of C or better in CHEM 118 or the equivalent.

- c. Indicate whether a required, elective, or selected elective

Required

6. Specific goals for the course

- a. Specific outcomes of instruction:

CHEM 222 is a 4 credit hour course which introduces the principles and practices of analytical chemistry. Often the more descriptive term quantitative analysis is given to this discipline. The focus of the course is on the separation and analysis of chemical substances. Analysis means to both identify a chemical substance and to determine its amount. The laboratory portion of the course is balanced by an in-depth review of general chemistry principles (in particular, stoichiometry, equilibrium, and titrimetry) as applied to the problem of separating and quantifying substances from mixtures. Your lab grade will be based upon the accuracy and precision of your laboratory results.

- b. Explicitly indicate which of the student outcomes are addressed by the course.

SLO # 1: An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

SLO # 6: An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.

7. **Brief list of topics to be covered:**

Lecture:

1. Statistics
2. Calibration Methods
3. Chemical Equilibria
4. Activity and Equilibria
5. Monoprotic and polyprotic acid/bases
6. Acid/Base titrations
7. EDTA titrations
8. Electrochemistry
9. Electrodes and Potentiometry
10. Redox titrations
11. Spectrophotometry and applications
12. Atomic Spectroscopy
13. Analytical separations
14. Chromatographic methods and electrophoresis

Lab:

Review, demonstrate proficiency on a balance, in titration

Exp. 1 Calibration of Volumetric Glassware Experimental Error

Exp. 2 - Gravimetric Determination of Ca as $\text{CaC}_2\text{O}_4 \times \text{H}_2\text{O}$

Exp. 3 - Gravimetric Determination of Fe as Fe_2O_3

Exp. 4 - Statistical Evaluation of Acid-Base Indicators

Exp. 5 - Preparing Standard Acid and Base

Exp. 6 - Using a pH Electrode for an Acid-Base Titration

Exp. 7 - Analysis of a Mixture of Carbonate and Bicarbonate

Exp. 8 - Determination of Glucose in Blood Serum (Calibration Curve)

Exp. 9 - Spectrophotometric Determination of Iron in Vitamin Tablets

Exp. 10 - Microscale Spectrophotometric Determination of Iron in Foods

Exp. 11 - Determination of Mn in Steel (Standard Addition)

Exp. 12 - Determination of Cu in Brass (Calibration Curve)

Exp. 13 - Determination of Nicotine in Tobacco (Nonaqueous Titration)

Exp. 14 - EDTA Titration of Ca^{2+} and Mg^{2+} in Natural Waters

Exp. 15 - Iodimetric Titration of Vitamin C

Exp. 16 - Determination of I^- in Salt (Ion-Selective Electrode, Standard Addition)

Exp. 17 - Testing Table Salt for Its Potassium Iodide Content

Exp. 18 - Ion-Exchange Separation and Spectrophotometric Determination of Nickel

Exp. 19 - Electrochemistry Separation of Fish Proteins by Gel Electrophoresis

Exp. 20 - DNA Fingerprinting with Gel Electrophoresis Potentiometry DNA Fingerprinting

CHEM 232

1. Course number and name

Organic Chemistry I

2. Credits, contact hours and categorization

4 credit hours, 150 min/week lecture; 1 hr./week discussion; Basic science

3. Instructor's or course coordinator's name

Department of Chemistry staff

4. Text book, title, author, and year

Organic Chemistry, Klein, 2nd Edition (2015) or 3rd Edition (2017)

a. other supplemental materials

WileyPLUS online HW access card

iClicker2 (www.iclicker.com)

Molecular model set (Prentice Hall, HGS, or anything similar)

5. Specific course information

a. Brief description of the content of the course (catalog description)

First semester of a one-year sequence. Structure, reactivity, and synthesis of organic molecules.

b. Prerequisites or co-requisites

Grade of C or better in CHEM 114; or Grade of C or better in CHEM 124 and Grade of C or better in CHEM 125; or Grade of C or better in CHEM

118.

c. Indicate whether a required, elective, or selected elective

Required

6. Specific goals for the course

a. Specific outcomes of instruction

The emphasis of the course is in understanding the structure, reactivity, and synthesis of organic molecules.

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcome # 1: An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

7. **Brief list of topics to be covered**

1. Molecular Representation
2. Acids & Bases
3. Alkanes & Cycloalkanes
4. Stereoisomerism
5. Infrared Spectroscopy
6. NMR Spectroscopy
7. Chemical Reactivity & Mechanism
8. Alkyl Halides (start)
9. Addition to Alkenes
10. Alkynes
11. Radical Reactions
12. Synthesis
13. Alcohols & Phenols
14. Alcohols & Phenols
15. Ethers & Epoxides

Online Video Lectures

CHEM232 is a “flipped classroom”. This means that classroom lecture time will be devoted to group problem solving together with the instructor and peer leaders. To successfully participate in each lecture, you will need to review the assigned online videos **before** you come to class.

CHEM 233

1. Course number and name

Organic Chemistry Laboratory I

2. Credits, contact hours and categorization

2 credit hours; two, 2 hour and 15 minute lab sessions per week; Basic science

3. Instructor's or course coordinator's name

Department of Chemistry staff

4. Text book, title, author, and year

Experimental Organic Chemistry: A Miniscale and Macroscale Approach, Gilbert & Martin, 6th Edition, 2016

- a. other supplemental materials

Organic Chemistry Laboratory Manual, Landrie, McQuade, Yermolina 9th Edition
Chemistry Lab Notebook – 100 pages (<http://www.hmpublishing.com/products/organic-chemistry-spiral-bound.html>), ISBN 978-1-930882-46-1).

5. Specific course information

- a. Brief description of the content of the course (catalog description)

Introductory organic chemistry laboratory. Basic techniques (distillation, crystallization, chromatography, MP and BP), reactions substitution, elimination, Diels-alder, oxidation-reduction), instrumentation (gas and liquid chromatography, IR.).

- b. Prerequisites or co-requisites

Grade of C or better in CHEM 230; or credit or concurrent registration in CHEM 232.

- c. Indicate whether a required, elective, or selected elective

Required

6. Specific goals for the course

- a. Specific outcomes of instruction:

CHEM 233 is divided into two complementary parts. In Part One, students will learn the theories and principles of isolation, purification, characterization, and compositional analysis of organic molecules through techniques such as thin-layer and column chromatography, gas chromatography, infrared spectroscopy, melting point, and distillation. These laboratories provide the foundation for Part Two of the course, in which students will learn the techniques of various chemical reactions that are relevant to organic synthesis. Each experiment in Part Two typically involves one functional group transformation.

- b. Explicitly indicate which of the student outcomes are addressed by the course.

SLO # 1: An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

SLO # 6: An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.

7. Brief list of topics to be covered:

Lab 1: Chromatographic Methods

Lab 2: Simple & Fractional Distillation GC Analysis

Lab 3: Steam Distillation of Carvone

Lab 4: Base Extraction & Recrystallization

Lab 5: Identification of Unknown Organic Compounds

Lab 6: Preparation of Alkyl Halides by SN1 & SN2 reactions

Lab 7: Preparation of Alkenes by Elimination

Lab 8: Preparation of Alcohols: Reduction of 9-Fluorenone

Lab 9: Oxidation of Alcohols

Lab 10: Williamson Ether Synthesis

Synthesis & Chemiluminescence of Luminol

CHEM 234

1. Course number and name

Organic Chemistry II: Reactivity and Synthesis of Organic Molecules

2. Credits, contact hours and categorization

4 credit hours, 150 min/week lecture; 1 hour/week discussion; Basic science

3. Instructor's or course coordinator's name

Department of Chemistry staff

4. Text book, title, author, and year

Organic Chemistry, Klein, 2nd Edition (2015)

b. other supplemental materials

iClicker2 (www.iclicker.com)

Molecular model set (Prentice Hall, HGS, or anything similar)

5. Specific course information

a. Brief description of the content of the course (catalog description)

Second semester of a one-year sequence. Structure, reactivity, and synthesis of organic molecules.

b. Prerequisites or co-requisites

Grade of C or better in CHEM 232 or the equivalent.

c. Indicate whether a required, elective, or selected elective

Required

6. Specific goals for the course

a. Specific outcomes of instruction

The emphasis of the course is in understanding the structure, reactivity, and synthesis of organic molecules.

b. Explicitly indicate which of the student outcomes are addressed by the course.

SLO # 1: An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

7. Brief list of topics to be covered

1. Mass Spectrometry
2. Organometallic Compounds
3. Alcohols and Epoxides
4. NMR and IR spectroscopy
5. Diels-Alder & Pericyclic Reactions
6. Aromaticity & Electrophilic Aromatic
7. Substitution
8. Cross Coupling Reactions
9. Aldehydes & Ketones
10. Carboxylic Acids & Derivatives
11. Enols & Enolates
12. Polymers
13. Radicals
14. Amino acids, Peptides and Proteins

CHEM 342

1. Course number and name

Physical Chemistry I

2. Credits, contact hours and categorization

3 credit hours, 150 min/week lecture; 1 hour/week discussion; Basic science

3. Instructor's or course coordinator's name

Petr Kral

4. Text book, title, author, and year

Principles of Physical Chemistry, L. M. Raff, 2001, Prentice Hall

a. other supplemental materials

Student Solutions Manual for this book, 2001, Prentice Hall

- Other useful books: Atkins, Physical Chemistry, Zemansky, Heat and Thermodynamics

5. Specific course information

a. Brief description of the content of the course (catalog description)

Thermodynamics of gases, solutions, reaction equilibria, and phase transitions.

b. Prerequisites or co-requisites

Grade of C or better in MATH 181; and Grade of C or better in CHEM 118 or CHEM 124 and CHEM 125; and Grade of C or better in PHYS 142; and grade of C or better or concurrent registration in MATH 210.

c. Indicate whether a required, elective, or selected elective

Required

6. Specific goals for the course

a. Specific outcomes of instruction

The emphasis of the course is in understanding thermodynamics of gases, solutions, reaction equilibria, and phase transitions.

b. Explicitly indicate which of the student outcomes are addressed by the course.

SLO # 1: An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

7. Brief list of topics to be covered

1. Properties of gases
2. The first law of thermodynamics
3. Thermochemistry

4. The second law of thermodynamics
5. Chemical equilibrium
6. Phase equilibrium
7. The thermodynamics of solids
8. Thermodynamics of nonelectrolytic solutions
9. Thermodynamics of electrolytic solutions

CHEM 346

1. Course number and name

Physical Chemistry II

2. Credits, contact hours and categorization

3 credit hours, 150 min/week lecture; 1 hour/week discussion; Basic science

3. Instructor's or course coordinator's name

Michael Trenary

4. Text book, title, author, and year

Principles of Physical Chemistry, L. M. Raff, 2001, Prentice Hall

a. other supplemental materials

Student Solutions Manual for this book, 2001, Prentice Hall

- Other useful books: Atkins, Physical Chemistry, Zemansky, Heat and Thermodynamics

5. Specific course information

a. Brief description of the content of the course (catalog description)

Kinetic and molecular theory of gases; introduction to the principles of quantum mechanics with application to model systems, multi-electron atoms, diatomic molecules, and bonding.

b. Prerequisites or co-requisites

Grade of C or better in CHEM 342 and Grade of C or better in MATH 210.

c. Indicate whether a required, elective, or selected elective

Required

6. Specific goals for the course

c. Specific outcomes of instruction

The emphasis of the course is in understanding kinetic and molecular theory of gases and the principles of quantum mechanics.

d. Explicitly indicate which of the student outcomes are addressed by the course.

SLO # 1: An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

7. **Brief list of topics to be covered**

1. Historical origins of quantum mechanics.
2. The Schrödinger equation and quantum mechanical postulates.
3. Particle in a box.
4. Commutation relations and the uncertainty principle.
5. Quantum mechanics of molecule vibration and rotation.
6. The Hydrogen Atom and other One-Electron Atoms; Many electron atoms.
7. Atomic spectroscopy.
8. Bonding in diatomic molecules.
9. Bonding and structure of polyatomic molecules.
10. Probability and the Boltzmann distribution.
11. Molecular partition functions.
12. Statistical thermodynamics.
13. Kinetic theory of gases.
14. Chemical kinetics.

MATH 180

1. Course number and name

Calculus I – MATH 180

2. Credits, contact hours, and categorization.

4 credit hours, 150 min/week lecture, 100 min/week discussion, Mathematics

3. Instructor's or course coordinator's name

Department of Mathematics staff

4. Text book, title, author, and year

Calculus: Early Transcendentals by William Briggs and Lyle Cochran, 3rd edition.

a. other supplemental materials

None

5. Specific course information

a. Brief description of the content of the course (catalog description)

Differentiation, curve sketching, maximum-minimum problems, related rates, mean-value theorem, antiderivative, Riemann integral, logarithm, and exponential functions.

b. Prerequisites or co-requisites

Grade of C or better in MATH 121 or appropriate performance on the department placement test.

c. Indicate whether a required, elective, or selected elective

Required

6. Specific goals for the course

a. Specific outcomes of instruction

Math 180 is the introductory calculus course in our standard three-semester calculus sequence. Students will develop advanced mathematical skills and will be able to apply calculus to other areas. As such, its goal is to introduce the study of calculus on the real line, which includes limits, differentiation, and basic integration techniques while also covering applications of said topics

b. Explicitly indicate which of the student outcomes are addressed by the course.

SLO # 1: An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

7. Brief list of topics to be covered

Functions, Limits and Continuity

Derivative, Techniques of differential Calculus

Applications: Mean Value Theorem, graphing

Max-Min Problems, Related rates
The Integral: Antiderivative, fundamental
Theorem of Calculus, Riemann Sum
Logarithm and exponential functions, applications

MATH 181

8. Course number and name

Calculus II – MATH 182

9. Credits, contact hours, and categorization.

4 credit hours, 150 min/week lecture, 100 min/week discussion, Mathematics

10. Instructor's or course coordinator's name

Department of Mathematics staff

11. Text book, title, author, and year

Calculus: Early Transcendentals by William Briggs and Lyle Cochran, 3rd edition.

a. other supplemental materials

None

12. Specific course information

a. Brief description of the content of the course (catalog description)

Techniques of integration, arc length, solids of revolution, applications, polar coordinates, parametric equations, infinite sequences and series, power series.

b. Prerequisites or co-requisites

Grade of C or better in MATH 180.

c. Indicate whether a required, elective, or selected elective

Required

13. Specific goals for the course

a. Specific outcomes of instruction

Math 181 is the second semester of our standard three-semester calculus sequence. As such, its goal is to continue the study of calculus on the real line, which was started in Math 180 (Calculus I), with a focus on integration, the basics of sequences and series, as well as parametric descriptions for sets in the plane.

b. Explicitly indicate which of the student outcomes are addressed by the course.

SLO # 1: An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

14. Brief list of topics to be covered

Techniques of Integration
Applications of the Integral
Polar Coordinates, parametric equations
Infinite Series
Power Series

MATH 210

1. Course number and name

Calculus III – MATH 210

2. Credits, contact hours, and categorization.

3 credit hours, 150 min/week lecture, 50 min/week discussion, Mathematics

3. Instructor's or course coordinator's name

Department of Mathematics staff

4. Text book, title, author, and year

Calculus: Early Transcendentals by William Briggs and Lyle Cochran, 3rd edition.

a. other supplemental materials

None

5. Specific course information

a. Brief description of the content of the course (catalog description)

Vectors in space, functions of several variables, partial differential and optimization, multiple integrals, vector fields, Green's Theorem, Stokes Theorem.

b. Prerequisites or co-requisites

Grade of C or better in MATH 181.

c. Indicate whether a required, elective, or selected elective

Required

6. Specific goals for the course

a. Specific outcomes of instruction

Math 210 is the third and the final part of our standard three-semester calculus sequence. The distinct feature of this part of the course is its focus on the multi-dimensional analysis, as opposed to one-dimensional analysis learned in Math 180 (Calculus I) and Math 181 (Calculus II). Students will learn concepts such as a vector, a vector field, a function of several variables, partial derivative, and a line-integral and multi-variable integrals.

b. Explicitly indicate which of the student outcomes are addressed by the course.

SLO # 1: An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

7. Brief list of topics to be covered

Vectors in the plane and space

Vector valued functions

Functions of several variables; partial derivatives, max-min problems

Double and Triple integrals

Green's Theorem

One hour of exercises in the computer laboratory each week. Exercises are based on the same topics that are covered each week in the lecture-discussion.

MATH 220

1. Course number and name

Differential Equations – MATH 220

2. Credits, contact hours, and categorization.

3 credit hours, 150 min/week lecture, 50 min/week discussion, Mathematics

3. Instructor's or course coordinator's name

Department of Mathematics staff

4. Text book, title, author, and year

Fundamentals of Differential Equations by Nagle, Saff, and Snider, 9th edition.

a. Other supplemental materials

Boyce and DiPrima, Elementary Differential Equations and Boundary Value Problems, 9th edition.

Edwards and Penney, Differential Equations and Boundary Value Problems, 4th edition

5. Specific course information

a. Brief description of the content of the course (catalog description)

Techniques and applications of differential equations, first and second order equations, Laplace transforms, series solutions, graphical and numerical methods, and partial differential equations.

b. Prerequisites or co-requisites

Grade of C or better in MATH 210.

c. Indicate whether a required, elective, or selected elective

Required

6. Specific goals for the course

a. Specific outcomes of instruction

Techniques and applications of differential equations, solving first and second order equations, Laplace transforms, series solutions, graphical and numerical methods, and partial differential equations.

b. Explicitly indicate which of the student outcomes are addressed by the course.

SLO # 1: An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

7. Brief list of topics to be covered

- a. Techniques and applications of differential equations
- b. First and second order equations

- c. Laplace transforms
- d. Series solution
- e. Graphical and numerical methods
- f. Partial differential equations

PHYS 141

1. Course number and name

General Physics I (Mechanics)

2. Credits, contact hours, and categorization.

4 credit hours, 200 min/week lecture/discussion, 110 min/week laboratory, Basic Science

3. Instructor's or course coordinator's name

Department of Physics staff

4. Text book, title, author, and year

University Physics, Young and Freedman, 15th Edition.

a. other supplemental materials

Lab documents will be provided electronically

5. Specific course information

a. Brief description of the content of the course (catalog description)

Kinematics, vectors, Newton's laws of motion; linear momentum, collisions; work and kinetic energy; potential energy, conservation of energy; rotational kinematics and energy; rotational dynamics, static equilibrium; simple harmonic motion.

b. Prerequisites or co-requisites

Grade of C or better or concurrent registration in MATH 180; or approval of the department; and Grade of C or better in PHYS 100 or adequate performance on the departmental placement test.

c. Indicate whether a required, elective, or selected elective

Required

6. Specific goals for the course

a. Specific outcomes of instruction

The course objective for Physics 141 is to teach students the major ideas and methods in the physics of mechanics.

b. Explicitly indicate which of the student outcomes are addressed by the course.

SLO # 1: An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

7. Brief list of topics to be covered

Introduction: Physical quantities, physical laws, units

Kinematics of motion in one dimension

Newton's Laws, one-dimensional application

Vectors, kinematics of motion in two and three dimensions, projectile motion, circular motion

Application of Newton's Laws to two and three dimensional problems

PHYS 142

1. Course number and name

General Physics II (Electricity and Magnetism)

2. Credits, contact hours, and categorization.

4 credit hours, 200 min/week lecture/discussion, 110 min/week laboratory, Basic Science

3. Instructor's or course coordinator's name

Department of Physics staff

4. Text book, title, author, and year

University Physics, Young and Freedman, 15th Edition.

a. other supplemental materials

Lab documents will be provided electronically

5. Specific course information

a. Brief description of the content of the course (catalog description)

Electrostatics; electric currents; d-c circuits; magnetic fields; magnetic media; electromagnetic induction; a-c circuits; Maxwell's equations; electromagnetic waves; reflection and refraction; interference; geometrical optics.

b. Prerequisites or co-requisites

Grade of C or better in PHYS 141; and Grade of C or better or concurrent registration in MATH 181.

c. Indicate whether a required, elective, or selected elective

Required

6. Specific goals for the course

a. Specific outcomes of instruction

The course objective for Physics 142 is to teach students the major ideas and methods in the physics of electricity and magnetism.

b. Explicitly indicate which of the student outcomes are addressed by the course.

SLO # 1: An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

7. Brief list of topics to be covered

- a. Electric Charge and Field
- b. Gauss Law
- c. Electric Potential
- d. Capacitance and Dielectrics

- e. Current, Resistance, and EMFs
- f. Direct Current Circuits
- g. Magnetic Field and Magnetic Forces
- h. Sources of Magnetic Field
- i. Electromagnetic Induction

A.4. English Courses

ENGL 160

1. Course number and name

ENGL 160, Academic Writing I: Writing in Academic and Public Contexts

2. Credits, contact hours and categorization

3 credit hours, 3 contact hours/week, other (non-engineering, non-science/math)

3. Instructor's or course coordinator's name

Department of English staff

4. Text book, title, author, and year

Feldman, Ann M., Nancy Downs, & Ellen McManus. In Context: Participating in Cultural Conversations. New York: Longman, 2003.

Anson, Chris and Robert Schweiger. The Longman Handbook for Writers and Readers. 3rd Edition. New York: Longman, 2003.

4a. other supplemental materials

Sample Sources and Resources from a recently taught section titled, "Writing on Location: Living, Working, and Workplaces" in which students wrote about neighborhood issues, the working life, and people in their workplaces.

5. Specific course information

5a. Brief description of the content of the course (catalog description)

Students write in a variety of genres with an emphasis on argument and sentence-level grammar. Topics vary by section. This class may be taught in a blended format. When that is the case, internet access will be required. A high-speed connection is strongly suggested. Please check the online class schedule for blended-online sections.

5b. Prerequisites or co-requisites

Eligibility as determined by performance on the Department placement test.

5c. Indicate whether a required, elective, or selected elective

Required

6. Specific goals for the course

6a. Specific outcomes of instruction

Joining an Academic Conversation: The ability to conceptualize, articulate, and craft a response or an argument in an appropriate genre.

Key Rhetorical Concepts: Students explore some version of these four terms:

Situation--histories, cultures, communities, and individual experiences.

Genre--the forms of reading and writing.

Language--syntax, lexicon, organization, and design-shape texts; and

Consequences of writing

Attention to Written English: Special focus on academic written English, including discourse and stylistic conventions.

Readings: Analyze texts related to an issue or set of issues from a wide variety of genres and contexts, considering how these texts contribute to academic work.

Writing Process: Effective approaches to planning, drafting, revising and editing.

Correctness: Focus on syntactic and rhetorical choices, standard English usage, and using the assigned handbook as a reference.

6b. Indicate which of the student outcomes are addressed by the course

SLO #3: An ability to communicate effectively with a range of audiences.

7. Brief list of topics to be covered

Students in Academic Writing I will produce at least 20 pages of finished writing and complete five writing projects in a variety of genres, including an argumentative essay. Although each section covers different topics, typical projects could include: argumentative essay, opinion piece/commentary, speech, interview, formal letter, or proposal. Class work is organized in the following way:

The projects listed below were assigned in a section titled, "Writing on Location: Living, Working, and Workplaces."

Introduction to Writing in Academic and Public Contexts

Opinion Piece/Commentary: 2 pages and a 1 page cover letter

Interview Paper: 3 pages

Dialogue/Symposium: 5 pages

Argumentative Essay: 4 pages and a 1 page cover letter

Feature Story: 4 pages and a 1 page cover letter

ENGL 161

1. Course number and name

ENGL 161, Academic Writing II: Writing for Inquiry and Research

2. Credits, contact hours and categorization

3 credit hours, 3 contact hours/week, other (non-engineering, non-science/math)

3. Instructor's or course coordinator's name

Department of English staff

4. Text book, title, author, and year

Anson, Chris and Robert Schwegler. 2005. The Longman Handbook for Writers and Readers. 3rd Edition. New York: Longman.

Graff, Gerald and Cathy Birkenstein. 2018. They Say / I Say: The Move That Matter in Academic Writing. 4th Edition. New York: W.W. Norton.

Greene, Stuart and April Lidinsky. 2018. From Inquiry to Academic Writing: A Practical Guide. 4th Edition. Boston: Bedford/St.Martin's.

4a. other supplemental materials

Sample Sources and Resources: from a section titled, "Makin' It: Identity, Resilience, and School Culture," in which students explored the dynamic relationships that contribute to the construction of self and success in school. Students read ethnographic reports, cultural theory, fiction, documents related to school reform, and stories of students' lives.

5. Specific course information

5a. Brief description of the content of the course (catalog description)

Students learn about academic inquiry and complete several writing projects including a documented research paper. Topics vary by section.

5b. Prerequisites or co-requisites

ENGL 160 or the equivalent. All students take the Writing Placement test. If students place into ENGL 060, ENGL 070, ENGL 071, and ENGL 160, the student must take the course (or courses) prior to enrolling in ENGL 161. Students with an ACT English subscore of 27 or higher, SAT Evidence-Based Critical Reading score of 630 or higher, AP English Language & Composition score of 3 or higher, or IB English Language A: Language & Literature score of 6 or higher, receive credit for ENGL 160 and permission to enroll in ENGL 161.

5c. Indicate whether a required, elective, or selected elective

Required

6. Specific goals for the course

6a. Specific outcomes of instruction

Objectives for English 161 fall into two categories that roughly coincide with the first and second parts of the semester.

Part 1: Meaning-Making in Intellectual Communities

Students examine a topic for depth and breadth in the following ways: exploring discipline-specific topics and contexts for research; examining discipline specific and public contexts; writing summaries; writing syntheses and analyses; writing arguments and developing research skills. Instruction includes quoting, paraphrasing as well as a focus on grammar and style.

Part 2: The Classroom as a Research Community

Students plan and write a research paper through the following process: proposing an inquiry for independent research; identifying the consequential nature of the topic; identifying source materials; working in small groups to solve ongoing research problems; applying and refining skills for summarizing, paraphrasing, quoting, and documenting sources.

6b. Indicate which of the student outcomes are addressed by the course

SLO #3: An ability to communicate effectively with a range of audiences.

7. Brief list of topics to be covered

Students complete 20 pages of writing through 4 projects that ask students to summarize, synthesize, develop an argument, propose a research project, and write a research paper. Each section covers different topics but adheres to the following framework:

Introduction to Academic Inquiry

Summarizing

Analyzing and Synthesizing Texts

Argument

Research Proposal

Research Paper/Portfolio

Appendix B – Faculty Vitae

Faculty Vita: Said Al-Hallaj

1. Education

B.Sc. Chemical Engineering, Jordan University of Science and Technology, Jordan, 1990
M.Sc. Chemical Engineering, Jordan University of Science and Technology, Jordan, 1994
PhD Chemical Engineering, Illinois Institute of Technology, Chicago, IL, USA, 1999

2. Academic Experience

Department of Chemical Engineering, University of Illinois at Chicago, Chicago, IL,
Visiting Research Professor, 2009-2020, Part-Time

Department of Chemical & Biological Engineering, Illinois Institute of Technology, Chicago, IL, Research Professor, Coordinator of IIT Renewable Energy Programs, 2000-2009, Full-Time

Department of Chemical & Biological Engineering, IIT, Sr Research Associate, 1999-2000, FT

Department of Applied Chemistry, College of Engineering, Sendai, Japan:
Associate Visiting Researcher, 1999, Full-Time

3. Non-Academic Experience

All Cell Technologies LLC, Chicago, IL, Chairman/CEO, 2008-present, Full-Time
NETenergy, Chicago, IL, Founder/CEO, 09/2014- present, Part-Time

4. Certification or professional registrations

None

5. Current membership in professional organizations

Electrochemical Society

6. Honors and Awards

- Shell-NREL Game Changer Battery Challenge, 2019
- Wells Fargo Innovation Incubator (IN2), USDOE National Renewable Energy Lab, 2016
- New Venture Challenge, 2nd place, Booth School of Business, University of Chicago, 2015
- Clean Energy Challenge, Clean Energy Trust 2015 1st place award,
- Chicago Magazine 2012 Green Award
- Embassy of France, Young Entrepreneur Initiative Award (YEI), 2007

7. Selected service activities

- Course development: developed and teaching a graduate level course entitled “ChE 451 Renewable Energy Technologies” and co-authored course textbook “Hybrid Hydrogen Systems”
- Engineering and Entrepreneurship: developed and teaching a course entitled “ChE 494 Entrepreneurship in Chemical Engineering”
- Technology Commercialization: currently serving as Commercialization Fellow for UIC Office of Technology Management and successfully spun off two university-based tech start-up companies

8. Selected Publications and presentations

- 1- “Energy Storage for Smart Grid Applications,” Said Al-Hallaj, Keynote Speaker: 2nd International Conference on Smart Grid and Renewable Energy, November 19-21, 2019, Doha, Qatar
- 2- “Thermal Management and Safety of Li-ion Batteries,” Said Al-Hallaj, Invited Speaker: AIChE Midwest Regional Conference, 2019, UIC, Chicago, IL USA
- 3- “Thermal Management and Safety Concerns for Li-ion Batteries in Electric Mobility and Autonomous Vehicles Applications,” Said Al-Hallaj, Keynote Speaker: IEEE Vehicle and Propulsion Conference, August 27-30, 2018, Chicago, IL, USA
- 4- Numerical modeling of transient heat transfer in a phase change composite thermal energy storage (PCC-TES) system for air conditioning applications, Aljehani, LC Nitsche, S Al-Hallaj, Applied Thermal Engineering 164, 114522, 2020
- 5- Development of the inorganic composite phase change materials for passive thermal management of Li-ion batteries: material characterization, Y Galazutdinova, S Al-Hallaj, M Grágeda, S Ushak, International Journal of Energy Research, 2019
- 6- Exploring the charge reactions in a Li-O₂ system with lithium oxide cathodes and nonaqueous electrolytes T Zhang, R Amine, X Bi, Y Qin, M Li, S Al-Hallaj, F Huo, J Lu, K Amine Journal of materials chemistry A 7 (26), 15615-15620, 2019
- 7- Liu, Y Yuan, J Wen, K He, X Bi, Y Guo, Nano Letters 17 (5), 2959-2966, 2017
- 8- Preventing thermal runaway propagation in lithium ion battery packs using a phase change composite material: An experimental study, S Wilke, B Schweitzer, S Khateeb, S Al-Hallaj, Journal of Power Sources 340, 51-59, 2017
- 9- Semi-Empirical Modeling of Capacity Fade: A Practical Approach for Battery Pack Manufacturers, S Wilke, B Schweitzer, S Khateeb, S Al-Hallaj, ECS Transactions 73 (1), 109-119, 2016

- 10- Modeling of compression curves of phase change graphite composites using Maxwell and Kelvin models, MF Alzoubi, S Khateeb, S Al-Hallaj, Journal of Composite Materials 50 (8), 1123-1135, 2016

9. Professional development activities

- Jordan Solar EV projects: working closely with Government of Jordan and international partners on developing and deploying a national solar powered electric vehicles infrastructure
- Community Service: founding and board member of Connect.Jo, a USAID funded initiative that focuses on connecting Jordanian diaspora professionals with researchers and startups from Jordan.

Faculty Vitae: Sanjay Kumar Behura

1. Education

B. Sc. Physics, Utkal University, Bhubaneswar, India, 2006

M.S. Physics, National Institute of Technology Rourkela, India, 2008

Ph.D. Materials Science, PDP, India (Work at U. Saskatchewan, Canada), 2014

Post-doc Chemical Engineering, University of Illinois, Chicago, IL, 2016

2. Academic Experience

Department of Chemical Engineering, University of Illinois, Chicago, IL,
Research Assistant Professor, 2016 – Present, PT

3. Non-Academic Experience

Gujarat Energy Research Institute, Gandhinagar, India, Research Scientist, Solar Energy,
2013-2014, Full-Time

CSIR-IMMT, Bhubaneswar, India, Research Assistant, Colloids and Materials Chemistry,
2008-2009, Full-Time

4. Certifications or professional registrations

None

5. Current membership in professional organizations

Materials Research Society, American Chemical Society, American Physical Society,
American Institute of Chemical Engineers, and Electrochemical Society.

6. Honors and awards

Not applicable

7. Service activities

- (1) UIC Freshman Engineering Success Program (FESP) Chemical Engineering project, “Dye-sensitized solar cells”, 2017-2019.
- (2) UIC Women In Engineering Summer Program project: “Printing graphene-based solar cells and working of dye-sensitized solar cells” 2017.
- (3) High School presentation: ‘Nano-carbon materials and energy generations’, George Westinghouse College Prep High School, 2017.
- (4) UIC Chemical Engineering Summer Workshop of Research Publications, 2018, 2019
- (5) UIC Chemical Engineering Chem-E Club demonstration, “Molecular solar cells”, 2019
- (6) UIC Chemical Engineering ChemE Forum panel discussion on nanotechnology, 2018
- (7) Referee for student posters in AIChE Nanoscience Session and Undergraduate Student Poster Competition, and UIC Research Forum, 2017-2019.
- (8) Editorial Board Member of Scientific Reports, a Nature Research Journal.

- (9) Symposium chair or co-chair for '2D Materials' session at MRS Fall Meetings and AIChE Meetings
- (10) Vice-chair and faculty adviser for UIC ECS Student Chapter.
- (11) Editorial board member; UIC Postdoctoral Association Newsletter.
- (12) Organized four graduate research and career symposiums in the fall of 2016, 2017, 2018, 2019, Department of Chemical Engineering, University of Illinois at Chicago.
- (13) Reviewer: Nature Communications, ACS Nano, Angewandte Chemie, Green Chemistry, Scientific Reports, 2D Materials, ACS Omega, Nanotechnology, Nanoscale, RSC Advances, Synthetic Metals, Sensors & Actuators A, & Solar Energy Materials & Solar Cells

8. Selected Publications

- (1) S. K. Behura*, C. Wang, Y. Wen, and V. Berry*, "Graphene-Semiconductor Heterojunction Sheds Light on Emerging Photovoltaics," *Nature Photonics*, Vol. 13, p. 312-318 (2019).
- (2) H. A. Maddah, V. Berry, and S. K. Behura*, "Biomolecular Photosensitizers for Dye-Sensitized Solar Cells: Recent Developments and Critical Insights," *Renewable and Sustainable Energy Reviews*, Vol. 121, p. 109678 (2020).
- (3) H. A. Maddah, V. Berry, and S. K. Behura*, "Cuboctahedral Stability in Titanium Halide Perovskites via Machine Learning," *Computational Materials Science*, Vol. 173, p. 109415 (2020).
- (4) S. K. Behura, P. Nguyen, R. Debbarma, S. Che, M. R. Seacrist, and V. Berry, "Chemical Interaction-Guided, Metal-Free Growth of Large-Area Hexagonal Boron Nitride on Silicon-Based Substrates," *ACS Nano*, Vol. 11, p. 4985-4994 (2017). [Highlighted in Nature India, doi:10.1038/nindia.2017.58]
- (5) S. K. Behura, P. Nguyen, S. Che, R. Debbarma, and V. Berry, "Large-Area, Transfer-Free Oxide-Assisted Synthesis of Hexagonal Boron Nitride Films and Their Heterostructures with MoS₂ and WS₂," *Journal of the American Chemical Society*, Vol. 137, p. 13060-13065 (2015).
- (6) S. K. Behura and V. Berry, "Interfacial Non-degenerate Doping of MoS₂ and Other Two-Dimensional Semiconductors," *ACS Nano*, Vol. 9, p. 2227-2230 (2015).
- (7) S. K. Behura*, S. Nayak, I. Mukhopadhyay, and O. Jani, "Junction Characteristics of Chemically-Derived Graphene/p-Si Heterojunction Solar Cell," *Carbon*, Vol. 67, p. 766-774 (2014).
- (8) R. Debbarma, S. K. Behura*, Y. Wen, S. Che, and V. Berry*, "WS₂-Induced Enhanced Optical Absorption and Efficiency in Graphene/Silicon Heterojunction Photovoltaic Cells," *Nanoscale*, Vol. 10, p. 20218-20225 (2018).

Patents (Filed)

- (1) "Boron-nitride nanotube membranes," S. Kim, J. Shan, V. Berry, S. Cetindag, S. K. Behura, A. Pendse, and R. Praino, WO2019014633A1, Priority Date: July 14, 2017, Publication Date: Jan. 17, 2019.
- (2) "Direct formation of boron nitride on silicon-based dielectrics," V. Berry, S. K. Behura, P. Nguyen, and M. R. Seacrist, WO2017196559A1, Priority Date: May 12, 2016, Publication Date: Nov. 16, 2017.

Faculty Vita: Vikas Berry

1. Education

B.S. Chemical Engineering, Indian Institute of Technology-Delhi, 1999
M.S. Chemical Engineering, University of Kansas, 2003
Ph.D. Chemical Engineering, Virginia Tech 2006

2. Academic Experience

University of Illinois at Chicago, Chicago, IL (Full-Time)

- Department Head, 2016 – Present
- Professor, 2019 – Present
- Interim Department Head, 2015 – 2016
- Associate Professor, 2014 – 2019

Kansas State University, Manhattan, KS (Full-Time)

- William H. Honstead Professor, 2011 – 2014
- Associate Professor, 2012 – 2014
- Assistant Professor, 2007 – 2012

3. Non-Academic Experience

Cadila Pharmaceuticals, India, Pharmaceutical Process Scientist, 1999 – 2000, Full-Time

4. Certifications or professional registrations

Not applicable.

5. Current membership in professional organizations

American Institute of Chemical Engineers, American Chemical Society, AAAS Member, Materials Research Society

6. Honors and awards

- Louis Stokes Alliances for Minority Participation-NSF International Center of Excellence (LSAMP-NICE) Faculty Advisor Award, 2019
- Rudolph A. Marcus Award, 2017
- NSF-CAREER Award, 2011
- William H. Honstead Professorship, 2011 – 2014
- Omega Chi Epsilon Honor Society of Chemical Engineering, Member, 2007
- Sigma Xi Scientific Research Society, Member, 2007
- Plenary and Keynote Lectures
 - Plenary: Graphene Conference (Rome, 2019), *Tsinghua University* (China 2011)
 - Keynote: TechConn (2020), RRR (2019), *ISFM* – (India 2018); *ICREA – Graphene Biosensors* (Spain - 2015); *IMST* (India, 2012)

7. Selected service activities

Editorial Service: Scientific Reports, Editorial Board Member, 2012 – Present; Science Advances Letters, Associate Editor, 2014 – Present; Advanced Carbon, Scientific Advisory Board; Journal of Nanomaterials and Molecular Nanotechnology, Scientific Advisory Board

Journal reviewer: Science, Nature, Nature Nanotechnology, Nature Materials, Nature Communications Scientific-Reports, Nature Protocols, Nano Letters, ACS-Nano, Angewandte Chemie, Journal of the American Chemical Society, Small, Advanced Materials, Nanoscale, Nanoresearch, IEEE-nano, The Analyst, Journal of Physical Chemistry, Chemistry of Materials

Served as a reviewer for several grant-review panels and individual proposals: NSF, UIC: POC grants, COE SEED, Chancellor's Award (Nov 2015), DoE, Czech Science Foundation and Ireland Science Office (March 2017)

Conference Chair: International Symposium on Functional Materials, India (April 2018), ICREA Workshop on Graphene Nanobiosensors – 2016, Recent Advanced in Graphene and Related materials: Chair; 2010 (Singapore), AIChE: Chair for “*Graphene and Carbon Nanotube Based Devices*”, (Nov 2009-2018, MRS: Chair 2017, 2012

University/College/Department: Department Head, Executive Committee Member, COE – UIC, Search Committee Member: Faculty, UIC, Search Committee Member: Dean, College of Engineering-KSU (2013), Core Facilities Committee Member, Undergraduate Advisor.

8. Selected Publications and presentations: (>27 from 2015-present)

1. Songwei Che, Sanjay K Behura, and Vikas Berry*, "*Photo-Organometallic, Nanoparticle-Nucleation on Graphene for Cascaded-Doping*", **ACS Nano**, 13 (11), 12929-12938, **2019**
2. Sanjay Behura*, Chen Wang, Yu Wen, and Vikas Berry*, "*Graphene/Semiconductor Heterojunction Sheds Light on Emerging Photovoltaics*", **Nature Photonics**, **13**, 312–318, **2019**
3. Hisham Middah, Vikas Berry and Sanjay Behura, “*Cuboctahedral Stability in Titanium Halide Perovskites via Machine Learning*”, **Computational Mat. Science**, Accepted, 2019
4. Hisham Middah, Vikas Berry and Sanjay Behura, “*Biomolecular Photosensitizers for Dye-Sensitized Solar Cells: Recent Developments and Critical Insights*”, **Renewable & Sustainable Energy Reviews**, Accepted, **2019**
5. Shikai Deng, Dongjoon Rhee, Won-Kyu Lee, Songwei Che, Bijentimala Keisham, Vikas Berry and Teri W. Odom, " *Graphene Wrinkles Enable Spatially-defined Chemistry*", **Nano Letters**, **19(8)**, 5640-5646, **2019**
6. Sungjoon Kim, Bijentimala Keisham, and Vikas Berry*, "*Cellular Nano-Transistor: An Electronic-Interface between Nanoscale Semiconductors and Biological Cells*", **Materials Today Nano**, 100063, **2019**

9. Professional development activities

Member: 2020 Faculty Academic Leadership Program, UIC.

Faculty Vitae: Betul Bilgin

1. Education

Ph.D. Chemical Engineering, Michigan State University, East Lansing, MI, 2014
M.S. Biomedical Engineering, University of Minnesota, Minneapolis, MN, 2008
B.S. Chemistry, Fatih University, Istanbul, Turkey, 2005

2. Academic Experience

Department of Chemical Engineering, University of Illinois at Chicago,
Clinical Assistant Professor, 2019-Present, Full-Time

Department of Chemical Engineering, University of Illinois at Chicago,
Visiting Clinical Assistant Professor, 2015-2019, Part-Time

Department of Chemical Engineering, University of Illinois at Chicago,
Visiting Lecturer, 2014-2015, Part-Time

3. Non-Academic Experience

Not applicable

4. Certifications or professional registrations

Not applicable

5. Current membership in professional organizations

American Institute of Chemical Engineers (AIChE), American Society for Engineering Education (ASEE)

6. Honors and awards

Teaching Scholar, Selected for UIC Teaching Scholar Program Spring 2017

American Institute of Chemical Engineers (AIChE) “35 Under 35” award

- Named as one of AIChE 35 under 35 awardees for 2017 August 2017

American Society for Engineering Education (ASEE) “20 Under 40” award

- Named as one of ASEE 20 under 40 awardees for 2018 June 2018

7. Service activities

UIC – Chemical Engineering Public Relations – Co Chair, 2016-present

UIC Chemical Engineering Industry Day organizer (2015-2017)

UIC Chemical Engineering Alumni Mixer organizer (2016-2018)

UIC Chemical Engineering ChemE Forum organizer (2017-2018)

UIC Open House College Recruitment event technical presentation, 2018-2019

Organized College of Engineering workshop on Scholarship of Teaching and Learning

Assisted in the coordination of events for the 2017 AIChE Midwest Regional Conference (MRC) at UIC for local high school students; responsible for logistics, programming, outreach to engineers.

Served on scientific committee for “2017 International Conference on Advances and

Innovations in Engineering”

Women’s Activities Coordinator of Niagara Foundation (2006-2014)

Judging

Judge for 2016 UIC Student Research Forum (April 2016)

Introduced chemical engineering discipline to middle and high school students via an art called water marbling; visited many different schools and presented at many different events (2016-2019)

8. Selected Publications

Bilgin, B., Nath, A., Chan, C., Walton, S.P. Characterization of transcription factor response kinetics in parallel. BMC Biotechnology, 2016:62.

Bilgin, B., Liu, L., Chan, C., Walton, S.P. Quantitative, solution-phase profiling of multiple transcription factors in parallel. Analytical and Bioanalytical Chemistry. 2013. 408(8): 2461-2468.

Cho, H., Wu, M., Bilgin, B., Walton, S.P., Chan, C. Latest developments in experimental and computational approaches to characterize protein-lipid interactions. Proteomics. 2012. 12(22): 3273-3285.

Cho, H., Mukherjee, S., Palasuberniam, P., Pillow, L., Bilgin, B., Nezich, C., Walton, S.P., Feig, M., Chan, C. Molecular mechanism by which palmitate inhibits PKR autophosphorylation. Biochemistry. 2011. 50(6): 1110-1119.

Xie, S.N., Moya, C., Bilgin, B., Jayaraman, A., Walton, S.P. Emerging affinity based techniques in proteomics. Expert Review of Proteomics. 2009. 6(5): 573-583.

Bozkurt, A., Karadedeli, B. Copolymers of 4(5)-vinylimidazole and ethyleneglycol methacrylate phosphate: Synthesis and proton conductivity properties. Reactive & Functional Polymers. 2007. 67(4): 348-354.

Karadedeli, B., Bozkurt, A., Baykal, A. Proton conduction in adipic acid/benzimidazole hybrid electrolytes. Physica B: Condensed Matter. 2005. 364(1-4): 279-284.

9. Professional Development

BASF Faculty Workshop on Process Safety July 23rd-26th 2018

Wyandotte, MI

ChemE Summer School July 29th-August 3rd 2017

NC State University, Raleigh, NC

National Effective Teaching Institute – 1 May 21st-23rd 2017

Philadelphia, PA

Faculty Vitae: Michael Caracotsios

1. Education

B.S. Chemical Engineering, University of Thessaloniki-Greece, 1979

M.S. Chemical Engineering, University of Kentucky-Lexington, 1981

Ph.D. Chemical Engineering, University of Wisconsin-Madison, 1986

2. Academic Experience

Clinical Associate Professor- Department of Chemical Engineering, University of Illinois at Chicago, 09/2015 to Present

Adjunct Professor- Department of Chemical and Biological Engineering, Northwestern University, Evanston IL, 01/2007 to 08/2015

Adjunct Professor- Department of Chemical Engineering, Illinois Institute of Technology, Chicago IL, 05/1986 to 12/2006

3. Non-Academic Experience

Research Fellow - UOP, Optimization and Advanced Control Applications, Process Information and Control Division, Solutions and Services Division, Modeling and Optimization Division, Gas Processing Division, 08/1996 to 09/2015

Senior Research Engineer - Amoco Chemical Company, Research and Development Department, Aromatics Oxidation Division, Polymers Catalysis and Process Division, Computer Aided Process Engineering Division, 03/1986 to 08/1996

Research Engineer, Chevron Oil Field Research Corporation, Reservoir Engineering Division. Developed software on Cray supercomputers for steam-flood simulations in oil exploration, 05/1984 to 09/1984

4. Certifications or professional registrations

Green belt six-sigma certification

5. Current membership in professional organizations

American Institute of Chemical Engineers

6. Honors and awards

Honoring your Professor UIC Award (2017)

Honeywell Team Performance Award, for designing state of the art catalyst regeneration systems (2010)

BP Helios Award for developing state of the art optimization technology (2000)

William M. Langdon Award for Excellence in Teaching, IIT, Chicago (1991-1992)

First prize award, National Competition in Mathematics, Greece (1974)

Five awards for outstanding performance (ranked # 1) in Chemical Engineering from National Fellowship Foundation, Greece (1974-79).

Three awards for outstanding performance (ranked # 1) in the entire College of Engineering, from the Technical Chamber, Greece (1974-79).

7. Service activities

Freshmen Engineering Success Program Committee, participant, 2017-2019
Chemical Engineering Advisory Committee, August 2016-present
Presenter, High School District 203 Career Night, Engineering, Industrial Technology,
College recruitment activities; (2014-2019)

8. Selected Publications

Kinetic Modelling of Water Inhibited Methane Oxidation Reactions over a Pd/Al₂O₃ Wash-coated Monolith using Spatially and Temporally Resolved Experimental Data, Ciaran Coney, Cristina Stere, Paul Millington, Agnes Raj, Sam Wilkinson, Geoffrey McCullough, Michael Caracotsios, Chris Hardacre, Kevin Morgan, David Thompsett, Alexandre Goguet, ACS Catalysis (2020)

Nonlinear mixed-effects models for kinetic parameter estimation with batch reactor data, Daniel A. Hickman, Michael J. Ignatowich, Michael Caracotsios, James D. Sheehan and Fabio D'Ottaviano, *Chemical Engineering Journal* (2019)

Robustness of coke burning from catalyst for light paraffin dehydrogenation process, US Patent, US 8,895,464 B. Egolf, L. Leonard, M. Caracotsios (2014)

Computer Aided Modeling of Chemically Reactive Systems, co-authored with W. E. Stewart, *John Wiley and Sons, Inc.* , 2008

9. Professional Development

Industrial Multivariable Process Control
Computational Fluid Dynamics and Combinatorial Chemistry.
Real Time On-Line Optimization with ASPEN RT-OPT
Listening, Interpersonal and Communication Skills, Influencing without Direct Authority
Concepts in Quality Manufacturing and Technology Transfer
Management Principles and Techniques and Managing Personal Growth
Artificial Intelligence and Expert Systems in Engineering
Polymer Reaction Engineering, Poly-olefins Manufacture Featuring Metallocene Catalysts
Advance Modeling and Control of Polymerization Reactors
Applications of Pinch Technology to Chemical Industries
Principles of Heterogeneous Catalysis and Organo-metallic Chemistry
Advanced Programming with Microsoft Visual Basic 3.0 for Windows, C and PASCAL

Faculty Vita: Brian P. Chaplin

1. Education

B.S. Civil Engineering, University of Minnesota, 1999

M.S. Civil Engineering, University of Minnesota, 2003

Ph.D. Environmental Engineering, University of Illinois at Urbana-Champaign, 2007

2. Academic Experience

Department of Chemical Engineering, University of Illinois at Chicago

Associate Professor, 2018–present. Full-Time

Department of Chemical Engineering, University of Illinois at Chicago

Assistant Professor, 2013–2018. Full-Time

Department of Civil and Environmental Engineering, Villanova University

Assistant Professor, 2010–2013. Full-Time

Department of Chemical and Environmental Engineering, University of Arizona,

Postdoctoral Research Associate, 2008–2010. Full-Time

Department of Civil and Environmental Engineering, University of Illinois at Urbana-

Champaign, Postdoctoral Research Associate, 2007–2008. Full-Time

3. Non-Academic Experience

U.S. Geological Survey, Mounds View, MN, Hydrologist, 1999–2003. Full-Time

4. Certifications or professional registrations

E.I.T. certified (Minnesota since 1999)

5. Current membership in professional organizations

American Institute of Chemical Engineers

Association of Environmental Engineering and Science Professors

6. Selected Honors and awards

- 2019 *Environmental Science & Technology and Environmental Science & Technology Letters Virtual Issue: Early Career Scientists*
- 2018 *Environmental Science & Technology Best Paper Award*
- *NSF Early CAREER Development Award* (2015–2020)
- *Environmental Science & Technology Excellence in Review Award* (2017)
- *Environmental Science & Technology Letters Excellence in Review Award* (2016)
- UIC College of Engineering Research Award (2015)
- Summer Research Fellowship/Research Support Grant, Villanova University (2011)
- Environmental Water Resources Institute's 2011 Samuel Arnold Greeley Award (Best Paper). **Chaplin** et al., (2009). *Journal of Environmental Engineering-ASCE*. **135(8)**, 666–676.

7. Selected service activities

- Member, *Chemical Engineering Departmental Advisory Board*, Department of Chemical Engineering 2018–present
- Chair, *Safety Committee*, Department of Chemical Engineering, 2018–present

- Chemical Engineering Department Seminar Series Coordinator, 2018-present.
- NSF CAREER panel reviewer, CBET-Env. Eng. Program, Nov 2015, Sept 2016
- NSF panel reviewer, CBET-Env. Eng. Program, March 2016, February 2017

8. Selected Publications from the last 5 years

1. Almassi, S.; Samonte, P.R.V.; Li, Z.; Xu, W.; **Chaplin, B.P.** Mechanistic Investigation of Haloacetic Acid Reduction using Carbon-Ti₄O₇ Composite Reactive Electrochemical Membranes. *Environmental Science & Technology* **2020**, <https://doi.org/10.1021/acs.est.9b06744>
2. Lin, M.-H, Mehraeen, S.; Cheng, G.; Rusinek, C.; **Chaplin, B.P.** Role of Near-Electrode Solution Chemistry on Bacteria Attachment and Poration at Low Applied Potentials. *Environmental Science & Technology* **2020**, 54, 1, 446-455.
3. Misal, S.N; Lin, M.-H.; Mehraeen, S.; **Chaplin, B.P.** Modeling Electrochemical Oxidation and Reduction of Sulfamethoxazole using Electrocatalytic Reactive Electrochemical Membranes. *Journal of Hazardous Materials* **2020**, 384, 121420.
4. Zoric, M.R.; Singh, V.; Warren, S.; Plunkett, S.; Khatmullin, R.; **Chaplin, B.P.**; Glusac, K.D. Electron Transfer Kinetics at Graphene Quantum Dot Assembly Electrodes. *ACS Applied Materials & Interfaces* **2019**, doi.org/10.1021/acsami.9b14161.
5. Pendse, A.; Cetindag, S.; Lin, M.-H.; Rackovic, A.; Debbarma, R.; Almassi, S.; **Chaplin, B.P.**; Berry, V.; Shan, J.W.; Kim, S. Charged Layered Boron Nitride-Nanoflake Membranes for Efficient Ion Separation and Water Purification. *Small* **2019**, doi.org/10.1002/sml.201904590.
6. Ge, J.; Zou, X.; Almassi, A.; Ji, L.; **Chaplin, B.P.**; Bard, A.J. Electrochemical Production of Si without Generation of CO₂ Based on the Use of a Dimensionally Stable Anode in Molten CaCl₂. *Angewandte Chemie* **2019**, doi.org/10.1002/anie.201905991.
7. T.X.H; Haflich, H.; Shah, A.D.; **Chaplin, B.P.** Energy-Efficient Electrochemical Oxidation of Perfluoroalkyl Substances Using a Ti₄O₇ Reactive Electrochemical Membrane Anode. *Environmental Science & Technology Letters* **2019**, 6, 8, 504-510.
8. **Chaplin, B. P.**, The Prospect of Electrochemical Technologies Advancing Worldwide Water Treatment. *Accounts of Chemical Research* **2019**, 52, (3), 596-604.
9. Y. Jing, S. Almassi, R.J. LeSuer, S. Mehraeen, **B.P. Chaplin**, “The roles of oxygen vacancies, electrolyte composition, lattice structure, and doping density on the electrochemical reactivity of Magnéli phase TiO₂ anodes.” *Journal of Materials Chemistry A*, Vol. 6, pp. 23828 – 23839, **2018**.
10. S. Nayak, **B.P. Chaplin**, “Fabrication and Characterization of Porous, Conductive, Monolithic Ti₄O₇ Electrodes.” *Electrochimica Acta*, Vol. 263, pp. 299-310, **2018**.

9. Professional development activities

Science Leadership Council, *Current*, Chicago, IL, 2017-present

Faculty Vitae: Gang Cheng

1. Education

Ph.D. Chemical Engineering, University of Washington Seattle, WA, 2009
M.S. Microbial Engineering, University of Minnesota Twin Cities, MN, 2005
B.S. Biochemical Engineering, Beijing University of Chemical Technology, 1999

2. Academic Experience

Department of Chemical Engineering, University of Illinois at Chicago, Chicago, IL,
Associate Professor, 2016-present, Full-Time
Department of Chemical and Biomolecular Engineering, University of Akron, Akron, OH,
Associate Professor, 2015-2016, Full-Time
Department of Chemical and Biomolecular Engineering, University of Akron, Akron, OH,
Assistant Professor, 2009-2015, Full-Time
Northeast Ohio Medical University, Adjunct Assistant Professor of Integrative Medical
Sciences, 2011-present, Part-Time

3. Non-Academic Experience

Not applicable

4. Certifications or professional registrations

Not applicable

5. Current membership in professional organizations

American Institute of Chemical Engineers

6. Honors and awards

Not applicable

7. Service activities

In 2012, 2013, 2015 and 2016 summer, I hosted a middle school teacher, Ms. Phoua Alayamini, to conduct an eight-week research project in my lab for two successive summers. In 2014 and 2015 summer, I hosted a high school biology instructor, Ms. Dori Hess, to conduct an eight-week research project in my lab. I participated in several high school student campus visitation events to introduce the Chemical Engineering department to students and their parents. I served as a judge in the Student Poster competition, Protein and Tissue Engineering Category at the AIChE Student Conference hosted at The University of Akron on April 21st, 2012

8. Selected Publications

1. Huifang Wang, Megan, Young, Dylan Lynch, and Gang Cheng*, Tunable Surface and Bulk Properties of Zwitterionic Polyurethane, ACS Applied Materials & Interfaces. 2018, DOI 10.1021/acsami.8b10450

2. Chen-Jung Lee, Huifang Wang, Megan, Young, Dylan Lynch, Hongbo Cong, and Gang Cheng*, Structure-function study of Poly(sulfobetaine 3,4-ethylenedioxythiophene) (PSBEDOT) and its derivatives, *Acta Biomaterialia*, 2018, 10.1016/j.actbio.2018.06.002
3. Chen-Jung Lee, Haiyan Wu, Yang Hu, Megan, Young, Dylan Lynch, Huifang Wang, Fujian Xu, Hongbo Cong, and Gang Cheng*, Ionic conductivity in polyelectrolyte hydrogels, *ACS Applied Materials & Interfaces*. 2018, 10 (6), pp 5845–5852
4. Cao B.; Lee, J. C.; Zeng, Z. P.; Cheng, F.; Xu, F. J.; Cong, H. B.; Gang Cheng*, Electroactive Zwitterionic Poly(sulfobetaine 3,4-ethylenedioxythiophene) (PSBEDOT) with Controllable Antimicrobial and Antifouling Properties, *Chemical Science* 2016, 7, 1976-1981.
5. Cao, B.; Tang, Q.; Li, L. L.; Lee, C. J.; Wang, H.; Zhang, Y. Q.; Castaneda, H.; Gang Cheng*, Integrated zwitterionic conjugated poly(carboxybetaine thiophene) as a new conducting biomaterial platform, *Chemical Science* 2014, 6 (1), 782 - 788.
6. Haiyan Wu, Chen-Jung Lee, Huifeng Wang, Yang Hu, Megan Young, Hongbo Cong and Gang Cheng*, Highly Sensitive and Stable Zwitterionic Poly(Sulfobetaine 3,4-Ethylenedioxythiophene) (PSBEDOT) Glucose Biosensor in Complex Media, *Chemical Science*, 2018, 9, 2540 - 2546
7. Chen-Jung Lee, Haiyan Wu, Yang Hu, Megan, Young, Dylan Lynch, Huifang Wang, Fujian Xu, Hongbo Cong, and Gang Cheng*, Ionic conductivity in polyelectrolyte hydrogels, *ACS Applied Materials & Interfaces*. 2018, 10 (6), pp 5845–5852
8. Cao, B.; Li, L. L.; Wu, H. Y.; Qiong, T.; Sun, B. B.; Dong, H.; Zhe, J.; Gang Cheng*, Zwitteration of dextran: a facile route to integrate antifouling, switchability and optical transparency into natural polymers, *Chemical Communications* 2014, 50 (24), 3234 - 3237.

9, Professional Development

Research collaborations:

Hongbo Cong, University of Akron, Ohio

Lingyun Liu, University of Akron

Bi-min Zhang Newby, University of Akron

Jie Zheng, University of Akron

Homero Castaneda-Lopez, University of Akron

Jiang Zhe, University of Akron

Faculty Vitae: Sangil Kim

1. Education

B.S. Polymer Science and Engineering, Chungnam National University South Korea, 1997
M.S. Polymer Science and Engineering, Chungnam National University South Korea, 1999
M.S. Chemical Engineering, University of Cincinnati, Cincinnati, OH, 2003
PhD Chemical Engineering, Virginia Tech, 2007

2. Academic Experience

Department of Chemical Engineering, University of Illinois, Chicago, Assistant Professor,
2015 – present, Full-Time
University of California Davis/Lawrence Livermore National Lab., Livermore, CA,
Post-doctoral Associate, 2007 – 2009, Full-Time

3. Non-Academic Experience

Lawrence Livermore National Lab, Livermore, CA, Guest Researcher, 2015 – present, Part-Time
Lawrence Livermore National Lab., Livermore, CA, Staff Scientist, 2011– 2015, Full-Time
Porifera Inc, Hayward, CA, Senior Researcher, 2009– 2011, Full-Time
Korea Institute of Science and Technology (KIST), Seoul, South Korea,
Researcher, Complex Fluid and Membrane Laboratory, Mar 2001 – Aug 2001, Full-Time

4. Certifications or professional registrations

Not applicable

5. Current membership in professional organizations

American Institute of Chemical Engineers

6. Honors and awards

Not applicable

7. Service activities

Not applicable

8. Selected Publications

- 1) T. Wang, J. Y. Jeon, J. Han, J. H. Kim, C. Bae, and S. Kim, "Poly(terphenylene) Anion Exchange Membranes with High Conductivity and Low Vanadium Permeability for Vanadium Redox Flow Batteries (VRFBs)", Journal of Membrane Science, In press, 2019
- 2) A. Pendse, S. Cetindag, A. Rackovic, R. Debbarma, S. Almassi, B. Chaplin, V. Berry, J. Shan, and S. Kim, "Highly Charged Layered Boron Nitride Nanoflake Membranes for Efficient Ion Separation and Water Purification", Small, 15, 1904590, 2019
- 3) T. Wang, S. J. Moon, D. S. Hwang, H. Park, J. Lee, S. Kim, Y.M. Lee, and S. Kim, "Selective Ion Transport for Vanadium Redox Flow Battery in Nano-Crack Regulated Nanocomposite Membranes", Journal of Membrane Science, 583, 16-22, 2019

- 4) S. Kim*, D. Dehlinger, J. Pena, M. Shusteff, N. M. Collette, M. El Sheikh, H. Seol, M. Davenport, P. Naraghi-Arani, and E. Wheeler, "Virus Concentration and Purification by a Microfluidic Filtering System with an Integrated PEGylated Antifouling Membrane", *Microfluidics and Nanofluidics*, 23:9, 2019
- 5) M.R.Cerón, L.S.Lai, A.Amiri, M.Monte, S.Katta, J.Kelly, M.Worsley, M. Merrill, S. Kim, and P. G. Campbell, "Surpassing the Conventional Limitations of CO₂ Separation Membranes with Hydroxide/Ceramic Dual-Phase Membranes, *Journal of Membrane Science*, 561, 191-198, 2018
- 6) S. Cetindag, B. Tiwari, D. Zhang, Y. K. Yap, S. Kim, and J. Shan, "Surface-Charge Effects on the Electro-Orientation of Insulating Boron-Nitride Nanotubes in Aqueous Suspension", *Journal of Colloid and Interface Science*, 505, 1185-1192, 2017
- 7) D. S. Hwang, T. Chung, T. Wang, and S. Kim*, "Recent Developments in Ion-Exchange Nanocomposite Membranes for Energy Applications", *Membrane Journal*, 26, 432-448, 2016

Faculty Vitae: Ying Liu

1. Education

B.S. Engineering Mechanics, Tsinghua University, 2001

M.A. Mechanical and Aerospace Engineering, Princeton University, 2004

Ph.D. Mechanical and Aerospace Engineering, Princeton University, 2007

2. Academic Experience

Department of Chemical Engineering, University of Illinois-Chicago

Assistant Professor, 2008-2015

Associate Professor, 2015-present

Department of Biopharmaceutical Sciences, University of Illinois-Chicago

Adjunct professor, 2008-present

Department of Bioengineering, University of Illinois-Chicago

Adjunct professor, 2018-present

Department of Chemistry, University of Chicago

Post Doctoral Researcher, July 2007- July 2008

3. Non-Academic Experience

Not applicable.

4. Certifications or professional registrations

Not applicable.

5. Current membership in professional organizations

American Institute of Chemical Engineers

American Physical Society

American Association of Pharmaceutical Science

Sino-American Pharmaceutical Association

6. Honors and awards

Chicago Biomedical Consortium, Catalyst Award, 2020

UIC Graduate Mentor Award, UIC, IL, 2017

UIC Researcher and Scholar of the Year, Rising Star Award, UIC, IL, 2016

CAREER Award, National Science Foundation, Jan. 2014 – Jan. 2019

College of Engineering Research Award, UIC, 2013

OTM POC Award, UIC, 2013

Chancellor Discovery Award, UIC, 2011

Technology for Developing Regions Fellowship, Princeton University, 2006

Howard Crathorne Phillips Graduate Fellowship, Princeton University, 2003

Engineering Fellowship, Princeton University, 2002

Rong Hong National Fellowship, Tsinghua University, 2000

7. Service activities

College of Engineering, Executive Committee, UIC, 2016-present

Member, Graduate Committee, Department of Chemical Engineering, University of Illinois at Chicago, 2008-2018

Student Sponsor for Illinois Math and Science Academy, 2010-2016

Mentor, Girls' Electronic Mentoring in Science, Engineering and Technology,
2009-present

Member, Editorial Board, Advances in Materials Science and Engineering, 2013-2019

Member, Editorial Board, Journal of Oncology and Biomarker Research, 2013-2018

8. Selected Publications Last 5 Years (Selected from 33 publications)

1. Pin Zhang, Veronica Villanueva, Joseph Kalkowski, Chang Liu, Tiep Pham, Ursula Perez-Salas, Wei Bu, Binhua Lin, Ying Liu*, Effects of Polyunsaturated Phospholipids on Membrane Degradation Catalyzed by a Secreted Phospholipase A2, *Langmuir*, 2019, 35, 36, 11643-11650.
2. Pin Zhang, Veronica Villanueva, Joseph Kalkowski, Chang Liu, Alexander J. Donovan, Wei Bu, Mark L. Schlossman, Binhua Lin, and Ying Liu, Molecular interactions of phospholipid monolayers with a model phospholipase, *Soft Matter* (featured and back cover), 2019, 15, 4068.
3. Joseph Kalkowski, Chang Liu, Paola Leon-Plata, Magdalena Szymusiak, Pin Zhang, Thomas Irvine, Weifeng Shang, Osman Bilsel, and Ying Liu*, In Situ Measurements of Polymer Micellization Kinetics with Millisecond Temporal Resolution, *Macromolecules*, 2019, 52, 3151-3157.
4. Paola Leo Plata, Ying Liu*, and Ludwig C. Nitsche*, Interaction of Multiple Drops and Formation of Toroidal-spiral Particles, *Physical Review Fluids*, 2018, 3, 093601.
5. Magdalena Szymusiak, Joseph Kalkowsk, Hanying Luo, Alexander J. Donovan, Pin Zhang, Chang Liu, Weifeng Sahng, Thomas Irving, Margarita Herrera-Alonso, and Ying Liu*, Core-shell Structure and Aggregation Number of Micelles Composed of Amphiphilic Block Copolymers and Amphiphilic Heterografted Polymer Brushes Determined by Small-angle X-ray Scattering, *Macro Letters*, 2017 6(9), 1005-1012.
6. Alexander J. Donovan, Joseph Kalkowski, Magdalena Szymusiak, Canhui Wang, Stephanie A. Smith, Robert F. Klie, James H. Morrissey, and Ying Liu, Artificial Dense Granules: A Procoagulant Liposomal Formulation Modeled after Platelet Polyphosphate Storage Pools, *Biomacromolecules*, 17(8), 2572-2581, 2016.
7. Magdalena Szymusiak, Xiaoyu Hu, Paulina Ciupinski, Zaijie Jim Wang, and Ying Liu, Bioavailability of Curcumin and Curcumin Glucuronide in the Central Nervous System of Mice after Oral Delivery of Nano-Curcumin, *International Journal of Pharmaceutics*, 511 (1), 415-423, 2016.

Faculty Vitae: Shafigh Mehraeen

1. Education

B.Sc. Mechanical Engineering, University of Tehran, Iran 2001

M.Sc. Mechanical Engineering, Stanford University, CA, 2006

PhD Mechanical Engineering, Stanford University, CA, 2011

2. Academic Experience

Department of Chemical Engineering, University of Illinois at Chicago,

Assistant Professor, 2016 – present, Full-Time

Massachusetts Institute of Technology, MA, Chemistry Research Fellow, 2013-2016, Full-Time

Georgia Institute of Technology, GA, Chemistry Research Associate, 2011-2013, Full-Time

3. Non-Academic Experience

Not applicable

4. Certifications or professional registrations

Not applicable

5. Current membership in professional organizations

6. Honors and awards

2014, Ascher Shapiro Award, MIT

7. Service activities

Undergraduate Research Mentor at UIC (2016- on going)

Graduate Admission Committee Member: Responsible for reviewing applications for M.S. and Ph.D. programs, reviewing instructional units and graduate courses concerning general policies for graduate education at UIC (2016- on going)

Reviewer: NSF panel reviewer in CBET program

Journal reviewer: Physical Review Letters, Physical Review E, Physical Review B, Organic Electronics, Physical Chemistry Letters, Physical Chemistry C, Langmuir (2016 - on going)

8. Selected Publications

1. V. Coropceanu, J-L. Bredas, and S. Mehraeen, Impact of active layer morphology on bimolecular recombination dynamics in organic solar cells, *Journal of Physical Chemistry C*, 121(45), 24954-24961 (2017).

2. M. Asbahi, S. Mehraeen, J. K.W. Yang, J. Cao, and M.C. Tan, Large area directed self-assembly of sub-10 nm particles with single particle positioning resolution, *Nano Letters*, 15, 6066-6070 (2015).

3. S. Mehraeen, M. Asbahi, J. K.W. Yang, J. Cao, and M.C. Tan, Sub-10 nm particles directed selfassembly: Role of driving forces and template geometry, *Langmuir*, 31, 8548-8557 (2015).

4. J. J. VanDersarl, S. Mehraeen, A. Schoen, S. Heilshorn, A. Spakowitz, and N. A. Melosh, Rheology and simulation of 2-dimensional clathrin protein network assembly, *Soft Matter*, 10, 6219-6227 (2014).
5. M. Asbahi, S. Mehraeen, K. T.P. Lim, F. Wang, J. Cao, M.C. Tan, and J. K.W. Yang, Template-induced structure transition in sub-10 nm self-assembling nanoparticles, *Nano Letters*, 14, 2642-2646 (2014).
6. S. Shoaee, S. Mehraeen, J. G. Labram, D. D.C. Bradley, T. D. Anthopoulos, J-L. Brédas, V. Coropceanu, and J. R. Durrant, Correlating non-geminate recombination with film microstructure: a comparison of polythiophene : fullerene bilayer and blend films, *Physical Chemistry Letters*, 5, 3669-3676 (2014).
7. N. Cordella, T. J. Lampo, S. Mehraeen, and A. J. Spakowitz, Membrane fluctuations destabilize clathrin protein lattice order, *Biophysical Journal*, 106, 7, 1476-1488 (2014).

Faculty Vitae: Vivek Sharma

1. Education

B. Tech. Textile Technology, Indian Institute of Technology Delhi, New Delhi, India, 2001
M.S. Polymer Science, University of Akron, Akron, OH, 2003
M.S. Chemical Engineering, Minor: Nonlinear Sciences, Georgia Institute of Technology, Atlanta, GA 2006
Ph.D. Polymer Science & Engineering, Minor: Physics, Georgia Institute of Technology Atlanta, GA, 2008
Post-Doc Mechanical Engineering, Massachusetts Institute of Technology Cambridge, MA 2012

2. Academic Experience

Department of Chemical Engineering, University of Illinois at Chicago, Associate Professor, 2019-present, Full-Time

Department of Chemical Engineering, University of Illinois at Chicago, Assistant Professor, 2012-2019, Full-Time

Institute for Molecular Engineering, University of Chicago, Chicago IL, Visiting Assistant Professor, 2017- Present, Part-Time

3. Non-Academic Experience

Not applicable

4. Certifications or professional registrations

Not applicable

5. Current membership in professional organizations

American Institute of Chemical Engineers, Society of Rheology, American Physical Society (APS), Adhesion Society, American Chemical Society

6. Honors and awards

UIC College of Engineering Teaching Award

7. Service activities

Chair, Membership Committee, APS – DSOF and Social Media
Chair, Membership committee, Society of Rheology (SOR)
Reviewer: Soft Matter, Journal of Fluid Mechanics, Physical Review Letters, Advanced Materials, Journal of Rheology, Rheologica Acta, Physical Review E, International Journal of Multiphase Flow, Langmuir, Applied Rheology, International Journal of Polymer Science, Journal of Material Chemistry, Macromolecules, Lab Chip, Polymer Chemistry, RSC Advances, and Experiments in Fluids
Transport Phenomena course instructor; Institute for Molecular Engineering, University of Chicago

Short course instructor; Structure and rheology of foams and emulsions; Society of Rheology meeting at Houston, TX, October 2018.

8. Selected Publications

1. Capillary Break-up and Extensional Rheology Response of Food Thickener Cellulose Gum (NaCMC) in Salt-Free and Excess Salt Solutions, L. N. Jimenez, C. Martinez Navararez, and V. Sharma, *Phys. Fluids*, 32, 012113 (2020).
2. J. Dinic and V. Sharma*, “Macromolecular Relaxation, Strain and Finite Extensibility Determine Visco-elastocapillary Thinning and Extensional Viscosity of Polymer Solutions”, *Proceedings of National Academy of Sciences*, 116(18), 8766-8774 (2019).
3. J. Dinic and V. Sharma*, “Computational analysis of capillary-driven thinning and pinch-off dynamics during dripping using the volume-of-fluids method”, *Physics of Fluids*, 31, 021211 (2019).
4. S. Yilixiati, E. Wojcik#, Y. Zhang and V. Sharma*, “Spinodal Stratification in Ultrathin Micellar Foam Films”, *Molecular Systems and Design Engineering*, 4, 626-638 (2019).
5. L. N. Jimenez, J. Dinic, N. Parsi^, and V. Sharma*, “Extensional relaxation time, pinch-off dynamics and printability of semi-dilute polyelectrolyte solutions”, *Macromolecules*, 51, 5191-5208 (2018).
6. S. Yilixiati, R. Rafeeq#, Y. Zhang, and V. Sharma*, “Influence of salt on supramolecular oscillatory structural forces and stratification in micellar freestanding films”, *ACS Nano*, 12, 1050-1061 (2018).
7. J. Dinic, L. N. Jimenez and V. Sharma*, “Pinch-off dynamics and dripping-onto-substrate (DoS) rheometry of complex fluids,” *Lab Chip*. 4, 804-808 (2017)
8. Y. Zhang, S. Yilixiati, C. Pearsall and V. Sharma*, “Nanoscopic terraces, mesas and ridges in stratifying foam films sculpted by supramolecular oscillatory surface forces”, *ACS Nano*, 10(4), 4678- 4683 (2016).
9. J. Dinic, Y. Zhang, L. N. Jimenez and V. Sharma*, “Extensional relaxation times of dilute, aqueous polymer solutions,” *ACS Macro Lett.* 4, 804-808 (2015)

9. Professional Development

Chaired /organized sessions (>30) at ACS Colloids, APS-DFD, AIChE, SOR, APS March, ACS, and Adhesion Society Meetings.

Faculty Vitae: Meenesh Singh

1. Education

PhD Chemical Engineering, Purdue University, 2013
M. Tech. Chemical Engineering, Indian Institute of Technology-Mumbai, 2008
B. E. Chemical Engineering, Sardar Patel University, India, 2005

2. Academic Experience

Department of Chemical Engineering, University of Illinois at Chicago, Assistant Professor,
2016 – present, Full-Time
University of California Berkeley, Berkeley, CA, Postdoctoral Fellow, 2015-2016, Full-Time

3. Non-Academic Experience

Lawrence Berkeley National Laboratory, Berkeley CA, Affiliate Scientist,
2016 – Present, Part-Time
Lawrence Berkeley National Laboratory, Berkeley CA, Chemist Postdoctoral Fellow,
2013 – 2015, Full-Time
Abbott Laboratories, Chicago IL, Research Intern,
05/2011 – 08/2011, Full-Time

4. Certifications or professional registrations

Not applicable

5. Current membership in professional organizations

American Institute of Chemical Engineers

6. Honors and awards

Not applicable

7. Service activities

Guest Editor of a special issue on “Modeling and Control of Crystallization” in Processes,
2019
Editor of Current Trends in Chemical Engineering and Process Technology, 2018.
Ad hoc Reviewer of > 50 articles from scientific journals such as Nature Comm., Energy
Environ. Sci., J. Phys. Chem. Lett., Chem. Eng. Sci., AIChE, J. Colloid Interf. Sci., Ind. Eng.
Chem. Res., Cryst. Growth Des. and many others.
Chair, “Modeling & Control of Crystallization”, AIChE annual meeting, 2017
Chair, “Electrocatalysis and Photoelectrocatalysis,” AIChE annual meeting, 2017
Chair, “Solid Form Selection” in AIChE annual meeting, 2016
Co-Chair, “Electrocatalysis and Photoelectrocatalysis,” AIChE annual meeting, 2016
Co-Chair, “Modeling & Control of Crystallization”, AIChE annual meeting, 2015
Co-Chair, “Fundamentals of Hydrogen Production”, AIChE annual meeting, 2015
Lead Speaker- Delivered > 40 talks in international conferences, and 7 invited talks.
Mentor- ~10 undergraduate researchers

8. Selected Publications

1. Harri Ali-Loytty, Mary W. Louie, Meenesh R. Singh, Lin Li, Hernan G. Sanchez Casalongue, Hirohito Ogasawara, Ethan J. Crumlin, Zhi Liu, Alexis T. Bell, Anders Nilsson, and Daniel Friebe, "Ambient-Pressure XPS Study of a Ni-Fe Electrocatalyst for the Oxygen Evolution Reaction," *J. Phys. Chem. C*, 120 (4), 2247-2253, 2016
 2. Anish V. Dighe, and Meenesh R. Singh, "Solvent Fluctuations in the Solvation-Shell Determines the Activation Barrier for Crystal Growth Rates," *Proc. Nat. Acad. Sci.*, 116 (48), 23954 – 23959, 2019
 3. Meenesh R. Singh, Jason D. Goodpaster, Adam Z. Weber, Martin Head-Gordon, and Alexis T. Bell, "Mechanistic Insights into Electrochemical Reduction of CO₂ over Ag using DFT and Transport Models," *Proc. Nat. Acad. Sci.*, 114 (42), E8812-E8821, 2017.
 4. Meenesh R. Singh, Youngkook Kwon, Yanwei Lum, Joel W. Ager III, and Alexis T. Bell, "Hydrolysis of Electrolyte Cations Enhances the Electrochemical Reduction of CO₂ over Ag and Cu," *J. Amer. Chem. Soc.*, 138 (39), 13006 – 13012, 2016
 5. Meenesh R. Singh, Ezra L. Clark, and Alexis T. Bell, "Thermodynamic and Achievable Efficiencies for Solar-Driven Electrochemical Reduction of Carbon Dioxide to Transportation Fuels," *Proc. Nat. Acad. Sci.*, 112 (45), E6111-E6118, 2015
- Other Significant Products
6. Meenesh R. Singh, Kimberly Papadantonakis, Chengxiang Xiang, and Nathan S. Lewis, "An Electrochemical Engineering Assessment of the Operational Conditions and Constraints for Solar-Driven Water-Splitting Systems at Near-Neutral pH," *Energy Environ. Sci.*, 8, 2760-2767, 2015
 7. Meenesh R. Singh, Ezra L. Clark, and Alexis T. Bell, "Effects of Electrolyte, Catalyst, and Membrane Composition and Operating Conditions on the Performance of Solar-Driven Electrochemical Reduction of Carbon Dioxide," *Phys. Chem. Chem. Phys.*, 17, 18924-18936, 2015.
 8. Meenesh R. Singh, Chengxiang Xiang, and Nathan S. Lewis, "Evaluation of Flow Schemes for Near-Neutral Electrolytes in Solar-Fuels Generators," *Sustainable Energy Fuels*, 1 (3), 458 - 466.
 9. Meenesh R. Singh, and Alexis T. Bell, "Design of an Artificial Photosynthetic System for Production of Alcohols in High Concentration from CO₂," *Energy Environ. Sci.*, 9, 193-199, 2016.
 10. Ezra L. Clark, Meenesh R. Singh, Youngkook Kwon, and Alexis T. Bell, "Differential Electrochemical Mass Spectrometer Cell Design for Online Quantification of the Products Produced during Electrochemical Reduction of CO₂," *Anal. Chem.*, 87 (15), 8013-8020, 2015

Faculty Vitae: Christos Takoudis

1. Education

Diploma Chemical Engineering, National Technical University of Athens, Greece, 1977
PhD Chemical Engineering, University of Minnesota, 1982

2. Academic Experience

Department of Chemical Engineering, University of Illinois at Chicago, Professor,
1996-2005, Full-Time, 2005-Present, Part-Time
Department of Bioengineering, University of Illinois at Chicago,
Professor, 2005-Present, Full-Time, Adjunct Professor, 2003 – 2005, Part-Time
Department of Restorative Dentistry, University of Illinois at Chicago, Adjunct Professor,
2014 – Present, Part-Time
School of Chemical Engineering, Purdue University,
Professor, 1995 – 1997, Full-Time
Associate Professor, 1987 – 1995, Full-Time
Assistant Professor, 1981 – 1987, Full-Time
Adjunct Professor, 1997 – 1998, Full-Time
Department of Chemical Engineering & Materials Science, University of Minnesota,
Instructor, 1/1981 – 12/1981, Full-Time
Other (Full-Time)
Universidade Estadual Paulista, Bauru, Brazil, Visiting Professor, 6/2015- 8/2015
Institute of Biomedical Research and Technology, Greece, Visiting Professor, 5/2008 -
8/2008
University of Cambridge, England, Dept. of Chem. Eng., Visiting Professor, 2/1990 - 6/1990
University of Leeds, England, School of Chemistry, Visiting Professor, 1/1990 - 2/1990

3. Non-Academic Experience

Applied Materials, Santa Clara, Epi Applications Lab,
Visiting Scientist, 1/1988 - 2/1988, Full-Time

4. Certifications or professional registrations

Not applicable

5. Current membership in professional organizations

American Institute of Chemical Engineers

6. Honors and awards

Not applicable

7. Service activities

Principal Investigator, "REU Site in Novel Materials and Processing with Applications in
Biomedical, Chemical and Electrical Engineering," NSF 2012 - 2015
“Area Selective Deposition 2016”, Invited Speaker, Leuven, Belgium 2016

8. Selected Publications

Radhe Agarwal, Yogesh Sharma, Siliang Chang, Krishna C. Pitike, Changhee Sohn, Serge M. Nakhmanson, Christos G. Takoudis, Ho Nyung Lee, Rachel Tonelli, Jonathan Gardner, James F. Scott, Ram S. Katiyar, and Seungbum Hong, "Room-temperature relaxor ferroelectricity and photovoltaic effects in SnTiO_x/Si thin film heterostructures," *Physical Review B* 97, 054109/1-7 (2018).

Siliang Chang, Christos G. Takoudis, "Growth behavior and structural analysis of atomic layer deposited Sn_xTi_{1-x}O_y films," *J. Vac. Sci. Technol. A* 36, 01A121-1/-8 (2018).

Sathees Kannan Selvaraj, Jaya Parulekar and Christos G Takoudis, "Selective atomic layer deposition of zirconia on copper patterned silicon substrates using ethanol as oxygen source as well as copper reductant", *Journal of Vacuum Science and Technology A*, 32, 010601 (2014)

Multi-metal films, alternating film multilayers, formation methods and deposition system, Takoudis, Christos G.; Singh, Manish; Selvaraj, Sathees Kannan, Patent US-1,021,4817 issued in February 2019

Sarah Hashemi Astaneh, Gregory Jursich, Cortino Sukotjo, Christos G. Takoudis, "Surface and Subsurface film growth of titanium dioxide on polydimethylsiloxane by atomic layer deposition," *Applied Surface Science* 493, 779-786 (2019)

5 Other Significant Products

Arghya K Bishal, Alvin G. Wee, Valentim A.R. Barão, Judy Chia-Chun Yuan, Richard Landers, Sukotjo Cortino, Christos G Takoudis "Color stability of maxillofacial prosthetics silicone functionalized with oxide nano-coating," *Journal of Prosthetic Dentistry* 121, 538-543 (2019).

Siliang Chang, Sathees K. Selvaraj, Yoon-Young Choi, Seungbum Hong, Serge M. Nakhmanson and Christos G. Takoudis "Atomic layer deposition of environmentally benign SnTiO_x as a potential ferroelectric material" *Journal of Vacuum Science and Techn. A* 34, 01A119.1-5 (2016)

Siliang Chang, Vijayan Sriram Aindow, Mark, Jursich, Gregory, Christos G. Takoudis, "Hydrogen annealing effects on local structures and oxidation states of atomic layer deposited SnO_x," *J. Vac. Sci. Technol. A* 36, 031519/1 – 031519/8 (2018)

Arghya K Bishal, Cortino Sukotjo, Jacob R Jokisaari, Robert F Klie, Christos G Takoudis, "Enhanced Bio-activity of Collagen Fiber Functionalized with Room Temperature Atomic Layer Deposited Titania," *ACS Applied Materials & Interfaces*, 10 (40), 34443-34454 2018 (<https://pubs.acs.org/doi/10.1021/acsami.8b05857>).

Sathees Kannan Selvaraj and Christos G Takoudis, "Scalable control program for multi-precursor flow-type atomic layer deposition system"; *Journal of Vacuum Science and Technology A*, 33, 013201 (2015)

Faculty Vitae: Lewis E. Wedgewood

1. Education

Ph.D. Chemical Engineering, University of Wisconsin-Madison, 1988,
Major Professor: R. Byron Bird,
Research Area: Macromolecular Fluid Mechanics
Thesis Title: *A Theory of Chainlike Polymers: This thesis investigated the effects of hydrodynamic interaction on flow material properties*

B.S. Chemical Engineering, Cum Laude with highest distinction, University of Illinois-Urbana, 1982

2. Academic Experience

Department of Chemical Engineering, University of Illinois-Chicago
Associate Professor, 1998-Present

Department of Chemical Engineering, University of Wisconsin-Madison,
Visiting Faculty, 1997-1998

Department of Chemical Engineering, University of Washington,
Assistant Professor, 1990-1998

Department of Chemical Engineering, Wisconsin-Madison,
Research Associate, 1989-1990

NATO/NSF Postdoctoral Fellow, Albert-Ludwigs University, Freiburg, West Germany,
1988-1989

3. Non-Academic Experience

None

4. Certifications or professional registrations

None

5. Current membership in professional organizations

American Institute of Chemical Engineers 1982-present
The Society of Rheology 1982-present

6. Honors and awards

Award for Excellence in Teaching / University of Illinois at Chicago, 2010-2011
2009 Faculty Teaching Award, UIC College of Engineering
Teaching Recognition Program Award 2008-2007
2008 Faculty Advising Award, UIC College of Engineering
2007 Faculty Teaching Award, UIC College of Engineering
2007 Faculty Advising Award, UIC College of Engineering
Harold Simon Teaching Award, 2004
Teaching Recognition Program Award 2000-2001
Outstanding Instructor Award / University of Wisconsin, 1998
Excellence Award - Exceptional Teacher/ University of Washington, 1990-1998
3M Company Non-Tenured Faculty Award, 1997
3M Company Non-Tenured Faculty Award, 1996
NATO/NSF Postdoctoral Fellowship, 1987

7. Service activities

Member of Advisory Committee, UIC Department of Chemical Engineering, 2003-present

Member of Awards Committee, UIC College of Chemical Engineering, 2008-2010, 2011-present

Co-Chair of Faculty Search, UIC Department of Chemical Engineering, 2018-2020

8. Selected Publications Last 5 Years

1. Dubina, Sean Hyun; Wedgewood, Lewis Edward "Application of nonuniform magnetic fields in a Brownian dynamics model of ferrofluids with an iterative constraint scheme to fulfill Maxwell's equations" PHYSICS OF FLUIDS Volume: 29 Issue: 9 Article Number: 092001 Published: SEP 2017
2. Dubina, Sean Hyun; Wedgewood, Lewis Edward "A Brownian dynamics study on ferrofluid colloidal dispersions using an iterative constraint method to satisfy Maxwell's equations" PHYSICS OF FLUIDS Volume: 28 Issue: 7 Article Number: 072001 Published: JUL 2016

Faculty Vitae: Alan D. Zdunek

1. Education

B.A. Chemistry, Knox College, 1982

M.S. Chemical Engineering, Illinois Institute of Technology, 1986

Ph.D. Chemical Engineering, Illinois Institute of Technology, 1990

2. Academic Experience

Department of Chemical Engineering, University of Illinois-Chicago
Clinical Assistant Professor, Director of Undergraduate Studies,
Director, Freshmen Engineering Success Program, 2014-present

Department of Chemical Engineering, University of Illinois-Chicago,
Adjunct Professor, 2010-2014

Department of Chemical Engineering, Illinois Institute of Technology
Research Associate Professor/Adjunct Faculty, 2007-2009

3. Non-Academic Experience

S&C Electric, Chicago, IL, Consultant. 5/14-9/14

Advanced Diamond Technologies, Romeoville IL, Consultant. 5/12–9/12

HOH Water Technology, Palatine IL, Consultant. 6/08–1/09

Argonne National Laboratory, CSE Division, Lemont IL, Consultant. 7/08–11/09

American Air Liquide, Countryside, IL, Group Expert. 3/03–8/06

American Air Liquide, Countryside IL, Senior Scientist. 3/99–2/03

American Air Liquide, Countryside IL, Scientist, 3/95–2/99

BIRL Industrial Research Laboratory, Evanston IL, Scientist, 11/90–11/94

4. Certifications or professional registrations

Project Management Certificate, Loyola University, 12/09.

Hazards Material Technician, 11/97

5. Current membership in professional organizations

American Institute of Chemical Engineers, The Electrochemical Society,
American Chemical Society

6. Honors and awards

College of Engineering 2018 Teaching Award

College of Engineering 2017 Teaching Award

College of Engineering 2016 Teaching Award

Keynote Speaker, Conference on International Exchange of Talents & Projects Discussion of
Yellow River Delta High-Efficiency Economic Zone, Dongying City, China, November
2011.

7. Service activities

College of Engineering Education Policy Committee: 2014-present

Freshmen Engineering Success Program Committee, Director, 2016-present

Faculty Advisor, AIChE Student Chapter, UIC. 9/11- 8/12, 9/14-present.
 Chair, Electrochemical Society Chicago Section, April 2015-August 2017
 Von Steuben HS student research project for CPS City Science Fair, 2016, 2018
 Editor, Journal of Bio- and Tribo-Corrosion, 2015-2018
 AIChE Midwest Regional Conference planning committee, 2016-2017
 UIC Electrochemistry Collaborative Group (2015-2017)
 Career Services Advisory Council, 2017-2019
 Reviewer, DPI Seed Funding Program, Summer 2019
 Chemical Engineering Advisory Committee, August 2016-present
 Chemical Engineering Safety Committee, 2015-2017
 Chemical Engineering Undergraduate Committee, 2014-present
 Chemical Engineering Public Relations Committee, 2015-present
 Chemical Engineering Classroom Upgrade Ad-Hoc Committee, 2015-2016
 Chemical Engineering Club Development, 2018-2018
 Initiated and Organized Chemical Engineering Industry Day, 2015-present
 ChemE Forum and Chemical Engineering Alumni Mixer, 2016-present
 Presenter, High School District 214 Career Night, Engineering, Industrial Technology,
 Business & Computer Career, Wheeling High School, November, 2014, 2016
 Booth organizer and attendee at ProFood Expo, Chicago, IL March 2019
 Chemical Engineering/Engineering Career Center Ad-Hoc Committee, 2014-2015
 College recruitment activities; (2014-2019)
 College of DuPage Explore Engineering Night, 2014-2019
 Women In Engineering Summer Program Lunch Presentation, Summer 2016
 HS Senior Project Mentor, North Lawndale College Prep High School, 2015-2016

8. Selected Publications

M. H. Mustafa, A. Zdunek, "Supercapacitor Nanofiber Electrodes Graphene-Based", Int. J. Electrochem. Sci, 14 (2017).
 X. Cui, A. Zdunek, Gregory Jursich, and C. Takoudis, "Nanostructured Ni-YSZ Anode by Atomic Layer Deposition", ECS Journal of Solid State Science and Technology, 4(12), P1-P7, 2015.
 R. Sparks, G. Jursich, A. Zdunek, Jorge Ivan Rossero, C. Takoudis, "Tunability and Electrical Properties of Yttria-doped-ceria Films Fabricated via Atomic Layer Deposition", The Journal of Undergraduate Research at the University of Illinois at Chicago, Vol.7 No. 1, (2014).

9. Professional Development

2017 ABET Symposium, Baltimore, MD April 2017
 Faculty Summer Institute, University of Illinois Urbana Champaign, Urbana, IL, May 2015

Appendix C – Equipment

Equipment List for Chemical Engineering Laboratory (ChE 381/382) Experimental Demonstration Apparatus:

1. Distillation
 - a. Continuous Distillation column: Armfield UOP3CC (10L, 8 tray)
 - b. Control Module with computer interface/data acquisition
 - c. ATAGO Digital Hand-held pocket Refractometer: PAL-RI, Analog Refractometer; Bausch & Lomb Model ABBE-3L
2. Liquid-Liquid Extraction
 - a. Liquid-liquid extractor; Denver Corp. Model SEM-SU
3. Fluid Flow in Pipe Systems
 - a. Hampden Model H-6920 Pipe Friction Demonstrator
4. Ultrafiltration
 - a. Ultrafiltration column and apparatus; UIC Designed/Built w/ Koch Membrane Systems PM50 filter column
5. Membrane Separation
 - a. Membrane Air Separation Apparatus; Perfected Experiments, Inc. Model ChE06
6. Heat Transfer
 - a. Constant Stirred Tank Reactor (CSTR); UIC Designed/Built
7. Viscosity of Liquids
 - a. Capillary Tube Viscometer; UIC Designed/Built
8. Fluidized Bed
 - a. Two-Column fluidized bed apparatus; UIC Designed/Built
9. Humidification
 - a. Falling Liquid Film humidification apparatus; UIC Designed/Built
10. Evaporation
 - a. Single-stage glass evaporator, vacuum pressure; UIC Designed/Built
11. Gas Absorption
 - a. Armfield Model UOP7 Gas Absorption Column
12. Process Control
 - a. Armfield PCT50-Level Control
 - b. Armfield PCT53-Pressure Control
 - c. Armfield PCT54- Industrial PID Controller
 - d. Armfield PCT55- Programmable Logic Controller

General Lab & Support Equipment

1. Laboratory space
 - a. Dedicated workspaces for chemical process demonstrators with utility hook-up

- b. 9 moveable lab benches with lab cabinet storage underneath, 5 with wood countertops, 4 with chemical-resistant countertops, overhead storage areas and electric power outlets
 - c. 2 wall-service chemical resistant countertop lab benches with overhead storage and cabinets
- 2. Classroom space
 - a. 4 conference tables/chairs for lab team collaboration and lab analysis
 - b. WiFi connectivity, ethernet connectivity
- 3. Chemical/gas storage
 - a. Two (2) flammable liquid cabinets (under fume hoods)
 - b. Acids/base storage (under fume hoods)
 - c. Compressed gas cylinder storage
 - d. 55-gallon drum storage for white mineral oil
 - e. 4, 10-gallon vessels for white mineral oil
- 4. Laboratory Utilities:
 - a. Two horizontal sash fume hoods with automatic face velocity control
 - b. Four exhaust snorkels
 - c. Two laboratory sinks (hot/cold water, purified water)
 - d. Steam boiler; Lattner Model 48 BF electric boiler, 100 psi max
 - e. Compressed air; house, several taps throughout lab, max pressure 120 psi
 - f. Industrial chilled water; various taps throughout lab
 - g. Industrial cooling water inlet/return (two locations)
 - h. Purified (deionized) water; three taps in lab, two in sink, one overhead
 - i. Hot purified water (Tube-in-tube steam heat exchanger)
 - j. 5 floor drains
 - k. Electrical; 120V and 220V, individual circuit breakers on specific equipment
 - l. Vacuum line; in fume hoods
- 5. Miscellaneous:
 - a. Three Digital balances, Ohaus Scout (300g), Valor 3000 Xtreme (100 g), Mettler PJ (12kg)
 - b. One mechanical scale; Fisher Scientific, 5 kg
 - c. Fisher Scientific Isotemp drying oven
 - d. pH meters (2)
 - e. Magnetic stirrers/burettes and burette holders
 - f. Flasks, beakers, buckets, pipettes, rulers, stopwatches
 - g. Digital tachometer, digital refractometer, digital hygrometer, digital barometer, analog refractometer, analog hygrometers, electric mixer (for viscosity experiment), Heating/cooling temperature bath, gas-diffusion apparatus
- 6. Safety equipment:
 - a. Two (2) Eyewash stations
 - b. One (1) Safety Shower
 - c. Safety gloves; heat-resistant, chemical-resistant, sharp-resistant, general nitrile gloves.

- d. Safety glasses; 30+ count
- e. 10 industrial hard hats
- f. Two (2) First aid kits
- g. Two (2) Chemical spill kits, oil-spill pads, drip-trays
- h. MSDS Binder, Laboratory Safety Plan

Appendix D – Institutional Summary

The Institution

a. University of Illinois at Chicago

2800 University Hall (m/c 102), 601 S. Morgan Street, Chicago, IL 60607-7128

b. *Chief Executive Officer:*

Michael D. Amiridis, Chancellor

University of Illinois at Chicago

2800 University Hall (m/c 102), Chicago, IL 60607-7128

c. Michael McNallan, Interim Associate Dean for Undergraduate Affairs

University of Illinois at Chicago

851 S. Morgan St. (m/c 154), Chicago, IL 60607-7128

d. The University of Illinois at Chicago is accredited by the Commission on Institutions of Higher Education of the North Central Association of Colleges and Schools (NCA), 30 N. LaSalle St., Suite 2400, Chicago, Illinois 60602-2504, (312) 263-0456. The NCA is recognized by the Commission on Recognition of Postsecondary Accreditation. The initial accreditation of UIC was in 1970. In 2007, the North Central Association of Colleges and Schools granted continued accreditation of the University of Illinois at Chicago for the maximum period of 10 years. The next comprehensive evaluation of UIC is scheduled for 2016-17. Verification of accreditation status is available in the Office of the Chancellor (M/C 102), University of Illinois at Chicago, 601 South Morgan Street, Chicago, Illinois 60607-7128; (312) 413-3350. The undergraduate academic degree programs at UIC have been approved by the Illinois Board of Higher Education, 431 East Adams, Second Floor, Springfield, Illinois 62701-1418, (217) 782-2551.

History of Institution

In 1946, an undergraduate division of the University of Illinois was established at Navy Pier. This facility, renamed the University of Illinois at Chicago Circle, moved to its present location in 1965, when it opened its doors as a four-year university. By 1982, it had grown to include eight academic colleges offering degree programs at both the undergraduate and graduate levels.

The University of Illinois at Chicago was formed by the consolidation, in the fall of 1982, of the two Chicago campuses (formerly known as the University of Illinois at the Medical Center and the University of Illinois at Chicago Circle) into a single institution of higher learning. At that time, the University set as a goal to become a quality alternative to the Big Ten campuses. The University's facilities for medical instruction date back to 1894, when the Chicago College of Pharmacy became the School of Pharmacy of the University of Illinois. In 1897, the independent College of Physicians and Surgeons of Chicago became the "Department of

Medicine" of the University; in 1901, the Columbian Dental College became the University School of Dentistry; and in 1925 the University Hospital opened. Programs in nursing education under University auspices began in the 1940s, becoming the School of Nursing in 1951 and, in 1959, the College of Nursing. Other health sciences units of the University of Illinois at Chicago include the College of Applied Health Sciences, the School of Public Health, and over 50 clinics and research facilities. A new \$60 million University of Illinois Hospital was completed in 1981.

Today the University of Illinois at Chicago has a total enrollment of approximately 33,000 students, including almost 11,000 graduate and professional students. Academic support services include eight libraries, extensive computer facilities with a 10,000-user network, and an instructional resources development office. The campus has a number of centers and institutes whose research activities complement classroom teaching. Other support services include tutoring programs; guidance in the improvement of reading, mathematics, and study skills; a writing center; academic and personal counseling; special instruction in English for international students; and financial aid.

The College of Engineering was one of four colleges at Navy Pier along with the Colleges of Commerce and Business Administration, The College of Liberal Arts and Science, and the College of Physical Education. The College of Engineering at Navy Pier was quite small, with about 400 students attending in an average year.

In 1965, when the University of Illinois at Chicago Circle was established, the College of Engineering reorganized into "functional" departments of Materials Engineering, Energy Engineering, Information Engineering, and Systems Engineering. Degree programs in the College were identified by functions that engineers perform, such as Mechanical Analysis and Design, Systems Analysis, or Communications Engineering. The College grew rapidly at Chicago Circle, reaching nearly 2,000 undergraduate students by 1968. In 1973, a number of the College's programs received their first accreditation from the Engineers Council for Professional Development (now ABET).

Although the non-traditional engineering degree programs administered by the College in the 1970's were accredited, they were not well understood by the public, and it was clear that our industrial constituents preferred to recruit students from traditional degree programs, so that the backgrounds and qualifications of their employees could be more easily predicted. In 1982, the University made the decision to reorganize the College into traditional departments and traditional degree programs, essentially as they are offered now.

Student Body

The nearly 33,000 students who study at the University of Illinois at Chicago come from the city of Chicago and its suburbs, and from all 50 states, three United States territories, and 100 foreign countries. The student body is rich in its diversity, its youth and maturity, and its cultural heritage. Of the 21,641 undergraduate students, 50.3 percent are female and 49.7

percent are male. Minority students comprise 32.7 percent of the undergraduate enrollment. Many full-time students also hold part-time jobs, both on and off campus. In addition, a large number find time to participate in one or more of approximately 233 campus student organizations. With a growing residential population, UIC houses students on the east, west, and south sides of campus. Approximately 3,800 students live on campus, including more than half of first-year undergraduate students.

University of Illinois enrollment figures for the past 10 years are shown below. The overall campus student population has been increasing from 27,000 to 33,000 students over the past 10 years.

Table D.1 Total Enrollment Fall 2010-2019

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
Undergraduate	16,790	16,911	16,671	16,660	16,718	17,575	17,959	19,448	20,783	21,641
Graduate	7,925	8,012	8,119	8,186	8,307	8,466	8,060	7,915	7,673	7,572
Professional	2,594	2,657	2,722	2,743	2,944	3,007	3,101	3,176	3,227	4,177
Total	27,309	27,580	27,512	27,589	27,969	29,048	29,120	30,539	31,683	33,390

Ethnic diversity among the undergraduate student population are shown below. UIC continues to have one of the most ethnically diverse student bodies of any American University. Apart for some small fluctuations from year to year, the overall numbers have been fairly stable over the past 10 years.

Table D.2 Ethnic diversity among the undergraduate student population

Race/Ethnicity	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
Am. Indian	24	22	21	24	18	15	12	11	8	9
	0.10%	0.10%	0.10%	0.10%	0.11%	0.09%	0.07%	0.06%	0.04%	0.04%
Afr Amer	1392	1,361	1,339	1,303	1,327	1,429	1,480	1,558	1,630	1,637
	8.30%	8.00%	8.00%	7.80%	7.94%	8.13%	8.24%	8.01%	7.84%	7.56%
Asian	3,576	3,605	3,656	3,737	3,815	3,904	3,967	4,164	4,339	4,444
	21.30%	21.30%	21.90%	22.40%	22.82%	22.21%	22.09%	21.41%	20.88%	20.54%
Hispanic	3,386	3,711	3,950	4,120	4,433	4,977	5,527	6,349	6,977	7,297
	20.20%	21.90%	23.70%	24.70%	26.52%	28.32%	30.72%	32.65%	33.57%	33.72%
Caucasian	7,451	7,173	6,664	6,304	6,061	5,991	5,801	5,791	5,822	5,671
	44.40%	42.40%	40.00%	37.80%	36.25%	34.09%	32.30%	29.78%	28.01%	26.20%
Foreign	241	236	265	350	285	448	473	840	1,270	1,576
	1.40%	1.40%	1.60%	2.10%	1.70%	2.55%	2.63%	4.32%	6.11%	7.28%
Nat HI/Pac Isl	76	86	7	69	45	35	16	16	13	8
	0.50%	0.50%	0.50%	0.40%	0.27%	0.20%	0.09%	0.08%	0.06%	0.04%
Two or More	344	371	407	410	419	471	551	575	627	637
	2.00%	2.20%	2.40%	2.50%	2.51%	2.68%	3.07%	2.96%	3.02%	2.94%
Unknown	300	346	290	343	315	305	142	144	97	362
	1.80%	2.00%	1.70%	2.10%	1.88%	1.74%	0.79%	0.74%	0.47%	1.67%
UIC Total	16,790	16,911	16,671	16,660	16,718	17,575	17,959	19,448	20,783	21,641

Beginning Fall 2010, new ethnicity and race categories were created in response to federal reporting requirements. The implementation of these new categories included a survey of all current students to verify ethnicity and race. Changes include the ability of respondents to indicate Hispanic ethnicity separately from racial origin and to identify more than one race category if applicable. The federal guidelines also require separating the category 'Asian or Pacific Islander' into two groups, one for 'Asian' and one for 'Native Hawaiian or Other Pacific Islander'. Students who are in the U.S. on a visa or temporary basis are defined as International.

UIC's undergraduate student body is largely drawn from local sources as shown in the following table showing the geographical origin of entering freshmen. Chicago and Cook County account for a majority of UIC's entering students, but have declined substantially as a percentage of entering students over the past 10 years, as UIC has become increasingly important as a statewide educational resource. The number of international students and non-resident U.S. students in the UIC undergraduate student body remains small.

Table D.3 The Geographical Location of entering freshmen

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
Chicago	932	991	1,025	1,070	1,094	1,326	1,356	1,703	1,744	1,793
	29.10%	31.80%	32.80%	34.50%	36.11%	38.05%	41.00%	42.05%	41.93%	40.69%
Cook County	1,993	1,987	2,029	2,038	921	1,010	898	967	980	1,124
	62.20%	63.80%	65.00%	65.70%	30.40%	28.98%	27.15%	23.88%	23.56%	25.50%
Illinois	3,097	2,980	2,992	2,928	NA	NA	NA	NA	NA	NA
	96.70%	95.70%	95.80%	94.30%	NA	NA	NA	NA	NA	NA
Out of State	57	93	71	110	127	148	160	198	200	184
	1.80%	3.00%	2.30%	3.50%	4.19%	4.25%	4.84%	4.89%	4.81%	4.18%
Foreign	50	42	60	66	67	104	91	204	273	257
	1.60%	1.30%	1.90%	2.10%	2.21%	2.98%	2.75%	5.04%	6.56%	5.83%

Admission to the University of Illinois at Chicago is competitive, with the average enrolled freshman student being in the top 25% of his or her high school class, as shown in the following table for freshmen enrolled over the past 10 years. While the high school rank has been steady over this period, the average ACT score of entering freshmen has gradually increased by 1 point over this period, rising from 23.1 to 24.1.

Table D.4 The Mean ACT and High School Percentile Rank for New Freshmen for Fall 2010-2019

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
ACT	23.8	23.7	24	24.1	27.4	27.4	27.4	26.3	27.2	27
HSPR	75.9	76.6	75	76.8	80.6	NA	NA	NA	NA	NA

As shown below in Table D.5, College of Engineering enrollments have fluctuated more substantially over the ten-year period (2010-2019) than the overall university enrollment. Historically, from 2000 to 2005, the College of Engineering experienced a gradual decrease in undergraduate enrollment, as the College focused on student quality and graduate programs.

Since 2005, the College implemented a recruiting drive aimed at increasing the undergraduate enrollments while maintaining the high student quality that had been obtained in the smaller classes.

Table D.5 College of Engineering Enrollment

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
COE Undergrad	2,098	2,158	2,311	2,524	2,724	2,955	3,094	3,434	3,714	4,003

Figure D-1 shows the College of Engineering enrollment since 2015 leading to undergraduate enrollment of approximately 4003 for Fall 2019. Figure D-2 shows the distribution of the enrollments among the College's programs during that period. The Bioengineering, Computer Science, and Mechanical Engineering programs have grown significantly during this time, while other programs have had stable or decreasing enrollment.

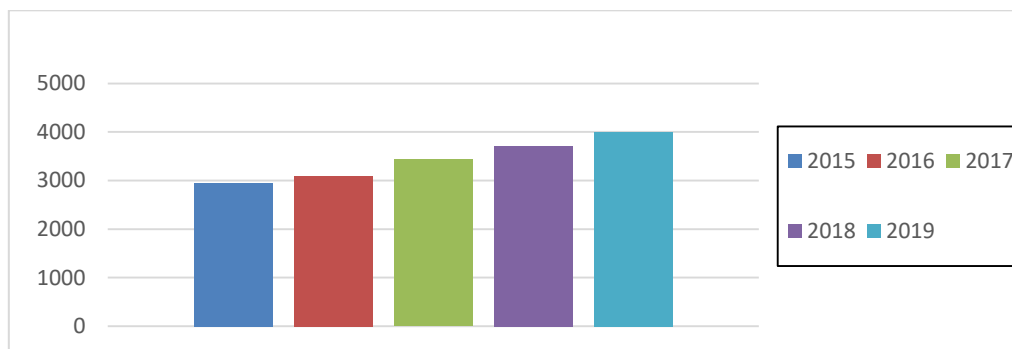


Figure D.1 College of Engineering Undergraduate Enrollment in Fall semester from 2015 to 2019

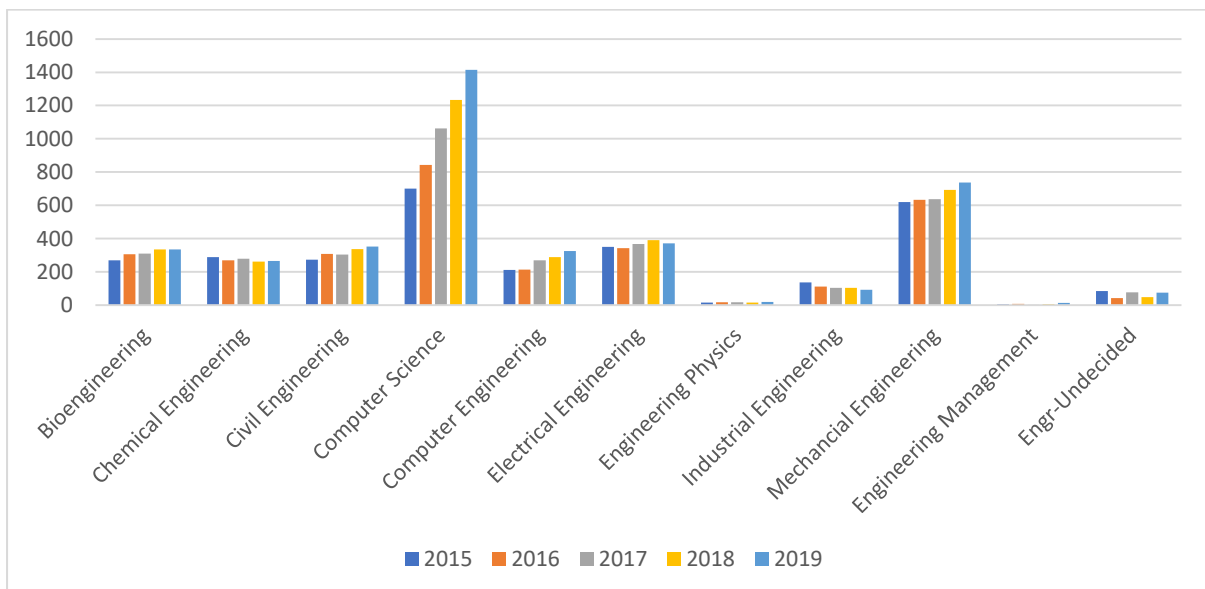


Figure D.2 College of Engineering Undergraduate Enrollment shown by program from 2015 to 2019

COE Freshmen Demographics

As shown below in Tables D.6 and D.7, for entering freshmen students in the 2010 to 2019 period, the ethnic and gender diversity of the undergraduate engineering student body is generally consistent with the overall University statistics. There was a small increase in the percentage of Caucasian students in the College as the initial recruiting drive led to the enrollment of a larger number of majority students from suburban high schools. However, since 2016 this trend has reversed and the percentage of Caucasian students is now fairly constant. Subsequently, there has been a notable increase in Hispanic and Asian students since 2016 due to recent recruiting efforts. African American students have also seen an increase during the past few years as well.

Table D.6 Ethnicity of Freshmen Engineering Students

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
Nat. Amer	1	0	0	2	0	0	0	0	0	2
	0	0	0.50%	1.50%	0.00%	0.00%	0.00%	0.00%	0.00%	0.30%
Afr. Amer	78	100	103	91	23	17	20	26	21	25
	5.00%	2.50%	3.10%	4.30%	6.76%	4.25%	5.10%	4.31%	4.24%	3.71%
Asian	79	78	100	103	105	125	93	175	139	185
	23.20%	23.90%	30.60%	27.80%	30.88%	31.25%	23.72%	29.02%	28.08%	27.49%
Hispanic	68	64	67	79	74	90	106	158	101	169
	19.90%	19.60%	20.50%	21.30%	21.76%	22.50%	27.04%	26.20%	20.40%	25.11%
Caucasian	157	152	125	140	105	126	141	132	107	161
	46.00%	46.6	38.20%	37.70%	30.88%	31.50%	35.97%	21.89%	21.62%	23.92%
Foreign	5	9	8	12	19	22	19	98	109	108
	1.50%	2.80%	2.40%	3.20%	5.59%	5.50%	4.85%	16.25%	22.02%	16.05%
Nat. HI/Pac Isl	2	2	2	2	0	1	1	0	0	0
	0.60%	0.60%	0.60%	0.50%	0.00%	0.25%	0.26%	0.00%	0.00%	0.00%
Two or More	11	8	11	14	12	16	12	14	16	15
	3.20%	2.50%	3.40%	3.80%	3.53%	4.00%	3.06%	2.32%	3.23%	2.23%
Unknown	1	5	4	3	2	3	0	0	2	8
	0.30%	1.50%	1.20%	0.80%	0.59%	0.75%	0.00%	0.00%	0.40%	1.19%
COE Total	341	326	327	371	340	400	392	603	495	673

Table Note: Beginning Fall 2010, new ethnicity and race categories were created in response to federal reporting requirements. The implementation of these new categories included a survey of all current students to verify ethnicity and race. Changes include the ability of respondents to indicate Hispanic ethnicity separately from racial origin and to identify more than one race category if applicable. The federal guidelines also require separating the category 'Asian or Pacific Islander' into two groups, one for 'Asian' and one for 'Native Hawaiian or Other Pacific Islander'. Students who are in the U.S. on a visa or temporary basis are defined as International.

Historically, the percentage of women in the undergraduate student body has been in the range of 15% since the 1990's, and fell slightly during the early recruiting drive. However, changes in the recruiting program to make the College more attractive to women and minority students

have been implemented and there has been a significant reverse in these trends. As shown in Table D.7, female student enrollment has steadily increased for the past six years, making up 25-27% of the undergraduate engineering student body. Several engineering departments, such as Chemical Engineering, have high percentages of total female students (30-40%) as shown in Table D.8.

Table D.7 COE Freshmen Gender Distribution

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
Female	42	34	53	61	70	87	108	148	134	177
	12%	10%	16%	16%	21%	22%	28%	25%	27%	26%
Male	299	292	274	310	270	313	284	455	361	496
	88%	90%	84%	84%	79%	78%	72%	75%	73%	74%
COE (Total)	341	326	327	371	340	400	392	603	495	673

Table D.8 Chemical Engineering Gender Distribution: All Students

	2013	2014	2015	2016	2017	2018	2019
Female	79	82	108	103	115	104	102
(%)	32%	31%	37%	38%	41%	40%	38%
Male	165	181	181	167	164	158	163
(%)	68%	69%	63%	62%	59%	60%	62%
CHE Total	244	263	289	270	279	262	265

Student quality, as measured by ACT and high school class rank, has increased every year since 2010, as shown in Table D.8.

Table D.8 ACT / HSPR Data for Entering Freshmen

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
ACT	26.2	26.5	27.1	27.3	27.4	27.4	27.4	26.3	27.2	27
HSPR	75.2	75.5	78.1	79.7	80.6	NA	NA	NA	NA	NA

2. Type of Control

The State of Illinois through a Board of Trustees has managerial control for the University of Illinois System. The University of Illinois at Chicago (UIC) is one of three campuses. The other two campuses are the University of Illinois at Urbana-Champaign and the University of Illinois at Springfield. Each campus has a Chancellor, and there is one President and one Board of Trustees for the three-campus system.

3. Educational Unit

The organizational structure of the College of Engineering at the University of Illinois at Chicago is shown schematically in Figure 1-2. The Directors of Undergraduate Study responsible for the B.S. programs presented in this document, report to the Department Heads

of their respective departments, who report, in turn, to the Dean of the College. The Dean is supported in his activity by Associate Deans with responsibilities for Research and Graduate Studies, Corporate Relations (also supervises the Engineering Career Center and Co-op Program), On-line Master of Engineering Program and International Programs (all graduate level programs), Administration (primarily budget and financial issues), and Advancement.

The authority lines for the Associate Dean for Undergraduate Affairs are shown in more detail in Figure 1-3, because these personnel address the programs reviewed in this report. The Associate Dean for Undergraduate Affairs supervises the Director of the Minority Recruiting and Retention Program (MERRP). The Assistant Dean is supported by Assistant Directors for both the recruiting program and the MERRP program. The recruiting activities are carried out throughout the Chicago region to continuously enhance the number and quality of students entering the College's programs. The MERRP program (described in more detail elsewhere in this report) is in place to recruit and support minority enrollments in the College of Engineering.

The Director of Engineering Admission and Records also reports to the Associate Dean for Undergraduate Affairs. The Director of Admission and Records reviews the records of student applicants to the College of Engineering and makes admission decisions. His office also maintains the records of continuing students and reviews them for compliance with College policies and graduation standards. In this, he is supported by three academic counselors who meet with individual students to advise them on policies and requirements, and also conduct orientation and advising sessions with new freshmen and transfer students. He is also supported by an Office Manager, who is usually the first responder for students with problems. Together with the Associate Dean for Undergraduate Affairs, he sets policies for special petitions and enforces probation and academic drop rules.

Organizational Structure

The authority lines for the Associate Dean for Undergraduate Affairs are shown in more detail in the figures below, because these personnel address the programs reviewed in this report. The Associate Dean for Undergraduate Affairs supervises the Assistant Dean for Undergraduate Recruiting, who is also the Director of the Minority Recruiting and Retention Program (MERRP). The Assistant Dean is supported by Assistant Directors for both the recruiting program and the MERRP program. The recruiting activities are carried out throughout the Chicago region to continuously enhance the number and quality of students entering the College's programs. The MERRP program (described in more detail elsewhere in this report) is in place to recruit and support minority enrollments in the College of Engineering. The following flowchart (Figure D.3) shows the organizational relationships among the offices of the College of Engineering.

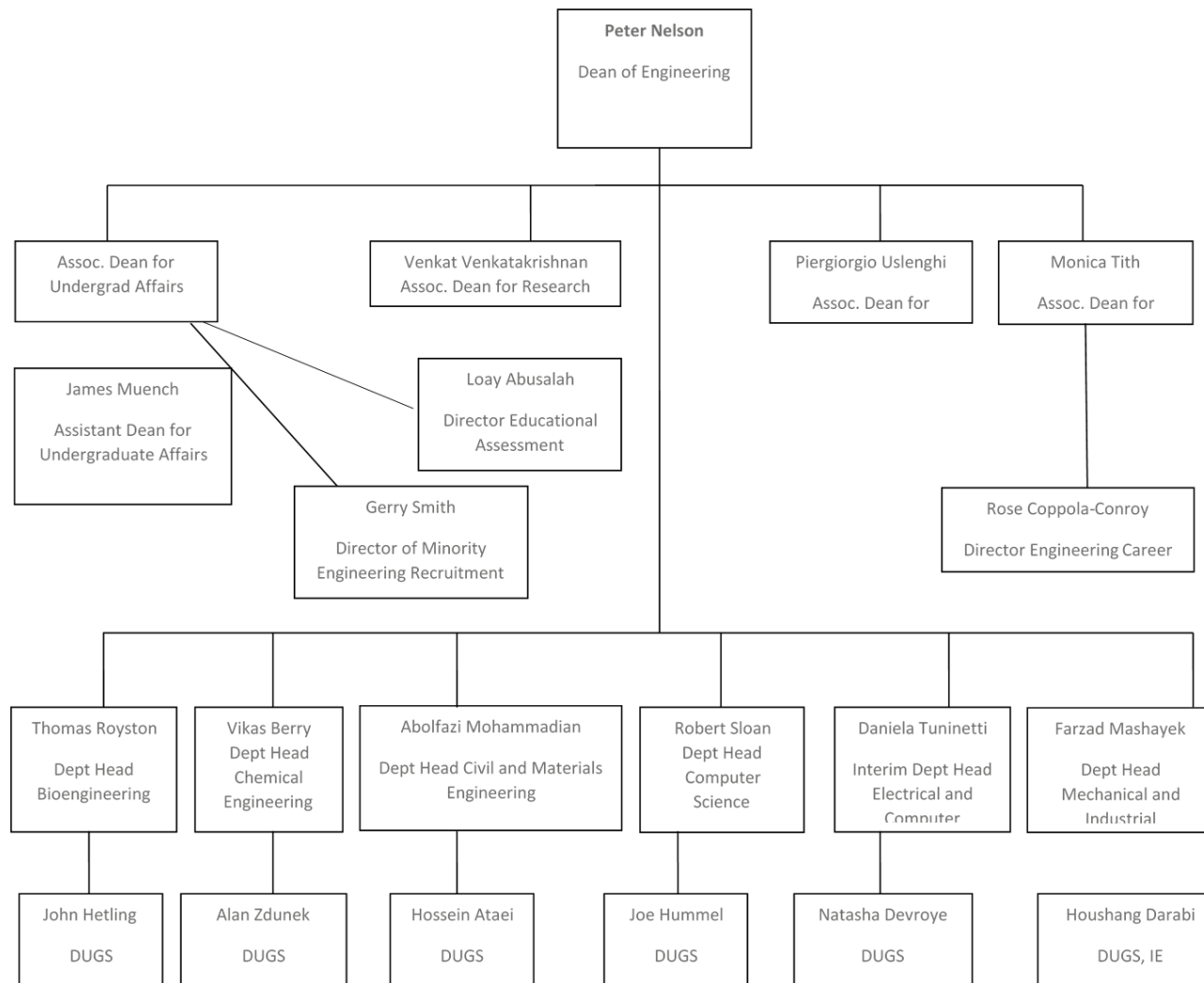
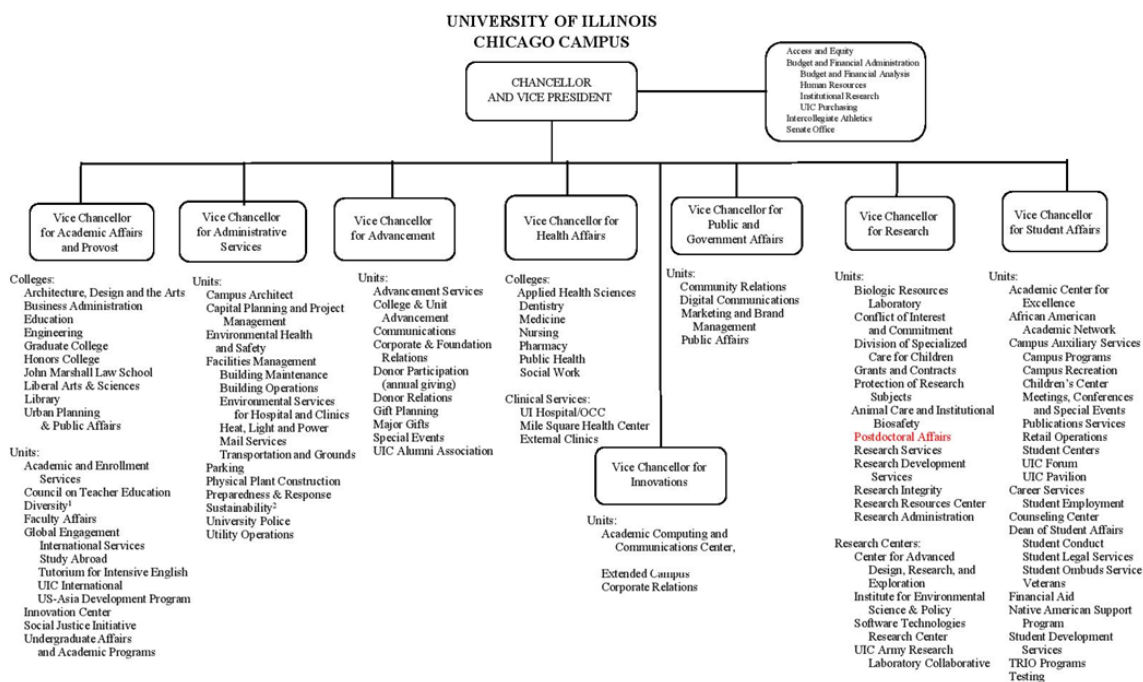


Figure D3. The flowchart shows the organizational relationships among the offices of the College of Engineering.

The Director of Engineering Admission and Records also reports to the Associate Dean for Undergraduate Affairs. The Director of Admission and Records reviews the records of student applicants to the College of Engineering and makes admission decisions. His office also maintains the records of continuing students and reviews them for compliance with College policies and graduation standards. In this, he is supported by three academic counselors who meet with individual students to advise them on policies and requirements, and also conduct orientation and advising sessions with new freshmen and transfer students. He is also supported by an Office Manager, who is usually the first responder for students with problems. Together with the Associate Dean for Undergraduate Affairs, he sets policies for special petitions and enforces probation and academic drop rules. The list of college of engineering associate deans and directors could also be found on our website <http://engineering.uic.edu/bin/view/COE/Staff>. The organizational structure for the University of Illinois at Chicago is shown in Figure D.4.



¹ Joint report to the Chancellor/Vice President and to the Provost/Vice Chancellor for Academic Affairs

² Joint report to the Chancellor/Vice President and to the Vice Chancellor for Administrative Services.

Note: Organizational chart updated October 2019.

Figure D4. The organizational structure for the University of Illinois at Chicago.

1. Academic Support Units

Required courses for College of Engineering programs are taught by faculty from several departments in the College of Liberal Arts and Sciences. These include:

Department of Mathematics, Statistics, and Computer Science

Department of Mathematics, Statistics, and Computer Science which teaches required mathematics courses. The MSCS Department is headquartered at:

Department of Mathematics, Statistics, and Computer Science
University of Illinois at Chicago
322 Science and Engineering Offices (M/C 249)
851 S. Morgan Street
Chicago, IL 60607-7045

Department Head: Brooke Shipley, 312 SEO; (312) 996-2171; shipleyb@uic.edu
The MSCS Department houses 85 faculty and 89 teaching assistants, who are responsible for offering mathematics courses taken by students in the College of Engineering and in other programs which rely on mathematics as a component of their knowledge base. Additional information is available at: <https://mscs.uic.edu>

Department of Chemistry

The Department of Chemistry, which teaches required chemistry courses. The Chemistry Department is headquartered at:

Department of Chemistry MC 111
4500 SES (Science and Engineering South)
University of Illinois at Chicago
845 West Taylor Street
Chicago, IL 60607

Department Head: Wonhwa Cho, 4508 SES; (312) 996-4883; wcho@uic.edu

The Department of Chemistry houses 25 faculty and a number of graduate teaching assistants who are responsible for offering chemistry courses taken by students in the College of Engineering and in other programs which rely on chemistry as a component of their knowledge base. Additional information is available at: <http://www.chem.uic.edu>

Department of Physics

The Department of Physics, which teaches the required physics courses. The Department of Physics is headquartered at:

Department of Physics
2244 SES (Science and Engineering South)
University of Illinois at Chicago
845 W. Taylor St. M/C 273
Chicago IL 60607-7059

Department Head: David J. Hofman, 2244 SES; (312)-413-2798; hofman@uic.edu

The Department of Physics houses 21 faculty and 52 graduate teaching assistants who are responsible for offering physics courses taken by students in the College of Engineering and in other programs which rely on physics as a component of their knowledge base. Additional information is available at: <https://phys.uic.edu>

Department of English

The Department of English, which teaches the required English composition courses. The Department of English is headquartered at:

Department of English
College of Liberal Arts and Sciences
2027 University Hall
601 South Morgan Street
Chicago, Illinois 60607

Department Head: Lisa A. Freeman, 2033 University Hall; (312) 413-2204; lfreeman@uic.edu

The Department of English houses 49 faculty and 52 lecturers, who are responsible for offering the English composition courses taken by students in the College of Engineering and by other undergraduate students at the University of Illinois at Chicago. Additional Information is available at: <https://engl.uic.edu>

In addition, several other academic departments, including the Department of Biological Sciences, <https://bios.uic.edu> and the Department of Earth and Environmental Sciences, <https://eaes.uic.edu> offer courses which are used in specific College of Engineering

programs. Courses that satisfy the Humanities and Social Sciences requirements for College of Engineering programs are selected from a list of courses that satisfy the General Education requirements of the University of Illinois at Chicago, <http://www.uic.edu/ucatalog/GE.shtml> and are offered by a variety of departments within the University.

5. Non-Academic Supporting Units

Library

The University of Illinois at Chicago is supported by an extensive library system with sites at several locations around the campus. The Library is administered by:

University Librarian: Mary Case E-mail: marycase@uic.edu
Richard J. Daley Library
801 S. Morgan, M/C 234
Chicago, IL 60607

Reference materials relevant to College of engineering courses are most commonly found at either:

The RICHARD J. DALEY LIBRARY, 801 S. Morgan Street, Chicago
The SCIENCE LIBRARY, 3500 SES; 845 W. Taylor Street –OR–
The LIBRARY OF THE HEALTH SCIENCES, 1750 W. Polk Street

In addition, the library maintains extensive collections of electronic resources and access to web based journals and abstracting services that are frequently used by College of Engineering students and faculty. More information is available at: <http://library.uic.edu>

Computing facilities

Computing facilities at the University of Illinois at Chicago are provided by: the Academic Computing and Communications Center (ACCC). The ACCC is directed by:

Cynthia E. Herrera Lindstrom, Chief Information Officer and Executive Director,
Academic Computing and Communications Center
114 BGRC MC 135
1940 West Taylor St
Chicago IL 60612-7352
cynthiar@uic.edu

The ACCC provides accounts for all students, faculty and staff at UIC. They also maintain an extensive set of public computer laboratories for student use, maintain software licenses for the University, and provide consulting services on computing issues. The ACCC coordinates frequent training courses and seminars on various aspects of computer usage and software. ACCC supports the Instructional Technology Laboratory (ITL), which provides computing services to support classroom use and teaching through electronic means. In addition to the ACCC, the College of Engineering departments all maintain individual computing facilities for their students, containing equipment and software appropriate to their specific programs. More information about the ACCC and computing resources at UIC is available at: <http://accc.uic.edu>

Tutoring and academic resources: Tutoring and other academic support is available to College of Engineering students through a number of organizations on campus. These include:

Academic Center for Excellence

The Academic Center for Excellence (ACE) is a multifaceted academic support program administered by the Office of the Vice Chancellor for Student Affairs designed to help UIC students accomplish their academic goals. ACE is open to all UIC students, from freshman through graduate level and provides assistance with learning styles, time management, and tutoring support. More information about ACE is available at: <http://www.uic.edu/depts/ace/index.shtml>

The Honors College

After they have been admitted to UIC, all incoming freshmen with an ACT composite score of at least 28 or an SAT total score of 1240, and who rank in the top 15 percent of their high school class, are invited to join the Honors College (membership is not automatic; students must submit an application). Members are required to carry out honors activity every semester.

One of the honors activity services is to provide tutoring to underclass students in a variety of subjects, including the math and science activities relevant to engineering. More information about Honors College tutoring is available at: <http://www.uic.edu/honors>

The Mathematical Sciences Learning Center (MSLC)

The Center hosts several activities focused on improving mathematics instruction and student performance. A major initiative of the Department will be the opening of the Mathematical Sciences Learning Center (MSLC) dedicated to supporting the study of the mathematical sciences for all UIC students. MSLC will offer a program of peer-led workshops and tutoring, as well as more traditional assistance from graduate students. One successful support activity, the Emerging Scholars Program (ESP), is a long-standing UIC effort derived from the work of Uri Treisman (Treisman 1985). Treisman showed the importance of creating opportunities for students, particularly those from underrepresented groups, to work in teams while mastering mathematics. ESP offers students the opportunity to work in groups on more challenging math problems for an additional 2.5 hours each week, supervised by TAs. An internal study has demonstrated the effectiveness of the program: $>1/2$ of ESP students receive an A or B in pre-Calculus compared to just over $1/3$ of non-ESP students; 65% of ESP students in multivariable calculus receiving an A or B while only about half of non-ESP students do this well. More information about the Mathematical Sciences Learning Center and the Emerging Scholars Program is available at: <http://www.math.uic.edu/undergrad/mslc/>

The Science Learning Center (SLC)

The Science Learning Center is a wonderfully designed common space at UIC shared by the sciences (biology, chemistry, earth and environmental science, and physics) and dedicated to enhancing the educational experience of undergraduates enrolled in UIC's introductory science courses. Although the main task of the learning center is to provide tutoring in the sciences, secondary goals are to demonstrate the interdisciplinary nature of science, to provide students with a nurturing learning environment, and to encourage team problem-solving. Two types of structured tutoring occur in the SLC: tutoring by TAs from the natural sciences and peer-led study groups for 100-level science courses, led by students who have recently completed the course. Peer leaders are trained in group dynamics and serve to promote a group's ability to work together toward the solution of a problem. More information about the Science Learning Center is available at: <http://www.chem.uic.edu/slc/>

The Writing Center

Students who are having difficulties with written communication can obtain tutoring and support services through UIC's Writing Center. The Writing Center at the University of Illinois at Chicago is committed to the improvement of writing across the campus through peer-centered education and tutor development. The Writing Center complements

classroom instruction by providing tutorials, resources, workshops, and publication opportunities for writers, educators, and writing groups from all disciplines and experience levels. More information about the Writing Center is available at:

<http://www.uic.edu/depts/engl/writing/>

In addition to these programs, the College of Engineering and the campus also provide specific support programs aimed at underrepresented and disadvantaged students.

Minority Engineering Recruitment and Retention Program (MERRP)

The Minority Engineering Recruitment and Retention Program (MERRP), housed in the College of Engineering, has, as its major effort, to increase in the number of African American, Latino, and Native American students earning engineering degrees. The Program staff is composed of professionals with backgrounds in higher education, engineering, and engineering education as follows.

Gerald A. Smith, Director

1232 SEO

gasmith@uic.edu

Elsa Soto, Assistant Director

1237 SEO

esoto3@uic.edu

MERRP is the foundation of the minority engineering student experience at UIC. Most minority engineering students are introduced to MERRP through the college orientation program and the engineering student orientation course, ENGR 100.

MERRP has supported the student chapters of the Society of Hispanic Professional Engineers (SHPE) and the National Society of Black Engineers (NSBE) and has been a source of experience and advice for both groups. The hallmark of MERRP is Supplemental Instruction (SI) and the “academic family” atmosphere.

As part of its minority student recruitment efforts, MERRP maintains summer programs for high school students including.

Preparation for Majoring in Engineering (PREP-ME)

This program is for UIC freshmen and community college students in engineering. Prep-ME Plus is a six week program designed to introduce new UIC engineering students to college-level math and engineering as a discipline and profession. The primary focus is

math instruction. Students also attend seminars on goal setting, study skills, and time management. Students gain an understanding of college requirements and expectations. Qualified students enroll in a freshman English course and a math workshop.

Women in Science & Engineering Program (WISE)

845 West Taylor Street (MC 180)
Office 205D Science Learning Center, SES
Chicago, IL 60607

The goal of the UIC Women in Science and Engineering Program, WISE, is to increase the number of women students pursuing and graduating in science, technology, engineering and math, STEM, disciplines, and to promote the recruitment, retention and advancement of women who have chosen academic careers. Nationally, retention rates for women STEM students fall far behind those of their male counterparts, and women consistently make up less than 20% of tenured STEM faculty. WISE supports women undergrad/graduate students and faculty in STEM by sponsoring activities that foster a positive educational and professional environment, and enable excellence in scholarship, teaching and service. More information about the WISE program is available at: <http://www.uicwise.org/>

African American Academic Network

Student Services 1200 West Harrison
Suite 2800
Chicago, IL 60607

AAAN's mission is to increase the recruitment, retention, and graduation rates of African American students. In keeping with that focus, AAAN is also committed to establishing an inclusive and supportive campus environment. AAAN sponsors social and cultural activities to encourage student involvement, and advocates for the interests of its participants. During the 2006-2007 academic year, the African American Academic Network advised 65% of African American first time freshmen. More information about AAAN is available at: <http://www.uic.edu/depts/aaan/index.shtml>

Latin American Recruitment and Educational Services Program (LARES)

Suite 2640, Student Services Building
1200 West Harrison Street
Chicago, Illinois 60607

LARES is a recruitment and academic assistance program at the University of Illinois at Chicago. It was established to recruit, advise and provide educational assistance to Latino students at both the high school and college levels. LARES was primarily designed to assist Latino students who are interested in pursuing higher education and who are in need of guidance and support. More information about LARES is available at: <http://lares.uic.edu>

Native American Support Program (NASP)

The Native American Support Program (NASP) strives to increase the enrollment, retention and graduation rates of Native American students. NASP fosters a climate supportive of positive academic experiences for Native American students at the University of Illinois at Chicago. More information about NASP is available at: <http://www.uic.edu/depts/nasp>

Disability Resource Center

As reflected in the University of Illinois' Nondiscrimination Statement and the UIC Chancellor's Statement of Commitment to Persons with Disabilities, UIC strives to maintain a barrier-free environment so that students with disabilities can fully access classes, programs, services and other campus activities. The Disability Resource Center facilitates access for students through consultation with faculty and campus departments, and the provision of reasonable accommodations. The Disability Resource Center recognizes various environments in which people function: physical, programmatic, informational and attitudinal. Some modifications to these environments are readily-achievable through direct consultation with faculty or staff. To be eligible for accommodations, students should apply for services through the Disability Resource Center. More information about the Disability Resource Center is available at: http://www.uic.edu/depts/oaa/disability_resources/index.html

Counseling Center

Counseling Services offices are located in Suite 2010, Student Services Building, on the corner of Racine and Harrison (1200 West Harrison Street). As one of the two units of the Counseling Center, the Services are staffed by licensed and board certified psychologists and a psychiatrist, clinical therapists, advanced doctoral psychology trainees, psychiatric residents, and undergraduate paraprofessional volunteers, all of whom are trained to help students with a wide range of personal problems, emotional and psychological difficulties, career questions, and relationship issues. The Center is accredited by the International Association of Counseling Services, and its internship is accredited by the American Psychological Association. The Counseling Services staff adheres to all relevant ethical

codes of conduct and state and federal laws and regulations. The staff is committed to the highest standards of competency in meeting the needs of individuals from diverse backgrounds, including differences of culture, race, ethnicity, national origin, class, gender, ability, age, and sexual orientation. More information about the Counseling Center is available at: <http://www.uic.edu/depts/counseling>

Student career placement and co-op activities are supported by office at both the College of Engineering and campus level.

The Cooperative Education and Internship Program

College of Engineering (M/C 160)
University of Illinois at Chicago
818 Science & Engineering Offices
851 South Morgan Street
Chicago, Illinois 60607-7050

Kate Kaplan, Director, Engineering Career Center, 312.996.2238 or email engrjobs@uic.edu

The Engineering Cooperative and Internship Program gives UIC engineering students the opportunity to gain practical work experience while they are still students. The Engineering Co-op/Internship Office helps students find full time or part time engineering employment before graduation. There are a substantial number of opportunities for qualified engineering students interested in obtaining relevant work experience. The Co-op Office matches employer requirements with student skills, abilities, interests, and background. More information about the Engineering Career Center and the Cooperative Education and Internship Program is available at: <http://www.ecc.uic.edu/ECC/WebHome>

UIC Office of Career Services

1200 W. Harrison
Room 3050 SSB (MC 099)
Chicago, IL 60607

The mission of the UIC Office of Career Services is to provide personalized services that assist UIC students and recent graduates in a process of self-assessment, career planning and preparation in order to facilitate lifetime career development and success.

Information is also provided about part-time employment opportunities to help make the connection between the academic world and real world work experiences while at UIC; full-time employment opportunities to ensure a successful transition from UIC to the world

of work; and graduate/professional school catalogues are available to assist students interested in continuing their education.

In addition, the office maintains and continues to develop mutually beneficial relationships with regional employers of all sizes to promote the employment of UIC students and graduates. More information about the UIC Office of Career Services is available at: <http://www.uic.edu/depts/ocs>

6. Credit Unit

A semester hour is the University's unit of academic credit. During the fall and spring semesters, a University semester hour represents one classroom period of fifty minutes weekly for one semester in lecture or discussion or a longer period of time in laboratory, studio, or other work. For example, a three-semester-hour lecture/discussion course could meet 3 times a week for 50 minutes each period or 2 times a week for 75 minutes each period. In either case, a student attends the lecture/discussion course for an equivalent amount of time each week during a 15-week semester. A minimum of two 50-minute periods each week per credit hour is required for lab, practicum, or clinical activity. During the eight-week summer session, the classroom period is 100 minutes for lecture/discussion. It is expected that students will spend at least the equivalent of two classroom periods of outside preparation for one classroom period per week of lecture or discussion. Those courses in which semester hours exceed contact hours may require additional readings, assigned papers, or other course work.

Instructional Modes

All of the programs in the College of Engineering are offered as traditional on-campus instruction. Students in the Co-op program may spend one or more semesters off campus in engineering internships, but their course work is all taken on campus in traditional classes.

Grade-Point Average

In order to calculate the grade point average: Take the grades for each course taken and determine the grade points per hour: A=4, B=3, C=2, D=1, F=0. Multiply the grade points per hour for each course to get the grade points for each course. Add the points for each course to get the total number of grade points for the semester. Add the semester hours taken for each course to get the total number of semester hours. Divide the total number of grade points for the semester by the total number of semester hours taken to get the grade point average.

In order to receive a degree from the College of Engineering at the University of Illinois at Chicago, a student must present a minimum grade point average of 2.00/4.00 in all work taken in the major. In addition, the student must satisfy the University requirement of a 2.00/4.00 grade point average in two categories: (1) all work taken at UIC; (2) all work taken at UIC and all other two- and four-year institutions combined.

Faculty Workload

The teaching load in most departments in the College of Engineering is four classroom courses per year or faculty who have some research activity (some graduate students and some research publications). Faculty who are active in research have reduced loads of three courses per year. Faculty who have no M.S. or PhD. Students should teach one additional course per semester in exchange for their lack of individual graduate instruction. Hence, faculty who have no research activities, (no graduate students, no publications, no funding) will be expected to teach six classroom courses per year. Departments have buy-out policies for reducing the teaching loads. A common model is for faculty to pay 1/9 of their academic year salary for every course reduction. In addition, faculty members receive course buy-outs for significant administrative loads, such as Department Heads, Director of Graduate Studies, Director of Undergraduate Studies, Associate Deans, etc.

7. Tables

Please find below are the two tables; Tables D.9: Program Enrollment and Degree Data, and Table D.10: Personnel.

Table D-9. Program Enrollment and Degree Data

Department of Chemical Engineering – Chemical Engineering Program

	Academic Year		Enrollment Year					Total Undergrad	Total Grad	Degrees Awarded			
			1st	2nd	3rd	4th	5th			Associates	Bachelors	Masters	Doctorates
Current Year		FT	32	43	63	106	X	244	66	X	60	17	5
		PT	0	1	5	15	X	21	7				
Fall 2018		FT	37	43	39	106	X	225	69	X	64	23	6
		PT	0	0	10	27	X	37	10				
Fall 2017		FT	35	29	57	113	X	234	82	X	72	17	5
		PT	1	3	7	34	X	45	4				
Fall 2016		FT	20	41	55	122	X	238	73	X	56	5	6
		PT	0	2	2	28	X	32	7				
Fall 2015		FT	28	37	69	128	X	262	62	X	49	9	4
		PT	0	0	1	15	X	27	5				

Give official fall term enrollment figures (head count) for the current and preceding four academic years and undergraduate and graduate degrees conferred during each of those years. The "current" year means the academic year preceding the on-site visit.

¹Degree data are from Summer XX, Fall XX, Spring XX+1 terms (Summer 2012, Fall 2012, Spring 2013)

²Current 2013-2014 only includes Summer 2013 and Fall 2013 data. Spring 2014 data available after June 13, 2014

FT--full time

PT--part time

Table D-10. Personnel

Name of the Program

Year¹: Fall 2019

	HEAD COUNT		FTE ²
	FT	PT	
Administrative ²	0.8		0.8
Faculty (tenure-track) ³	8	0.45	8.45
Other Faculty (excluding student Assistants)	3	0.95	3.95
Student Teaching Assistants ⁴		8.25	8.25
Technicians/Specialists	1		1
Office/Clerical Employees	3	0.5	3.5
Others ⁵	2	11.2	13.2

1. Data on this table should be for the fall term immediately preceding the visit. Updated tables for the fall term when the ABET team is visiting are to be prepared and presented to the team when they arrive.
2. Persons holding joint administrative/faculty positions or other combined assignments should be allocated to each category according to the fraction of the appointment assigned to that category.
3. For faculty members, 1 FTE equals what your institution defines as a full-time load
4. For student teaching assistants, 1 FTE equals 20 hours per week of work (or service). For undergraduate and graduate students, 1 FTE equals 15 semester credit-hours (or 24 quarter credit-hours) per term of institutional course work, meaning all courses — science, humanities and social sciences, etc.
5. Specify any other category considered appropriate, or leave blank.

Signature Attesting to Compliance

By signing below, I attest to the following:

That the Chemical Engineering program has conducted an honest assessment of compliance and has provided a complete and accurate disclosure of timely information regarding compliance with ABET's *Criteria for Accrediting Engineering Programs* to include the General Criteria and any applicable Program Criteria, and the ABET *Accreditation Policy and Procedure Manual*.

Peter C. Nelson
Dean's Name (As indicated on the RFE)

Signature _____

June 30, 2020
Date